FRM Part II

Recall + Priority Map Workbook

Volume 1: Tier A

Everything you saw in August 2026, plus the core nine. Revision notes, formulas, traps and GARP-style questions with full solutions.

32 sections · 136 questions

Built from the GARP 2026 source readings and SchweserNotes Book 5 where no GARP text exists. Retake: 25 November 2026.

Contents

Priority table 3

Specifics coverage index 4

Syllabus map (2026 LOs) 7

Chapters by book 10

Tier A: August 2026 Recall and Core Topics 12

A1 Unexpected Loss, UL Contribution and Economic Capital 13

A2 RAROC, Adjusted RAROC and Hurdle Rate 19

A3 Volatility Smiles, Skews and Put-Call Parity 30

A4 Fundamental Review of the Trading Book (FRTB) 36

A5 Merton Model, Distance to Default and Risk-Neutral PD 43

A6 CVA, DVA, BCVA and CVA as a Spread 51

A7 Ho-Lee, Time-Dependent Drift and Volatility 58

A8 Correlation Swaps, Quanto Options and Concentration Ratio 65

A9 Securitisation: Tranches, Waterfall, Prepayment and Performance Ratios 72

A10 Altman Z-Score and Linear Discriminant Analysis 82

A11 Liquidity-Adjusted VaR and Cost of Liquidation 87

A12 Incremental, Marginal and Component VaR 94

A13 Lognormal VaR 101

A14 Artificial Intelligence and GenAI 107

A15 Hazard Rates, CDS Spreads and Conditional vs Unconditional PD 113

A16 Duration Gap and Interest-Sensitive Gap 121

A17 Extreme Value Theory: GEV vs POT 127

A18 Performance Measures and Attribution 132

A19 Non-Parametric VaR: Weighted HS, Bootstrap and FHS **140**

A20 Wrong-Way and Right-Way Risk **147**

A21 Collateral, Two-Way CSA, Margin Period of Risk and Netting **152**

A22 Three Lines of Defence and Operational Risk Governance **161**

A23 Cyber Risk and Operational Resilience **167**

A24 Hedge Funds: Strategies, Survivorship and Other Biases, and Madoff **172**

A25 Gauss+ and Vasicek **178**

A26 Monitoring Liquidity: TSECF, TSAA and Liquidity Stress Testing **185**

A27 Non-Deposit Liabilities, Pooled Cost of Funds and Liquidity Transfer Pricing **192**

A28 Private Credit, Private Markets and Their Drivers **198**

A29 Money Laundering and Financing of Terrorism **205**

A30 Asset vs Borrowed Liquidity Management and the Investment Function **209**

A31 Basel IRB Capital **214**

A32 Net Stable Funding Ratio, LCR and Leverage Ratio **221**

Answer key 226

Formula index 227

Index **228**

Priority table

Every section in this volume in reading order. **Aug** = you saw it in the August 2026 sitting. **Map** = Cross-Source Priority Map entry and how many of 16 sources name it. **TG** = a Telegram recall cites it.

ID	Section	Aug	Map	TG	Qs	Page
A1	Unexpected Loss, UL Contribution and Economic Capital	✓	#1, 10/16	✓	4	13
A2	RAROC, Adjusted RAROC and Hurdle Rate	✓	#3, 9/16	✓	9	19
A3	Volatility Smiles, Skews and Put-Call Parity	✓	#4, 9/16	✓	4	30
A4	Fundamental Review of the Trading Book (FRTB)	✓	#5, 8/16	✓	4	36
A5	Merton Model, Distance to Default and Risk-Neutral PD	✓	#2, 8/16	✓	6	43
A6	CVA, DVA, BCVA and CVA as a Spread	✓	#6, 8/16	✓	5	51
A7	Ho-Lee, Time-Dependent Drift and Volatility	✓	#7, 7/16	✓	5	58
A8	Correlation Swaps, Quanto Options and Concentration Ratio	✓	#8, 7/16	✓	5	65
A9	Securitisation: Tranches, Waterfall, Prepayment and Performance Ratios	✓	#9, 7/16	✓	7	72
A10	Altman Z-Score and Linear Discriminant Analysis	✓	#15, 6/16	✓	3	82
A11	Liquidity-Adjusted VaR and Cost of Liquidation	✓	#11, 6/16	✓	5	87
A12	Incremental, Marginal and Component VaR	✓	#13, 6/16	✓	4	94
A13	Lognormal VaR	✓	#14, 6/16	✓	4	101
A14	Artificial Intelligence and GenAI	✓	#20, 6/16	✓	3	107
A15	Hazard Rates, CDS Spreads and Conditional vs Unconditional PD	✓	#17, 5/16	✓	7	113
A16	Duration Gap and Interest-Sensitive Gap	✓	#24, 5/16	✓	4	121
A17	Extreme Value Theory: GEV vs POT	✓	#25, 5/16	✓	2	127
A18	Performance Measures and Attribution	✓	#36, 5/16	✓	6	132
A19	Non-Parametric VaR: Weighted HS, Bootstrap and FHS	✓	#28, 4/16	✓	5	140
A20	Wrong-Way and Right-Way Risk	✓	#26, 4/16	✓	2	147
A21	Collateral, Two-Way CSA, Margin Period of Risk and Netting	✓	#27, 4/16	✓	6	152
A22	Three Lines of Defence and Operational Risk Governance	✓	#29, 4/16	✓	3	161
A23	Cyber Risk and Operational Resilience	✓	#31, 4/16	✓	2	167
A24	Hedge Funds: Strategies, Survivorship and Other Biases, and Madoff	✓	#34, 3/16		4	172
A25	Gauss+ and Vasicek	✓	#40, 2/16		5	178
A26	Monitoring Liquidity: TSECF, TSAA and Liquidity Stress Testing	✓	#64, 2/16		4	185
A27	Non-Deposit Liabilities, Pooled Cost of Funds and Liquidity Transfer Pricing	✓	#47, 2/16	✓	3	192
A28	Private Credit, Private Markets and Their Drivers	✓	#49, 2/16	✓	4	198
A29	Money Laundering and Financing of Terrorism	✓	#50, 2/16	✓	1	205
A30	Asset vs Borrowed Liquidity Management and the Investment Function	✓	#63, 1/16		2	209
A31	Basel IRB Capital	✓	not in map		5	214
A32	Net Stable Funding Ratio, LCR and Leverage Ratio	✓	not in map		3	221

Specifics coverage index

Every reported shape (S-number) and the questions that drill it. Notes-only sections list their shapes for revision.

ID	S	Reported shape	Questions
A1	S1	UL for a single component: standalone UL from EA, PD, LR, σ_{LR} and the binomial σ^2_{PD}	A1.1, A1.2
A1	S2	ULC with all loans of the same value, ULC $=$ UL$\sqrt{\rho}$, plain form	A1.2
A1	S3	portfolio UL for n identical loans, UL$_p$ $=$ UL$\sqrt{n+\rho(n^2-n)}$, forward and reverse-solved	A1.3, A1.4
A1	S4	economic capital EC $=$ UL$_p$ $\times$ CM: the omitted capital multiplier	A1.3
A1	S5	intermediate-result trap: the single-asset UL planted when ULC is asked (Q2; standalone UL in Q1 and Q3)	A1.2
A2	S1	compute after-tax RAROC, in plain form and with red-herring inputs	A2.1, A2.2, A2.3, A2.4
A2	S2	two on adjusted RAROC: value plus accept/reject decision, and the break-even beta	A2.6, A2.7
A2	S3	direction of the effect on RAROC of changes in economic capital, income and expense	A2.9
A2	S4	find the hurdle rate: market-value-weighted cost of common equity (CAPM) and preferred equity	A2.5, A2.8
A2	S5	weighted cost of capital given break-even cost (Q8; break-even revenue at the hurdle in Q3)	A2.3, A2.8
A3	S1	equity skew: effect of changing strike and maturity on implied volatility, plain form	A3.1
A3	S2	two questions on FX and equity smiles	A3.2, A3.3
A3	S3	implied volatility shape for FX options	A3.2, A3.3
A3	S4	lognormal vols versus implied vols for FX and index options	A3.3
A3	S5	equity options volatility smile property	A3.1, A3.3
A3	S6	sibling swap: FX explanation attributed to equities or vice versa	A3.2
A3	S7	put-call parity: a European call and put with the same strike and maturity share one implied volatility, tested in its smile context (Q4; distractor in Q3)	A3.3, A3.4
A4	S1	stressed ES at 97.5% replaces 10-day VaR at 99%, and why	A4.2
A4	S2	liquidity horizons by asset class and the cascade ES	A4.1
A4	S3	standardized vs internal models approach, with the desk-level backtesting and P&L attribution tests that decide which applies	A4.3, A4.4
A4	S4	trading book / banking book boundary	A4.3
A4	S5	FRTB vs Basel I (1996 Amendment), Basel II, Basel 2.5: overall changes	A4.4
A4	S6	curvature risk charge, from the GARP text (Q4)	A4.4
A5	S1	distance to default, then assign a rating band from the score	A5.4, A5.6
A5	S2	back out V_0 and σ_V from the simultaneous equations using the solver's converged values	A5.5
A5	S3	risk-neutral PD $N(-d_2)$ versus the real-world PD with μ	A5.1, A5.3
A5	S4	Merton PD converted into an average hazard rate	A5.1
A5	S5	leverage ratio to return on equity through the delta $N(d_1)$	A5.2
A5	S6	short-form DtD $\ln(V/F)/\sigma_V$ versus the full form	A5.4, A5.6
A5	S7	scope trap: Merton over horizons beyond one year (Q6; two- and three-year horizons in Q1, Q2)	A5.1, A5.2, A5.6
A6	S1	UCVA from an EPE profile and a PD term structure, $\sum$ LGD $\times$ EPE $\times$ PD	A6.1
A6	S2	CVA as a running spread: $-$average EPE $\times$ spread, and upfront CVA divided by the risky annuity	A6.3, A6.4
A6	S3	BCVA $=$ CVA $+$ DVA with the sign convention CVA negative, DVA positive	A6.2, A6.5
A6	S4	BCVA with survival (first-to-default) adjustments	A6.5
A6	S5	BCVA as a spread, plain form as seen in August	A6.3
A6	S6	change from UCVA to BCVA, with the BCVA itself planted as a distractor (Q5)	A6.5
A7	S1	interest rate after 2 months in the top node of a monthly Ho-Lee tree	A7.2
A7	S2	Ho-Lee and time-dependent volatility models: rate in the upper nodes	A7.1, A7.2, A7.4
A7	S3	Ho-Lee and Model 3 two-step plug and play	A7.2, A7.4
A7	S4	no-arbitrage model, node-2 Δr and the drift read back from the nodes	A7.2, A7.3
A7	S5	two questions on the Ho-Lee short-rate tree	A7.1, A7.2, A7.3
A7	S6	sibling swap against Vasicek and Models 1 and 2	A7.5
A7	S7	August plain forms: Ho-Lee plug values (Q1); time-dependent drift and volatility theory (Q5)	A7.1, A7.5
A8	S1	correlation swap payoff to the fixed-rate payer, plain numerical as seen in August	A8.1
A8	S2	the same swap from the seller's (fixed-rate receiver's) side, with irrelevant volatilities	A8.2
A8	S3	correlation swap return per unit notional, reverse-solved for the strike	A8.3
A8	S4	quanto options: what correlation risk they carry and in which direction it moves the price	A8.5
A8	S5	concentration ratio concept and calculation, tied to the joint default probability (Q4)	A8.4
A9	S1	delinquency ratio, plain numerical as seen in August	A9.1
A9	S2	cash flow into the equity tranche after the interim waterfall, with defaults and fees	A9.4
A9	S3	overcollateralisation account balance at a year end	A9.4
A9	S4	shares paid to the senior and equity tranches in the final-year waterfall	A9.5
A9	S5	CPR, SMM and PSA conversion, with the deal's overcollateralisation as the twist	A9.6
A9	S6	prepayment rate theory and forecasting methodology	A9.7
A9	S7	closed-book formulas CPR, DSCR and WAL (Q6, Q7)	A9.6, A9.7
A10	S1	direct plug: compute Z from a set of filings	A10.1
A10	S2	the five ratios and their weights recalled closed-book, none given in the stem	A10.1, A10.2, A10.3
A10	S3	cut-off zones used to classify a firm	A10.2, A10.3
A10	S4	'calculate z score (LDA)': Z as Altman's linear discriminant function assigning a firm to a group	A10.2
A10	S5	the two LDAs: linear discriminant analysis for scoring versus the loss distribution approach for operational capital (Q2; the loss distribution approach itself is in D19)	A10.2

ID	S	Reported shape	Questions
A11	S1	LVaR numerical with the normal-market (constant-spread) cost, the 'CAR, LVaR' pairing (CAR itself: see D3)	A11.1
A11	S2	stressed (exogenous-spread) cost of liquidation $0.5\,V(\mu + z\sigma)$, with the constant-spread input as the sibling-swap distractor	A11.1, A11.2
A11	S3	cost of liquidation (CoL) as its own number, plain form of the August recall	A11.1, A11.2
A11	S4	LVaR theory: normal vs stressed spread, perfect spread correlation, liquidation horizon	A11.5
A11	S5	Malz expected transactions cost and 99% spread risk factor	A11.3
A11	S6	Malz multi-day liquidation adjustment, with time to escape from trading volume (Q4)	A11.4
A12	S1	incremental VaR of including a platinum position in a silver book, full revaluation versus the marginal VaR shortcut	A12.1
A12	S2	allocation using marginal VaR: where to add (Q2) and which position to grow for a better return-to-risk trade-off	A12.2, A12.4
A12	S3	marginal VaR, volatility and the risk-free rate chained into a Sharpe ratio	A12.4
A12	S4	closed-book formulas: marginal VaR $= (\text{VaR}/W)\beta$; component VaR = individual VaR $\times\ \rho$	A12.3, A12.4
A12	S5	shortcut: when ranking marginal VaRs only beta matters	A12.2, A12.4
A12	S6	intermediate-result trap between marginal and component VaR (Q3; tangent versus curve in Q1)	A12.1, A12.3
A13	S1	choose between two volatilities, 16% rising to 18% six weeks ago: which regime governs the VaR horizon	A13.3
A13	S2	direct lognormal VaR from the formula, the plain form seen in August	A13.2, A13.4
A13	S3	lognormal form $1 - e^{\mu - z\sigma}$ kept distinct from the normal arithmetic form $-\mu + \sigma z$	A13.1, A13.2, A13.3, A13.4
A14	S1	AI vs GenAI: the IMF's two model groups, what GenAI adds, and its signature failure mode	A14.1
A14	S2	disadvantages of GenAI: fraud, deepfakes, cyber attack vectors, the short-term edge of bad actors	A14.2
A14	S3	application of AI in market risk: volatility and liquidity models, risk scenarios, dynamic risk sensing	A14.3
A14	S4	financial-stability axis: herding and correlated models, third-party concentration, explainability and model risk	A14.2, A14.3
A15	S1	hazard rate and PD: from a constant hazard rate to cumulative and year-t default probabilities	A15.1
A15	S2	hazard rate from PD by the direct formula $\bar{\lambda} = -\ln(1-Q)/T$	A15.4, A15.7
A15	S3	hazard rate from a CDS spread, $\bar{\lambda} = s/(1-R)$	A15.5, A15.6, A15.7
A15	S4	marginal PD as the difference of successive cumulative PDs	A15.1, A15.2, A15.3
A15	S5	sibling swap: $1 - e^{-\lambda t}$ is the cumulative PD, not the hazard rate	A15.1, A15.7
A15	S6	conditional versus unconditional default probability (Q1, Q2, Q3)	A15.1, A15.2, A15.3
A16	S1	duration gap, plain form of the August recall	A16.2
A16	S2	impact on net worth given asset and liability durations, with direction	A16.1
A16	S3	duration of assets and liabilities: the liability duration that immunises net worth	A16.4
A16	S4	IS gap and ISR: cumulative gap and the change in net interest income	A16.3
A16	S5	polarity of the leverage-adjusted gap against a rate move, including a sign reversal from the leverage adjustment (Q1, Q2)	A16.1, A16.2
A17	S1	GEV theory: Fisher-Tippett, three parameters, the Frechet, Gumbel and Weibull cases	A17.1
A17	S2	GEV versus POT: block maxima versus exceedances, data use, threshold	A17.1
A17	S3	peaks over threshold: the GPBdH theorem and the GP distribution of excess losses	A17.2
A17	S4	POT threshold set too low: more data, more bias, less extreme	A17.2
A17	S5	shape parameter $\xi > 0$ means fat tails	A17.1
A17	S6	operational-risk framing of EVT and block maxima	A17.1
A18	S1	Sharpe ratio, the harder form: Sharpe paired with M^2 (Q4); Sharpe as the wrong lens for one of many subportfolios	A18.3, A18.4
A18	S2	marginal VaR, volatility and the risk-free rate chained into a Sharpe ratio (A12.4, see also)	A12.4
A18	S3	Jensen's alpha and the information ratio	A18.1, A18.2
A18	S4	attribution split into asset allocation and security selection effects	A18.5
A18	S5	Grinold's fundamental law and the maximum attainable information ratio	A18.6
A18	S6	which measure suits which use: Sharpe, Treynor, information ratio (Q2, Q3, Q6)	A18.2, A18.3, A18.6
A19	S1	age-weighted HS (BRW): single weights, cumulative weights and the weighted VaR read-off	A19.1, A19.2, A19.4
A19	S2	volatility-weighted HS (HW), with the polarity trap: returns scaled up when current volatility is above the historical level	A19.3, A19.5
A19	S3	bootstrap HS property question	A19.5
A19	S4	differences among the non-parametric approaches	A19.5
A19	S5	expected shortfall in filtered historical simulation	A19.5
A20	S1	classify trades as wrong-way or right-way, plain form as seen in August	A20.1
A20	S2	the test: is the exposure linked to the counterparty's credit quality, and in which direction	A20.1
A20	S3	wrong-way collateral	A20.2
A20	S4	the effect of WWR on CCPs	A20.2
A20	S5	modelling approaches: hazard rate (intensity), structural, parametric, jump (Q2)	A20.2
A21	S1	collateral required from the MtM value, threshold, MTA, rounding and margin already held	A21.1, A21.2
A21	S2	the threshold-plus-MTA test that fails, so no transfer is made	A21.2
A21	S3	overcollateralisation: initial margin and haircuts (Q5); the securitisation overcollateralisation account is drilled in A9	A21.5
A21	S4	features and terms of a collateral agreement, CR-17 g (LO 35.g)	A21.4
A21	S5	disputes and reconciliation, CR-17 f (LO 35.f)	A21.4, A21.6

ID	S	Reported shape	Questions
A21	S6	market, operational and liquidity risk through collateralisation, with role misassignment, CR-17 j (LO 35.j)	A21.6
A21	S7	termination features and trade compression, CR-16 d (Reading 34)	A21.6
A21	S8	two-way vs one-way CSA and rating-linked terms, as seen in August	A21.1, A21.2, A21.4
A21	S9	margin period of risk: components and effect on collateralised exposure and CVA, as seen in August (Q3, Q5)	A21.3, A21.5
A22	S1	three lines of operational risk management: who does what, with every distractor moving an activity across a line	A22.1
A22	S2	Basel event-type taxonomy, Level 1, Level 2 and the Level 3 examples	A22.3
A22	S3	committees versus the board, and who owns risk appetite and limits (Q2)	A22.2
A23	S1	cyber risk framework, the one cyber risk question: supervisory practice, frameworks and the Equifax lessons	A23.1
A23	S2	operational resiliency as a regulatory concept: UK first, then BCBS, ECB, MAS	A23.2
A23	S3	applied vulnerability comparison: payment bank versus community bank during COVID (Q2)	A23.2
A24	S1	hedge fund strategies: equity, macro, arbitrage, event-driven, distressed	A24.1
A24	S2	hedge fund risks, systemic risk, regulation and structure	A24.2
A24	S3	survivorship bias against its siblings: selection, reporting, backfill	A24.3
A24	S4	Madoff red flags as a case↔lesson matching item against other cases (Q4)	A24.4
A25	S1	closed-book $E[r_T] = r_0 e^{-kT} + \theta(1 - e^{-kT})$, with θ built from r_∞ and the risk premium	A25.4
A25	S2	closed-book half-life $\ln 2/k$	A25.3, A25.5
A25	S3	Vasicek model calculation: rate change and standard deviation	A25.4, A25.5
A25	S4	LO 14.g expected rate in T years	A25.4
A25	S5	Gauss+ multi-factor structure: no shock on r, the $r \to m \to l \to \mu$ cascade, two sources of risk	A25.1, A25.2, A25.3
A25	S6	August plain form: Gauss+ factor changes (Q1 short rate, Q2 medium-term factor, Q3 long-term factor)	A25.1, A25.2, A25.3
A26	S1	TSECF and TSAA numerical, plain form of the August recall: which term structures move and by how much	A26.3
A26	S2	liquidity requirement built from trend, seasonal and cyclical components, then the liquidity deficit or surplus	A26.4
A26	S3	repo of an available asset: cash into the TSCLGC, haircut and accrued interest	A26.1
A26	S4	TSECCF with initial cash and the asset sale that keeps the TSL_e non-negative (Q2)	A26.2
A27	S1	pooled-funds approach (forward looking hurdle rate) against the historical average cost approach, numerical	A27.1
A27	S2	sources of non-deposit liabilities and the factors in choosing among them, theory as seen in August	A27.2
A27	S3	pooled average cost of funds against matched-maturity marginal cost in LTP	A27.3
A27	S4	LTP's role in charging liquidity costs back to business lines, including the liquidity cushion (Q3)	A27.3
A28	S1	growth drivers of private credit and the BIS regression results	A28.3
A28	S2	bank disintermediation: private credit vs bank loans, cost of equity, cost of debt and leverage	A28.3
A28	S3	valuation opacity: stale marks, deferred losses, NAV-based metrics	A28.1, A28.2
A28	S4	financial-stability concerns and the IMF's policy fixes	A28.2
A28	S5	private equity performance measures: IRR vs TVPI and DPI vs PME	A28.1
A28	S6	private markets structures, co-investing and intermediated vehicles	A28.4
A29	S1	ML/FT governance role trap: who files suspicious transaction reports and contacts supervisors and FIUs, and what the chief AML/CFT officer may and may not combine	A29.1
A29	S2	customer due diligence, including reliance on another bank's CDD	A29.1
A29	S3	correspondent banking risk	A29.1
A29	S4	the three stages of laundering: placement, layering, integration (Q1)	A29.1
A30	S1	asset liquidity (asset conversion) vs borrowed liquidity (liability management) vs balanced strategy: pros, cons and who uses which, as seen in August	A30.1
A30	S2	reserve-management strategy: the net liquidity position as the starting point	A30.1
A30	S3	investment instruments available to banks and their advantages and disadvantages	A30.2
A30	S4	factors affecting the choice of securities, including tax exposure (Q2)	A30.2
A31	S1	IRB capital for a single corporate loan under Advanced IRB, with the correlation formula and maturity adjustment: the plain form	A31.1
A31	S2	retail IRB: fixed correlation, no maturity adjustment, reported as RWA	A31.2
A31	S3	Foundation versus Advanced IRB inputs: supervisory LGD and the PD floor	A31.3
A31	S4	December 2017 IRB revisions: A-IRB removed for certain asset classes, equities out of IRB, input floors	A31.4
A31	S5	the 72.5% output floor and how input floors and the output floor bind (Q5)	A31.5
A32	S1	NSFR from a balance sheet with ASF and RSF factors given: the plain form	A32.1
A32	S2	NSFR shortfall: new funding needed to reach 100%, with both the funding leg and the deployment leg	A32.2
A32	S3	NSFR versus LCR with both parameter sets in one table: pick the right factors (Q3)	A32.3

Syllabus map (2026 LOs)

Each section's official 2026 reading ID, reading title, source text and chapter, and the LO letters it covers, per GARP's 2026 FRM Part II Learning Objectives.

ID	Topic	Reading	Official 2026 title	Source and chapter	LOs
A1	Unexpected Loss, UL Contribution and Economic Capital	**CR-4**	Capital Structure in Banks	*Schroeck, Risk Management and Value Creation in Financial Institutions, Chapter 5 (pp. 170-186 only)*	a to i
A2	RAROC, Adjusted RAROC and Hurdle Rate	**ORR-18**	Risk Capital Attribution and Risk-Adjusted Performance Measurement	*Crouhy, Galai & Mark, The Essentials of Risk Management 2e, Chapter 17*	a to h
		CR-5	Introduction to Credit Risk Modeling and Assessment	*Doumpos, Lemonakis, Niklis & Zopounidis, Analytical Techniques in the Assessment of Credit Risk, Chapter 1*	g
A3	Volatility Smiles, Skews and Put-Call Parity	**MR-17**	Volatility Smiles and Volatility Surfaces	*Hull, Options, Futures, and Other Derivatives 11e, Chapter 20*	a to i
A4	Fundamental Review of the Trading Book (FRTB)	**MR-18**	Fundamental Review of the Trading Book	*Hull, Risk Management and Financial Institutions 6e, Chapter 18*	a to c
		ORR-21	Capital Regulation Before the Global Financial Crisis	*Mark Carey, GARP Risk Institute, April 2019*	c
		ORR-22	Solvency, Liquidity, and Other Regulation After the Global Financial Crisis	*Mark Carey, GARP Risk Institute, April 2019*	a to c
A5	Merton Model, Distance to Default and Risk-Neutral PD	**CR-5**	Introduction to Credit Risk Modeling and Assessment	*Doumpos, Lemonakis, Niklis & Zopounidis, Analytical Techniques in the Assessment of Credit Risk, Chapter 1*	e, f
		CR-9	Estimating Default Probabilities	*Hull, Risk Management and Financial Institutions 6e, Chapter 17*	k to n
A6	CVA, DVA, BCVA and CVA as a Spread	**CR-20**	CVA	*Gregory, Chapter 17*	a to i
A7	Ho-Lee, Time-Dependent Drift and Volatility	**MR-14**	The Art of Term Structure Models: Drift	*Tuckman & Serrat 3e, Chapter 9*	a to e
		MR-15	The Art of Term Structure Models: Volatility and Distribution	*Tuckman & Serrat 3e, Chapter 10*	a to c
A8	Correlation Swaps, Quanto Options and Concentration Ratio	**MR-8**	Correlation Basics: Definitions, Applications, and Terminology	*Meissner, Correlation Risk Modeling and Management 2e, Chapter 1*	a to g
A9	Securitisation: Tranches, Waterfall, Prepayment and Performance Ratios	**CR-22**	Structured Credit Risk	*Malz, Chapter 9*	a to m
		CR-23	An Introduction to Securitisation	*Choudhry, Structured Credit Products 2e, Chapter 12*	a to h
A10	Altman Z-Score and Linear Discriminant Analysis	**CR-9**	Estimating Default Probabilities	*Hull, Risk Management and Financial Institutions 6e, Chapter 17*	a to c
		CR-5	Introduction to Credit Risk Modeling and Assessment	*Doumpos, Lemonakis, Niklis & Zopounidis, Analytical Techniques in the Assessment of Credit Risk, Chapter 1*	d
A11	Liquidity-Adjusted VaR and Cost of Liquidation	**LTR-1**	Liquidity Risk	*Hull, Risk Management and Financial Institutions 5e, Chapter 24*	a
		LTR-2	Liquidity and Leverage	*Malz, Financial Risk Management: Models, History, and Institutions, Chapter 12*	e, f
A12	Incremental, Marginal and Component VaR	**IM-5**	Portfolio Risk: Analytical Methods	*Jorion, Value-at-Risk 3e, Chapter 7*	a to e
A13	Lognormal VaR	**MR-1**	Estimating Market Risk Measures: An Introduction and Overview	*Dowd, Measuring Market Risk 2e, Chapter 3*	a to f
		MR-2	Non-Parametric Approaches	*Dowd, Chapter 4*	c

ID	Topic	Reading	Official 2026 title	Source and chapter	LOs
A14	Artificial Intelligence and GenAI	**CI-1**	Advances in Artificial Intelligence: Implications for Capital Markets Activities	*IMF, October 2024*	a to d
		CI-2	The Financial Stability Implications of Artificial Intelligence	*FSB, November 2024*	a to c
A15	Hazard Rates, CDS Spreads and Conditional vs Unconditional PD	**CR-9**	Estimating Default Probabilities	*Hull, Risk Management and Financial Institutions 6e, Chapter 17*	d to k
		CR-10	Credit Value at Risk	*Hull, RMFI 6e, Chapter 19*	cross-reference
A16	Duration Gap and Interest-Sensitive Gap	**LTR-17**	Risk Management for Changing Interest Rates: Asset-Liability Management and Duration Techniques	*Rose & Hudgins, Chapter 7*	a to d
A17	Extreme Value Theory: GEV vs POT	**MR-3**	Parametric Approaches (II): Extreme Value	*Dowd, Chapter 7*	a to f
		ORR-4	Risk Measurement and Assessment	*GARP, Chapter 4*	f
A18	Performance Measures and Attribution	**IM-7**	Portfolio Performance Evaluation	*Bodie, Kane & Marcus, Investments 12e, Chapter 24*	a to i
		IM-3	Alpha (and the Low-Risk Anomaly)	*Ang, Chapter 10*	b to d
A19	Non-Parametric VaR: Weighted HS, Bootstrap and FHS	**MR-2**	Non-Parametric Approaches	*Dowd, Chapter 4*	a to d
A20	Wrong-Way and Right-Way Risk	**CR-20**	CVA	*Gregory, Chapter 17*	k to p
		CR-17	Margin (Collateral) and Settlement	*Gregory, Chapter 7*	e, j
A21	Collateral, Two-Way CSA, Margin Period of Risk and Netting	**CR-16**	Netting, Close-out and Related Aspects	*Gregory, Chapter 6*	a to f
		CR-17	Margin (Collateral) and Settlement	*Gregory, Chapter 7*	a to k
		CR-20	CVA	*Gregory, Chapter 17*	j
A22	Three Lines of Defence and Operational Risk Governance	**ORR-2**	Risk Governance	*GARP, Chapter 2*	a to d
		CR-2	Governance	*Bouteille & Coogan-Pushner, Chapter 2*	a
		ORR-1	Introduction to Operational Risk and Resilience	*GARP, Operational Risk and Resilience, Chapter 1*	b
		ORR-3	Risk Identification	*GARP, Chapter 3*	c, d
A23	Cyber Risk and Operational Resilience	**ORR-8**	Cyber-resilience: Range of Practices	*Basel Committee on Banking Supervision, December 2018*	a to e
		ORR-9	Case Study: Cyberthreats and Information Security Risks	*GARP, Chapter 9*	a, b
		ORR-1	Introduction to Operational Risk and Resilience	*GARP, Operational Risk and Resilience, Chapter 1*	a, c, d
A24	Hedge Funds: Strategies, Survivorship and Other Biases, and Madoff	**IM-8**	Hedge Fund Investment Strategies	*Stowell & Stowell, Investment Banks, Hedge Funds, and Private Equity 4e, Chapter 12*	a to d
		IM-9	Risk, Regulation and Organizational Structure	*Stowell & Stowell, Chapter 14*	a to d
		IM-14	Madoff: A Riot of Red Flags	*Gregoriou & Lhabitant, EDHEC Publication, 2009*	a, b
		IM-12	Performing Due Diligence on Specific Managers and Funds	*Mirabile, Hedge Fund Investing 2e, Chapter 12*	a to g
		IM-17	Illiquid Assets	*Ang, Asset Management, Chapter 13*	c
A25	Gauss+ and Vasicek	**MR-16**	The Vasicek and Gauss+ Models	*Tuckman & Serrat 4e, Chapter 9*	a to d
		MR-14	The Art of Term Structure Models: Drift	*Tuckman & Serrat 3e, Chapter 9*	f to h
A26	Monitoring Liquidity: TSECF, TSAA and Liquidity Stress Testing	**LTR-7**	Monitoring Liquidity	*Castagna & Fede, Measuring and Managing Liquidity Risk, Chapter 6*	a to e
		LTR-9	Liquidity Stress Testing	*Venkat & Baird, Chapter 3*	a to c
		LTR-10	Liquidity Risk Reporting and Stress Testing	*Choudhry, The Principles of Banking, Chapter 14*	a to c
		LTR-5	Liquidity and Reserves Management: Strategies and Policies	*Rose & Hudgins, Chapter 11*	c
A27	Non-Deposit Liabilities, Pooled Cost of Funds and Liquidity Transfer Pricing	**LTR-12**	Managing Non-Deposit Liabilities	*Rose & Hudgins, Chapter 13*	a to d

ID	Topic	Reading	Official 2026 title	Source and chapter	LOs
		LTR-14	Liquidity Transfer Pricing: A Guide to Better Practice	*Joel Grant, Occasional Paper, Financial Stability Board / BIS, 2011*	a to d
A28	Private Credit, Private Markets and Their Drivers	**IM-10**	The Rise and Risks of Private Credit	*"The Last Mile: Financial Vulnerabilities and Risks", IMF Global Financial Stability Report, Chapter 2, April 2024*	a to d
		IM-11	Private Markets Investing	*Demaria, Pedergnana, He, Rissi & Debrand, Asset Allocation and Private Markets, Chapter 3 (up to section 3.3.10)*	a to f
		CI-3	The Global Drivers of Private Credit	*BIS, February 2025*	a to c
		IM-13	Distress Symptoms and Remedies	*Ippolito, Private Capital Investing, Chapter 20*	cross-reference
A29	Money Laundering and Financing of Terrorism	**ORR-10**	Sound Management of Risks related to Money Laundering and Financing of Terrorism	*Basel Committee on Banking Supervision, revised July 2020 (read through p.16, paragraph 83)*	a to c
		ORR-11	Case Study: Financial Crime and Fraud	*GARP, Chapter 11*	a, b
A30	Asset vs Borrowed Liquidity Management and the Investment Function	**LTR-5**	Liquidity and Reserves Management: Strategies and Policies	*Rose & Hudgins, Chapter 11*	a to c
		LTR-4	The Investment Function in Financial-Services Management	*Rose & Hudgins, Bank Management & Financial Services 9e, Chapter 10*	a to c
A31	Basel IRB Capital	**CR-10**	Credit Value at Risk	*Hull, RMFI 6e, Chapter 19*	d
		ORR-21	Capital Regulation Before the Global Financial Crisis	*Mark Carey, GARP Risk Institute, April 2019*	e, f
		ORR-23	High-Level Summary of Basel III Reforms	*Basel Committee on Banking Supervision, December 2017*	a to c
		ORR-22	Solvency, Liquidity, and Other Regulation After the Global Financial Crisis	*Mark Carey, GARP Risk Institute, April 2019*	h
A32	Net Stable Funding Ratio, LCR and Leverage Ratio	**ORR-22**	Solvency, Liquidity, and Other Regulation After the Global Financial Crisis	*Mark Carey, GARP Risk Institute, April 2019*	f
		ORR-23	High-Level Summary of Basel III Reforms	*Basel Committee on Banking Supervision, December 2017*	b

Chapters by book

Every 2026 reading this volume teaches, grouped by book, with the sections where it appears.

Book 1 · Market Risk Measurement and Management — 9 readings

MR-1	Estimating Market Risk Measures: An Introduction and Overview	A13
MR-2	Non-Parametric Approaches	A13, A19
MR-3	Parametric Approaches (II): Extreme Value	A17
MR-8	Correlation Basics: Definitions, Applications, and Terminology	A8
MR-14	The Art of Term Structure Models: Drift	A7, A25
MR-15	The Art of Term Structure Models: Volatility and Distribution	A7
MR-16	The Vasicek and Gauss+ Models	A25
MR-17	Volatility Smiles and Volatility Surfaces	A3
MR-18	Fundamental Review of the Trading Book	A4

Book 2 · Credit Risk Measurement and Management — 10 readings

CR-2	Governance	A22
CR-4	Capital Structure in Banks	A1
CR-5	Introduction to Credit Risk Modeling and Assessment	A2, A5, A10
CR-9	Estimating Default Probabilities	A5, A10, A15
CR-10	Credit Value at Risk	A31
CR-16	Netting, Close-out and Related Aspects	A21
CR-17	Margin (Collateral) and Settlement	A20, A21
CR-20	CVA	A6, A20, A21
CR-22	Structured Credit Risk	A9
CR-23	An Introduction to Securitisation	A9

Book 3 · Operational Risk and Resilience — 12 readings

ORR-1	Introduction to Operational Risk and Resilience	A22, A23
ORR-2	Risk Governance	A22
ORR-3	Risk Identification	A22
ORR-4	Risk Measurement and Assessment	A17
ORR-8	Cyber-resilience: Range of Practices	A23
ORR-9	Case Study: Cyberthreats and Information Security Risks	A23
ORR-10	Sound Management of Risks related to Money Laundering and Financing of Terrorism	A29
ORR-11	Case Study: Financial Crime and Fraud	A29
ORR-18	Risk Capital Attribution and Risk-Adjusted Performance Measurement	A2
ORR-21	Capital Regulation Before the Global Financial Crisis	A4, A31
ORR-22	Solvency, Liquidity, and Other Regulation After the Global Financial Crisis	A4, A31, A32
ORR-23	High-Level Summary of Basel III Reforms	A31, A32

Book 4 · Liquidity and Treasury Risk Measurement and Management — 10 readings

LTR-1	Liquidity Risk	A11
LTR-2	Liquidity and Leverage	A11
LTR-4	The Investment Function in Financial-Services Management	A30
LTR-5	Liquidity and Reserves Management: Strategies and Policies	A26, A30
LTR-7	Monitoring Liquidity	A26
LTR-9	Liquidity Stress Testing	A26
LTR-10	Liquidity Risk Reporting and Stress Testing	A26
LTR-12	Managing Non-Deposit Liabilities	A27
LTR-14	Liquidity Transfer Pricing: A Guide to Better Practice	A27
LTR-17	Risk Management for Changing Interest Rates: Asset-Liability Management and Duration Techniques	A16

Book 5 · Risk Management and Investment Management — 10 readings

IM-3	Alpha (and the Low-Risk Anomaly)	A18
IM-5	Portfolio Risk: Analytical Methods	A12
IM-7	Portfolio Performance Evaluation	A18
IM-8	Hedge Fund Investment Strategies	A24
IM-9	Risk, Regulation and Organizational Structure	A24
IM-10	The Rise and Risks of Private Credit	A28
IM-11	Private Markets Investing	A28
IM-12	Performing Due Diligence on Specific Managers and Funds	A24
IM-14	Madoff: A Riot of Red Flags	A24
IM-17	Illiquid Assets	A24

Book 5 · Current Issues in Financial Markets — 3 readings

CI-1	Advances in Artificial Intelligence: Implications for Capital Markets Activities	A14
CI-2	The Financial Stability Implications of Artificial Intelligence	A14
CI-3	The Global Drivers of Private Credit	A28

Tier A: August 2026 Recall and Core Topics

Every topic you saw in August 2026, plus the map's nine most corroborated topics. Ordered by source count.

A1 Unexpected Loss, UL Contribution and Economic Capital

Readings: CR-4 a, b, c, d, e, f, g, h, i · Capital Structure in Banks (Schroeck, Ch.5, pp. 170 to 186): Expected Losses; Unexpected Losses (UL-Standalone); Unexpected Loss Contribution (ULC); Economic Capital for Credit Risk; Problems with the Quantification of Credit Risk

AUG 2026 RECALL | MAP TIER A · #1 · 10/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: UL contribution straight forward numerical (identical loans).
Map #1: "Recall notes specify 'UL for a single component in a portfolio (numerical)' and 'Unexpected Loss Contribution: all loans with same value'. One source reports a candidate got economic capital wrong by omitting the capital multiplier. Closed-book formulas: $\text{UL} = \text{EAD} \cdot \sqrt{\text{PD} \cdot \sigma^2_{\text{LGD}} + \text{LGD}^2 \cdot \sigma^2_{\text{PD}}}$; $\text{ULC} = \text{UL} \cdot \sqrt{\rho}$; $\text{UL}_p = \text{UL} \cdot \sqrt{n + \rho(n^2 - n)}$; $\text{EC} = \text{UL}_p \cdot \text{CM}$. The intermediate-result trap is live here: GARP plants the single-asset UL as a distractor when ULC is asked."
Shapes covered here:
S1 UL for a single component: standalone UL from EA, PD, LR, σ_{LR} and the binomial σ^2_{PD} (Q1, Q2)
S2 ULC with all loans of the same value, $\text{ULC} = \text{UL}\sqrt{\rho}$, plain form (Q2)
S3 portfolio UL for n identical loans, $\text{UL}_p = \text{UL}\sqrt{n + \rho(n^2 - n)}$, forward and reverse-solved (Q3, Q4)
S4 economic capital $\text{EC} = \text{UL}_p \times \text{CM}$: the omitted capital multiplier (Q3)
S5 intermediate-result trap: the single-asset UL planted when ULC is asked (Q2; standalone UL in Q1 and Q3).

Revision notes

- **Economic capital** (CR-4 a, g): "an estimate of the overall capital reserve needed to guarantee the solvency of a bank for a given confidence level", set consistent with the bank's *target credit rating*. It rises with (i) the confidence level (a higher CM) and (ii) the riskiness of the assets (a higher UL). Holding less capital frees it to earn returns elsewhere: EC is how the bank balances risk and return. See also D3 for regulatory versus economic capital.
- **Three components of EL** (CR-4 b): PD (also labelled expected default frequency, EDF), the exposure amount EA at the time of default (including all promised interest), and the loss rate LR (also called severity, LGD or LIED; LR = 1− recovery rate). PD is borrower-specific (linked to the rating); EA and LR reflect the *facility* (collateral, amortisation, limit versus utilisation).
- **Expected loss** (CR-4 c): the actuarial-type long-run average loss "across a range of typical economic conditions", *not* next year's cyclical forecast. It is the foreseeable cost of doing business: covered by loan loss reserves (standard risk costs) and reimbursed through loan pricing. "By definition, EL does not itself constitute risk."
- **Unexpected loss** (CR-4 d): the *standard deviation* of credit losses around EL. ULs "cannot be anticipated and hence cannot be adequately priced for in a loan's interest rate", so they need a cushion of economic capital. UL would be zero only if there were no uncertainty in default *and* none in the recovery rate.
- **Independence assumptions:** PD is independent of EA and LR because default is binary; EA and LR are usually treated as independent. If they were not, a covariance cross-term would be needed, but its impact on UL is small in practice.
- **Variance of PD** (CR-4 e): default is Bernoulli, a binomial $B(1; \text{PD})$, so $\sigma^2_{\text{PD}} = \text{PD}(1 - \text{PD})$. For LR, the binomial *overstates* the variance (all or nothing), the uniform makes any loss from 0% to 100% equally likely, and the *normal* is the most reasonable assumption "because of the lack of better knowledge".
- **Additivity** (CR-4 f): EL is linear and additive across a portfolio; UL is *not* (diversification). Portfolio UL needs ρ_{ij}, the correlation that default or a credit migration (in the same direction) of assets i and j occurs over the same horizon.
- **ULC** (CR-4 f): the marginal contribution $\text{ULMC}_i = \partial \text{UL}_p / \partial \text{UL}_i$ times UL_i. The ULCs sum exactly to UL_p ("the sum of the parts equals the whole"). The marginal contribution depends only on the UL weights of the loans, not on the size of the portfolio.
- **Identical-loans case** (S2): with equal size $1/n$ and constant ρ, $\text{ULC}_i = \text{UL}_p/n$, which reduces for large

n to $\text{UL}_i\sqrt{\rho}$; ρ is then the (weighted) average correlation between loans. Default correlations are small but positive, so the benefits of diversification are considerable.

- **Limits of the shortcut:** the full approach needs $n(n-1)/2$ pairwise default correlations, impossible for thousands of credits and hard to observe (they can be estimated from asset correlations). The identical-loan equation ignores different loan *sizes* and different *correlations* (industry, geography), so it does *not* reveal concentrations; it can be decomposed by segment. The analytical approach is cumbersome, so banks use numerical procedures (Monte Carlo).
- **Capital multiplier** (CR-4 g): $\text{EC}_p = \text{UL}_p \times \text{CM}$ and $\text{EC}_i = \text{ULC}_i \times \text{CM}$: capital at the transaction level is proportional to the loan's contribution to portfolio risk, not to its standalone UL. CM is the distance from EL_p to the confidence level in units of UL_p. At 99.97% (AA target), CM is between 7.0 and 7.5, "far higher than the capital multiples for the normally distributed events in market risk".
- **Loss distribution** (CR-4 h): credit losses are not normal but highly skewed (upside limited to the promised payments). The *beta distribution* on $[0,1]$ is fully characterised by EL_p and UL_p; tail fitting combines it with Monte Carlo. For high-quality portfolios ($\text{EL}_p > \text{UL}_p$) the fitted beta has too fat a tail and overestimates EC; for lower-quality ones it has too thin a tail.
- **Challenges** (CR-4 i): the approach assumes credits are *illiquid* (internal "betas", not market-priced correlations; a value approach would suit traded credit); almost all internal models use a *one-year* horizon because multi-period modelling is cumbersome; and market and operational risk are assumed to be measured and managed separately.
- **Economic versus regulatory view:** the regulatory approach "assigns roughly 8% equity capital to credits on a standalone basis"; the economic approach differentiates capital by borrower quality and credits diversification.
- **Country (transfer) risk:** same three components, plus an "overlap" adjustment because the bank can only lose its money once. It is "lumpy": higher correlations and fewer diversification benefits.

Formulas

Expected loss (eq. 4.1). $\text{EL} = \text{PD} \times \text{EA} \times \text{LR}$
Standalone unexpected loss (eq. 4.2) and variance of default (eq. 4.3).

$$\text{UL} = \text{EA} \times \sqrt{\text{PD} \cdot \sigma^2_{\text{LR}} + \text{LR}^2 \cdot \sigma^2_{\text{PD}}}, \qquad \sigma^2_{\text{PD}} = \text{PD}(1-\text{PD})$$

If LR is known with certainty ($\sigma_{\text{LR}} = 0$): $\text{UL} = \text{EA} \times \text{LR} \times \sqrt{\text{PD}(1-\text{PD})}$.
Portfolio EL and UL (eqs. 4.4, 4.5).

$$\text{EL}_p = \sum_i \text{EA}_i\,\text{PD}_i\,\text{LR}_i, \qquad \text{UL}_p = \sqrt{\sum_i \sum_j \rho_{ij}\,\text{UL}_i\,\text{UL}_j}$$

UL contribution (eq. 4.9).

$$\text{ULC}_i = \text{ULMC}_i \times \text{UL}_i = \frac{\text{UL}_i \sum_j \text{UL}_j \rho_{ij}}{\text{UL}_p}, \qquad \sum_i \text{ULC}_i = \text{UL}_p$$

n **identical loans, constant pairwise** ρ **(eqs. 4.13 to 4.15).**

$$\text{UL}_p = \text{UL}_i\sqrt{n + \rho(n^2-n)}, \qquad \text{ULC}_i = \frac{\text{UL}_p}{n} = \text{UL}_i\sqrt{\frac{1}{n} + \rho\left(1-\frac{1}{n}\right)} \xrightarrow{\ n \text{ large}\ } \text{UL}_i\sqrt{\rho}$$

Reverse-solved: $\rho = \dfrac{(\text{UL}_p/\text{UL}_i)^2 - n}{n^2 - n}$.
Economic capital (eqs. 4.17, 4.18). $\text{EC}_p = \text{UL}_p \times \text{CM}, \qquad \text{EC}_i = \text{ULC}_i \times \text{CM}$
EC covers UL only; EL is covered by reserves and pricing.

Exam traps

- **Omitted capital multiplier (dropped term):** UL_p and ULC_i are volatilities, not capital. Economic capital needs $\times$ CM; the credit CM (7.0 to 7.5 at 99.97%) is not the normal quantile used for market risk.
- **Standalone versus contribution (intermediate result):** when ULC is asked, the single-asset UL is the planted distractor; capital for a loan in a portfolio is $\text{ULC}_i \times \text{CM}$, not $\text{UL}_i \times \text{CM}$; $n \times \text{UL}_i$ is the undiversified ($\rho = 1$) portfolio UL.
- **$\sqrt{\rho}$ versus ρ (correlation misuse):** ULC tends to $\text{UL}\sqrt{\rho}$, but inside $\text{UL}_p = \text{UL}\sqrt{n + \rho(n^2 - n)}$ the correlation enters *unrooted*.
- **Wrong input twin:** the recovery rate where LR belongs, or PD where $\sigma^2_{\text{PD}} = \text{PD}(1 - \text{PD})$ belongs; forgetting the square root of the binomial variance.
- **EL in capital:** economic capital excludes EL. Adding EL double-counts what reserves and pricing already cover; subtracting EL again from $\text{UL}_p \times \text{CM}$ removes it twice, because CM is already measured from EL_p.

Question A1.1 CR-4 b, c, d, e · S1 · [Map #1]

A junior credit analyst at a Singapore commercial bank is computing the standalone risk of a single term loan for the bank's economic capital report. The data are:

- Exposure amount (EA): SGD 20 million
- One-year PD: 2.0%; default is a Bernoulli event
- Expected loss rate (LR): 40%; standard deviation of the loss rate: 25%
- EA and LR are independent of each other and of default

Which of the following is closest to the standalone unexpected loss (UL) of the loan?

A. SGD 1.12 million

B. SGD 1.32 million

C. SGD 1.48 million

D. SGD 1.82 million

Solution **Answer: B**

Step 1 – Variance of the default indicator

$$\sigma^2_{\text{PD}} = \text{PD}(1 - \text{PD}) = 0.02 \times 0.98 = 0.0196, \qquad \sigma_{\text{PD}} = 0.14$$

$\Rightarrow$ *Default is a 0/1 event with a 2% chance of a 1; its standard deviation is 14 percentage points.*

Step 2 – Loss-rate uncertainty term

$$\text{PD} \cdot \sigma^2_{\text{LR}} = 0.02 \times 0.25^2 = 0.00125$$

Step 3 – Default uncertainty term

$$\text{LR}^2 \cdot \sigma^2_{\text{PD}} = 0.40^2 \times 0.0196 = 0.003136$$

$\Rightarrow$ *Default uncertainty dominates: about 72% of the loss variance per unit of EA comes from not knowing whether default happens.*

Step 4 – Standalone UL

$$\text{UL} = 20 \times \sqrt{0.00125 + 0.003136} = 20 \times 0.06623 = \text{SGD } 1.32 \text{ million}$$

Interpretation

EL is only $0.02 \times 20 \times 0.40 = \text{SGD } 0.16$ million, which is priced into the loan. The UL of SGD 1.32 million is the one-standard-deviation surprise around that average; it is the input to the loan's UL contribution and hence to its economic capital, after diversification.

Why the other options are wrong

A is incorrect. Ignored loss-rate volatility (dropped the PD · σ_{LR}^2 term): $20 \times 0.40 \times 0.14 = 1.12$.
C is incorrect. Added EL to UL, treating the capital cushion as covering expected loss too: $1.32 + 0.16 = 1.48$.
D is incorrect. Used the recovery rate (60%) instead of the loss rate in the default term: $20 \times \sqrt{0.00125 + 0.36 \times 0.0196} = 1.82$.

LO: CR-4 d: Define and calculate unexpected loss (UL). CR-4 e: Estimate the variance of default probability assuming a binomial distribution. · Reference: Schroeck, Ch.5, Unexpected Losses (UL-Standalone), eqs. 4.2 and 4.3

Question A1.2

CR-4 d, e, f · S1, S2, S5 · [Aug-2026] [Map #1]

A credit risk analyst at a Canadian bank is attributing portfolio risk to individual loans in a book of 1,000 small-business loans. Every loan has the same exposure and the same internal rating:

- Exposure amount per loan: CAD 5 million
- One-year PD: 1.0%; default is a Bernoulli event
- Loss rate: 50%, known with certainty
- Pairwise default correlation between any two loans: 0.05
- Weighted average remaining maturity: 3.2 years

Which of the following is closest to the unexpected loss contribution (ULC) of one loan to the portfolio's unexpected loss?

A. CAD 5,500

B. CAD 56,000

C. CAD 118,000

D. CAD 249,000

Solution

Answer: B

Step 1 – Standard deviation of the default indicator

$$\sigma_{\text{PD}} = \sqrt{\text{PD}(1-\text{PD})} = \sqrt{0.01 \times 0.99} = \sqrt{0.0099} = 0.09950$$

Step 2 – Standalone UL per loan ($\sigma_{\text{LR}} = 0$)

$$\text{UL}_i = \text{EA} \times \text{LR} \times \sigma_{\text{PD}} = 5 \times 0.50 \times 0.09950 = \text{CAD } 0.2487 \text{ million}$$

$\Rightarrow$ *On its own, each loan has a loss standard deviation of about CAD 248,700.*

Step 3 – UL contribution for identical loans

$$\text{ULC}_i \approx \text{UL}_i\sqrt{\rho} = 0.2487 \times \sqrt{0.05} = 0.2487 \times 0.2236 = \text{CAD } 0.0556 \text{ million}$$

$\Rightarrow$ *With 1,000 loans the exact form gives the same answer:* $UL_i\sqrt{1/n + \rho(1-1/n)} = 0.2487 \times 0.2257 = 0.0561$. *Either way, about CAD 56,000.*

Interpretation

Inside the portfolio each loan contributes only about 23% of its standalone UL, because default correlation (0.05) is small but positive. The 1,000 ULCs add back to $\text{UL}_p \approx$ CAD 56.1 million. The 3.2-year maturity is irrelevant: the model uses a one-year horizon.

Why the other options are wrong

A is incorrect. Forgot the square root of the binomial variance: $\text{UL}_i = 2.5 \times 0.0099 = 0.02475$, then $0.02475 \times 0.2236 = 0.0055$.
C is incorrect. Correlation misuse: put $\sqrt{\rho}$ instead of ρ inside the portfolio formula, $0.2487 \times \sqrt{1{,}000 + 0.2236 \times 999{,}000}/1{,}000 = 0.118$.
D is incorrect. Reported the standalone UL of one loan (the intermediate result GARP plants when ULC is asked): 0.249.

LO: CR-4 f: Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size. CR-4 e: Estimate the variance of default probability assuming a binomial distribution. · Reference: Schroeck, Ch.5, Unexpected Loss Contribution (ULC), eqs. 4.14 and 4.15

Question A1.3 CR-4 f, g · S3, S4 · [Map #1]

A senior risk analyst at an Australian bank targeting a AA rating is preparing the economic capital figures for a portfolio of 400 near-identical mid-market loans. The data are:

- Each loan: EA AUD 2.5 million; PD 1.5%; LR 45%; standard deviation of LR 20%
- Average pairwise default correlation: 0.04
- Capital multiplier of the credit loss distribution at the 99.97% confidence level: 7.0
- The market-risk desk's normal-distribution multiple at 99.97%: 3.43

Which of the following is closest to the economic capital for the portfolio and the economic capital attributed to one loan?

A. Portfolio AUD 12.3 million; one loan AUD 0.031 million

B. Portfolio AUD 42.3 million; one loan AUD 0.106 million

C. Portfolio AUD 79.6 million; one loan AUD 0.199 million

D. Portfolio AUD 86.4 million; one loan AUD 0.216 million

Solution **Answer: D**

Step 1 – Standalone UL

$$\text{PD}\cdot\sigma_{\text{LR}}^2 = 0.015\times 0.04 = 0.000600, \qquad \text{LR}^2\cdot\sigma_{\text{PD}}^2 = 0.2025\times 0.014775 = 0.002992$$
$$\text{UL}_i = 2.5\times\sqrt{0.003592} = \text{AUD } 0.1498 \text{ million}$$

Step 2 – Portfolio UL

$$\text{UL}_p = \text{UL}_i\sqrt{n+\rho(n^2-n)} = 0.1498\times\sqrt{400+0.04\times 159{,}600} = 0.1498\times 82.37 = \text{AUD } 12.34 \text{ million}$$

$\Rightarrow$ *The 400 loans behave like 82 standalone loans: diversification removes most of the undiversified* $400\times 0.1498 = 59.9$.

Step 3 – UL contribution of one loan

$$\text{ULC}_i = \frac{\text{UL}_p}{n} = \frac{12.34}{400} = \text{AUD } 0.03085 \text{ million}$$

Step 4 – Apply the capital multiplier

$$\text{EC}_p = \text{UL}_p\times\text{CM} = 12.34\times 7.0 = \text{AUD } 86.4 \text{ million}$$
$$\text{EC}_i = \text{ULC}_i\times\text{CM} = 0.03085\times 7.0 = \text{AUD } 0.216 \text{ million}$$

$\Rightarrow$ *Each loan carries AUD 216,000 of capital, about 8.6% of its exposure; the 400 attributions add back exactly to AUD 86.4 million.*

Interpretation

CM converts a one-standard-deviation measure into the distance from EL_p to the 99.97% point of a skewed credit loss distribution. Portfolio EL ($400\times 2.5\times 0.015\times 0.45 =$ AUD 6.75 million) is covered by reserves and pricing, not by this capital.

Why the other options are wrong

A is incorrect. Omitted the capital multiplier: reported $\text{UL}_p = 12.3$ and $\text{ULC}_i = 0.031$ as if volatility were capital.

B is incorrect. Wrong input twin: used the normal multiple 3.43 instead of the credit CM, $12.34\times 3.43 = 42.3$ and $0.03085\times 3.43 = 0.106$. Schroeck: the credit CM is far higher than normal-distribution multiples because credit losses are skewed.

C is incorrect. Subtracted EL from the capital a second time: $86.4 - 6.75 = 79.6$ and $0.216 - 0.0169 = 0.199$. CM is already measured from EL_p.

LO: CR-4 g: Describe how economic capital is derived. CR-4 f: Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size. · Reference: Schroeck, Ch.5, Economic Capital for Credit Risk, eqs. 4.13, 4.17 and 4.18

Question A1.4 CR-4 f · S3 · [Map #1]

An internal model validator at a UK building society is backing out the default correlation implied by the credit portfolio model for a homogeneous pool of 200 equal-sized loans. The model reports:

- Standalone UL of each loan: GBP 0.30 million
- Portfolio UL: GBP 10.0 million

Using the exact identical-loan formula for portfolio UL, which of the following is closest to the implied pairwise default correlation?

A. 0.1667

B. 0.0458

C. 0.0278

D. 0.0229

Solution **Answer: D**

Step 1 – Formula

$$\mathrm{UL}_p = \mathrm{UL}_i\sqrt{n + \rho(n^2 - n)} \;\Rightarrow\; \rho = \frac{(\mathrm{UL}_p/\mathrm{UL}_i)^2 - n}{n^2 - n}$$

Step 2 – Squared ratio

$$\left(\frac{10.0}{0.30}\right)^2 = 33.33^2 = 1{,}111.1$$

$\Rightarrow$ *The portfolio behaves like* $\sqrt{1{,}111} = 33.3$ *standalone loans out of 200.*

Step 3 – Remove the variance terms

$$1{,}111.1 - 200 = 911.1$$

$\Rightarrow$ *This is the part of portfolio variance coming from the* $n^2 - n = 39{,}800$ *covariance terms.*

Step 4 – Implied correlation

$$\rho = \frac{911.1}{39{,}800} = 0.0229$$

Interpretation

An implied default correlation of about 2.3% is small but positive, as Schroeck expects. The validator would compare it with correlations estimated from observable asset correlations for the segment.

Why the other options are wrong

A is incorrect. Forgot to square: $\mathrm{ULC}/\mathrm{UL} = 0.05/0.30 = 0.1667$ is (approximately) $\sqrt{\rho}$, not ρ.
B is incorrect. Counted each of the $n(n-1)/2$ distinct pairs once, without the factor 2: $\rho = 911.1/19{,}900 = 0.0458$.
C is incorrect. Used the large-n approximation $\mathrm{ULC} = \mathrm{UL}\sqrt{\rho}$: $\rho = (10/(200 \times 0.30))^2 = 0.0278$, which ignores the $1/n$ variance term.

LO: CR-4 f: Calculate UL for a credit asset portfolio and the UL contribution of each asset under various scenarios of portfolio composition, asset characteristics and size. · Reference: Schroeck, Ch.5, Unexpected Loss Contribution (ULC), eqs. 4.13 to 4.16

A2 RAROC, Adjusted RAROC and Hurdle Rate

Readings: ORR-18 a, b, c, d, e, f, g, h · Risk Capital Attribution and Risk-Adjusted Performance Measurement (Crouhy, Galai and Mark, Essentials 2e, Ch.17; GARP sections 18.1 to 18.6, Boxes 18.1 to 18.5); CR-5 g · Introduction to Credit Risk Modeling and Assessment (Doumpos et al., Ch.1, 5.7)

AUG 2026 RECALL | MAP TIER A · #3 · 9/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: RAROC - 1; Hurdle rate - 1.
Map #3: "Reported variants: '2 on adjusted RAROC'; 'effect on RAROC by changing its components (economic capital, income, expense)': a sensitivity/direction question, not a plug-in; 'find hurdle rate'; 'weighted cost of capital given break-even cost'. Hurdle rate = value-weighted cost of common equity (via CAPM) and preferred equity. The ARAROC adjustment is $\alpha - (\beta_E \times$ excess market return$)$. The polarity of each component's effect on RAROC is the corruption operator of choice."
Shapes covered here:
S1 compute after-tax RAROC, in plain form and with red-herring inputs (Q1, Q2, Q3, Q4)
S2 two on adjusted RAROC: value plus accept/reject decision, and the break-even beta (Q6, Q7)
S3 direction of the effect on RAROC of changes in economic capital, income and expense (Q9)
S4 find the hurdle rate: market-value-weighted cost of common equity (CAPM) and preferred equity (Q5, Q8)
S5 weighted cost of capital given break-even cost (Q8; break-even revenue at the hurdle in Q3).

Revision notes

- **Risk capital** (ORR-18 a): the cushion that protects against the firm's risks so it remains a going concern even in a near-catastrophic worst-case scenario. Usually treated as identical to economic capital, but the chapter later defines *economic capital* = *risk capital* + *strategic risk capital*. Risk capital covers the worst-case loss *minus* EL at the required confidence level over one year (Figure 18.1: 165 bps − 15 bps = 150 bps of unexpected loss at 99%).
- **Regulatory capital versus risk capital** (ORR-18 a): regulatory capital applies only to a few regulated industries (banks, insurers), is set by industrywide rules and formulas, and sets only a *minimum*; it rarely captures the true level of risk, and may differ from risk capital business line by business line even when the firm totals are similar. Some practitioners allocate the *higher* of the two. Under Basel III, regulatory capital may end up much higher than economic capital for activities such as securitisation, which then tend to migrate to the shadow banking sector.
- **Who sets the parameters:** the confidence level and horizon for risk capital are key policy parameters set by *senior management* (or the senior risk management committee) and usually endorsed by the board; the confidence level is tied to the target credit rating and the stated risk appetite.
- **Why economic capital matters to banks (Box 18.1):** capital absorbs risk (debt-to-equity up to 20 to 1, and off-balance-sheet activities draw on risk capital); target solvency is part of the product sold to liability holders; banks are opaque, so capital reduces agency costs; capital is expensive, so neither too much nor too little. Uses: performance measurement and incentive compensation, entry/exit decisions (with the diversification effect), and risk-based pricing via the marginal economic capital requirement. One measure cannot serve all four purposes.
- **RAROC** (ORR-18 b): proposed by Bankers Trust in the late 1970s; after-tax expected risk-adjusted net income / economic capital. It adjusts the *numerator* (subtracts EL) *and* the *denominator* (economic for accounting capital). RORAC adjusts only the denominator; RAROA uses risk-adjusted assets. A business unit with RAROC above the cost of the bank's equity is deemed to add shareholder value.
- **Capital budgeting numerator** (ORR-18 b, c): expected revenues − costs − expected losses − taxes + return on risk capital ± transfers. EL is priced into the spread over funding cost, so *no risk capital* is held for it, yet it is still *subtracted* in the numerator. The return on risk capital assumes the capital is invested in risk-free securities. Transfers include treasury funding charges, hedging costs and head-office overhead.
- **Strategic risk capital** (ORR-18 b): goodwill (the premium paid above the replacement value of net assets in an acquisition) + burned-out capital (capital spent starting up a business that may not be kicked off), both amortised over time. Some banks also allocate capital to unused risk limits.

- **EVA or NIACC** (Box 18.2): after-tax adjusted net income less a capital charge equal to economic capital times the after-tax cost of equity. An activity whose RAROC is above the hurdle rate is also EVA positive.
- **Ex ante versus ex post** (ORR-18 c): RAROC was first an anticipatory capital budgeting tool on *expected* revenues and losses; for ex post performance evaluation, *realized* revenues and losses may be used.
- **Horizon** (ORR-18 d): all inputs on one horizon; practitioners use *one year* (business planning cycle; time to recapitalise after a major unexpected loss). A longer horizon does not necessarily raise capital, because the required confidence in solvency *decreases* as the horizon is extended. Market risk (Box 18.3): scale to the *core risk level* over the *time to reduce*, assuming an orderly unwind, not $\sqrt{252}$ (Figure 18B.1: 52.8% of annualised VaR).
- **PIT versus TTC PD** (ORR-18 d): point-in-time PD (KMV, structural) suits near-term EL and pricing; through-the-cycle PD (rating agencies) suits economic capital, current profitability and strategic decisions, and *reduces* the volatility of economic capital. **Confidence level:** AA target implies a one-year PD of 3 to 5 bps, i.e. 99.95% to 99.97%; a *lower* confidence level can significantly *reduce* risk capital, especially where operational, credit and settlement risks dominate.
- **Hurdle rate** (ORR-18 e): most firms use a *single* hurdle rate, the after-tax weighted-average cost of equity capital (common and preferred, at *market* values; common via CAPM; preferred = yield on preferred shares). Reset periodically, say every six months, or when it changes by more than 10%. Flaw: one hurdle can lead the firm to accept high-risk projects (oil exploration) that lower value and reject low-risk ones (well-managed retail banking) that raise it.
- **Adjusted RAROC** (ORR-18 f, Box 18.5): ARAROC = RAROC $-\beta_E(R_M - r_f)$, stripping out the compensation for nondiversifiable systematic risk (shareholders assumed well diversified). Accept if ARAROC $> r_f$: the hurdle becomes the *risk-free rate*, the same for all business lines.
- **Diversification** (ORR-18 g): firm risk capital $<$ sum of stand-alone capital. No fully integrated VaR model exists, so banks run bottom-up models at a lower business-level confidence (e.g. 99.5% versus 99.97%); the aggregate lies between the zero-correlation root sum of squares (USD 787 in the book) and the perfect-correlation simple sum (USD 1,200). The portfolio effect is usually allocated *pro rata* to stand-alone capital (USD 46 and 54 in Figure 18.2). Stand-alone capital for incentive compensation; fully diversified for solvency and minimum risk pricing; marginal for active portfolio management and business mix. Marginal capitals sum to *less* than the BU total (70 versus 100).
- **Best practices** (ORR-18 h): (1) senior management commitment (CEO sponsorship; "not how much income is generated, but how well the firm is compensated for the risks"); (2) communication and education; (3) ongoing consultation via a "parameter review group" (credit parameters over the cycle; market volatilities and correlations at least monthly; core risk level and time to reduce annually); (4) integrity of the process (the RAROC group owns collection, calculation and reporting; business units and finance own the data they feed in); (5) combine RAROC with qualitative factors on the strategic grid ("managed growth" = high return, low strategic importance; "investment" = low return, high growth potential); (6) a quarterly active capital management process, disagreements arbitrated by the senior executive team. Business units complain capital is too high (never too low) and too unstable; the acrimony is over *parameters*, not the VaR methods.
- **Loan RAROC in Doumpos** (CR-5 g): RAROC = loan revenues / capital at risk, with loan revenues $(s+f-l-c)(1-x)$. Capital at risk is either the duration-based value change $\Delta L = -LD\,\Delta i/(1+i)$ (market-based approach) or the unexpected loan loss $\alpha \times$ LGD $\times$ EAD, with $\alpha = 2.6\sigma$ at 99.5% under normality (5 to 6 for skewed losses). The loan is profitable if RAROC exceeds the bank's cost of capital. No return-on-capital term: do not mix it with the ORR-18 form.
- **See also** A1 (unexpected loss and economic capital via the capital multiplier) and D3 (regulatory versus economic capital).

Formulas

Generic RAROC. $\text{RAROC} = \dfrac{\text{after-tax expected risk-adjusted net income}}{\text{economic capital}}$

Capital budgeting form (18.4).

$$\text{RAROC} = \frac{\text{expected revenues} - \text{costs} - \text{expected losses} - \text{taxes} + \text{return on risk capital} \pm \text{transfers}}{\text{economic capital}}$$

Book's worked form. $\text{RAROC} = \dfrac{(\text{Rev} - \text{Cost} - \text{Funding} - \text{EL} + r_f\,\text{EC})(1-t)}{\text{EC}}$. Book check: $(90 - 9 - 60 - 10 + 3.75)(0.7)/75 = 14\%$.
Economic capital. EC = risk capital + strategic risk capital, strategic risk capital = goodwill + burned-out capital; risk capital = worst-case loss − EL.
EVA (NIACC). NIACC = after-tax adjusted net income − EC $\times\, h_{AT}$ (positive exactly when RAROC $> h_{AT}$).
Hurdle rate (Box 18.4). $h_{AT} = \dfrac{CE \times r_{CE} + PE \times r_{PE}}{CE + PE}$, $r_{CE} = r_f + \beta_{CE}(R_M - r_f)$, r_{PE} = yield on preferred shares; CE, PE at market value. No debt term: it is a cost of *equity*.
Adjusted RAROC (Box 18.5). ARAROC = RAROC − $\beta_E(R_M - r_f)$; accept if ARAROC $> r_f$. Equivalently RAROC $> r_f + \beta_E(R_M - r_f)$. Break-even beta: $\beta^* = (\text{RAROC} - r_f)/(R_M - r_f)$.
Reverse-solves. Break-even revenue: $\text{Rev}^* = \dfrac{h\,\text{EC}}{1-t} + \text{Cost} + \text{Funding} + \text{EL} - r_f\text{EC}$. Implied cost of common equity at break-even (RAROC $= h$): $r_{CE} = \dfrac{h(CE + PE) - PE\,r_{PE}}{CE}$.
Diversification bounds (ORR-18 g). Zero correlation: $\sqrt{\sum_i \text{EC}_i^2}$; perfect correlation: $\sum_i \text{EC}_i$. Marginal capital of X = EC(BU) − EC(BU without X).
CR-5 g loan form. $\text{RAROC} = \dfrac{(s + f - l - c)(1-x)}{\text{capital at risk}}$, $\Delta L = -L\,D\,\dfrac{\Delta i}{1+i}$ or $\alpha \times \text{LGD} \times \text{EAD}$. Book check: $2{,}000/26{,}786 = 7.47\%$.

Exam traps

- **Return on risk capital (dropped term or sign):** it is *added* (capital sits in risk-free securities). Dropping it or subtracting it each moves the answer by one option; the book taxes it with the rest of the numerator.
- **Wrong denominator twin:** regulatory capital, a short-horizon trading VaR, or risk capital without goodwill and burned-out capital instead of economic capital; EL never goes into the denominator.
- **Polarity of components (the map's favourite):** higher costs, funding charges or EL lower RAROC; a lower confidence level means less capital and a *higher* RAROC for a profitable unit; "EL has no effect because no capital is held for it" is a scope error.
- **Hurdle construction:** book instead of market weights, a simple average, βR_M instead of $\beta(R_M - r_f)$, adding debt (a WACC), or grossing up for tax (the hurdle is already after tax).
- **ARAROC benchmark:** compare ARAROC with r_f, not with the old hurdle and not with zero. Subtracting the whole CAPM cost of equity gives RAROC minus the hurdle, which is compared with zero.
- **Horizon mismatch:** multi-year deal totals over one year of capital. Annualise every numerator item.

Question A2.1

ORR-18 b, c · S1 · [Aug-2026] [Map #3]

A senior risk analyst at a Singapore commercial bank is preparing the capital-budgeting committee's review of a corporate loan portfolio. The committee wants both the after-tax RAROC and the economic value added (NIACC) of the portfolio. The data are:

- Loan portfolio: SGD 800 million, headline yield 8.5% per year; direct operating cost: SGD 7 million per year
- Funding: SGD 800 million of deposits, transfer-priced at 5.5% per year
- Expected loss: 0.8% of the portfolio per year
- Economic capital: 6% of the portfolio, invested in government securities yielding 4%
- Effective tax rate: 25%; the bank's after-tax hurdle rate: 12%

Which of the following is closest to the portfolio's after-tax RAROC and its NIACC?

A. RAROC 13.6%; NIACC SGD 0.75 million

B. RAROC 16.6%; NIACC SGD 2.19 million

C. RAROC 19.6%; NIACC SGD 3.63 million

D. RAROC 19.6%; NIACC SGD 5.55 million

Solution **Answer: C**

Step 1 – Revenue, funding charge and expected loss

$$\text{Rev} = 8.5\% \times 800 = 68.0, \qquad \text{Funding} = 5.5\% \times 800 = 44.0, \qquad \text{EL} = 0.8\% \times 800 = 6.4$$

⇒ *EL is the average annual default cost, priced into the spread; no capital is held for it, but it is still deducted.*

Step 2 – Economic capital and its return

$$\text{EC} = 6\% \times 800 = 48.0, \qquad r_f\,\text{EC} = 4\% \times 48 = 1.92$$

⇒ *The capital buffer sits in government securities and earns SGD 1.92 million, which is added to the numerator.*

Step 3 – After-tax risk-adjusted income

$$(68.0 - 7.0 - 44.0 - 6.4 + 1.92)(1 - 0.25) = 12.52 \times 0.75 = \text{SGD } 9.39 \text{ million}$$

Step 4 – RAROC

$$\text{RAROC} = \frac{9.39}{48.0} = 19.6\%$$

⇒ *Each SGD of economic capital earns about 19.6 cents a year after tax.*

Step 5 – NIACC (EVA)

$$\text{NIACC} = 9.39 - 48.0 \times 12\% = 9.39 - 5.76 = \text{SGD } 3.63 \text{ million}$$

⇒ *After charging the shareholders' required return on the capital tied up, the portfolio creates SGD 3.63 million of value a year.*

Interpretation

The two measures agree, as Box 18.2 says they must: RAROC of 19.6% is above the 12% hurdle, so the portfolio is EVA positive. The hurdle is the full after-tax cost of equity; the risk-free return on capital is already credited in the numerator.

Why the other options are wrong

A is incorrect. Subtracted the return on economic capital as a cost: $(68 - 7 - 44 - 6.4 - 1.92)(0.75) = 6.51$, RAROC 13.6%, NIACC $6.51 - 5.76 = 0.75$.

B is incorrect. Dropped the return on economic capital: $(68 - 7 - 44 - 6.4)(0.75) = 7.95$, RAROC 16.6%, NIACC $7.95 - 5.76 = 2.19$.

D is incorrect. Correct RAROC, but charged capital at the hurdle net of the risk-free rate, $48 \times (12\% - 4\%) = 3.84$, giving $9.39 - 3.84 = 5.55$. This credits the risk-free return twice.

LO: ORR-18 b: Describe the RAROC (risk-adjusted return on capital) methodology and its use in capital budgeting. ORR-18 c: Calculate and interpret the RAROC for a project, loan, or loan portfolio and use RAROC to compare business unit performance. · Reference: Crouhy, Galai and Mark, Essentials 2e, Ch.17, 18.4 and Box 18.2

Question A2.2

ORR-18 a, c · S1 · [Aug-2026] [Map #3]

A risk consultant at a Japanese trust bank is reviewing the performance-measurement pack of the bank's structured-lending unit. For the coming year the unit reports (JPY billion):

Item	JPY bn
Expected revenues	14.0
Direct operating costs	2.5
Funding transfer charge from treasury	8.0
Expected credit loss	1.2
Economic capital allocated	12.0
Regulatory capital requirement	16.0
10-day 99% trading VaR of the unit's hedge book	0.9

The risk-free rate on JGBs is 1.0% and the effective tax rate is 30%. Which of the following is closest to the unit's after-tax RAROC?

A. 10.6%

B. 13.1%

C. 13.4%

D. 14.1%

Solution **Answer: D**

Step 1 – Return on economic capital

$$r_f \,\mathrm{EC} = 1.0\% \times 12.0 = \text{JPY } 0.12 \text{ bn}$$

Step 2 – Pre-tax and after-tax risk-adjusted income

$$14.0 - 2.5 - 8.0 - 1.2 + 0.12 = 2.42, \qquad 2.42 \times (1 - 0.30) = \text{JPY } 1.694 \text{ bn}$$

Step 3 – RAROC on economic capital

$$\mathrm{RAROC} = \frac{1.694}{12.0} = 14.1\%$$

⇒ *The unit returns about 14.1% after tax on the economic capital it consumes.*

Interpretation

The denominator is *economic* capital. Regulatory capital is the wrong twin (an industrywide minimum that rarely captures true risk), and the hedge book's 10-day VaR is a short-horizon market-risk number already reflected in the unit's one-year economic capital (Box 18.3).

Why the other options are wrong

A is incorrect. Used regulatory capital as the denominator: $1.694/16.0 = 10.6\%$.
B is incorrect. Added the 10-day trading VaR to economic capital: $1.694/(12.0+0.9) = 13.1\%$, double counting a risk already inside EC.
C is incorrect. Dropped the return on economic capital: $(14 - 2.5 - 8 - 1.2)(0.7)/12 = 13.4\%$.

LO: ORR-18 c: Calculate and interpret the RAROC for a project, loan, or loan portfolio and use RAROC to compare business unit performance. ORR-18 a: Define, compare, and contrast risk capital, economic capital, and regulatory capital, and explain methods and motivations for using economic capital approaches to allocate risk capital. · Reference: Crouhy, Galai and Mark, Essentials 2e, Ch.17, 18.1, 18.4 and Box 18.3

Question A2.3 ORR-18 c, e · S1, S5 · [Map #3]

A treasury analyst at a Canadian bank is pricing a new CAD 500 million mid-market loan book. The book will incur direct costs of CAD 4 million per year, will be funded at a transfer price of 4.5%, carries an expected loss of 0.6% per year, and requires economic capital of CAD 40 million invested in Government of Canada bonds yielding 3.5%. The effective tax rate is 28% and the bank's hurdle rate is 12%. Which of the following is closest to the minimum annual expected revenue at which the book's after-tax RAROC just equals the hurdle rate?

A. CAD 32.90 million

B. CAD 34.77 million

C. CAD 36.17 million

D. CAD 37.57 million

Solution **Answer: B**

Step 1 – Required after-tax income

$$h \times \mathrm{EC} = 12\% \times 40 = \text{CAD } 4.80 \text{ million}$$

Step 2 – Gross up to pre-tax income

$$\frac{4.80}{1 - 0.28} = \text{CAD } 6.667 \text{ million}$$

⇒ *Tax takes 28%, so the book must earn CAD 6.667 million before tax to keep CAD 4.8 million after tax.*

Step 3 – Components to add back

$$\text{Funding} = 4.5\% \times 500 = 22.50, \qquad \text{EL} = 0.6\% \times 500 = 3.00, \qquad r_f\,\text{EC} = 3.5\% \times 40 = 1.40$$

Step 4 – Solve for revenue

$$\text{Rev} - 4 - 22.5 - 3.0 + 1.4 = 6.667 \;\Rightarrow\; \text{Rev} = \text{CAD } 34.77 \text{ million}$$

$\Rightarrow$ *The book must yield at least* $34.77/500 = 6.95\%$ *to be value-neutral.*

Interpretation

This is RAROC used for risk-based pricing: the break-even yield at which the activity neither creates nor destroys shareholder value under the traditional rule.

Why the other options are wrong

A is incorrect. No tax gross-up: $4.8 + 4 + 22.5 + 3 - 1.4 = 32.90$.
C is incorrect. Forgot the return on economic capital: $6.667 + 4 + 22.5 + 3 = 36.17$.
D is incorrect. Treated the return on economic capital as a cost to recover: $6.667 + 4 + 22.5 + 3 + 1.4 = 37.57$.

LO: ORR-18 c: Calculate and interpret the RAROC for a project, loan, or loan portfolio and use RAROC to compare business unit performance. ORR-18 e: Calculate the hurdle rate and apply this rate in making business decisions using RAROC. · Reference: Crouhy, Galai and Mark, Essentials 2e, Ch.17, 18.4 and Hurdle Rate and Capital Budgeting Decision Rule

Question A2.4

ORR-18 b, c, d · S1 · [Map #3]

The head of performance measurement at a eurozone bank is evaluating a two-year acquisition-finance deal. Totals over the full two-year life are: revenues EUR 104 million, direct costs EUR 36 million, expected losses EUR 12 million and funding transfer charges EUR 30 million, all accruing evenly. Economic capital is held constant over the life and comprises risk capital of EUR 60 million (measured at the bank's 99.97% target confidence level), goodwill of EUR 8 million from the acquired origination team, and burned-out capital of EUR 4 million. Capital is invested at a risk-free rate of 3% per year and the tax rate is 25%. Using a one-year horizon, which of the following is closest to the deal's annual after-tax RAROC?

A. 15.8%

B. 18.5%

C. 21.1%

D. 29.3%

Solution

Answer: A

Step 1 – Economic capital

$$\text{EC} = \text{risk capital} + \text{goodwill} + \text{burned-out capital} = 60 + 8 + 4 = \text{EUR } 72 \text{ million}$$

$\Rightarrow$ *Strategic risk capital is part of economic capital, not an optional extra.*

Step 2 – Annualise the deal quantities

$$\text{Rev} = 52, \quad \text{Cost} = 18, \quad \text{EL} = 6, \quad \text{Funding} = 15 \quad (\text{EUR million per year})$$

Step 3 – Annual after-tax risk-adjusted income

$$(52 - 18 - 6 - 15 + 3\% \times 72)(1 - 0.25) = 15.16 \times 0.75 = 11.37$$

Step 4 – RAROC

$$\text{RAROC} = \frac{11.37}{72} = 15.8\%$$

Interpretation

The deal earns 15.8% per year after tax on the full economic capital it ties up. The 99.97% confidence level is already embedded in the risk capital figure; it is not a further input.

Why the other options are wrong

B is incorrect. Used risk capital alone in the denominator and the capital return: $(13 + 1.8)(0.75)/60 = 18.5\%$.
C is incorrect. Forgot taxes: $15.16/72 = 21.1\%$, a pre-tax figure.
D is incorrect. Horizon mismatch: two-year totals over one year of capital, $(104-36-12-30+2.16)(0.75)/72 = 29.3\%$.

LO: ORR-18 d: Explain challenges that arise when using RAROC for performance measurement, including choosing a time horizon, measuring default probability, and choosing a confidence level. ORR-18 c: Calculate and interpret the RAROC for a project, loan, or loan portfolio and use RAROC to compare business unit performance. · Reference: Crouhy, Galai and Mark, Essentials 2e, Ch.17, 18.4 and 18.5 (RAROC Horizon)

Question A2.5

ORR-18 e · S4 · [Aug-2026] [Map #3]

The CFO of a Canadian bank is resetting the single firm-wide hurdle rate used in RAROC-based capital budgeting. The data are:

Item	Value
Market value of common equity	CAD 6.0 billion
Book value of common equity	CAD 3.0 billion
Market value of preferred equity	CAD 1.5 billion
Yield on the bank's preferred shares	7.0%
Beta of the bank's common equity	1.2
Risk-free rate	4.0%
Expected return on the market portfolio	10.0%
Effective tax rate	25%

Which of the following is closest to the bank's hurdle rate?

A. 10.36%

B. 11.20%

C. 13.81%

D. 14.20%

Solution

Answer: A

Step 1 – Cost of common equity (CAPM)

$$r_{CE} = r_f + \beta_{CE}(R_M - r_f) = 4.0\% + 1.2 \times 6.0\% = 11.2\%$$

$\Rightarrow$ *Shareholders require 11.2% a year for holding the bank's common stock.*

Step 2 – Cost of preferred equity

$$r_{PE} = \text{yield on preferred shares} = 7.0\%$$

Step 3 – Market-value weights

$$CE + PE = 6.0 + 1.5 = 7.5: \qquad 80\% \text{ common},\ 20\% \text{ preferred}$$

Step 4 – Hurdle rate

$$h_{AT} = \frac{6.0 \times 11.2\% + 1.5 \times 7.0\%}{7.5} = \frac{0.672 + 0.105}{7.5} = 10.36\%$$

$\Rightarrow$ *Any activity with after-tax RAROC above 10.36% is deemed to add value under the traditional rule.*

Interpretation

Book equity and the tax rate are red herrings: Box 18.4 weights by *market* values, and the cost of equity is already an after-tax return to shareholders.

Why the other options are wrong

B is incorrect. Reported the cost of common equity alone, ignoring the preferred equity in the weighted average: 11.20%.

C is incorrect. Grossed the hurdle up for tax: $10.36\%/(1-0.25) = 13.81\%$. The hurdle is the *after-tax* cost of equity and is compared with after-tax RAROC.
D is incorrect. Multiplied beta by the market return instead of the premium: $r_{CE} = 4\% + 1.2 \times 10\% = 16\%$, giving $(6 \times 16\% + 1.5 \times 7\%)/7.5 = 14.20\%$.

LO: ORR-18 e: Calculate the hurdle rate and apply this rate in making business decisions using RAROC. · Reference: Crouhy, Galai and Mark, Essentials 2e, Ch.17, Box 18.4

Question A2.6

ORR-18 f · S2 · [Map #3]

A senior risk analyst at a UK bank is reviewing a proposed GBP trade-finance business line for the risk committee. Its expected after-tax risk-adjusted net income is GBP 21 million per year and it requires economic capital of GBP 150 million. The bank's equity beta is 1.3, the expected return on the market portfolio is 9.0%, and the three-month Treasury bill rate is 3.5%. Using the adjusted RAROC, which of the following is the correct value and decision?

A. ARAROC of 2.30%; reject, because it is below the risk-free rate

B. ARAROC of 3.35%; reject, because it is below the risk-free rate

C. ARAROC of 6.85%; reject, because it is below the bank's cost of equity of 10.65%

D. ARAROC of 6.85%; accept, because it is above the risk-free rate

Solution

Answer: D

Step 1 – Traditional RAROC

$$\text{RAROC} = \frac{21}{150} = 14.0\%$$

Step 2 – Systematic risk adjustment

$$\beta_E(R_M - r_f) = 1.3 \times (9.0\% - 3.5\%) = 1.3 \times 5.5\% = 7.15\%$$

$\Rightarrow$ *Shareholders need 7.15% above the risk-free rate for the nondiversifiable risk of this activity.*

Step 3 – Adjusted RAROC

$$\text{ARAROC} = 14.0\% - 7.15\% = 6.85\%$$

Step 4 – Decision rule

$$6.85\% > r_f = 3.5\% \;\Rightarrow\; \text{accept}$$

Interpretation

After the adjustment the hurdle is the risk-free rate, the same for every business line. Equivalently, RAROC of 14.0% beats the beta-specific cost of equity $3.5\% + 7.15\% = 10.65\%$.

Why the other options are wrong

A is incorrect. Multiplied beta by the market return instead of the premium: $14.0\% - 1.3 \times 9.0\% = 2.30\%$.
B is incorrect. Subtracted the full CAPM cost of equity (10.65%) rather than only the risk premium: $14.0\% - 10.65\% = 3.35\%$, then compared it with r_f. That figure is RAROC minus the hurdle and belongs against zero.
C is incorrect. Correct value, wrong benchmark: once returns are adjusted for systematic risk, ARAROC is compared with r_f, not with the cost of equity, which would remove the risk premium twice.

LO: ORR-18 f: Calculate the adjusted RAROC for a project to determine its viability. · Reference: Crouhy, Galai and Mark, Essentials 2e, Ch.17, Box 18.5

Question A2.7

ORR-18 e, f · S2 · [Map #3]

A risk consultant at an Australian bank is advising on a new AUD equipment-leasing activity. The activity's expected after-tax RAROC is 12.4%. The bank's traditional single hurdle rate is 10.5%, the risk-free rate is 3.0%, and the expected return on the market portfolio is 8.0%. The consultant wants to know how much systematic risk the activity can carry before it fails the adjusted-RAROC test. Which of the following is closest to the maximum

equity beta at which the activity would still be accepted?

A. 1.18

B. 1.55

C. 1.88

D. 2.48

Solution **Answer: C**

Step 1 – Acceptance condition

$$\text{ARAROC} = \text{RAROC} - \beta_E(R_M - r_f) > r_f$$

Step 2 – Market risk premium

$$R_M - r_f = 8.0\% - 3.0\% = 5.0\%$$

Step 3 – Break-even beta

$$12.4\% - \beta_E \times 5.0\% = 3.0\% \;\Rightarrow\; \beta_E = \frac{9.4\%}{5.0\%} = 1.88$$

$\Rightarrow$ *At a beta of 1.88 the activity earns exactly r_f after compensating shareholders for systematic risk; any higher beta fails.*

Interpretation

The traditional rule accepts the activity ($12.4\% > 10.5\%$) whatever its beta; the adjusted rule accepts it only if $\beta_E < 1.88$. The 10.5% hurdle is irrelevant once ARAROC is used.

Why the other options are wrong

A is incorrect. Divided by the market return instead of the premium: $9.4\%/8.0\% = 1.18$.
B is incorrect. Used $\beta_E R_M$ and compared ARAROC with zero: $12.4\%/8.0\% = 1.55$.
D is incorrect. Compared ARAROC with zero instead of r_f: $12.4\%/5.0\% = 2.48$.

LO: ORR-18 f: Calculate the adjusted RAROC for a project to determine its viability. ORR-18 e: Calculate the hurdle rate and apply this rate in making business decisions using RAROC. · Reference: Crouhy, Galai and Mark, Essentials 2e, Ch.17, Box 18.5

Question A2.8 ORR-18 c, e · S4, S5 · [Map #3]

The CFO of a US bank is explaining the firm-wide hurdle rate to the board. She notes that the bank's trade-finance unit is exactly at break-even: its expected after-tax RAROC equals the bank's single hurdle rate. Data (USD):

- Unit, per year: expected revenues 42 million; direct costs 9 million; funding transfer charge 24 million; expected loss 3 million
- Unit economic capital: 60 million, invested at the risk-free rate of 4%; effective tax rate 25%
- Bank: market value of common equity 9 billion; market value of preferred equity 3 billion; yield on preferred shares 6%
- Expected return on the market portfolio: 9%

Which of the following is closest to the cost of common equity implied by the break-even, and to the common equity beta consistent with it under the CAPM?

A. Cost of common equity 10.5%; beta 1.30

B. Cost of common equity 12.0%; beta 1.60

C. Cost of common equity 12.0%; beta 1.33

D. Cost of common equity 16.7%; beta 2.53

Solution **Answer: B**

Step 1 – The unit's RAROC (the break-even cost of capital)

$$\text{RAROC} = \frac{(42 - 9 - 24 - 3 + 4\% \times 60)(1 - 0.25)}{60} = \frac{8.4 \times 0.75}{60} = \frac{6.3}{60} = 10.5\%$$

⇒ *At break-even, the hurdle rate h_{AT} is 10.5%: this is the bank's after-tax weighted cost of equity capital.*

Step 2 – Back out the cost of common equity from the weighted hurdle

$$h_{AT} = \frac{CE\, r_{CE} + PE\, r_{PE}}{CE + PE} \;\Rightarrow\; r_{CE} = \frac{10.5\% \times 12 - 3 \times 6\%}{9} = \frac{1.26 - 0.18}{9} = 12.0\%$$

⇒ *Preferred equity is cheaper (6%), so common equity must cost more than the 10.5% average.*

Step 3 – Implied CAPM beta

$$r_{CE} = r_f + \beta_{CE}(R_M - r_f) \;\Rightarrow\; \beta_{CE} = \frac{12.0\% - 4\%}{9\% - 4\%} = \frac{8.0\%}{5.0\%} = 1.60$$

Interpretation

The unit just covers a 10.5% value-weighted cost of equity. Check: $(9 \times 12\% + 3 \times 6\%)/12 = 10.5\%$. Any rise in the bank's beta would lift the hurdle and push the unit into value destruction under the traditional rule.

Why the other options are wrong

A is incorrect. Treated the break-even RAROC as the cost of common equity, ignoring the preferred weight: $r_{CE} = 10.5\%$, $\beta = (10.5\% - 4\%)/5\% = 1.30$.

C is incorrect. Correct cost of equity, but divided by the market return instead of the premium: $\beta = 12\%/9\% = 1.33$.

D is incorrect. Used the pre-tax RAROC ($8.4/60 = 14\%$) as the hurdle: $r_{CE} = (14\% \times 12 - 0.18)/9 = 16.7\%$, $\beta = 2.53$. The hurdle is an after-tax rate.

LO: ORR-18 e: Calculate the hurdle rate and apply this rate in making business decisions using RAROC. ORR-18 c: Calculate and interpret the RAROC for a project, loan, or loan portfolio and use RAROC to compare business unit performance. · Reference: Crouhy, Galai and Mark, Essentials 2e, Ch.17, 18.4 and Box 18.4

Question A2.9 ORR-18 b, c, d · S3 · [Map #3]

The head of the RAROC group at an Australian bank is presenting to business heads how a unit's after-tax RAROC responds to changes in its inputs. The unit's current figures (AUD million) are: expected revenues 14.0, direct costs 2.5, funding transfer charge 8.0, expected loss 1.2, economic capital 12.0 invested at a risk-free rate of 1%, tax rate 30%, giving a RAROC of 14.1%. Each change below is considered alone, all else equal. Which of the following statements should the head include?

A. Lowering the confidence level for economic capital from 99.97% to 99.5% would lower RAROC, because a smaller capital allocation earns less return on risk capital.

B. A higher funding charge transferred from treasury would raise RAROC, because transfer-pricing charges enter the numerator alongside the return on risk capital.

C. A one-million rise in direct costs would lower RAROC by exactly as much as a one-million fall in expected revenues, both effects reduced by the factor $(1 - t)$.

D. A one-million rise in expected loss would leave RAROC unchanged, because expected loss is priced into the spread and no risk capital is held against it.

Solution **Answer: C**

Step 1 – Write RAROC as a function of its components

$$\text{RAROC} = \frac{(\text{Rev} - \text{Cost} - \text{Funding} - \text{EL})(1 - t)}{\text{EC}} + r_f(1 - t) = \frac{2.3 \times 0.7}{12} + 0.7\% = 14.1\%$$

⇒ *Every pre-tax item in the bracket moves RAROC by $(1 - t)/EC$ per unit; economic capital moves only the first term.*

Step 2 – Costs versus revenues

$$\text{Cost}+1: \ \frac{(14-3.5-8-1.2+0.12)(0.7)}{12}=8.3\%; \qquad \text{Rev}-1: \ \frac{(13-2.5-8-1.2+0.12)(0.7)}{12}=8.3\%$$

$\Rightarrow$ *Both cut RAROC by* $1\times 0.7/12=5.8$ *points: identical, and scaled by* $(1-t)$.

Step 3 – The other components

Funding $+1$ and EL $+1$ give the same 8.3%. Lowering the confidence level cuts economic capital; for example EC of 9.0 gives $(2.3+0.09)(0.7)/9=18.6\%$, a *higher* RAROC.

Interpretation

Direction is the whole question: costs, funding charges and EL lower RAROC; less capital raises it for a profitable unit. The book stresses that the choice of confidence level "can materially affect risk-adjusted performance measures and the resulting capital allocation decisions".

Why the other options are wrong

A is incorrect. Polarity flip. A lower confidence level significantly reduces risk capital (18.5, Confidence Level). For a unit with positive risk-adjusted income the smaller denominator dominates the lost risk-free return, so RAROC rises (14.1% to 18.6% in the illustration).

B is incorrect. Polarity flip on a transfer. The funding charge from treasury is a cost charged to the business unit; it is subtracted, so a higher charge lowers RAROC (to 8.3%).

D is incorrect. Scope error. "No risk capital for EL" is a statement about the denominator; EL is still subtracted in the numerator, so a higher EL lowers RAROC (to 8.3%).

LO: ORR-18 b: Describe the RAROC (risk-adjusted return on capital) methodology and its use in capital budgeting. ORR-18 d: Explain challenges that arise when using RAROC for performance measurement, including choosing a time horizon, measuring default probability, and choosing a confidence level. · Reference: Crouhy, Galai and Mark, Essentials 2e, Ch.17, 18.4 and 18.5 (Confidence Level)

A3 Volatility Smiles, Skews and Put-Call Parity

Readings: MR-17 a, b, c, d, e, f, g, h, i · Volatility Smiles and Volatility Surfaces (Hull, OFOD 11e, Ch.20; 17.1 to 17.8 and Appendix)

AUG 2026 RECALL | MAP TIER A · #4 · 9/16 SOURCES | MAP TIER D · #71 · 2/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: Volatility skew 1Q - effect of changing maturity and strike (equity option)
Map #4: "Explicit recalls: '2 questions from volatility smiles on FX and equity options'; 'implied volatility: foreign exchange options shape'; 'lognormal vols vs implied vols for fx and index options'; 'equity options volatility smile property'. Also flagged: put-call parity's implication that a call and put at the same strike/maturity share one implied volatility (LO 17.b), reported as a standalone theory question. The sibling swap here is attributing the FX explanation to equities or vice versa."
Map #71: "Reported: 'put call parity.' Within Part II its home is LO 17.b: the implication that a European call and put with the same strike and maturity must share one implied volatility, which is why the smile is measured identically from either. Do not treat this as a Part I recap; the exam tested it in its smile context."
Shapes covered here:
S1 equity skew: effect of changing strike and maturity on implied volatility, plain form (Q1)
S2 two questions on FX and equity smiles (Q2, Q3)
S3 implied volatility shape for FX options (Q2; FX tails in Q3)
S4 lognormal vols versus implied vols for FX and index options (Q3)
S5 equity options volatility smile property (Q1, Q3)
S6 sibling swap: FX explanation attributed to equities or vice versa (Q2)
S7 put-call parity: a European call and put with the same strike and maturity share one implied volatility, tested in its smile context (Q4; distractor in Q3).

Revision notes

- **The specific: the equity skew** (MR-17 a, e). Since the October 1987 crash, the smile for equity options (individual stocks and indices) slopes downward and is called a volatility *skew*: implied volatility *decreases* as the strike price *increases*. Low-strike options (deep out-of-the-money puts, deep in-the-money calls) carry significantly higher implied volatilities than high-strike options (deep in-the-money puts, deep out-of-the-money calls). Before October 1987 there was no marked smile.
- **The specific: maturity** (MR-17 g). Implied volatility tends to be an *increasing* function of maturity when short-dated volatilities are historically *low* (volatilities are expected to rise) and a *decreasing* function when they are historically *high*. Separately, the smile becomes less pronounced as option maturity increases (observed for currencies and for options on most other assets).
- **Smile and surface defined** (MR-17 a). A volatility smile is the implied volatility of options with a certain life plotted against strike price; a volatility surface is implied volatility as a function of both strike price and time to maturity. Traders use Black-Scholes-Merton (BSM), but not as originally intended: they let the volatility depend on strike and maturity.
- **Put-call parity implication** (MR-17 b). $p + S_0 e^{-qT} = c + Ke^{-rT}$ rests on a simple no-arbitrage argument with *no* assumption about the distribution of the asset price: true whether or not it is lognormal. It holds for BSM prices and for market prices, so $p_{BS} - p_{mkt} = c_{BS} - c_{mkt}$ (17.2): the dollar pricing error is the same for a European put and call with the same strike and maturity, so their implied volatilities are equal (put at 22% implies call at 22%). The smile and the surface are therefore the same for calls and puts; approximately true for American options. Example 17.1: FX call at strike 0.59 with implied vol 14.5%; parity with q = foreign rate gives put price 0.0419, also 14.5%.
- **Smile shape versus implied distribution** (MR-17 c). Read the tails through deep out-of-the-money options: if the implied (risk-neutral) distribution puts more probability in a tail than a lognormal with the same mean and standard deviation, the option paying off in that tail is priced relatively high, so its implied volatility is relatively high; less probability means a relatively low price and implied volatility.
- **FX smile** (MR-17 d, e). Implied volatility is relatively low for at-the-money options and becomes progressively higher as an option moves either into or out of the money: U-shaped. The implied distribution has

heavier tails than the lognormal (kurtosis) and is more peaked: small and large moves more likely, intermediate moves less likely. Deep OTM calls (high K_2) and deep OTM puts (low K_1) are both priced relatively high.

- **FX evidence** (MR-17 d). Table 17.1 (10 currencies against the USD, 2005 to 2015): daily moves exceed 3 SD on 1.30% of days versus 0.27% under the lognormal model; 4, 5 and 6 SD on 0.49%, 0.24% and 0.13% versus almost never. Business Snapshot 17.1: buy deep OTM calls and puts on many currencies and wait; the mid-1980s traders who knew about heavy tails made money until, by the late 1980s, the smile was priced in.
- **Why FX is not lognormal** (MR-17 d, e). Lognormality needs (1) constant volatility and (2) smooth price changes with no jumps. Exchange rate volatility is far from constant and rates frequently jump, sometimes in response to central bank actions; both make extreme outcomes more likely. As maturity increases, the percentage impact of non-constant volatility on *prices* becomes more pronounced but on *implied volatility* usually less pronounced; jumps' impact on both becomes less pronounced (they are averaged out). Net: the smile flattens with maturity.
- **Equity implied distribution** (MR-17 c, e). Heavier left tail and less heavy right tail than the lognormal. A deep OTM call (K_2) is priced relatively low (low implied vol); a deep OTM put (K_1) relatively high (high implied vol). Hull's summary: traders assume equity distributions have a heavier left tail and less heavy right tail, and exchange rate distributions a heavier right *and* left tail.
- **Why the equity skew** (MR-17 e). Negative correlation between equity prices and volatility: as prices fall, volatility rises, making further declines possible; as prices rise, volatility falls. Reasons: (1) *leverage*: falling equity raises leverage and so volatility; (2) *volatility feedback*: an external rise in volatility makes investors require a higher return, so the price declines; (3) *crashophobia* (Rubinstein): fear of another October 1987, supported by the skew steepening when the S&P 500 falls and flattening when it rises.
- **Alternative ways to characterise the smile** (MR-17 f). Against K the curve shifts right (left) as the asset price rises (falls), so implied volatility is often plotted against K/S_0; a refinement uses K/F_0, and traders often define at-the-money as $K = F_0$ (the risk-neutral expected price at maturity). Against delta: at-the-money is a call delta of 0.5 or a put delta of -0.5 ("50-delta options"), which lets smiles apply to options other than European and American calls and puts.
- **Volatility surfaces in pricing** (MR-17 g). Table 17.2 combines smiles and the term structure; entries with reliable market data come from market prices, the rest are interpolated. A 9-month option at $K/S_0 = 1.05$ interpolates 13.4% and 14.0% to 13.7%; a 1.5-year option at $K/S_0 = 0.925$ uses bilinear interpolation to give 14.525%. That volatility goes into BSM or a binomial tree. Plotting against $\frac{1}{\sqrt{T}}\ln(K/F_0)$ makes the smile much less dependent on maturity.
- **Greeks and the smile** (MR-17 h). Standard Greeks assume implied volatility is unchanged when the asset price changes, which is not what is expected. The minimum variance delta $\Delta_{MV} = \Delta_{BSM} + \nu_{BSM}\,\partial E(\sigma_{imp})/\partial S$; for equities vega is positive and $\partial E(\sigma_{imp})/\partial S$ negative, so $\Delta_{MV} < \Delta_{BSM}$.
- **Role of the model** (MR-17 g). BSM can be argued to be a sophisticated interpolation tool: switching to another plausible model would change the surface but arguably not the dollar prices. Greek letters, and so hedging, do depend on the model; an unrealistic model leads to poor hedging. Models matter most for derivatives that do not trade actively (many exotics).
- **A single large jump** (MR-17 i). A stock at USD 50 that will move to 42 or 58 on news (takeover, lawsuit): the true distribution is bimodal (a mixture of two lognormals). With $r = 12\%$, $p = 0.5314$: implied volatilities 58.8% at 44, 69.5% at 48, 69.2% at 50, 49.0% at 56, 0% at 42 and 58. The pattern is a *frown*, the opposite of the currency smile; the volatility implied at strike 50 overprices options at 44 or 56.
- **Implied distribution from the smile** (Appendix). Breeden-Litzenberger: $g(K) = e^{rT}\,\partial^2 c/\partial K^2 \approx e^{rT}(c_1 + c_3 - 2c_2)/\delta^2$, the value of a butterfly spread. Example 17A.1 (stock at USD 10, vols 30% at strike 6 falling to 22% at 14): probability of 6 to 7 is 0.0057 versus 0.0031 lognormal, of 13 to 14 is 0.0113 versus 0.0167: heavier left tail, less heavy right tail.

Formulas

Put-call parity (17.1). $p + S_0 e^{-qT} = c + Ke^{-rT}$ (FX: q = foreign risk-free rate)
Equal pricing errors (17.2). $p_{BS} - p_{mkt} = c_{BS} - c_{mkt} \;\Rightarrow\; \sigma_{imp}^{call}(K,T) = \sigma_{imp}^{put}(K,T)$

Smile axes. K, K/S_0, K/F_0, delta; maturity-normalised $\frac{1}{\sqrt{T}}\ln\left(\frac{K}{F_0}\right)$

Minimum variance delta (17.6). $\Delta_{MV} = \Delta_{BSM} + \nu_{BSM}\frac{\partial E(\sigma_{imp})}{\partial S}$

Implied density (17A.1, 17A.2). $g(K) = e^{rT}\frac{\partial^2 c}{\partial K^2} \approx e^{rT}\frac{c_1 + c_3 - 2c_2}{\delta^2}$

Shape map. Heavier tail ⇒ OTM option in that tail priced high ⇒ high implied vol. FX: both tails heavy, U-shaped smile. Equity: heavy left, thin right, downward skew. Single anticipated jump: bimodal, frown.

Exam traps

- **Polarity flip (strike):** for equity options implied volatility *falls* as strike rises; the low-strike (OTM put) volatilities are highest.
- **Polarity and condition (maturity):** the smile flattens as maturity increases; the term structure slopes up only when short-dated volatilities are historically low, and down when they are historically high.
- **Sibling swap (shape and reason):** FX is U-shaped with both tails heavy, explained by non-constant volatility and jumps; equity is skewed with a heavy left and thin right tail, explained by leverage, volatility feedback and crashophobia. Attributing one market's explanation to the other is the map's flagged trap.
- **Lognormal price versus market:** with one at-the-money volatility, deep OTM index puts and both FX wings are priced *below* market; deep OTM index calls are priced *above* market.
- **Scope error (put-call parity):** parity equates implied volatilities of a call and a put with the *same* strike and maturity, with no distributional assumption; it says nothing about equality across strikes, and a gap between them is an arbitrage even if prices are not lognormal.
- **Jump shape:** a single large anticipated jump gives a frown, not a smile.

Question A3.1

MR-17 a, e, g · S1, S5 · [Aug-2026] [Map #4]

A senior equity derivatives analyst at a Hong Kong bank is presenting implied volatilities of European options on a stock index to a group of newly hired analysts. Short-dated implied volatilities on the index are currently at historically low levels. Which of the following statements should the analyst include in the presentation?

A. Implied volatility rises as the strike price rises, and at-the-money implied volatility is expected to increase with option maturity.

B. Implied volatility is lowest at the money and rises toward both lower and higher strikes, and at-the-money implied volatility is expected to increase with option maturity.

C. Implied volatility falls as the strike price rises, and at-the-money implied volatility is expected to decrease with option maturity.

D. Implied volatility falls as the strike price rises, and at-the-money implied volatility is expected to increase with option maturity.

Solution

Answer: D

Step 1 – Strike: the equity skew

For equity options (stocks and indices) the volatility decreases as the strike price increases (Hull 17.3). Low-strike options, deep OTM puts and deep ITM calls, carry the highest implied volatilities.

Step 2 – Maturity: the term structure

Implied volatility tends to be an increasing function of maturity when short-dated volatilities are historically low, because volatilities are then expected to increase (Hull 17.5). The stem sets that condition.

Step 3 – The second maturity effect

Across a surface, the skew itself becomes less pronounced as maturity increases: jumps are averaged out over long horizons and the impact of non-constant volatility on implied volatility usually shrinks.

Interpretation

D is the only statement with both directions right. If short-dated volatilities were historically high instead, the same skew would sit on a term structure that *decreases* with maturity.

Why the other options are wrong

A is incorrect. Polarity flip on strike: for equities implied volatility falls, not rises, as the strike increases.
B is incorrect. Sibling swap: a U-shape lowest at the money is the foreign currency smile, not the equity skew.
C is incorrect. Scope/condition error: a term structure that decreases with maturity applies when short-dated volatilities are historically *high*; the stem says they are low.

LO: MR-17 e: Describe the volatility smile for equity options and foreign currency options and provide possible explanations for its shape. MR-17 g: Describe volatility term structures and volatility surfaces and how they may be used to price options. · Reference: Hull, OFOD 11e, Ch.20, 17.3 and 17.5, Table 17.2

Question A3.2

MR-17 d, e · S2, S3, S6 · [Map #4]

A risk consultant is reviewing the options desk of a German bank that trades EUR/USD options and options on a European equity index. The desk prices the two books with differently shaped volatility smiles, and the consultant is preparing a report explaining why. Which of the following statements is correct to include in the report?

A. The FX smile is U-shaped because exchange rate volatility is not constant and rates jump, while the equity skew slopes downward because equity prices and volatility are negatively correlated.

B. The FX smile is U-shaped because of the leverage effect and crashophobia, while the equity skew slopes downward because volatility is not constant and prices jump on central bank actions.

C. The FX smile is U-shaped because exchange rate volatility is not constant and rates jump, while the equity skew slopes downward because equity prices and volatility are positively correlated.

D. The FX smile is U-shaped because exchange rate volatility is not constant and rates jump, while the equity skew slopes downward in a pattern that was equally marked before the October 1987 crash.

Solution

Answer: A

Step 1 – FX shape and reason

Implied volatility is relatively low at the money and progressively higher into or out of the money (Hull 17.2). Two lognormal conditions fail: volatility is far from constant and exchange rates frequently jump, sometimes on central bank actions. Both make extreme outcomes, in either direction, more likely.

Step 2 – Equity shape and reason

Volatility decreases as the strike increases (Hull 17.3). There is a negative correlation between equity prices and volatility, explained by leverage, the volatility feedback effect and crashophobia, so declines bring higher volatility (heavy left tail) and rises bring lower volatility (thin right tail).

Interpretation

The consultant should explain each smile by its own market's mechanism; the two shapes are not interchangeable even though both depart from the lognormal.

Why the other options are wrong

B is incorrect. Sibling swap: the explanations are exchanged. Leverage and crashophobia explain the equity skew; non-constant volatility and jumps (including central bank actions) explain the FX smile.
C is incorrect. Polarity flip: the equity correlation between prices and volatility is negative; a positive correlation would thicken the right tail instead.
D is incorrect. Scope error: there was no marked volatility smile for equity options before October 1987; the skew has existed only since the crash, which is why crashophobia is one explanation.

LO: MR-17 d: Describe characteristics of foreign exchange rate distributions and their implications on option prices and implied volatility. MR-17 e: Describe the volatility smile for equity options and foreign currency options and provide possible explanations for its shape. · Reference: Hull, OFOD 11e, Ch.20, 17.2 and 17.3, Business Snapshot 17.2

Question A3.3

MR-17 b, c, d, e · S2, S3, S4, S5, S7 · [Map #4]

A junior trader at a Canadian dealer prices every European option on a stock index, and every European option on USD/CAD, with a single lognormal volatility per underlying equal to that underlying's at-the-money implied volatility. A market risk analyst compares the resulting model prices with market prices of options with the same maturity. Which of the following conclusions should the analyst reach?

A. Deep out-of-the-money index puts are priced below market, because the implied distribution has a heavier left

tail than the lognormal.

B. Deep out-of-the-money index calls are priced below market, because the implied distribution has a heavier right tail than the lognormal.

C. Deep out-of-the-money USD/CAD puts are priced above market, because the implied distribution has a thinner left tail than the lognormal.

D. Options at every strike are priced at market, because put-call parity forces one implied volatility on every option with the same maturity.

Solution **Answer: A**

Step 1 – What a single lognormal volatility assumes

BSM with one volatility assumes a lognormal distribution. The market prices with a smile, which corresponds to the implied distribution (Hull 17.2, 17.3 and Appendix).

Step 2 – Index options

The equity implied distribution has a heavier left tail and a less heavy right tail than the lognormal. A deep OTM put pays off only in the left tail, so the market prices it relatively high, at an implied volatility above the at-the-money level. The trader's at-the-money volatility therefore prices it *below* market. A deep OTM call pays off only in the thin right tail and is priced relatively low by the market, so the trader prices it *above* market.

Step 3 – FX options

The FX implied distribution has heavier tails on both sides. Both deep OTM puts and deep OTM calls carry implied volatilities above the at-the-money level, so the trader prices both wings below market. Table 17.1 supports the market: 3 SD daily moves occur on 1.30% of days versus 0.27% under the lognormal model.

Interpretation

The trader is selling tail protection too cheaply on index puts and on both FX wings: the trade Business Snapshot 17.1 describes, from the wrong side.

Why the other options are wrong

B is incorrect. Sibling swap: a heavier right tail is the FX property, not the equity one. The index implied distribution has a less heavy right tail, so deep OTM index calls are priced *above* market by the lognormal.
C is incorrect. Polarity flip: the FX implied distribution has a heavier, not thinner, left tail, so deep OTM USD/-CAD puts are priced *below* market.
D is incorrect. Scope error: put-call parity makes a call and a put with the same strike and maturity share one implied volatility; it does not equalise implied volatilities across strikes.

LO: MR-17 c: Compare the shape of the volatility smile (or skew) to the shape of the implied distribution of the underlying asset price and to the pricing of options on the underlying asset. MR-17 d: Describe characteristics of foreign exchange rate distributions and their implications on option prices and implied volatility. · Reference: Hull, OFOD 11e, Ch.20, 17.1 to 17.3, Table 17.1, Business Snapshot 17.1

Question A3.4 MR-17 b · S7 · [Map #4] [Map #71]

An options trader at a Swiss bank observes European call and put options on a non-dividend-paying stock with the same strike price and the same maturity. Using BSM, the call's implied volatility is 24% and the put's implied volatility is 27%. The head of market risk asks what the discrepancy means. Which of the following is the most appropriate response?

A. No action is needed, because the equity skew assigns higher implied volatilities to puts than to calls.

B. An arbitrage exists only if the stock price is lognormally distributed, because the implied volatilities come from BSM.

C. The call is expensive relative to the put, so the trader should sell the call, buy the put and buy the stock, borrowing at the risk-free rate.

D. The put is expensive relative to the call, so the trader should sell the put, buy the call and short the stock, investing the proceeds at the risk-free rate.

Solution **Answer: D**

Step 1 – What parity implies

Put-call parity $p + S_0 = c + Ke^{-rT}$ holds for BSM prices and for market prices, so $p_{BS} - p_{mkt} = c_{BS} - c_{mkt}$ (17.2). At any volatility the dollar pricing errors are equal, so a European call and put with the same strike and maturity must have the same implied volatility.

Step 2 – Which leg is rich

A higher implied volatility means a higher price relative to BSM. At 27% versus 24%, the put is priced too high relative to the call: $p_{mkt} + S_0 > c_{mkt} + Ke^{-rT}$.

Step 3 – The arbitrage

Sell the put, buy the call and short the stock: the cash received $p - c + S_0$ exceeds Ke^{-rT}. Invest it at the risk-free rate. At T the long call and short put together buy the stock for K whatever happens, which closes the short; the invested proceeds exceed K.

Interpretation

Parity is a no-arbitrage relation with no assumption about the asset price distribution, so the profit is locked in whether or not prices are lognormal. This is why the smile is the same for calls and puts.

Why the other options are wrong

A is incorrect. Sibling swap: the skew relates implied volatility to the strike, across strikes. At the same strike and maturity the call and put must share one implied volatility.

B is incorrect. Scope/condition error: put-call parity does not require any assumption about the probability distribution; it is true both when the distribution is lognormal and when it is not.

C is incorrect. Polarity flip: the put, with the higher implied volatility, is the expensive leg. Selling the call and buying the put would buy the overpriced side.

LO: MR-17 b: Explain the implications of put-call parity on the implied volatility of call and put options. · Reference: Hull, OFOD 11e, Ch.20, 17.1 eqs. 17.1 and 17.2, Example 17.1; Practice Questions 17.8 and 17.17

A4 Fundamental Review of the Trading Book (FRTB)

Readings: MR-18 a, b, c · Fundamental Review of the Trading Book (Hull, RMFI 6e Ch.18); ORR-21 c · Capital Regulation Before the Global Financial Crisis; ORR-22 a, b, c · Solvency, Liquidity, and Other Regulation After the Global Financial Crisis

AUG 2026 RECALL | MAP TIER A · #5 · 8/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: FRTB - 2Q.
Map #5: "Reported angles: the ES-replaces-VaR shift and why; liquidity horizons by asset class; the standardised approach vs. internal models approach; trading-book / banking-book boundary; and comparative questions framed as 'FRTB vs Basel 1, Basel 2, Basel 2.5: overall changes'. One source flags 'refer GARP for Curve Risk' specifically, i.e. the curvature risk charge is not well covered in condensed notes."
Shapes covered here:
S1 stressed ES at 97.5% replaces 10-day VaR at 99%, and why (Q2)
S2 liquidity horizons by asset class and the cascade ES (Q1)
S3 standardized vs internal models approach, with the desk-level backtesting and P&L attribution tests that decide which applies (Q3, Q4)
S4 trading book / banking book boundary (Q3)
S5 FRTB vs Basel I (1996 Amendment), Basel II, Basel 2.5: overall changes (Q4)
S6 curvature risk charge, from the GARP text (Q4).

Revision notes

The specifics first: ES replaces VaR, horizons, two approaches (MR-18 a, b)

- **Timeline.** Consultative document May 2012; Quantitative Impact Studies (QISs: banks estimate the effect of proposed rules); final rules January 2019; agreed implementation January 2023, with many jurisdictions a year or two later.
- **Measure change (S1).** Basel I and Basel 2.5 used 10-day VaR at 99%. FRTB uses **stressed ES at 97.5%**: calculations are based on how market variables moved during *stressed* market conditions, as with stressed VaR. Basel I VaR was "current": based on the immediately preceding one to four years.
- **Why 97.5% ES.** For normal losses with mean μ and standard deviation σ: 99% VaR $= \mu + 2.326\sigma$, 97.5% ES $= \mu + 2.338\sigma$, almost exactly the same. For non-normal losses they are not equivalent: with a **heavier tail than the normal**, 97.5% ES can be **considerably greater** than 99% VaR. So the change bites exactly where tail risk is fat.
- **Liquidity horizons (S2).** The fixed 10-day horizon is replaced by horizons reflecting the liquidity of each market variable ("risk factor"); changes over the horizon in stressed markets are "shocks". Five horizons: **10, 20, 40, 60, 120 days** = categories 1 to 5.
- **Table 18.1 by asset class.** 10: equity price large cap. 20: sovereign IG credit spread; equity price small cap; large-cap equity volatility; energy price; precious metal price. 40: sovereign non-IG spread; corporate IG spread; FX volatility. 60: interest rate volatility; corporate non-IG spread; small-cap equity volatility; equity other; other commodities price; energy price volatility; precious metal volatility. 120: credit spread other; credit spread volatility; other commodities price volatility; commodity other. Ranges: interest rate 10 to 60 and FX rate 10 to 40 (currency dependent).
- **Two approaches, both always computed (S3).** Even a bank approved for the internal models approach must also calculate standardized capital, consistent with standardized approaches acting as a floor (December 2017: total capital at least 72.5% of standardized, reached by 2028 after a five-year phase-in). Trend since 2008: less reliance on internal models.
- **Desk level.** Most calculations are done per trading desk, and internal models permission is granted **desk by desk**: the FX desk can be on internal models while the equity desk is not.

Internal models approach and the cascade (MR-18 b)

- **Overlapping 10-day changes.** All ES calculations use 10-day changes from the stressed period (Day 0 to 10, Day 1 to 11, …), not one-day changes scaled by $\sqrt{10}$ as in Basel I and Basel 2.5. Overlap means trials

are not independent: **no bias**, but a smaller effective sample, so noisier results. Method is not prescribed; historical simulation is likely.

- **Cascade.** ES_1: shock all risk factors. ES_2: shock categories 2 to 5, hold category 1 constant. ES_3: categories 3 to 5. ES_4: categories 4 and 5. ES_5: category 5 only. Combine with weights $(LH_j - LH_{j-1})/10 = 1, 1, 2, 2, 6$ (independence of successive periods: an extension of the square-root rule). Also usable for VaR. Six desks need $5 \times 6 = 30$ ES runs. If a desk has no factors in category 1, holding category 1 constant changes nothing, so $ES_2 = ES_1$.
- **Stressed period search.** Back as far as 2007. A subset of risk factors may be used if it covers at least 75% of current factors; ES is then scaled up by (ES, last 12 months, all factors) / (ES, last 12 months, subset). This can double the runs from 30 to 60.
- **Weighted ES and capital.** Partial ES per desk (shock that desk's factors only); the sum of partial ES always exceeds total-portfolio ES. $WES = \lambda ES_T + (1 - \lambda) \sum ES_{Pj}$, $\lambda = 0.5$. **Non-modelable risk factors**: fewer than 24 observations a year or more than one month between observations; handled by stress tests (NMC). Capital uses 60-day averages and $m_c \geq 1.5$.

Standardized approach, including curvature (MR-18 a)

- **Three components.** (1) Sensitivity-based risk charge: delta, vega and curvature for each of 7 risk classes (general interest rate, FX, commodity, equity, three credit spread classes); (2) **default risk charge**; (3) **residual risk add-on**. No diversification across risk classes, nor between delta and vega within a class. Correlations tested at 1.25, 1.00 and 0.75 times the Basel values; the greatest charge is used.
- **Deltas and risk weights.** Equity, FX, commodity deltas are per percentage change (1% move gives USD 3,000, so $\delta = 300{,}000$); rate and spread deltas are per absolute change (1 bp costs USD 200, so $\delta = -2{,}000{,}000$). Risk weight = stressed daily volatility $\times\sqrt{LH} \times 2.338$, e.g. $0.02 \times \sqrt{20} \times 2.338 = 0.209$. Each vertex of a term structure (e.g. 10 rate vertices from 3 months to 30 years) is a separate risk factor.
- **Curvature risk charge (S6).** The standardized charge for **gamma** risk. Shock factor i up and down by its risk weight W_i; for each direction take the delta-predicted change minus the actual change: $W_i\delta_i - \Delta V^+$ and $-W_i\delta_i - \Delta V^-$. The charge is the **greater** of the two; if negative, it counts as **zero**. Short options (Figure 18.1a) attract it; long options (Figure 18.1b) give zero, since curvature raises value relative to delta for both moves. Term structures are shifted in parallel by the **largest** W_i on the curve, with each point's own δ_i removing delta.
- **Default risk charge and residual add-on.** Jump-to-default: exposure × LGD × default risk weight (senior debt LGD 75%; A-rated weight 3%; equity LGD 100%). Residual add-on: notional × risk weight (1% for exotic options not decomposable into vanillas). The **simplified approach** for smaller banks (June 2017, in the 2019 rules) drops vega and gamma.

Boundary, backtesting, P&L attribution, credit, securitizations (MR-18 c)

- **Trading book vs banking book (S4).** Trading book gets market risk capital; banking book (held to maturity) gets credit risk capital. Banks exploited the gap (credit-dependent instruments parked in the trading book for lower capital). FRTB: "intent to trade" is **no longer sufficient**; the bank must be **able** to trade and manage the risks on a trading desk, and day-to-day value changes should affect equity and pose solvency risk. Assignment is at **initiation**; transfers only in extraordinary circumstances (desk closure, change in fair-value accounting standards), and any capital benefit from moving is **disallowed**.
- **Backtesting.** Stressed ES is **not** backtested (ES is harder to backtest than VaR; a stressed measure cannot be backtested at all). Each desk backtests **1-day VaR at 99% and 97.5%** on the most recent 12 months: more than **12** exceptions at 99% or more than **30** at 97.5% sends the desk to the standardized approach until neither condition holds. Regulators may also test that daily p-values are uniform.
- **P&L attribution.** U = actual minus model P&L; V = actual P&L ("actual" = hypothetical, no intraday trading). Mean(U)/SD(V) within -10% to $+10\%$; Var(U)/Var(V) below 20%. **Four or more** situations outside these ranges in 12 months: standardized approach.
- **Credit risk and securitizations.** Credit spread risk is treated like market risk (horizons 20 to 120 days; spread volatility 120). Jump-to-default risk is treated like banking-book default risk: internal models use **1-year 99.9% VaR**. Securitizations: the Basel 2.5 comprehensive risk measure allowed own models, but capital varied too much across banks, so FRTB requires the **standardized approach**.

FRTB vs earlier Basel market risk rules (S5; ORR-21 c, ORR-22 a, b, c)

- **Basel I, 1996 Amendment.** Standardized or internal models approach; internal models: 10-day 99% VaR from at least one year of data, usually 1-day VaR $\times\sqrt{10}$; $MR = \max(VaR_{t-1}, m \times VaR_{avg})$, 60-day average, $m \geq 3$ (specific risk multiplier 4). Daily backtest of 1-day 99% VaR over 250 days: 10 or more exceptions forces $m = 4$. Tier 3 capital introduced.
- **Basel II.** Retained much of Basel I: its innovations were IRB credit risk weights, operational risk capital, the three pillars and repeated QISs. Market risk capital stayed on the 1996 Amendment basis.
- **Basel 2.5 (by end-2011).** Three changes: **stressed VaR** (250-day most stressful period in the last seven years), so $MR_{2.5} = \max(VaR_{t-1}, m_c VaR_{avg}) + \max(SVaR_{t-1}, m_s SVaR_{avg})$, at least twice the 1996 charge with equal multipliers; **incremental risk charge** (99.9%, one-year, defaults and downgrades, liquidity horizon at least three months); **comprehensive risk measure** for the correlation book. FRTB then replaces VaR plus stressed VaR with one stressed ES, adds liquidity horizons, desk-level approval, the SA floor and a firmer book boundary.

Formulas

Normal benchmark: $VaR_{99\%} = \mu + 2.326\sigma, \qquad ES_{97.5\%} = \mu + 2.338\sigma$
Cascade (liquidity-adjusted ES), $LH_1 = 10$:

$$ES = \sqrt{ES_1^2 + \sum_{j=2}^{5}\left(ES_j\sqrt{\frac{LH_j - LH_{j-1}}{10}}\right)^2} = \sqrt{ES_1^2 + ES_2^2 + 2ES_3^2 + 2ES_4^2 + 6ES_5^2}$$

Weighted ES: $WES = \lambda\, ES_T + (1-\lambda)\sum_j ES_{Pj},\ \lambda = 0.5$
IMA capital (day t): $\max\left(WES_{t-1} + NMC_{t-1},\ m_c \times WES_{avg} + NMC_{avg}\right),\ m_c \geq 1.5$, averages over 60 days
SA delta (or vega) charge: $\sqrt{\sum_i \sum_j \rho_{ij}\delta_i\delta_j W_i W_j}$; risk weight e.g. $W = 0.02 \times \sqrt{20} \times 2.338 = 0.209$
Curvature charge, factor i: $\max\left(W_i\delta_i - \Delta V_i^+,\ -W_i\delta_i - \Delta V_i^-,\ 0\right)$
Illustration (source deltas and weight, illustrative revaluations): $\delta = 300{,}000$, $W = 0.209$, so $W\delta = 62{,}700$. Short options: $\Delta V^+ = 41{,}200$, $\Delta V^- = -88{,}900$, terms 21,500 and 26,200, charge USD 26,200. Long options: $\Delta V^+ = 80{,}000$, $\Delta V^- = -48{,}000$, terms $-17{,}300$ and $-14{,}700$, charge 0.
Default risk charge: exposure × LGD × default risk weight; **Residual add-on:** notional × risk weight (1% exotics)
P&L attribution: $-10\% \leq \dfrac{\text{Mean}(U)}{\text{SD}(V)} \leq 10\%, \qquad \dfrac{\text{Var}(U)}{\text{Var}(V)} < 20\%$; four or more breaches in 12 months: SA
Backtesting: 1-day VaR, last 12 months: more than 12 exceptions at 99% or more than 30 at 97.5%: SA
History: 1996: $MR = \max(VaR_{t-1}, mVaR_{avg}),\ m \geq 3$; Basel 2.5: $MR_{2.5} = \max(VaR_{t-1}, m_c VaR_{avg}) + \max(SVaR_{t-1}, m_s SVaR_{avg})$

Exam traps

- **Cascade weights (unit/time conversion):** scaling by $\sqrt{LH_j/10}$ (weights 1, 2, 4, 6, 12) instead of the increment $\sqrt{(LH_j - LH_{j-1})/10}$ (weights 1, 1, 2, 2, 6); or adding the scaled ES linearly (perfect correlation) instead of root-sum-of-squares (independence).
- **Missing category (dropped term):** a desk with no 10-day factors still has an ES_2 term, and it equals ES_1. Setting it to zero understates ES.
- **Polarity flip on tails:** heavy tails make 97.5% ES *greater* than 99% VaR, not smaller; under normality they nearly coincide (2.338 vs 2.326).
- **Sibling swap on backtesting:** FRTB backtests 1-day VaR at 99% and 97.5% (thresholds: more than 12, more than 30), never the stressed ES; P&L attribution needs *four or more* breaches, and both ratio tests apply. The 1996 rule (10 or more exceptions in 250 days: multiplier 4) is a different test.
- **Absolute claims and scope:** internal models approval never removes the standardized calculation; approval is desk by desk, not bank-wide; intent to trade alone no longer qualifies for the trading book; re-booking is for extraordinary circumstances only and gives no capital benefit.
- **Polarity flip on curvature:** short options attract the curvature charge; long options give zero. Wrong accord: stressed VaR, IRC and CRM are Basel 2.5, not Basel II.

Question A4.1 MR-18 b · S2 · [Aug-2026] [Map #5]

A market risk analyst at a Canadian bank is producing the FRTB internal models approach figures for the bank's credit trading desk. The desk's only risk factors are:

- sovereign investment-grade credit spreads
- corporate investment-grade credit spreads
- corporate non-investment-grade credit spreads
- credit spread volatilities

Using overlapping 10-day changes from the stressed period, the analyst has computed stressed 97.5% ES (CAD million): ES_1 (all risk factors shocked) $= 18.0$; $ES_3 = 12.0$; $ES_4 = 7.5$; $ES_5 = 3.0$. The ES_2 run was dropped from the overnight batch. The desk's 1-day 99% VaR used for backtesting is CAD 4.6 million. Which of the following is closest to the desk's liquidity-adjusted ES?

A. CAD 27.9 million

B. CAD 33.2 million

C. CAD 44.6 million

D. CAD 70.9 million

Solution **Answer: B**

Step 1 – Map each risk factor to its liquidity horizon (Table 18.1)

Sovereign IG spread: 20 days (category 2). Corporate IG spread: 40 days (category 3). Corporate non-IG spread: 60 days (category 4). Credit spread volatility: 120 days (category 5).

⇒ *The desk has no category 1 (10-day) risk factors at all.*

Step 2 – Infer the missing ES_2

ES_2 shocks categories 2 to 5 and holds category 1 constant. With no category 1 factors, holding them constant changes nothing, so $ES_2 = ES_1 = 18.0$.

⇒ *The missing run is not a zero: every factor on this desk still moves in the second 10-day block, because every factor has a horizon of at least 20 days.*

Step 3 – Cascade weights

$\text{weight}_j = (LH_j - LH_{j-1})/10$: $(10-0)/10 = 1$, $(20-10)/10 = 1$, $(40-20)/10 = 2$, $(60-40)/10 = 2$, $(120-60)/10 = 6$.

⇒ *Each weight counts how many extra independent 10-day blocks that category is shocked for.*

Step 4 – Squared, weighted terms

$1 \times 18.0^2 = 324.0$; $1 \times 18.0^2 = 324.0$; $2 \times 12.0^2 = 288.0$; $2 \times 7.5^2 = 112.5$; $6 \times 3.0^2 = 54.0$.

Step 5 – Liquidity-adjusted ES

$ES = \sqrt{324.0 + 324.0 + 288.0 + 112.5 + 54.0} = \sqrt{1102.5} = 33.20$, so CAD 33.2 million.

⇒ *About 1.84 times the 10-day all-factor ES of CAD 18.0 million: the long horizons of spreads and spread volatility add capital that a single 10-day horizon would miss.*

Interpretation

This is the desk's stressed ES feeding the weighted ES and the capital formula. The 1-day 99% VaR is a red herring: it is used only for the desk-level backtest, and scaling it by $\sqrt{10}$ is the Basel I / Basel 2.5 practice that FRTB abandons.

Why the other options are wrong

A is incorrect. Dropped term: treats the missing ES_2 as zero, $\sqrt{324.0 + 288.0 + 112.5 + 54.0} = \sqrt{778.5} = 27.90$.

C is incorrect. Unit/time conversion error: scales each ES_j by $\sqrt{LH_j/10}$ (weights 1, 2, 4, 6, 12) instead of the increment, $\sqrt{324 + 648 + 576 + 337.5 + 108} = \sqrt{1993.5} = 44.65$.

D is incorrect. Adds the scaled ES linearly (perfect correlation between blocks) instead of root-sum-of-squares (independence): $18.0 + 18.0 + 12.0\sqrt{2} + 7.5\sqrt{2} + 3.0\sqrt{6} = 70.93$.

LO: MR-18 b: Compare the various liquidity horizons proposed by the FRTB for different asset classes and explain how a bank can calculate its expected shortfall using the various horizons. · Reference: Hull, RMFI 6e, Ch.18, Section 18.3 and Table 18.1

Question A4.2 MR-18 a · S1 · [Aug-2026] [Map #5]

The CRO of a Japanese bank is preparing a board risk committee briefing on why FRTB replaces 10-day 99% VaR with 97.5% expected shortfall for market risk capital under the internal models approach. One desk head argues the change is cosmetic, since for his desk, whose losses are close to normally distributed, the two measures give almost the same JPY figure. Which of the following statements should the CRO include as correct?

A. With loss distributions whose tails are heavier than the normal, 97.5% ES can be considerably lower than 99% VaR, so the change mainly reduces capital for fat-tailed desks.

B. Like the Basel I VaR measure, the 97.5% ES is a current measure, calibrated to market movements over the one to four years immediately preceding the calculation date.

C. Because 97.5% ES and 99% VaR are almost exactly equal for any loss distribution, the switch affects capital only through the new liquidity horizons.

D. For a normal loss distribution the two measures nearly coincide, but with heavier tails 97.5% ES can be considerably greater, and FRTB calibrates the ES to stressed market conditions.

Solution **Answer: D**

Step 1 – Compare the two measures under normality

For normal losses with mean μ and standard deviation σ: 99% VaR $= \mu + 2.326\sigma$ and 97.5% ES $= \mu + 2.338\sigma$.
$\Rightarrow$ *The desk head is right for his desk: the multipliers differ by only 0.012σ.*

Step 2 – Drop the normality condition

For non-normal losses the measures are not equivalent; when the loss distribution has a heavier tail than the normal, 97.5% ES can be considerably greater than 99% VaR.
$\Rightarrow$ *ES averages the losses beyond the quantile, so it sees the shape of the tail that VaR ignores. The change is not cosmetic for fat-tailed desks.*

Step 3 – The calibration

The measure is *stressed* ES at 97.5%: calculated from how market variables moved during stressed market conditions, as for Basel 2.5 stressed VaR.

Interpretation

D is correct. The CRO's answer to the desk head: near-equality is a property of the normal case only; the capital consequences come from fat tails, stressed calibration and (separately) liquidity horizons.

Why the other options are wrong

A is incorrect. Polarity flip: heavier tails make 97.5% ES considerably *greater* than 99% VaR, raising capital for fat-tailed desks.
B is incorrect. Sibling swap: "current" calibration over the preceding one to four years describes the Basel I VaR; the FRTB measure is stressed ES.
C is incorrect. Scope/condition error: near-equality holds for normal distributions only; extending it to "any" distribution drops the condition.

LO: MR-18 a: Describe the changes to the Basel framework for calculating market risk capital under the Fundamental Review of the Trading Book (FRTB) and the motivations for these changes. · Reference: Hull, RMFI 6e, Ch.18, Section 18.1

Question A4.3 MR-18 a, c · S3, S4 · [Map #5]

The head of market risk at an Australian bank is reviewing the FRTB eligibility pack for four trading desks, all of which currently hold internal models approval. Results for the most recent 12 months:

Desk	1-day 99% VaR exceptions	1-day 97.5% VaR exceptions	P&L attribution situations outside ranges
Rates	11	34	1
FX	7	22	2
Equity	10	27	3

The credit desk holds a portfolio of illiquid loans that it has documented an intent to trade but is not able to trade or hedge on the desk. Which of the following conclusions is correct?

A. The rates desk must calculate capital using the standardized approach, even though its 99% VaR exceptions are within the permitted number.

B. The equity desk must calculate capital using the standardized approach, because its P&L attribution ratios fell

outside the ranges in three situations.

C. The credit desk may keep the loans in the trading book, because its documented intent to trade them meets the FRTB trading book test.

D. The FX desk's internal models approval fully replaces the standardized approach, so no standardized capital needs to be calculated for it.

Solution **Answer: A**

Step 1 – Backtesting test, desk by desk

Rule: more than 12 exceptions at 99% or more than 30 at 97.5% sends the desk to the standardized approach. Rates: $11 \leq 12$ passes at 99%, but $34 > 30$ fails at 97.5%. FX: 7, 22: passes. Equity: 10, 27: passes.
$\Rightarrow$ *Either condition alone is enough. The rates desk stays on the standardized approach until neither condition holds.*

Step 2 – P&L attribution test

Rule: four or more situations in 12 months where Mean(U)/SD(V) is outside $\pm 10\%$ or Var(U)/Var(V) is not below 20%. Rates 1, FX 2, Equity 3: all below four.
$\Rightarrow$ *No desk loses approval on P&L attribution this year; the equity desk is one breach from the threshold.*

Step 3 – Trading book boundary

Under FRTB, intent to trade is no longer sufficient: the bank must be able to trade and manage the risks on a trading desk. The credit desk's loans fail this test.

Interpretation

A is correct. The recommendation: move the rates desk to standardized capital, keep FX and equity on internal models while watching equity's attribution, and review the credit desk's loans for banking book treatment. Because approval is desk by desk, the rates result does not affect other desks. (Note the 1996 Amendment backtest: 10 or more exceptions in 250 days forced a multiplier of 4. That rule would have penalised the equity desk; FRTB's desk-level thresholds do not.)

Why the other options are wrong

B is incorrect. Scope/condition error: the trigger is four or more situations in 12 months; three does not trigger the standardized approach.
C is incorrect. Sibling swap: substitutes the pre-FRTB "intent to trade" test for the FRTB test, which requires the ability to trade and manage the risks on a desk.
D is incorrect. Absolute claim: even with internal models approval, banks must calculate capital under both approaches; the standardized approach provides a floor.

LO: MR-18 c: Explain the FRTB revisions to Basel regulations in the following areas: classification of positions in the trading book compared to the banking book; backtesting, profit and loss attribution, credit risk, and securitizations. · MR-18 a: Describe the changes to the Basel framework for calculating market risk capital under the Fundamental Review of the Trading Book (FRTB) and the motivations for these changes. · Reference: Hull, RMFI 6e, Ch.18, Sections 18.1, 18.3.1, 18.3.2, 18.4

Question A4.4 MR-18 a; ORR-21 c; ORR-22 a · S3, S5, S6 · [Map #5]

A risk consultant is preparing a one-page comparison of market risk capital under the 1996 Amendment to Basel I, Basel II, Basel 2.5 and FRTB for the chief compliance officer of a UK bank. The FRTB section discusses two GBP options desks under the standardized approach: one is short a book of at-the-money options and the other is long a similar book. Which of the following statements is correct to include?

A. FRTB moves from 99% to 97.5% confidence and from one 10-day horizon to liquidity horizons, but still lets banks scale one-day changes to 10 days with a square-root-of-10 multiplier.

B. Basel II overhauled market risk capital by adding a stressed VaR term to the current VaR term, alongside the introduction of its three pillars.

C. Under FRTB's standardized approach, the long-option desk attracts the curvature risk charge, because its gamma raises value relative to delta for both up and down moves.

D. Under FRTB's standardized approach, the short-option desk's curvature charge is the larger of its two shocked losses net of the delta effect, with negative results set to zero.

Solution **Answer: D**

Step 1 – Curvature mechanics (Section 18.2.2)

Shock risk factor i up and down by its risk weight W_i. Compute $W_i\delta_i - \Delta V_i^+$ and $-W_i\delta_i - \Delta V_i^-$; the charge is the greater, and a negative result counts as zero.

$\Rightarrow$ *Each term is the extra loss from curvature after the linear (delta) effect has been removed.*

Step 2 – Apply to the two desks

Short options (Figure 18.1a): both moves are worse than delta predicts, so both terms are positive and the charge is the larger (AB in the figure). Long options (Figure 18.1b): curvature increases value relative to delta for both moves, both terms are negative, so the charge is zero.

$\Rightarrow$ *Illustration from the formula box: short book terms 21,500 and 26,200 give USD 26,200; long book terms* $-17{,}300$ *and* $-14{,}700$ *give 0.*

Step 3 – Place the accords

1996 Amendment: 10-day 99% VaR, usually 1-day VaR $\times\sqrt{10}$, $m \geq 3$. Basel II: IRB credit risk, operational risk capital, three pillars; market risk retained. Basel 2.5: stressed VaR, IRC, CRM. FRTB: stressed 97.5% ES from overlapping 10-day changes, liquidity horizons, desk-level approval, standardized approach with delta, vega and curvature.

Interpretation

D is correct. The curvature charge is the standardized approach's gamma charge, and it falls on the short-gamma desk.

Why the other options are wrong

A is incorrect. Sibling swap: $\sqrt{10}$ scaling of one-day changes is the Basel I / Basel 2.5 practice; FRTB requires actual overlapping 10-day changes from the stressed period.

B is incorrect. Sibling swap (wrong accord): stressed VaR is a Basel 2.5 change (implemented by end-2011); Basel II's innovations were credit risk weights, operational risk capital and the three pillars.

C is incorrect. Polarity flip: the reason given is why a long option book receives a *zero* curvature charge; it is the short book that attracts it.

LO: MR-18 a: Describe the changes to the Basel framework for calculating market risk capital under the Fundamental Review of the Trading Book (FRTB) and the motivations for these changes. · ORR-21 c: Describe measures introduced in the 1995 and 1996 amendments, including guidelines for netting of credit exposures and methods for calculating market risk capital for assets in the trading book. · ORR-22 a: Describe and calculate the stressed VaR introduced in Basel 2.5 and calculate the market risk capital charge. · Reference: Hull, RMFI 6e, Ch.18, Sections 18.1, 18.2.2, 18.3; Carey (GARP), Ch.21 (1996 Amendment, Basel II) and Ch.22 (Section 22.2 Basel 2.5)

A5 Merton Model, Distance to Default and Risk-Neutral PD

Readings: CR-5 e, f · Introduction to Credit Risk Modeling and Assessment (Doumpos et al., Ch.1, 5.6.3); CR-9 k, l, m, n (and e for the hazard conversion) · Estimating Default Probabilities (Hull, RMFI Ch.17, 9.7.2 and 9.8)

AUG 2026 RECALL | MAP TIER A · #2 · 8/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: Merton model.
Map #2: "Specific variants reported: 'finding distance to default and assign a rating based on the score'; 'Merton model, calculation of V_0 and sigma(V)' (the simultaneous-equation form); 'Merton model risk-neutral prob'; 'hazard rate and PD based on Merton model'; 'leverage ratio → ROE'. Your formula card carries both the full form and the short form DtD = ln(V/F)/σ: the short form is the shortcut GARP rewards when drift ≈ 0. Watch the scope trap: 'Merton cannot be applied beyond one year' is a false absolute."
Shapes covered here:
S1 distance to default, then assign a rating band from the score (Q6; KMV DD in Q4)
S2 back out V_0 and σ_V from the simultaneous equations using the solver's converged values (Q5)
S3 risk-neutral PD $N(-d_2)$ versus the real-world PD with μ (Q1, Q3)
S4 Merton PD converted into an average hazard rate (Q1)
S5 leverage ratio to return on equity through the delta $N(d_1)$ (Q2)
S6 short-form DtD $\ln(V/F)/\sigma_V$ versus the full form (Q6; full form in Q4)
S7 scope trap: Merton over horizons beyond one year (Q6; two- and three-year horizons in Q1, Q2).

Revision notes

- **Equity as a call option** (CR-9 l, CR-5 e): with one zero-coupon bond repaying D at T, $E_T = \max(V_T - D, 0)$. If $V_T < D$ it is rational to default and equity is worth zero. Doumpos: the shareholders are the *buyers* of the option and the creditors are the *writers*.
- **Two unknowns, two equations** (CR-9 l, S2): V_0 and σ_V are not observable. E_0 (market capitalisation) and σ_E (from historical equity data) are. Equation 9.4 (the BSM equity value) and equation 9.5 ($\sigma_E E_0 = N(d_1)\sigma_V V_0$, from Ito's lemma with equity delta $N(d_1)$) are solved together: Hull uses Excel Solver minimising $[F]^2 + [G]^2$; Doumpos uses "a simple iterative Newton-type algorithm".
- **Inputs in Doumpos:** face value of debt L is "usually taken as the short-term debt of the firm"; T is the time in years up to the repayment of the loan; r is the T-bill yield. Doumpos writes A, L, σ_A where Hull writes V_0, D, σ_V.
- **Leverage multiplier** (S5): equation 9.5 rearranges to $\sigma_E = N(d_1)(V_0/E_0)\sigma_V$. The same delta gives $\Delta E/E_0 \approx N(d_1)(V_0/E_0)(\Delta V/V_0)$: a higher leverage ratio V_0/E_0 amplifies both the volatility and the return on equity relative to the assets.
- **Debt and expected loss** (CR-9 l): market value of debt $= V_0 - E_0$; PV of the promised payment $= De^{-rT}$; expected loss = (PV promised − market value)/PV promised. Hull Example 9.4: 9.51 versus 9.40, about 1.2%.
- **Risk-neutral PD** (CR-9 m, S3): the company defaults when the option is not exercised; the probability is $N(-d_2)$. It is risk-neutral "because it is calculated from an option pricing model" with assets growing at r.
- **Real-world PD** (CR-5 e, CR-9 k): replace r with the expected return on assets μ (Doumpos eq. 5.3); μ is estimated as the rate of change in A over past periods after solving 5.1 and 5.2. Since real-world asset growth is usually higher than r, risk-neutral PDs are *higher* than real-world PDs.
- **Which PD to use** (CR-9 k): risk-neutral for valuation (pricing, credit spreads, CVA); real-world for scenario analysis (Hull 9.7.2). Either can come from Merton "provided that the model is calibrated appropriately".
- **Distance to default** (CR-9 m, S1): "the number of standard deviations the asset price must change for default to be triggered T years in the future"; one definition is d_2. As DD declines, default becomes more likely. Example 9.4: one-year DD $= 1.14$. The horizon is T years: nothing in either reading restricts Merton to one year (S7).
- **Short-form DD** (S6): with a one-year horizon and a small drift, DD $\approx \ln(V/F)/\sigma_V$. Use it when the question says "approximation" or drift ≈ 0; use the full d_2 when r or μ is given and asked for.

- **Moody's KMV EDF** (CR-5 e, f): the best-known implementation of Merton. Two differences: the default point is short-term debt plus half of long-term debt, and DD is mapped to PD through an *empirical* distribution of historical default frequencies, not the normal distribution.
- **Quality of Merton PDs** (CR-9 n): Merton and its extensions produce a good *ranking* of default probabilities, so a monotonic transformation converts the output to a real-world or risk-neutral PD. Moody's KMV and Kamakura sell that transformation into a real-world PD; the assumption is that the rankings of risk-neutral, real-world and Merton PDs are all the same. These market-based estimates do not target stability and respond faster than agency ratings.
- **Limitations and extensions** (CR-5 e): basic Merton assumes one zero-coupon bond, default only at T, constant σ_V, unobservable V_0, σ_V. Extensions: default whenever assets fall below a barrier; debt payments at more than one time; implied volatilities (Hull et al. 2004). Evidence is mixed: Hillegeist et al. find structural models beat empirical models; Agarwal and Taffler, Campbell et al. find little added information; reduced form approaches were found superior in most cases.
- **Comparison of approaches** (CR-5 f): structural models treat default as endogenous (assets versus debt); reduced form models treat it as a random event driven by an exogenous Poisson jump process calibrated to bond and credit-derivative prices. CreditMetrics is a mark-to-market model on a rating transition matrix, recovery rates and yield spreads (not a rating tool or pricing model). CreditRisk+ (Credit Suisse Financial Products, 1997) is default-mode, Poisson (mean equals variance), does not use the capital structure and needs minimal data.
- **Hazard conversion** (CR-9 e, S4): a cumulative PD $Q(T)$ from Merton converts to an average hazard rate $\bar{\lambda} = -\ln(1-Q)/T$ (Hull eq. 9.2), not Q/T.

Formulas

Equity value (Hull 9.4; Doumpos 5.1).

$$E_0 = V_0 N(d_1) - De^{-rT} N(d_2), \qquad d_1 = \frac{\ln(V_0/D) + (r + \sigma_V^2/2)T}{\sigma_V \sqrt{T}}, \qquad d_2 = d_1 - \sigma_V \sqrt{T}$$

Equity volatility (Hull 9.5; Doumpos 5.2). $\sigma_E E_0 = N(d_1)\,\sigma_V V_0 \;\Rightarrow\; \sigma_E = N(d_1)\dfrac{V_0}{E_0}\sigma_V$

Back-solve once the solver reports $N(d_1)$, $N(d_2)$. $V_0 = \dfrac{E_0 + De^{-rT}N(d_2)}{N(d_1)}, \qquad \sigma_V = \dfrac{\sigma_E E_0}{N(d_1)V_0}$

Leverage to return on equity (from delta $N(d_1)$). $\dfrac{\Delta E}{E_0} \approx N(d_1)\dfrac{V_0}{E_0}\dfrac{\Delta V}{V_0}$

Debt. Market value $= V_0 - E_0$; PV promised $= De^{-rT}$; expected loss $= \dfrac{De^{-rT} - (V_0 - E_0)}{De^{-rT}}$

PD. Risk-neutral $= N(-d_2)$. Real-world (Doumpos 5.3): PD $= N\left(-\dfrac{\ln(A/L) + (\mu - \sigma_A^2/2)T}{\sigma_A\sqrt{T}}\right)$

Distance to default (Hull 9.8.3). Full: DD $= \dfrac{\ln V_0 - \ln D + (r - \sigma_V^2/2)T}{\sigma_V\sqrt{T}}$. Short form ($T = 1$, drift ≈ 0): DD $\approx \dfrac{\ln(V/F)}{\sigma_V}$

Moody's KMV default point. DP = short-term debt + $0.5 \times$ long-term debt; DD to EDF mapping is empirical.

Average hazard rate from a cumulative PD (Hull 9.2). $\bar{\lambda}(T) = -\dfrac{1}{T}\ln\big(1 - Q(T)\big)$

Book check (Example 9.4). $E_0 = 3, \sigma_E = 0.80, r = 5\%, T = 1, D = 10$: $V_0 = 12.40, \sigma_V = 0.2123, d_2 = 1.1408$, PD $= 12.7\%$, debt $= 9.40$, PV promised $= 9.51$, expected loss $\approx 1.2\%$.

Exam traps

- r **versus** μ **(wrong-input twin):** $N(-d_2)$ built on r is risk-neutral and *higher*. A "real-world" PD for scenario analysis needs μ; a "risk-neutral" PD for valuation or CVA keeps r.
- d_1 **versus** d_2**, and the sign:** PD is $N(-d_2)$, not $N(-d_1)$ (too low) and not $N(d_2)$ (the probability of *no* default). In equation 9.5 the multiplier is $N(d_1)$, not $N(d_2)$ and not 1.
- σ_E **versus** σ_V**:** equity volatility only calibrates σ_V. It never goes into d_2. A stem that lists both is planting a

red herring.

- **Hazard conversion:** the average hazard rate is $-\ln(1-Q)/T$. Dividing the cumulative PD by T understates it; forgetting $/T$ overstates it.
- **Default point:** KMV uses short-term debt plus *half* of long-term debt, not total debt (DD too low) and not short-term debt alone (DD too high).
- **Scope (absolute claim):** "Merton cannot be applied beyond one year" is false: DD is defined for default "T years in the future" and the drift scales with T, volatility with $\sqrt{T}$.

Question A5.1

CR-9 m; CR-9 e · S3, S4, S7 · [Aug-2026] [Map #2]

A risk analyst at a Singapore bank is estimating the risk-neutral default probability of a corporate borrower using Merton's model. The bank's CVA desk stores each counterparty's default risk as a constant average hazard rate, so the analyst must also convert the Merton output. The asset value and asset volatility have already been solved from the equity data:

- Value of assets V_0: SGD 150 million; asset volatility σ_V: 25%
- Zero-coupon debt due in 2 years: SGD 120 million
- Risk-free rate: 3% per annum, continuously compounded

Which of the following is closest to the two-year risk-neutral probability of default and the corresponding average hazard rate per year?

A. PD 16.4%; average hazard rate 9.0%

B. PD 26.6%; average hazard rate 15.5%

C. PD 26.6%; average hazard rate 13.3%

D. PD 33.0%; average hazard rate 20.0%

Solution

Answer: B

Step 1 – d_1

$$d_1 = \frac{\ln(V_0/D) + (r + \sigma_V^2/2)T}{\sigma_V\sqrt{T}} = \frac{\ln(150/120) + (0.03 + 0.25^2/2)(2)}{0.25\sqrt{2}} = \frac{0.2231 + 0.1225}{0.3536} = 0.9776$$

Step 2 – d_2 (distance to default)

$$d_2 = d_1 - \sigma_V\sqrt{T} = 0.9776 - 0.3536 = 0.6241$$

$\Rightarrow$ *Assets need to fall about 0.62 two-year standard deviations for default to be triggered.*

Step 3 – Risk-neutral PD

$$Q(2) = N(-d_2) = N(-0.6241) = 0.2663$$

$\Rightarrow$ *A 26.6% risk-neutral probability that $V_T < D$ at year 2, so the equity call option is not exercised.*

Step 4 – Average hazard rate

$$\bar{\lambda} = -\frac{1}{T}\ln\big(1 - Q(T)\big) = -\frac{1}{2}\ln(1 - 0.2663) = \frac{0.3096}{2} = 0.1548$$

$\Rightarrow$ *A constant default intensity of 15.5% per year reproduces a 26.6% two-year cumulative default probability.*

Interpretation

Because the PD comes from an option pricing model with assets growing at r, both numbers are risk-neutral, which is the right type for a CVA desk (valuation). The two-year horizon is no obstacle: the drift is multiplied by T and the volatility by $\sqrt{T}$.

Why the other options are wrong

A is incorrect. Used $N(-d_1) = N(-0.9776) = 0.164$ instead of $N(-d_2)$; the hazard rate $-\ln(1-0.164)/2 = 0.090$ inherits the error.

C is incorrect. Correct PD, but divided the cumulative PD by T: $0.2663/2 = 0.133$. This ignores survival compounding in $Q = 1 - e^{-\bar{\lambda}T}$ and understates the hazard rate.
D is incorrect. Scaled volatility by T instead of $\sqrt{T}$: $d_2 = (0.2231 - 0.0025)/0.50 = 0.4413$, PD $= N(-0.4413) = 0.330$, hazard $-\ln(0.670)/2 = 0.200$.

LO: CR-9 m: Calculate distance to default and default probability using the Merton model. CR-9 e: Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities. · Reference: Hull, RMFI 6e, Ch.17, 9.8 and 9.8.3; 9.2.1 eq. 9.2

Question A5.2

CR-9 I · S5, S7 · [Map #2]

A fixed-income portfolio manager at a European insurer is valuing a block of a company's zero-coupon debt using Merton's model. The CRO also wants to know how leverage transmits asset shocks to the shareholders. Inputs:

- Value of assets: EUR 50 million; asset volatility: 30%
- Face value of debt due in 3 years: EUR 40 million
- Risk-free rate: 2% per annum, continuously compounded

Which of the following is closest to the market value of the company's debt today, and to the approximate instantaneous return on the company's equity if the value of its assets falls by 6%?

A. Debt EUR 37.7 million; return on equity -14.4%

B. Debt EUR 35.0 million; return on equity -15.8%

C. Debt EUR 33.6 million; return on equity -14.4%

D. Debt EUR 33.6 million; return on equity -18.3%

Solution

Answer: C

Step 1 – d_1 and d_2

$$d_1 = \frac{\ln(50/40) + (0.02 + 0.30^2/2)(3)}{0.30\sqrt{3}} = \frac{0.2231 + 0.1950}{0.5196} = 0.8047$$
$$d_2 = 0.8047 - 0.5196 = 0.2851, \qquad N(d_1) = 0.7895, \quad N(d_2) = 0.6122$$

Step 2 – Equity value

$$E_0 = V_0 N(d_1) - De^{-rT}N(d_2) = 50(0.7895) - 40e^{-0.06}(0.6122) = 39.475 - 37.671(0.6122) = 16.41$$

$\Rightarrow$ *The shareholders' call on the assets is worth EUR 16.41 million.*

Step 3 – Debt value

$$\text{Debt} = V_0 - E_0 = 50 - 16.41 = 33.59$$

$\Rightarrow$ *Versus a risk-free value of* $40e^{-0.06} = 37.67$*: an expected loss of* $(37.67 - 33.59)/37.67 = 10.8\%$.

Step 4 – Leverage ratio and multiplier

$$\frac{V_0}{E_0} = \frac{50}{16.41} = 3.046, \qquad N(d_1)\frac{V_0}{E_0} = 0.7895 \times 3.046 = 2.405$$

$\Rightarrow$ *Each 1% move in asset value moves equity by about 2.4%. The same multiplier gives* $\sigma_E = 2.405 \times 30\% = 72.2\%$ *(eq. 9.5).*

Step 5 – Return on equity

$$\frac{\Delta E}{E_0} \approx N(d_1)\frac{V_0}{E_0}\frac{\Delta V}{V_0} = 2.405 \times (-6\%) = -14.4\%$$

Interpretation

Debt is worth EUR 33.6 million; creditors are effectively short a put on the assets. The leverage ratio of about 3 is damped by the delta $N(d_1)$ because part of any asset loss is absorbed by the creditors' put; a 6% asset loss becomes roughly a 14.4% equity loss. (The $\Delta E/E_0$ relation is the first-order use of the delta $\partial E/\partial V = N(d_1)$ from Hull's derivation of eq. 9.5.)

Why the other options are wrong

A is incorrect. Reported the risk-free PV of the promised payment, $40e^{-0.06} = 37.67$, as the debt value, ignoring default risk.

B is incorrect. Did not discount the strike: $E_0 = 50(0.7895) - 40(0.6122) = 14.99$, so debt $= 35.01$ and the multiplier becomes $0.7895 \times 50/14.99$, giving -15.8%.

D is incorrect. Used the raw leverage ratio without the delta: $3.046 \times (-6\%) = -18.3\%$, treating equity as if it absorbed every asset loss one-for-one in value terms.

LO: CR-9 l: Calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model. · Reference: Hull, RMFI 6e, Ch.17, 9.8, eqs. 9.4 and 9.5, Example 9.4

Question A5.3

CR-5 e; CR-9 k · S3 · [Map #2]

A senior risk analyst at a Japanese bank is preparing a one-year stress scenario and needs the *real-world* default probability of a corporate borrower. Merton's model has been calibrated as follows (JPY billion):

- Value of assets: 800; asset volatility: 20%; equity volatility: 45%
- Face value of debt due in 1 year: 600
- Risk-free rate: 4%; expected return on assets: 10% (both continuously compounded)

Which of the following is closest to the probability of default to use?

A. 2.1%

B. 3.3%

C. 6.2%

D. 26.2%

Solution

Answer: B

Step 1 – Choose the drift

Scenario analysis uses real-world estimates (Hull 9.7.2), so the drift is $\mu = 10\%$, not r (Doumpos eq. 5.3).

Step 2 – Real-world distance to default

$$\text{DD} = \frac{\ln(800/600) + (0.10 - 0.20^2/2)(1)}{0.20\sqrt{1}} = \frac{0.2877 + 0.0800}{0.20} = 1.8384$$

Step 3 – Real-world PD

$$\text{PD} = N(-1.8384) = 0.0330$$

⇒ A 3.3% chance, in the real world, that assets end the year below the JPY 600 billion owed.

Interpretation

The risk-neutral value with r would be 6.2%: higher, because assets grow only at r in the risk-neutral world (Hull 9.8.2). The 45% equity volatility is not an input here: it was used to calibrate σ_V.

Why the other options are wrong

A is incorrect. Correct drift μ, but used $N(-d_1)$: $d_1 = 1.8384 + 0.20 = 2.0384$, $N(-2.0384) = 0.021$.

C is incorrect. Used $r = 4\%$ instead of μ: DD $= (0.2877 + 0.02)/0.20 = 1.5384$, PD $= 0.062$, a risk-neutral PD.

D is incorrect. Plugged equity volatility 45% in place of asset volatility: DD $= (0.2877 + 0.10 - 0.1013)/0.45 = 0.637$, PD $= 0.262$.

LO: CR-9 k: Describe the difference between real-world and risk-neutral default probabilities and determine which one to use in the analysis of credit risk. CR-5 e: Apply the Merton model to calculate default probability and the distance to default and describe the limitations of using the Merton model. · Reference: Doumpos et al., Ch.1, 5.6.3.1 eq. 5.3; Hull, RMFI 6e, Ch.17, 9.7.2 and 9.8.2

Question A5.4 CR-5 e, f; CR-9 m · S1, S6 · [Map #2]

A credit analyst at a US asset manager is replicating the Moody's KMV approach for a borrower. Balance sheet and market data (USD million):

- Value of assets: 2,400; asset volatility: 18%; equity volatility: 40%
- Short-term debt: 900; long-term debt: 1,000
- Risk-free rate: 4% (continuous); horizon: 1 year

Using the Moody's KMV default point and the full-form distance to default defined as d_2, which of the following is closest to the distance to default?

A. 1.43

B. 3.13

C. 3.31

D. 5.58

Solution **Answer: B**

Step 1 – Default point

$$\text{DP} = \text{STD} + 0.5 \times \text{LTD} = 900 + 0.5(1{,}000) = 1{,}400$$

$\Rightarrow$ *KMV treats only half of long-term debt as part of the trigger, which "better approximates the actual loan obligations of a firm".*

Step 2 – Numerator

$$\ln(2{,}400/1{,}400) + (0.04 - 0.18^2/2)(1) = 0.5390 + 0.0238 = 0.5628$$

Step 3 – Distance to default

$$\text{DD} = 0.5628/(0.18\sqrt{1}) = 3.13$$

Interpretation

Assets sit 3.13 standard deviations above the default point. KMV would then map this DD to an EDF through its empirical database of historical default frequencies, rather than reading $N(-3.13) = 0.09\%$ off the normal distribution. The short form $\ln(2{,}400/1{,}400)/0.18 = 2.99$ is close, but the question asks for the full form, so the drift stays in.

Why the other options are wrong

A is incorrect. Used total debt (1,900) as the default point: $(\ln(2{,}400/1{,}900) + 0.0238)/0.18 = 1.43$.
C is incorrect. Used d_1 instead of d_2, adding $\sigma_V^2/2$: $(0.5390 + 0.04 + 0.0162)/0.18 = 3.31$.
D is incorrect. Used short-term debt alone (900) as the default point: $(\ln(2{,}400/900) + 0.0238)/0.18 = 5.58$.

LO: CR-5 f: Compare and contrast different approaches to credit risk modeling, such as those related to the Merton model, Credit Risk Plus (CreditRisk+), CreditMetrics, and the Moody's-KMV model. CR-9 m: Calculate distance to default and default probability using the Merton model. · Reference: Doumpos et al., Ch.1, 5.6.3.1; Hull, RMFI 6e, Ch.17, 9.8.3

Question A5.5 CR-9 l · S2 · [Map #2]

A treasury analyst at a UK bank is calibrating Merton's model for a listed borrower. Market and balance sheet data (GBP million):

- Market value of equity: 25; equity volatility: 80%
- Zero-coupon debt due in 1 year: 100
- Risk-free rate: 3% per annum, continuously compounded; expected return on assets: 8%

The analyst's Newton-type solver was seeded with $V_0 = E_0 + De^{-rT} = 122.04$ and $\sigma_V = \sigma_E E_0/V_0 = 16.39\%$. After convergence, it reports $N(d_1) = 0.9018$ and $N(d_2) = 0.8661$. Which of the following is closest to the converged value of the firm's assets and its asset volatility?

A. $V_0 =$ GBP 120.9 million; $\sigma_V = 18.3\%$

B. $V_0 =$ GBP 120.9 million; $\sigma_V = 16.5\%$

C. $V_0 =$ GBP 122.0 million; $\sigma_V = 16.4\%$

D. $V_0 =$ GBP 123.8 million; $\sigma_V = 17.9\%$

Solution **Answer: A**

Step 1 – PV of the promised payment

$$De^{-rT} = 100e^{-0.03} = 97.045$$

Step 2 – Asset value from the equity equation (Hull 9.4)

$$E_0 = V_0 N(d_1) - De^{-rT}N(d_2) \;\Rightarrow\; V_0 = \frac{E_0 + De^{-rT}N(d_2)}{N(d_1)} = \frac{25 + 97.045(0.8661)}{0.9018} = \frac{109.050}{0.9018} = 120.93$$

⇒ The market values the firm's assets at GBP 120.9 million, so the leverage ratio V_0/E_0 is about 4.8.

Step 3 – Asset volatility from the volatility equation (Hull 9.5)

$$\sigma_E E_0 = N(d_1)\sigma_V V_0 \;\Rightarrow\; \sigma_V = \frac{\sigma_E E_0}{N(d_1)V_0} = \frac{0.80(25)}{0.9018(120.93)} = \frac{20.0}{109.05} = 0.1834$$

⇒ An 80% equity volatility corresponds to only 18.3% volatility of the underlying assets: leverage magnifies asset risk into equity risk.

Interpretation

At the converged pair, $d_2 = 1.108$ and the one-year risk-neutral PD is $N(-1.108) \approx 13.4\%$. The 8% expected return is a red herring for calibration: it matters only if a real-world PD is later required (Doumpos eq. 5.3).

Why the other options are wrong

B is incorrect. Correct V_0, but dropped the delta in eq. 9.5: $\sigma_V = \sigma_E E_0/V_0 = 20/120.93 = 16.5\%$.
C is incorrect. Reported the solver's starting values (iteration zero) as the answer: $V_0 = 25 + 97.045 = 122.04$, $\sigma_V = 20/122.04 = 16.4\%$.
D is incorrect. Did not discount the debt in eq. 9.4: $V_0 = (25 + 100 \times 0.8661)/0.9018 = 123.76$, then $\sigma_V = 20/(0.9018 \times 123.76) = 17.9\%$.

LO: CR-9 l: Calculate the value of a firm's debt and equity, the volatility of firm value, and the volatility of firm equity using the Merton model. · Reference: Hull, RMFI 6e, Ch.17, 9.8, eqs. 9.4 and 9.5, footnote 21; Doumpos et al., Ch.1, 5.6.3.1 eqs. 5.1 and 5.2

Question A5.6 CR-5 e, f; CR-9 m, n · S1, S6, S7 · [Map #2]

A senior credit analyst at a Singapore bank is presenting the bank's internal scoring step to a group of newly hired analysts. The step converts a Merton distance to default into a rating band:

- Internal bands: DD of 4.0 or more: A; 3.0 to below 4.0: BBB; 2.0 to below 3.0: BB; below 2.0: B
- Borrower data (SGD million): value of assets 900; default point 600; asset volatility 20%; risk-free rate 4%; horizon 1 year

Which of the following statements should the analyst include in the presentation?

A. The short-form distance to default is about 2.03, placing the borrower in band BB; adding the drift term $(r - \sigma_V^2/2)T$ raises it by about 0.10 and leaves the band unchanged.

B. The short-form distance to default is about 2.03, placing the borrower in band BB; the same scoring step could not be used for its three-year notes, because Merton's model cannot be applied beyond one year.

C. The short-form distance to default is about 2.03, placing the borrower in band BB; a rise in asset volatility to 25% would increase the distance to default and support an upgrade toward BBB.

D. The short-form distance to default is about 2.03, placing the borrower in band BB; Moody's KMV would convert the same score into a default probability by reading $N(-\text{DD})$ off the standard normal distribution.

Solution **Answer: A**

Step 1 – Short-form distance to default

$$\text{DD} \approx \frac{\ln(V/F)}{\sigma_V} = \frac{\ln(900/600)}{0.20} = \frac{0.4055}{0.20} = 2.03 \;\Rightarrow\; \text{band BB}$$

Step 2 – Full form check

$$\text{DD} = \frac{0.4055 + (0.04 - 0.20^2/2)(1)}{0.20} = \frac{0.4055 + 0.0200}{0.20} = 2.13 \;\Rightarrow\; \text{still BB}$$

$\Rightarrow$ *The drift contributes* $0.02/0.20 = 0.10$ *standard deviations: small over one year, which is why the short form is the accepted shortcut. Both forms agree on the band, which is what the rating step needs: Merton is judged on its* ranking *of default risk (Hull 9.8.1).*

Interpretation

A is correct: the shared first clause is true, and the second clause is the only one consistent with the readings.

Why the other options are wrong

B is incorrect. Scope/absolute-claim error. Hull defines distance to default for default "T years in the future", Doumpos sets T as the time up to repayment of the loan, and Hull's own end-of-chapter problem uses two-year debt. The drift scales with T and volatility with $\sqrt{T}$; nothing limits Merton to one year.

C is incorrect. Polarity flip. Higher asset volatility *lowers* DD: $\ln(1.5)/0.25 = 1.62$, which moves the borrower down to band B. As DD declines, the company becomes more likely to default.

D is incorrect. Sibling swap. Mapping through the normal distribution is the plain Merton/Doumpos eq. 5.3 approach; Moody's KMV uses historical data to derive an empirical distribution of default frequencies.

LO: CR-9 m: Calculate distance to default and default probability using the Merton model. CR-9 n: Assess the quality of the default probabilities produced by the Merton model, the Moody's KMV model, and the Kamakura model. CR-5 e: Apply the Merton model to calculate default probability and the distance to default and describe the limitations of using the Merton model. · Reference: Hull, RMFI 6e, Ch.17, 9.8.1 and 9.8.3; Doumpos et al., Ch.1, 5.6.3.1

A6 CVA, DVA, BCVA and CVA as a Spread

Readings: CR-20 a, b, c, d, e, f, g, h, i · CVA (Gregory, The xVA Challenge 4e, Ch.17: 20.2 Credit Value Adjustment; 20.2.3 CVA as a Spread; 20.2.5 Credit Spread Effects; 20.2.6 Loss Given Default; 20.3 Debt Value Adjustment; 20.3.3 Bilateral CVA Formula; 20.3.4 Close-Out and Default Correlation; 20.4 CVA Allocation)

AUG 2026 RECALL | MAP TIER A · #6 · 8/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: BCVA spread numerical.
Map #6: "Reported: '2 numericals on BCVA, CVA (couldn't do them)'; 'UCVA as a spread: calculation (formula)', listed twice in the same consolidated sheet, so the spread form specifically is the tested one. Closed-book: UCVA $= \sum(1-\text{RR})\cdot\text{EE}\cdot\text{PD}\cdot\text{DF}$; UCVA spread $= -\text{EPE}\times$ credit spread; BCVA $=$ CVA $+$ DVA with survival adjustments and the sign convention CVA negative, DVA positive. Intermediate-result trap: BCVA planted as a distractor when the change from UCVA is asked."
Shapes covered here:
S1 UCVA from an EPE profile and a PD term structure, $\sum$ LGD $\times$ EPE $\times$ PD (Q1)
S2 CVA as a running spread: $-$average EPE $\times$ spread, and upfront CVA divided by the risky annuity (Q3, Q4)
S3 BCVA $=$ CVA $+$ DVA with the sign convention CVA negative, DVA positive (Q2, Q5)
S4 BCVA with survival (first-to-default) adjustments (Q5)
S5 BCVA as a spread, plain form as seen in August (Q3)
S6 change from UCVA to BCVA, with the BCVA itself planted as a distractor (Q5).

Revision notes

- **The specific: CVA as a spread** (CR-20 c, i; S2). For pricing, an upfront CVA is often charged as a running spread. One simple way is to divide by the risky annuity (duration) of the maturity. This ignores the "CVA of the CVA": charging CVA raises the trade value, which raises the CVA. The exact answer solves recursively; an xVA desk charging the first-order CVA *always* loses a little when the trade is booked. Gregory's example: divide by risky annuity -1.92 bps, exact (recursive) -1.96, EPE approximation -2.01.
- **The average-EPE approximation** (eq. 20.4): assuming constant EPE, UCVA $=$ $-$average EPE $\times$ spread, in the units of the spread (for the maturity of the instrument). It uses the *undiscounted* EPE and has *no LGD input*: with $\text{LGD}_{\text{actual}} = \text{LGD}_{\text{mkt}}$ the LGDs cancel to first order. It separates the credit component (spread) from the market risk component (average EPE).
- **Why CVA is hard** (CR-20 a): a bond's cash flow is fully at risk in default, but a swap's cash flow is only partly at risk because it is partially offset by opposing flows; the fraction at risk depends on curve shape, forward rates and volatilities. CVA is now computed with market-implied (risk-neutral) parameters as an exit price (IFRS 13), although market-implied PDs are much higher than real-world ones, rarely hedgeable, and banks mostly warehouse credit risk.
- **CVA** (CR-20 b): driven by three questions: does the counterparty default (PD), what is the exposure then (EPE), how much is lost (LGD $= 1 - R$). With independence (no wrong-way risk) only a single average LGD is needed and CVA becomes a weighted average of the EPE profile: market risk (EPE) times credit risk (PD). CVA is a *negative* adjustment. The PD in eq. 20.3 is the *marginal* PD for each interval.
- **Direct vs path-wise:** the direct approach simulates default times and values the portfolio only at the default date; in the book's 10-year swap it had a standard deviation six times smaller, a speed-up of $6 \times 6 = 36$ times in valuations (Amdahl's law limits the overall gain). It is less obviously applied to path-dependent (collateralised, exotic) portfolios.
- **Credit curve shape** (CR-20 d): with the same five-year spread (150 bps), the *upward-sloping* curve gives the *largest* CVA on a 10-year receiver swap (-24.6 vs flat -20.0 vs inverted -15.7 bps), mainly from the largest extrapolated 10-year spread; defaults are back-loaded. CVA generally rises with the spread, but as default approaches it converges to the exposure (zero for a par trade): large gamma and jump-to-default risk.
- **LGD** (CR-20 d): LGD_{mkt} calibrates the PD from spreads; $\text{LGD}_{\text{actual}}$ is the loss actually expected. When equal, a higher LGD lowers the implied PD but raises the loss: 60% to 50% changes CVA by under 2%. A different $\text{LGD}_{\text{actual}}$ is justified by seniority (e.g. project finance, about 75% recovery vs about 40% for corporates),

credit enhancements, or work-out experience. Avoid double counting: ratings mapped to spreads should reflect PD only (S&P definitions are PD-based; Moody's may include recovery).

- **DVA** (CR-20 e): the mirror of CVA using ENE, the party's *own* PD and own LGD; it is *positive* because ENE is negative. Accounting drove it: FAS 157 (2006), IFRS 13 (2013). "My CVA is your DVA" ($\text{CVA}_A = -\text{DVA}_B$) creates *price symmetry*. DVA is a double-edged sword: monetising it means defaulting, unwinds and novations (monolines, MBIA), close-out, or hedging by selling CDS on oneself (not possible), so banks sold protection on correlated names (some on Lehman). Basel derecognises DVA from capital; the FASB puts DVA in other comprehensive income; shareholders get nothing in default while bondholders may see DVA as real. Market practice treats DVA as a funding benefit (FVA).
- **UCVA vs BCVA** (CR-20 f; S4): bilateral CVA and DVA include the survival probability of the *other* party up to the start of each interval (the "first-to-default" effect); UCVA and UDVA do not. The survival term makes CVA sensitive to one's own credit quality (seen as unnatural; regulation may prohibit it). Default dependency and close-out (2002 ISDA lets the claim reflect the determining party's creditworthiness) complicate it; Gregory and German (2013) find the formulas *without* survival probabilities probably the best simple approximation. Survey practice is split between contingent and non-contingent.
- **BCVA pricing intuition** (CR-20 g; S5): with average EPE $=$ $-$average ENE, $\text{BCVA} = -\text{EPE} \times (\text{Spread}_C - \text{Spread}_P)$: the weaker party pays the stronger. In Table 20.3 the receive-fixed swap's BCVA is close to zero (large ENE, so large DVA) and the OTM receive-fixed swap's BCVA is positive ("paying through mid").
- **Netting** (CR-20 h): netting can only reduce CVA: $\text{CVA}_{NS} \geq \sum_i \text{CVA}_i^{\text{SA}}$ (less negative). It is captured by replacing standalone EPE with netting-set EPE, which needs Monte Carlo at the netting-set level; benefits are strong for balanced portfolios, weak for directional ones.
- **Incremental CVA** (CR-20 i): netting-set CVA after the trade minus before. It is never more negative than standalone CVA, depends on trade *order* but not on later trades, and approaches standalone CVA as the new trade's relative size grows (Figure 20.12: -6.2 bps small, towards -17.9 bps large). It also prices unwinds, restructurings and changes to margin terms. Incremental EPE (and DVA) can change sign for offsetting trades; with equal spreads BCVA has no incremental effect.
- **Marginal CVA** (CR-20 i): uses marginal EPE; contributions *sum to total CVA* and are "fairer" (Table 20.5: IRS -17.7, XCCY -19.2, total -36.9). Not used to price new trades (it would re-price existing ones) but suited to trades executed together and to accounting allocation.
- **See also** A21 for collateralised CVA (MPoR, thresholds, initial margin, CR-20 j), A20 for wrong-way risk (CR-20 k to p), and C8 for EPE and ENE profiles.

Formulas

Unilateral CVA, discrete (eq. 20.3).

$$\text{UCVA} = -\text{LGD} \sum_{i=1}^{m} \text{EPE}(t_i) \times \text{PD}(t_{i-1}, t_i)$$

$\text{EPE}(t_i)$: discounted EPE (risk-free discounting applied in the EPE; with explicit discount factors this is the map's $\sum(1 - \text{RR}) \cdot \text{EE} \cdot \text{PD} \cdot \text{DF}$); $\text{PD}(t_{i-1}, t_i)$: *marginal* default probability (cumulative at t_i minus cumulative at t_{i-1}); $\text{LGD} = 1 - R$. Gregory's sign convention: CVA negative (a cost).

Bilateral CVA (eqs. 20.7a, 20.8a, 20.8b).

$$\begin{aligned}
\text{BCVA} &= \text{CVA} + \text{DVA} \\
\text{CVA} &= -\text{LGD}_C \sum_i \text{EPE}(t_i) \times \text{PD}_C(t_{i-1}, t_i) \times [1 - \text{PD}_P(0, t_{i-1})] \\
\text{DVA} &= -\text{LGD}_P \sum_i \text{ENE}(t_i) \times \text{PD}_P(t_{i-1}, t_i) \times [1 - \text{PD}_C(0, t_{i-1})]
\end{aligned}$$

C: counterparty; P: party making the calculation. ENE is negative, so DVA is positive. Dropping the bracketed survival terms gives UCVA and UDVA.

Change from UCVA to BCVA. $\Delta = \text{BCVA} - \text{UCVA} = \text{DVA} + (\text{CVA} - \text{UCVA})$: the DVA benefit plus the (small) survival reduction of the CVA leg.

Spread approximations (eqs. 20.4, 20.9).

$$\text{UCVA} \approx -\text{average EPE} \times \text{Spread}_C, \qquad \text{BCVA} \approx -\text{average EPE} \times \text{Spread}_C - \text{average ENE} \times \text{Spread}_P$$

Output in the units of the spread (bps per annum when EPE is a fraction of notional). No LGD. Undiscounted EPE.

Upfront to running. CVA (running) $\approx \dfrac{\text{CVA (upfront)}}{\text{notional} \times \text{risky annuity}}$ (first order; ignores the CVA of the CVA).

Incremental CVA (eqs. 20.11, 20.12). $\text{CVA}_{NS,\text{incr}} = \text{CVA}_{NS'} - \text{CVA}_{NS} = -\text{LGD} \sum_i \text{EPE}^{\text{incr}}(t_i) \times \text{PD}(t_{i-1}, t_i)$

Exam traps

- **Cumulative vs marginal PD (intermediate result):** the CVA sum uses the PD for each interval, not the cumulative PD to each date. The *survival* bracket, by contrast, uses the other party's *cumulative* PD to the *start* of the interval.
- **Wrong-input twin:** recovery rate instead of LGD; the counterparty's PD in the DVA leg (DVA uses the party's *own* PD with ENE); the own spread on the EPE leg of the spread formula.
- **Sign convention:** ENE is negative, so DVA is positive and BCVA = CVA + DVA. Subtracting DVA, or reporting CVA as a gain, flips the result.
- **The change is not the BCVA (intermediate result):** when the move from UCVA to BCVA is asked, the answer is BCVA − UCVA; the BCVA itself and the DVA leg alone are the planted distractors.
- **Spread formula has no LGD:** multiplying EPE × spread by LGD double counts it.
- **Incremental vs standalone vs total:** the price of a new trade under netting is the incremental CVA, not its standalone CVA, not the whole netting set's CVA, and not the netting benefit.

Question A6.1

CR-20 b, c · S1 · [Map #6]

An xVA analyst at a US dealer is computing the unilateral CVA on an uncollateralised 5-year interest rate swap with a corporate counterparty, notional USD 200 million. The counterparty's recovery rate is 40%; the dealer's own credit spread is 90 bps. The simulated profile is:

Year	1	2	3	4	5
Discounted EPE (% of notional)	2.0	3.2	3.6	3.0	1.6
Counterparty *cumulative* PD (%)	1.5	3.2	5.0	6.9	8.9

Assuming no wrong-way risk, which of the following is closest to the UCVA?

A. −USD 0.794 million

B. −USD 0.476 million

C. −USD 0.286 million

D. −USD 0.191 million

Solution

Answer: C

Step 1 – Marginal PDs from cumulative PDs

$$\text{PD}(t_{i-1}, t_i) = \text{PD}(0, t_i) - \text{PD}(0, t_{i-1}): \quad 1.5,\ 1.7,\ 1.8,\ 1.9,\ 2.0\ (\%)$$

$\Rightarrow$ *The chance of the counterparty defaulting* within *each year, rising slowly from 1.5% to 2.0%.*

Step 2 – EPE × marginal PD per year (% × %)

$$2.0(1.5) = 3.00,\ 3.2(1.7) = 5.44,\ 3.6(1.8) = 6.48,\ 3.0(1.9) = 5.70,\ 1.6(2.0) = 3.20; \quad \text{sum} = 23.82$$

$\Rightarrow$ *Sum* $= 23.82 \times 10^{-4} = 0.002382$ *of notional: the probability-weighted exposure before applying LGD.*

Step 3 – LGD

$$\text{LGD} = 1 - R = 1 - 0.40 = 0.60$$

$\Rightarrow$ *60% of the exposure is lost if the counterparty defaults.*

Step 4 – UCVA

$$\text{UCVA} = -0.60 \times 0.002382 \times 200 = -\text{USD } 0.286 \text{ million}$$

Interpretation

The swap is worth USD 0.286 million less to the dealer than its risk-free value (about 14.3 bps of notional upfront); this is the charge the xVA desk levies on the trader. The dealer's own spread is a red herring for UCVA (it would enter DVA), and the CVA is negative in Gregory's convention.

Why the other options are wrong

A is incorrect. Used cumulative instead of marginal PDs: $\sum \text{EPE} \times \text{PD}_{\text{cum}} = 66.18 \times 10^{-4}$, so $-0.60 \times 0.006618 \times 200 = -0.794$.
B is incorrect. Dropped the LGD term, treating the whole exposure as lost: $-0.002382 \times 200 = -0.476$.
D is incorrect. Used the recovery rate (40%) in place of LGD: $-0.40 \times 0.002382 \times 200 = -0.191$.

LO: CR-20 b: Describe credit value adjustment (CVA). CR-20 c: Calculate CVA and CVA as a spread with no wrong-way risk, netting, or collateralization. · Reference: Gregory, The xVA Challenge 4e, Ch.17, 20.2.2, eq. 20.3

Question A6.2

CR-20 e, f, g · S3 · [Map #6]

A valuation control officer at a French bank is computing BCVA for an uncollateralised 5-year swap with a BBB corporate, notional EUR 150 million. Both LGDs are 60%. Survival adjustments are ignored (the bank reports UCVA + UDVA). The profiles are:

Year	1	2	3	4	5
Discounted EPE (% of notional)	2.0	3.2	3.6	3.0	1.6
Discounted ENE (% of notional)	−1.0	−1.8	−2.4	−2.6	−2.0
Counterparty marginal PD (%)	1.5	1.7	1.8	1.9	2.0
Bank's own marginal PD (%)	0.6	0.7	0.8	0.8	0.9

Which of the following is closest to the BCVA?

A. −EUR 0.145 million

B. −EUR 0.097 million

C. −EUR 0.054 million

D. +EUR 0.069 million

Solution

Answer: A

Step 1 – CVA leg (counterparty PD with EPE)

$$\sum \text{EPE} \times \text{PD}_C = 3.00 + 5.44 + 6.48 + 5.70 + 3.20 = 23.82 \ (\times 10^{-4})$$

$$\text{CVA} = -0.60 \times 0.002382 \times 150 = -\text{EUR } 0.2144 \text{ million}$$

$\Rightarrow$ *The expected loss from the counterparty defaulting while the bank is owed money.*

Step 2 – DVA leg (own PD with ENE)

$$\sum \text{ENE} \times \text{PD}_P = -0.60 - 1.26 - 1.92 - 2.08 - 1.80 = -7.66 \ (\times 10^{-4})$$

$$\text{DVA} = -0.60 \times (-0.000766) \times 150 = +\text{EUR } 0.0689 \text{ million}$$

$\Rightarrow$ *The "gain" from the bank's own default while it owes the counterparty money; positive because ENE is negative.*

Step 3 – BCVA

$$\text{BCVA} = \text{CVA} + \text{DVA} = -0.2144 + 0.0689 = -\text{EUR } 0.145 \text{ million}$$

Interpretation

Recognising DVA cuts the net adjustment by about a third. Basel derecognises the EUR 0.069 million DVA from capital, and many banks pass only part of it into pricing. Subtracting DVA instead of adding it would give −0.283: the sign error the convention is designed to prevent.

Why the other options are wrong

B is incorrect. Used the recovery rate (40%) in place of LGD in both legs: $-0.40 \times 0.002382 \times 150 + 0.40 \times 0.000766 \times 150 = -0.1429 + 0.0460 = -0.097$.
C is incorrect. Used the counterparty's PD in the DVA leg: $\sum \text{ENE} \times \text{PD}_C = -17.82 \times 10^{-4}$, DVA $= +0.160$, BCVA $= -0.054$.
D is incorrect. Reported the DVA leg alone (an intermediate result): $+0.069$.

LO: CR-20 e: Describe debt value adjustment (DVA) and bilateral CVA (BCVA). CR-20 f: Explain the differences between unilateral CVA (UCVA) and BCVA, and between unilateral DVA (UDVA) and BCVA. CR-20 g: Calculate DVA, BCVA, and BCVA as a spread. · Reference: Gregory, The xVA Challenge 4e, Ch.17, 20.3.3, eqs. 20.7a to 20.8b

Question A6.3

CR-20 c, g · S2, S5 · [Aug-2026] [Map #6]

A risk consultant reviewing pricing at an Indian bank wants quick estimates, as running spreads, of the counterparty risk adjustments on a 5-year swap with a weaker counterparty. The data are:

- Undiscounted EPE profile (% of notional), years 1 to 5: 1.8, 2.6, 3.0, 2.8, 1.8
- Undiscounted ENE profile (% of notional), years 1 to 5: -1.2, -2.0, -2.6, -2.4, -1.8
- Counterparty 5-year credit spread: 180 bps; bank's own 5-year credit spread: 120 bps
- LGD for both parties: 60%

Using Gregory's average-exposure approximations, which of the following is closest to the UCVA and the BCVA, each in bps per annum?

A. UCVA -4.32 bps; BCVA -1.92 bps

B. UCVA -4.32 bps; BCVA $+2.40$ bps

C. UCVA -2.88 bps; BCVA $+0.72$ bps

D. UCVA -2.59 bps; BCVA -1.15 bps

Solution

Answer: A

Step 1 – Average EPE and average ENE

$$\overline{\text{EPE}} = \frac{1.8 + 2.6 + 3.0 + 2.8 + 1.8}{5} = 2.4\%, \qquad \overline{\text{ENE}} = \frac{-1.2 - 2.0 - 2.6 - 2.4 - 1.8}{5} = -2.0\%$$

$\Rightarrow$ *On average the bank is owed 2.4% of notional and owes 2.0%.*

Step 2 – UCVA as a spread (eq. 20.4)

$$\text{UCVA} \approx -\overline{\text{EPE}} \times \text{Spread}_C = -0.024 \times 180 = -4.32 \text{ bps p.a.}$$

$\Rightarrow$ *Ignoring its own default, the bank would charge about 4.3 bps per annum for counterparty risk.*

Step 3 – DVA leg (eq. 20.9)

$$-\overline{\text{ENE}} \times \text{Spread}_P = -(-0.020) \times 120 = +2.40 \text{ bps p.a.}$$

Step 4 – BCVA as a spread

$$\text{BCVA} = -4.32 + 2.40 = -1.92 \text{ bps p.a.}$$

Interpretation

Because the counterparty is weaker (180 vs 120 bps) it pays the bank about 1.9 bps per annum to trade; with equal spreads and symmetric exposure the bilateral charge would be close to zero. The LGD is a red herring: the approximation has no LGD input because the spread already embeds it.

Why the other options are wrong

B is incorrect. Correct UCVA, but reported the DVA leg alone as the BCVA (an intermediate result): $+2.40$.
C is incorrect. Swapped the parties' spreads, applying the bank's own spread to the EPE and the counterparty's spread to the ENE: $-0.024 \times 120 = -2.88$; $-2.88 + 0.020 \times 180 = +0.72$.
D is incorrect. Multiplied by LGD as well, double counting it: $0.60 \times (-4.32) = -2.59$ and $0.60 \times (-1.92) = -1.15$.

LO: CR-20 c: Calculate CVA and CVA as a spread with no wrong-way risk, netting, or collateralization. CR-20 g: Calculate DVA, BCVA, and BCVA as a spread. · Reference: Gregory, The xVA Challenge 4e, Ch.17, 20.2.3 eq. 20.4; 20.3.3 eq. 20.9

Question A6.4

CR-20 h, i · S2 · [Map #6]

A trader at a German bank is pricing a new 5-year EUR swap, notional EUR 150 million, with a counterparty under an existing ISDA netting agreement. The xVA desk reports:

- CVA of the existing netting set: $-$EUR 1.84 million
- CVA of the netting set including the new swap: $-$EUR 2.27 million
- Standalone CVA of the new swap: $-$EUR 0.96 million
- 5-year risky annuity: 4.5; counterparty LGD: 60%

Which of the following is closest to the CVA charge for the new swap, quoted as a running cost in bps per annum on its notional?

A. 6.37 bps p.a.

B. 7.85 bps p.a.

C. 14.22 bps p.a.

D. 33.63 bps p.a.

Solution

Answer: A

Step 1 – Incremental CVA (upfront)

$$\text{CVA}_{\text{incr}} = \text{CVA}_{NS'} - \text{CVA}_{NS} = -2.27 - (-1.84) = -\text{EUR } 0.43 \text{ million}$$

⇒ The new swap partly offsets existing trades, so it adds only EUR 0.43 million of CVA, less than its standalone EUR 0.96 million, as netting guarantees.

Step 2 – Annuity value of notional

$$150 \times 4.5 = 675 \text{ (EUR million-years)}$$

Step 3 – Running spread

$$\frac{-0.43}{675} = -0.000637 = -6.37 \text{ bps p.a.}$$

⇒ In Gregory's convention the CVA is -6.37 bps; the counterparty pays a running cost of 6.37 bps per annum.

Interpretation

The trader can quote about 6.4 bps per annum instead of 14.2 bps standalone, which is why an existing netting relationship lets a bank offer better terms (Cooper and Mello). Dividing by the annuity ignores the CVA of the CVA, so the exact recursive charge is slightly larger. The LGD is a red herring: it is already inside the CVA figures, and applying it again would give 3.82 bps.

Why the other options are wrong

B is incorrect. Charged the netting benefit (standalone minus incremental) instead of the incremental CVA: $(0.96 - 0.43)/675 = 7.85$ bps.
C is incorrect. Used the standalone CVA, ignoring the netting benefit: $0.96/675 = 14.22$ bps.
D is incorrect. Charged the entire post-trade netting set CVA to the new swap: $2.27/675 = 33.63$ bps.

LO: CR-20 h: Explain how netting can be incorporated into the CVA calculation. CR-20 i: Define and calculate incremental CVA and marginal CVA and explain how to convert CVA into a running spread. · Reference: Gregory, The xVA Challenge 4e, Ch.17, 20.2.3; 20.4.1 eqs. 20.10 to 20.12

Question A6.5

CR-20 e, f, g · S3, S4, S6 · [Map #6]

The head of the CVA desk at a Canadian bank is moving the bank's reported counterparty risk adjustment on an uncollateralised 3-year swap (notional CAD 250 million) from UCVA to a contingent BCVA that includes survival

adjustments. The inputs are:

Year	1	2	3
Discounted EPE (% of notional)	3.0	4.0	3.0
Discounted ENE (% of notional)	−2.0	−3.5	−4.5
Counterparty marginal PD (%)	6.0	7.0	8.0
Bank's own marginal PD (%)	3.0	3.5	4.0

The counterparty's recovery rate is 40% and the bank's is 35%. Which of the following is closest to the change in the reported adjustment when the bank moves from UCVA to BCVA?

A. −CAD 0.475 million

B. +CAD 0.539 million

C. +CAD 0.575 million

D. +CAD 0.589 million

Solution **Answer: C**

Step 1 – UCVA (no survival term)

$$\text{UCVA} = -0.60 \times [3.0(6.0) + 4.0(7.0) + 3.0(8.0)] \times 10^{-4} \times 250 = -0.60 \times 0.0070 \times 250 = -1.0500$$

⇒ *The currently reported charge: CAD 1.050 million, assuming the bank cannot default.*

Step 2 – Survival probabilities to the start of each year

Bank: $1 - \text{PD}_P(0, t_{i-1}) = 1.000,\ 0.970,\ 0.935$. Counterparty: $1 - \text{PD}_C(0, t_{i-1}) = 1.000,\ 0.940,\ 0.870$.

⇒ *A default only counts if the other party has not defaulted first.*

Step 3 – Contingent CVA leg (eq. 20.8a)

$$\text{CVA} = -0.60 \times [18(1.000) + 28(0.970) + 24(0.935)] \times 10^{-4} \times 250 = -0.60 \times 0.00676 \times 250 = -1.0140$$

Step 4 – Contingent DVA leg (eq. 20.8b), own LGD $= 1 - 0.35 = 0.65$

$$\begin{aligned}\text{DVA} &= -0.65 \times [-6.0(1.000) - 12.25(0.940) - 18.0(0.870)] \times 10^{-4} \times 250 \\ &= -0.65 \times (-0.0033175) \times 250 = +0.5391\end{aligned}$$

⇒ *The benefit from the bank's own default while it owes the counterparty money: positive because ENE is negative.*

Step 5 – BCVA and the change

$$\text{BCVA} = \text{CVA} + \text{DVA} = -1.0140 + 0.5391 = -0.4749$$

$$\Delta = \text{BCVA} - \text{UCVA} = -0.4749 - (-1.0500) = +\text{CAD } 0.575 \text{ million}$$

⇒ *The reported adjustment becomes CAD 0.575 million less negative: the DVA of 0.539 plus the survival reduction of the CVA leg, 0.036.*

Interpretation

Moving to BCVA books a CAD 0.575 million gain, most of it DVA. Basel would derecognise the DVA from the CVA capital charge, and the survival term makes the bank's CVA depend on its own credit quality, which is why practice is split between contingent and non-contingent BCVA.

Why the other options are wrong

A is incorrect. Reported the BCVA itself (-0.475) as the change: the intermediate result planted when the move from UCVA is asked.

B is incorrect. Reported the DVA leg alone $(+0.539)$, forgetting that the survival term also shrinks the CVA leg from -1.050 to -1.014.

D is incorrect. Dropped the survival adjustments, so the change equals the UDVA: $-0.65 \times (-0.003625) \times 250 = +0.589$; the non-contingent BCVA would be -0.461.

LO: CR-20 e: Describe debt value adjustment (DVA) and bilateral CVA (BCVA). CR-20 f: Explain the differences between unilateral CVA (UCVA) and BCVA, and between unilateral DVA (UDVA) and BCVA. CR-20 g: Calculate DVA, BCVA, and BCVA as a spread. · Reference: Gregory, The xVA Challenge 4e, Ch.17, 20.3.3 eqs. 20.8a, 20.8b; 20.3.4

A7 Ho-Lee, Time-Dependent Drift and Volatility

Readings: MR-14 a, b, c, d, e · The Art of Term Structure Models: Drift (Tuckman & Serrat 3e, Ch.9; 14.1 Model 1, 14.2 Model 2, 14.3 The Ho-Lee Model, 14.4 Desirability of Fitting); MR-15 a, b, c · The Art of Term Structure Models: Volatility and Distribution (Tuckman & Serrat 3e, Ch.10; 15.1 Time-Dependent Volatility: Model 3)

AUG 2026 RECALL | MAP TIER A · #7 · 7/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: Ho-Lee numerical - plug values; time-dependent drift and vol - theory question.
Map #7: "Highly specific recalls: 'calculate interest rate after 2 months in the top node in Ho-Lee model'; 'Ho Lee and time-dependent volatility models: calculate value of rate in upper nodes'; 'Ho Lee Model & Model 3, 2-step plug and play calculation'; 'no-arbitrage model (Node 2 Δr)'; '2 on IR term structure (Ho Lee)'. Node formula: $r_0 + \sum \lambda\, dt \pm m\sigma\sqrt{dt}$. Sibling swap risk against Vasicek and Model 1/2."
Shapes covered here:
S1 interest rate after 2 months in the top node of a monthly Ho-Lee tree (Q2)
S2 Ho-Lee and time-dependent volatility models: rate in the upper nodes (Q1, Q2, Q4)
S3 Ho-Lee and Model 3 two-step plug and play (Q2, Q4)
S4 no-arbitrage model, node-2 Δr and the drift read back from the nodes (Q2, Q3)
S5 two questions on the Ho-Lee short-rate tree (Q1, Q2; reverse-solve in Q3)
S6 sibling swap against Vasicek and Models 1 and 2 (Q5; Model 1 distractor in Q1)
S7 August plain forms: Ho-Lee plug values (Q1); time-dependent drift and volatility theory (Q5).

Revision notes

- **The specific: the Ho-Lee tree** (MR-14 d). $dr = \lambda(t)\, dt + \sigma\, dw$: the drift may change from date to date (the book's illustration: an annualised drift of -20 bps over the first month and 20 bps over the second). Date 1: $r_0 + \lambda_1 dt \pm \sigma\sqrt{dt}$. Date 2: $r_0 + (\lambda_1 + \lambda_2)dt + 2\sigma\sqrt{dt}$, $r_0 + (\lambda_1 + \lambda_2)dt$, $r_0 + (\lambda_1 + \lambda_2)dt - 2\sigma\sqrt{dt}$. The top node after two months is reached by two up moves, so the change from today is $\Delta r = (\lambda_1 + \lambda_2)dt + 2\sigma\sqrt{dt}$. The tree recombines because σ is constant.
- **Calibration of Ho-Lee** (MR-14 d). With $dt = 1/12$, set r_0 equal to the one-month rate; find λ_1 so the model produces the market two-month spot rate; then λ_2 for the three-month spot rate; and so on until the tree ends. The free parameters match the prices of securities with fixed cash flows, so the rate curves match all the rates that are input into the model: Ho-Lee is *arbitrage-free*.
- **What Ho-Lee does not change.** Just as adding a constant drift to Model 1 does not affect the shape of the term structure of volatility nor the parallel-shift characteristic, adding a time-dependent drift does not change these features either. The time-dependent drift over each period is still some combination of risk premium and expected changes in the short rate.
- **Model 1, no drift** (MR-14 a, b). $dr = \sigma\, dw$, with σ the annual basis-point volatility and dw normal with mean 0 and standard deviation $\sqrt{dt}$. Book example: $r_0 = 6.18\%$, $\sigma = 1.13\%$, $dw = 0.15$ gives $dr = 0.17\%$; drift is zero and the standard deviation of the rate is $\sigma\sqrt{dt} = 32.6$ bps per month. Tree check: $E[dr] = 0$, $V[dr] = \sigma^2 dt$. Date-2 nodes 6.832%, 6.180%, 5.528%.
- **Effectiveness of Model 1** (MR-14 a). It is a normal (Gaussian) model; the curve slopes down only because of convexity (Table 14.1: -18.8 bps at 10 years, -135.3 bps at 30 years). The term structure of volatility is flat at 113 bps while the market's is humped; the only factor is the short rate, so it is a model of parallel shifts. Matching the 10-year par rate leaves every other point off: not enough flexibility for the simplest term structure shapes.
- **Negative rates** (MR-14 c). In a normal model the probability of a negative short rate increases with the horizon (10-year standard deviation $1.13\%\sqrt{10} = 3.573\%$; a shock of only 1.73 standard deviations pushes 6.18% below zero). Coupon bonds depend on the average path, so negative rates rarely disqualify a model; options asymmetrically sensitive to low rates (a 10-year call on a bond at a 0% yield) are badly mispriced. Remedies: change the distribution (lognormal, which may give unacceptable volatilities), or set negative rates in the tree to zero: *shadow* rates versus *observed* rates, a change localised around zero.
- **Model 2, constant drift** (MR-14 a). $dr = \lambda\, dt + \sigma\, dw$. Book example $r_0 = 5.138\%$, $\lambda = 0.229\%$, $\sigma = 1.10\%$: drift 1.9 bps a month, standard deviation 31.75 bps a month; 10-year terminal mean 7.428%,

standard deviation 347.9 bps. The drift raises the mean so negative rates are less likely, but the term structure of volatility, the convexity effect and the parallel-shift nature are unchanged. Calibrated to the 2- and 10-year par rates it fits much better than Model 1, but $\lambda = 0.229\%$ implies a Sharpe ratio of about 0.21 and the curve overshoots the 30-year rate by about 25 bps.

- **Equilibrium versus arbitrage-free** (MR-14 e). Models 1 and 2 are equilibrium models: no effort is made to match the initial term structure. Ho-Lee is arbitrage-free. The choice depends on the purpose of building the model.
- **Uses of arbitrage-free models** (MR-14 e). (1) Quoting securities that are not actively traded from more liquid ones, e.g. a swap to three years and four months from three- and four-year swaps: interpolation based on economic reasoning (expectations, risk premium and convexity) rather than an arbitrary functional form, but only as good as the volatility assumption. (2) Valuing and hedging derivatives for market making or proprietary trading, taking the underlying (e.g. the 10-year bond under a bond option) as fairly priced.
- **Fitting warnings** (MR-14 e). First, a mediocre or bad model cannot be rescued by calibrating it to market prices (a parallel-shift model with a fitted drift is no better for an application needing non-parallel moves). Second, fitting assumes prices are fair, but supply and demand imbalances, taxes and liquidity distort them: a bank dumping seven-year swaps does not mean expected rates in seven years have risen. A fitted drift that changes dramatically from one year to the next is likely attributing non-interest-rate effects to the rate process.
- **Relative value needs equilibrium models** (MR-14 e). Arbitrage-free models conclude by construction that all input bond or swap prices are fair relative to one another, so investors choosing among securities, market makers selecting hedges and traders hunting temporary mispricings must rely on equilibrium models. In practice the two are blended (a deterministic drift for a few years then a constant drift; or taking 2-, 5-, 10- and 30-year rates as given).
- **Model 3: time-dependent volatility** (MR-15 a). $dr = \lambda(t)dt + \sigma(t)dw$ (15.1); if $\sigma(1) = 1.26\%$ and $\sigma(2) = 1.20\%$, the short rate's volatility is 126 bps per year in one year and 120 bps in two. Model 3 is the special case $dr = \lambda(t)dt + \sigma e^{-\alpha t}dw$ (15.2): volatility starts at σ and declines exponentially to zero. Just as a time-dependent drift fits many bond or swap rates, a time-dependent volatility fits many option prices: caplets expiring on many dates. Popular with market makers in interest rate options.
- **Standard deviation in Model 3** (MR-15 b). Each step's perturbation is $\pm\sigma(t)\sqrt{dt}$; with $\sigma = 126$ bps and $\alpha = 0.025$ the standard deviation of the terminal distribution rises rapidly with horizon at first, then more slowly (Figure 15.1). With the initial volatility and decay rate equal to the Vasicek volatility and mean reversion speed, the terminal standard deviations are identical; if the drift also matches the Vasicek average path, the terminal distributions are identical.
- **Model 3 is still a parallel-shift model** (MR-15 a, c). As for any model without mean reversion, its term structure of volatility is flat, at levels that change over time, but always flat. Efficacy: fine for interpolating unobservable option prices from known ones. For valuing and hedging, mean reversion is preferred because (1) time-dependent volatility relies on the difficult argument that the market forecasts short-term volatility in the distant future (fix: time-dependent near term, then constant) and (2) the downward-sloping factor structure and volatility term structure of mean reversion capture rate behaviour better than parallel shifts. If Vasicek is not good enough for a purpose, a parallel-shift model calibrated to caplets is unlikely to be better.
- **See also** A25 for Vasicek (MR-14 f, g, h) and Gauss+; D1 for CIR, lognormal, Salomon Brothers and Black-Karasinski models (MR-15 d, e, f).

Formulas

Notation. r_0 = today's short rate (continuously compounded, annualised); σ = annual basis-point volatility; dt = time step in years (one month $= 1/12$, $\sqrt{1/12} = 0.28868$); dw normal with mean 0 and standard deviation $\sqrt{dt}$; each branch has probability $\frac{1}{2}$.
Model 1 (14.1). $dr = \sigma\, dw$; $E[dr] = 0$; SD of the rate change $= \sigma\sqrt{dt}$. Date 2: $r_0 \pm 2\sigma\sqrt{dt}$, r_0.
Model 2 (14.5). $dr = \lambda\, dt + \sigma\, dw$. Date 2: $r_0 + 2\lambda dt \pm 2\sigma\sqrt{dt}$, $r_0 + 2\lambda dt$. Terminal mean after T: $r_0 + \lambda T$; SD $\sigma\sqrt{T}$.
Ho-Lee (14.7). $dr = \lambda(t)\, dt + \sigma\, dw$.

$$\text{Date 1: } r_0 + \lambda_1 dt \pm \sigma\sqrt{dt} \qquad \text{Date 2: } r_0 + (\lambda_1 + \lambda_2)dt + \{2, 0, -2\}\,\sigma\sqrt{dt}$$

Reverse-solve. $\lambda_1 = \left[\frac{1}{2}(r_u + r_d) - r_0\right]/dt$; $\sigma = (r_u - r_d)/(2\sqrt{dt})$.

Model 3 (15.1, 15.2). $dr = \lambda(t)\,dt + \sigma(t)\,dw$, with $\sigma(t) = \sigma e^{-\alpha t}$. Step i: $\pm\sigma(t_i)\sqrt{dt}$. Upper node after two steps: $r_0 + (\lambda_1 + \lambda_2)dt + [\sigma(t_1) + \sigma(t_2)]\sqrt{dt}$. Two-step SD: $\sqrt{dt}\,\sqrt{\sigma(t_1)^2 + \sigma(t_2)^2}$.
Properties. Drift scales with dt; the volatility perturbation scales with $\sqrt{dt}$. Adding a constant or time-dependent drift (Model 2, Ho-Lee) or a time-dependent volatility (Model 3) leaves the model parallel-shift with a flat term structure of volatility; all are normal models, so negative rates are possible.

Exam traps

- $\sqrt{dt}$ **versus** dt **(unit conversion):** drift scales with dt, the volatility perturbation with $\sqrt{dt}$. $\sigma\,dt$ shrinks the branches; $\lambda\sqrt{dt}$ inflates the drift.
- **Wrong drift sum at date 2 (dropped term):** the date-2 nodes carry $(\lambda_1+\lambda_2)dt$, not $2\lambda_1 dt$ and not $\lambda_2 dt$ alone. Forgetting $\lambda_1 dt$ altogether turns Ho-Lee into Model 1.
- **Outer nodes on date 2:** the top and bottom nodes sit $2\sigma\sqrt{dt}$ from the centre, so the date-2 range is $4\sigma\sqrt{dt}$; a one-σ top node, or a $2\sigma\sqrt{dt}$ range, is the date-1 picture.
- **Intermediate result:** in a reverse-solve, $\lambda_1 dt$ (the monthly shift) and σ (from the node spread) are legitimate intermediates; the answer is the annualised λ_1.
- **Constant versus decaying volatility (Model 3):** use $\sigma(t) = \sigma e^{-\alpha t}$ step by step; reusing the initial σ is Ho-Lee. Standard deviations add in quadrature, not linearly.
- **Sibling swap:** time-dependent *drift* fits bond and swap rates; time-dependent *volatility* fits option (caplet) prices. Neither introduces mean reversion, a humped volatility structure (Gauss+) or a floor at zero (CIR, lognormal).

Question A7.1

MR-14 d · S2, S5, S7 · [Aug-2026] [Map #7]

A junior risk analyst at an Australian bank is building a monthly Ho-Lee short-rate tree for AUD swaps. The current one-month rate is 3.20%, annual basis-point volatility is 120 basis points, the calibrated drift for the first month is $\lambda_1 = 0.60\%$ and for the second month is $\lambda_2 = 0.90\%$. Which of the following is closest to the short rate in the *up* state at the end of the first month?

A. 2.904%

B. 3.350%

C. 3.546%

D. 3.596%

Solution

Answer: D

Step 1 – Time step

$$dt = \tfrac{1}{12} = 0.08333, \qquad \sqrt{dt} = 0.28868$$

⇒ *One monthly step, measured in years.*

Step 2 – Drift over the first step

$$\lambda_1\,dt = 0.60\% \times \tfrac{1}{12} = 0.050\%$$

⇒ *The risk-neutral drift lifts the rate by 5 basis points over the month in both states.*

Step 3 – Volatility perturbation

$$\sigma\sqrt{dt} = 1.20\% \times 0.28868 = 0.3464\%$$

⇒ *One standard deviation of the monthly rate change is about 34.6 basis points.*

Step 4 – Up-state node

$$r_u = r_0 + \lambda_1 dt + \sigma\sqrt{dt} = 3.20\% + 0.050\% + 0.3464\% = 3.596\%$$

⇒ *If the shock is up, the one-month rate one month from now is about 3.60%.*

Interpretation

λ_2 is irrelevant until date 2. The node is the date-1 building block from which λ_2 is later chosen so the model reprices the market three-month spot rate.

Why the other options are wrong

A is incorrect. Down-state node: $3.20\% + 0.050\% - 0.3464\% = 2.904\%$.
B is incorrect. dt instead of $\sqrt{dt}$ on volatility: $3.20\% + 0.050\% + 1.20\%/12 = 3.350\%$.
C is incorrect. Dropped the drift (used Model 1): $3.20\% + 0.3464\% = 3.546\%$.

LO: MR-14 d: Construct a short-term rate tree under the Ho-Lee Model with time-dependent drift. · Reference: Tuckman & Serrat, Fixed Income Securities 3e, Ch.9, 14.3 eq. 14.7

Question A7.2

MR-14 d · S1, S2, S3, S4, S5 · [Map #7]

A rates quant at a Canadian pension fund is extending a monthly Ho-Lee tree for CAD to the end of the second month, to check how wide the range of short rates becomes. Inputs:

- Current one-month rate r_0: 1.50%; annual basis-point volatility σ: 160 basis points
- Calibrated drift: $\lambda_1 = 0.60\%$ for month 1 and $\lambda_2 = 1.20\%$ for month 2 (annualised)

Which of the following is closest to the short rate in the top node after two months, and to the gap between the top and bottom nodes on that date?

A. Top node 2.112%; gap 0.924%

B. Top node 2.524%; gap 1.848%

C. Top node 2.574%; gap 0.924%

D. Top node 2.574%; gap 1.848%

Solution

Answer: D

Step 1 – Cumulative drift to date 2

$$(\lambda_1 + \lambda_2)\, dt = (0.60\% + 1.20\%) \times \tfrac{1}{12} = 0.150\%$$

$\Rightarrow$ *The two monthly drifts together shift every date-2 node up by 15 basis points.*

Step 2 – Volatility perturbation per step

$$\sigma\sqrt{dt} = 1.60\% \times 0.28868 = 0.4619\%$$

$\Rightarrow$ *Each monthly up or down move is about 46 basis points.*

Step 3 – Top node (two up moves) and node-2 Δr

$$r_{uu} = r_0 + (\lambda_1 + \lambda_2)dt + 2\sigma\sqrt{dt} = 1.50\% + 0.150\% + 0.9238\% = 2.574\%, \qquad \Delta r = +107.4 \text{ bps}$$

$\Rightarrow$ *After two up shocks the one-month rate is about 2.57%.*

Step 4 – Bottom node and the gap

$$r_{dd} = 1.650\% - 0.9238\% = 0.726\%, \qquad r_{uu} - r_{dd} = 4\sigma\sqrt{dt} = 1.848\%$$

$\Rightarrow$ *The date-2 nodes are 2.574%, 1.650% and 0.726%. The drift cancels in the gap; only volatility sets the width.*

Interpretation

The range still stays above zero at two months, but in a normal model the probability of negative rates rises with the horizon (MR-14 c): the width grows with $\sqrt{t}$ while the drift grows with t.

Why the other options are wrong

A is incorrect. Placed the date-2 top node only one $\sigma\sqrt{dt}$ above the centre (the date-1 picture): $1.650\% + 0.4619\% = 2.112\%$, gap $2\sigma\sqrt{dt} = 0.924\%$.
B is incorrect. Used λ_1 for both months ($2\lambda_1 dt = 0.10\%$) instead of $(\lambda_1 + \lambda_2)dt$: $1.50\% + 0.10\% + 0.9238\% = 2.524\%$.
C is incorrect. Correct top node, but measured the gap as $2\sigma\sqrt{dt} = 0.924\%$, the spread between the two date-1 nodes, not the date-2 range.

LO: MR-14 d: Construct a short-term rate tree under the Ho-Lee Model with time-dependent drift. · Reference: Tuckman & Serrat, Fixed Income Securities 3e, Ch.9, 14.3

Question A7.3

MR-14 d · S4, S5 · [Map #7]

A model validator at a Singapore bank is reviewing a monthly Ho-Lee tree for SGD. The documentation is missing λ_1, but the tree shows $r_0 = 2.50\%$ and date-1 nodes of 2.8887% (up) and 2.3113% (down). The model's two-year par rate is 2.95%. Which of the following is closest to the annualised drift λ_1?

A. 0.10%

B. 1.00%

C. 1.20%

D. 4.66%

Solution

Answer: C

Step 1 – Average of the date-1 nodes

$$\tfrac{1}{2}(r_u + r_d) = \tfrac{1}{2}(2.8887\% + 2.3113\%) = 2.6000\%$$

$\Rightarrow$ *The $\pm\sigma\sqrt{dt}$ shocks cancel in the average, leaving $r_0 + \lambda_1 dt$.*

Step 2 – Drift over the step

$$\lambda_1\, dt = 2.6000\% - 2.5000\% = 0.1000\%$$

$\Rightarrow$ *The month's drift raises the expected short rate by 10 basis points.*

Step 3 – Annualise

$$\lambda_1 = \frac{0.1000\%}{1/12} = 1.20\%$$

$\Rightarrow$ *λ_1 is quoted per year, like every parameter in $dr = \lambda(t)dt + \sigma dw$.*

Check – Implied volatility (not asked)

$$\sigma = \frac{r_u - r_d}{2\sqrt{dt}} = \frac{0.5774\%}{2 \times 0.28868} = 1.00\%$$

Interpretation

A 1.20% annualised drift for month 1 is what makes the model reprice the market two-month spot rate. The two-year par rate is irrelevant to this step.

Why the other options are wrong

A is incorrect. Intermediate result: the monthly shift $\lambda_1 dt = 0.10\%$, not annualised.
B is incorrect. Sibling quantity: computed $\sigma = 1.00\%$ from the node spread instead of the drift.
D is incorrect. Used the up node alone, treating its volatility shock as drift: $(2.8887\% - 2.50\%) \times 12 = 4.66\%$.

LO: MR-14 d: Construct a short-term rate tree under the Ho-Lee Model with time-dependent drift. · Reference: Tuckman & Serrat, Fixed Income Securities 3e, Ch.9, 14.3

Question A7.4

MR-15 a, b · S2, S3 · [Map #7]

An interest rate options market maker at a Japanese bank quotes caplets off Model 3, $dr = \lambda(t)dt + \sigma e^{-\alpha t} dw$, on a quarterly tree ($dt = 0.25$). For each quarter the desk uses the volatility at the start of that quarter, $\sigma(0)$ for the first step and $\sigma(0.25)$ for the second. Inputs (annualised):

- $r_0 = 2.40\%$; $\lambda_1 = 0.80\%$; $\lambda_2 = 1.00\%$
- Initial basis-point volatility $\sigma = 150$ bps; decay rate $\alpha = 0.60$

Which of the following is closest to the short rate in the upper node after two quarters, and to the standard deviation of the short rate after two quarters?

A. Upper node 4.246%; standard deviation 0.990%

B. Upper node 4.246%; standard deviation 1.396%

C. Upper node 4.350%; standard deviation 1.061%

D. Upper node 4.696%; standard deviation 0.990%

Solution **Answer: A**

Step 1 – Volatility for each step

$$\sigma(0) = 1.50\%, \qquad \sigma(0.25) = 1.50\% \times e^{-0.60\times 0.25} = 1.50\% \times 0.8607 = 1.2911\%$$

⇒ *Volatility decays from 150 to about 129 bps per year over the first quarter.*

Step 2 – Perturbations per step

$$\sigma(0)\sqrt{dt} = 1.50\% \times 0.5 = 0.7500\%, \qquad \sigma(0.25)\sqrt{dt} = 1.2911\% \times 0.5 = 0.6455\%$$

Step 3 – Drift over two quarters

$$(\lambda_1 + \lambda_2)\, dt = (0.80\% + 1.00\%) \times 0.25 = 0.450\%$$

Step 4 – Upper node

$$r_{uu} = 2.40\% + 0.450\% + 0.7500\% + 0.6455\% = 4.246\%$$

⇒ *Two up moves, each sized by the volatility in force during that quarter.*

Step 5 – Standard deviation after two quarters

$$\text{SD} = \sqrt{dt}\,\sqrt{\sigma(0)^2 + \sigma(0.25)^2} = 0.5 \times \sqrt{1.50^2 + 1.2911^2}\,\% = 0.5 \times 1.9791\% = 0.990\%$$

⇒ *The two independent steps add in variance. Because volatility decays, the SD rises more slowly with horizon than $\sigma\sqrt{t}$ (MR-15 b).*

Interpretation

A constant-volatility Ho-Lee tree with the same σ gives a wider distribution (1.061%); the decay is what lets the desk match caplets that expire later at lower volatilities. With σ changing between steps the up-down and down-up nodes no longer coincide (2.954% and 2.746%), and Model 3 remains a parallel-shift model with a flat (time-varying) volatility term structure.

Why the other options are wrong

B is incorrect. Correct node, but added the standard deviations linearly, $0.7500\% + 0.6455\% = 1.396\%$: this is the upper node's distance from r_0 + drift, not a standard deviation.

C is incorrect. Ignored the decay and used $\sigma = 1.50\%$ in both steps (Ho-Lee): $2.85\% + 2 \times 0.75\% = 4.350\%$, SD $1.50\%\sqrt{0.5} = 1.061\%$.

D is incorrect. Scaled the drift by $\sqrt{dt}$ instead of dt: $2.40\% + 1.80\% \times 0.5 + 1.3955\% = 4.696\%$.

LO: MR-15 a: Describe the short-term rate process under a model with time-dependent volatility. MR-15 b: Calculate the short-term rate change and determine the behavior of the standard deviation of the rate change using a model with time-dependent volatility. · Reference: Tuckman & Serrat, Fixed Income Securities 3e, Ch.10, 15.1 eqs. 15.1 and 15.2, Figure 15.1

Question A7.5 MR-14 d, e; MR-15 a, c · S6, S7 · [Aug-2026] [Map #7]

A senior risk analyst at a UK insurer is reviewing a GBP rates desk that uses the Ho-Lee model to quote odd-maturity swaps and Model 3 (time-dependent volatility, $dr = \lambda(t)dt + \sigma e^{-\alpha t}dw$) to quote caplets. Which of the following statements is correct to include in the review?

A. Because its drift varies with time, the Ho-Lee model produces a humped term structure of volatility that matches market implied volatilities.

B. Because $\lambda(t)$ is calibrated to market spot rates, the Ho-Lee model rules out negative short-term rates.

C. The Ho-Lee drift can match all input spot rates, but the model remains a parallel-shift model with a flat term structure of volatility.

D. The exponential decay in Model 3's volatility introduces mean reversion, so Model 3 is no longer a parallel-shift model.

Solution **Answer: C**

Step 1 – What time-dependent drift buys

The free parameters $\lambda_1, \lambda_2, \ldots$ are chosen one at a time to match the two-, three-, …-period spot rates, so the rate curves match all the rates input into the model (MR-14 d, e).

Step 2 – What it does not change

Adding a constant drift (Model 2) or a time-dependent drift (Ho-Lee) does not affect the shape of the term structure of volatility nor the parallel-shift characteristic of the model. Volatility is σ at every term.

Step 3 – Model 3 contrast

Time-dependent volatility fits caplet prices across expiries, but as for any model without mean reversion, Model 3 is a parallel-shift model with a flat term structure of volatility, flat at levels that change over time (MR-15 a, c).

Interpretation

Calibration buys fit, not structure. If the desk's hedges rely on non-parallel moves, neither model fixes that: a mediocre model cannot be rescued by calibrating it to market prices.

Why the other options are wrong

A is incorrect. Sibling swap: a humped volatility structure is a Gauss+ feature (no shock on r); drift has no effect on the volatility shape in Ho-Lee.

B is incorrect. Scope/condition error: calibrating the drift fits prices but leaves the distribution normal, so negative rates remain possible; ruling them out needs a different distribution (CIR or lognormal, see D1) or setting negative shadow rates to zero.

D is incorrect. Sibling swap: Model 3 can mimic the Vasicek *terminal distribution*, but it has no mean reversion and stays a parallel-shift model.

LO: MR-14 e: Describe uses and benefits of the arbitrage-free models and assess the issue of fitting models to market prices. MR-15 c: Assess the efficacy of time-dependent volatility models. · Reference: Tuckman & Serrat, Fixed Income Securities 3e, Ch.9, 14.3 and 14.4; Ch.10, 15.1

A8 Correlation Swaps, Quanto Options and Concentration Ratio

Readings: MR-8 a, b, c, d, e, f, g · Correlation Basics: Definitions, Applications, and Terminology (Meissner, Correlation Risk Modeling and Management 2e, Ch.1)

AUG 2026 RECALL | MAP TIER A · #8 · 7/16 SOURCES | MAP TIER D · #76 · 2/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: Correlation swap payoff.
Map #8: 'Reported as '1 on correlation swap return' (numerical), 'correlation swap' (twice, separate sittings), 'swap correlation', and 'quanto options'. The correlation-swap payoff is realised average pairwise correlation minus the fixed strike, times notional. The handwritten sheet also flags concentration ratio concept and calculation from the same chapter.'
Map #76: 'LO 8.g links correlation risk to systemic and concentration risk. The 2007-09 narrative (LO 8.b) is the case-study layer. Low recall evidence, but the sheet is specific enough that the calculation is worth one pass.'
Shapes covered here: S1 correlation swap payoff to the fixed-rate payer, plain numerical as seen in August (Q1); S2 the same swap from the seller's (fixed-rate receiver's) side, with irrelevant volatilities (Q2); S3 correlation swap return per unit notional, reverse-solved for the strike (Q3); S4 quanto options: what correlation risk they carry and in which direction it moves the price (Q5); S5 concentration ratio concept and calculation, tied to the joint default probability (Q4).

Revision notes

The specifics first: correlation swaps and quantos (MR-8 c, d)

- **Correlation swap structure** (MR-8 d). A fixed (known) correlation is exchanged for the correlation that actually occurs, the realised or stochastic (unknown) correlation. Realised correlation is the average of the pairwise Pearson correlations of the assets' log-returns over the swap life, using only the $i > j$ cells of the matrix (Equation 8.6). Payoff to the fixed-rate payer at maturity: $N(\rho_{\text{realised}} - \rho_{\text{fixed}})$ (Equation 8.7). Example 8.1: pairs 0.5, 0.3, 0.1 give $\rho_{\text{realised}} = 0.3$; fixed 10%, USD 1,000,000 notional, payer receives USD 200,000.
- **Buyer and seller.** Paying fixed is "buying correlation": the present value rises for the correlation buyer if realised correlation increases. The fixed-rate receiver is "selling correlation".
- **Uses.** When stock prices decrease, correlation between the stocks typically increases, so a fixed correlation payer protects themselves indirectly against a stock market decline. Another way of buying correlation: buy puts on an index (e.g., the Dow) and sell puts on its individual stocks, because correlation and volatility are positively related. The London Whale (Bruno Iksil, 2012) ran the mirror trade in credit: sold CDS on the CDX.NA.IG.9 index and bought single-name CDS; a short squeeze cost JP Morgan over USD 2 billion.
- **Valuation caveat.** There is no industry-standard valuation model for correlation swaps; there is no term structure of correlation; traders use historical data or emerging stochastic correlation models.
- **Quanto option** (MR-8 c). Lets a domestic investor exchange a foreign-currency option payoff back into the home currency at a fixed exchange rate, so it protects against currency risk (an American buying a Nikkei call but worried about a falling yen). The seller does not know (1) how deep in the money the call will be, i.e., how much yen must be converted, nor (2) the exchange rate at maturity. With X = domestic per unit foreign (USD per 1 yen): a *positive* correlation between the Nikkei and X means a rising Nikkei comes with a rising yen, which favours the seller (fewer yen buy the dollar payoff), so the *more positive* the correlation, the *lower* the quanto price; a negative correlation makes it more expensive.
- **Other multi-asset (rainbow, mountain-range) options** (MR-8 c). Better of two, worse of two, call on the maximum of two, exchange option, spread call, better of two or cash, dual strike call, basket option. Except for the option on the *worse* of two and the *basket* option, the *lower* the correlation, the *higher* the price. The exchange option price is close to 0 at high correlation and USD 15.08 at -1; its implied volatility of S_2/S_1 takes the covariance as input (Equation 8.5a).

The specific: concentration ratio (MR-8 g)

- **Concentration risk and the concentration ratio.** Concentration risk is the risk of financial loss due to a concentrated exposure to a specific group of counterparties. Concentration ratio $= 1/n$ for n loans of equal size (ten equal loans: 0.1; one loan: 1). The lower the ratio, the more diversified the default risk, *assuming*

default correlation is smaller than 1. Sector concentration: lending to more sectors reduces default risk because intra-sector defaults are more highly correlated.

- **Example 8.3.** One USD 10 million loan to W with PD 10%: ratio 1, worst case EL USD 1,000,000. Two USD 5 million loans to X and Y (PD 10% each): ratio $1/2$; the worst case, joint default, shrinks if $0 < \rho < 1$. Joint default of two binomial events: $P(X \cap Y) = \rho_{XY}\sqrt{P_X(1-P_X)}\sqrt{P_Y(1-P_Y)} + P_X P_Y$ (Equation 8.11): 10% at $\rho = 1$ (no benefit from the lower ratio), 5.5% at 0.5, 1% at 0, 0.1% at -0.1. Both a lower concentration ratio and a lower correlation reduce the creditor's worst case; higher copula correlation raises credit VaR.
- **Systemic risk** (MR-8 g). The risk that a financial market or an entire financial system collapses (credit markets frozen in September 2008; the Dow fell 53.54% from 14,093 to 6,547 by March 2009; only Walmart of the Dow 30 rose). In a systemic decline correlations between stocks increase sharply: Dow correlation rose from a non-crisis average of 27% to over 50% (96.97% at the February 2009 peak), so "well diversified" portfolios suffered unexpected losses.

Surrounding material (MR-8 a, b, e, f)

- **Financial correlation risk** (MR-8 a). The risk of financial loss due to adverse movements in correlation between two or more variables, financial or non-financial (Mexican and Greek bonds in 2012; commodities and interest rates; sovereign debt and the euro hurting US exporters). Static correlations measure association at a point in or within a period (VaR, copula for CDOs, Lucas binomial); dynamic correlations measure co-movement in time (pairs trading, Heston, stochastic correlation). Five areas: investments, trading, risk management, the GFC, regulation.
- **Wrong-way risk.** Investor hedges a Spanish bond with a CDS bought from BNP Paribas: positive default correlation between reference entity and counterparty is wrong-way risk. For correlation above about -0.3, the higher the correlation, the lower the CDS spread; at 1 the CDS is worthless. From -0.3 to -1 the spread also falls slightly: the relationship is non-monotonic.
- **Investments.** Lower (preferably negative) correlation raises the portfolio return/risk ratio, one of the few "free lunches" (Figure 8.3: close to 250% at high negative correlation, 50% at high positive). Table 8.1: $\rho_{XY} = -0.7403$, $\sigma_P = 16.66\%$.
- **Correlation and VaR** (MR-8 e). $\text{VaR}_P = \sigma_P\, \alpha\sqrt{x}$ with $\sigma_P = \sqrt{\beta_h C \beta_v}$. Example 8.2: USD 10 million and 5 million, daily vols 2% and 1%, $\rho = 0.7$, 99%, 10 days: $\sigma_P = \sqrt{0.0565} = 0.2377$ (USD million), VaR $= 0.2377 \times 2.3264 \times \sqrt{10} =$ USD 1.7486 million. The lower the correlation, the lower the VaR: about USD 1.1 million at $\rho = -1$, close to USD 1.9 million at $+1$.
- **The GFC** (MR-8 b). Causes: benign 2003 to 2006 environment, speculation and bubbles, new structured products, naive trust in the copula model (7,750 pairs in a 125-name CDO), rating agency moral hazard, lowered standards. May 2005: GM (BB) and Ford (BB+) downgraded to junk, CDO asset correlation *decreased*; hedge funds long equity and short mezzanine lost on *both* legs (equity spread up, mezzanine spread down). 2007 to 2009: default correlation *increased* (helps the equity tranche) but rising PDs swamped it; tranche correlation rose and super-senior tranches lost up to 20%, worse for leveraged super-senior (LSS) positions. See also D15.
- **Market risk and credit risk** (MR-8 f). VaR and ES take the covariance matrix as input, so they implicitly incorporate correlation risk. Credit risk = migration risk + default risk. Default correlations (Table 8.3) are higher within sectors than between them (energy is the exception, often negative), so a sector-diversified loan book is advised. Default term structure rises over time for investment grade (A-rated) and is typically *inverse* for distressed (CC-rated) names. See also B3 (joint PD and copulas) and D11 (empirical correlation behaviour).

Formulas

Realised correlation (Equation 8.6) and correlation swap payoff (Equation 8.7).

$$\rho_{\text{realised}} = \frac{2}{n^2 - n}\sum_{i>j}\rho_{i,j}, \qquad \text{Payoff}_{\text{payer}} = N\left(\rho_{\text{realised}} - \rho_{\text{fixed}}\right), \qquad \text{Payoff}_{\text{receiver}} = N\left(\rho_{\text{fixed}} - \rho_{\text{realised}}\right)$$

For $n = 3$, $2/(n^2 - n) = 1/3$: the simple average of the three off-diagonal pairs. Return per unit notional $=$ Payoff$/N = \rho_{\text{realised}} - \rho_{\text{fixed}}$ for the payer.

Concentration ratio and joint default (Equations 8.11, 8.12).

$$\text{Concentration ratio} = \frac{1}{n}\ (n \text{ equal loans}), \qquad P(X \cap Y) = \rho_{XY}\sqrt{P_X(1-P_X)}\sqrt{P_Y(1-P_Y)} + P_X P_Y$$

VaR with correlation (Equations 8.8, 8.9; two-asset variance, Equation 8.5).

$$\text{VaR}_P = \sigma_P\,\alpha\,\sqrt{x}, \qquad \sigma_P = \sqrt{\beta_h\,C\,\beta_v}, \qquad \sigma_P^2 = w_X^2\sigma_X^2 + w_Y^2\sigma_Y^2 + 2w_X w_Y \text{COV}_{XY}$$

β: vector of invested amounts; C: covariance matrix; α: standard normal quantile (2.3264 at 99%); x: horizon in days.

Exam traps

- **Wrong normaliser or wrong cells:** averaging all n^2 cells (diagonal 1s included), using $2/n^2$ or $1/(n^2-n)$, or summing both triangles and still applying $2/(n^2-n)$. Only the $i > j$ cells, once each.
- **Party flip (role misassignment):** the fixed-rate *payer* buys correlation and gains when realised exceeds fixed; the receiver sells correlation and has the mirror payoff. Read whose payoff the stem asks for. Volatilities and maturity are red herrings in Equation 8.7.
- **Dropped strike:** $N \times \rho_{\text{realised}}$ is not the payoff; the fixed correlation must be subtracted.
- **Quanto polarity:** the *more positive* the correlation between the underlying and the exchange rate (domestic per foreign), the *cheaper* the quanto; positive correlation favours the *seller*. Rainbow-option exceptions: worse of two and basket options get cheaper as correlation falls.
- **Concentration ratio direction and scope:** the ratio is $1/n$, not n; a lower ratio helps only if default correlation is below 1 (at $\rho = 1$ the two loans default together and the worst case is unchanged). In Equation 8.11 do not drop the $P_X P_Y$ term or use variances in place of standard deviations.
- **GFC direction:** in May 2005 correlation *fell* and the long equity, short mezzanine trade lost on both legs; in 2007 to 2009 correlations *rose* and hurt the super-senior tranches.

Question A8.1

MR-8 d · S1 · [Aug-2026] [Map #8]

A junior risk analyst at a USD-based hedge fund is presenting the settlement of a 1-year correlation swap to a group of newly hired analysts. The fund *pays* a fixed correlation of 25% on a notional of USD 5,000,000 on a basket of three equities. The realised pairwise Pearson correlations of daily log-returns at maturity are:

	Asset 1	Asset 2	Asset 3
Asset 1	1	0.6	0.2
Asset 2	0.6	1	0.4
Asset 3	0.2	0.4	1

Which of the following is closest to the payoff to the fund at maturity?

A. USD 750,000

B. USD 1,750,000

C. USD 2,000,000

D. USD 2,750,000

Solution

Answer: A

Step 1 – Identify the pairs with $i > j$

Only the three cells below the diagonal enter Equation 8.6: $\rho_{2,1} = 0.6$, $\rho_{3,1} = 0.2$, $\rho_{3,2} = 0.4$.

$$\sum_{i>j} \rho_{i,j} = 0.6 + 0.2 + 0.4 = 1.2$$

$\Rightarrow$ *The diagonal 1s (each asset with itself) carry no information about co-movement and are excluded.*

Step 2 – Normaliser

$$\frac{2}{n^2-n} = \frac{2}{9-3} = \frac{1}{3}$$

⇒ *With three assets there are three distinct pairs, so the formula is the simple average of the pairs.*

Step 3 – Realised correlation

$$\rho_{\text{realised}} = \tfrac{1}{3} \times 1.2 = 0.40$$

⇒ *The basket's average pairwise correlation over the year was 40%, above the 25% the fund locked in.*

Step 4 – Payoff to the fixed-rate payer

$$\text{Payoff}_{\text{payer}} = N(\rho_{\text{realised}} - \rho_{\text{fixed}}) = 5{,}000{,}000 \times (0.40 - 0.25) = \text{USD } 750{,}000$$

⇒ *The fund bought correlation at 25% and correlation came in at 40%, so it receives 15 correlation points times the notional.*

Interpretation

A positive payoff confirms the fund was long correlation. Since correlations tend to rise in sell-offs, this position is the indirect equity-crash hedge described in MR-8. Had the fund received fixed, the same numbers would give USD $-750{,}000$.

Why the other options are wrong

B is incorrect. Diagonal included: averaged all nine cells, $(3 + 2 \times 1.2)/9 = 0.60$, giving $5{,}000{,}000 \times (0.60 - 0.25) = 1{,}750{,}000$.

C is incorrect. Dropped term: paid out the realised correlation times the notional, $5{,}000{,}000 \times 0.40 = 2{,}000{,}000$, forgetting to subtract the fixed strike.

D is incorrect. Double counting: summed both triangles (2.4) and still applied $2/(n^2 - n)$, so $\rho = 0.80$ and $5{,}000{,}000 \times (0.80 - 0.25) = 2{,}750{,}000$.

LO: MR-8 d: Describe the structure, uses, and payoffs of a correlation swap. · Reference: Meissner, Correlation Risk Modeling and Management 2e, Ch.1, 8.5 Trading and Correlation, Equations 8.6 and 8.7, Example 8.1

Question A8.2

MR-8 d · S2 · [Map #8]

A treasury risk analyst at a European bank is preparing a report on a 6-month correlation swap in which the bank *receives* a fixed correlation of 35% on a notional of EUR 8,000,000 on a three-stock basket. Annualised volatilities of the three stocks over the period were 22%, 31% and 18%. The realised pairwise correlations at maturity are:

	Stock X	Stock Y	Stock Z
Stock X	1	0.7	0.5
Stock Y	0.7	1	0.3
Stock Z	0.5	0.3	1

Which of the following is closest to the bank's payoff at maturity?

A. EUR $-1{,}200{,}000$

B. EUR 133,333

C. EUR 800,000

D. EUR 1,200,000

Solution

Answer: A

Step 1 – Sum of pairwise correlations ($i > j$)

$$\sum_{i>j} \rho_{i,j} = 0.7 + 0.5 + 0.3 = 1.5$$

⇒ *Three distinct pairs; the 1s on the diagonal and the duplicate upper triangle are ignored.*

Step 2 – Realised correlation

$$\rho_{\text{realised}} = \frac{2}{3^2 - 3} \times 1.5 = \frac{1.5}{3} = 0.50$$

⇒ *The basket's average pairwise correlation was 50%.*

Step 3 – Payer's payoff (Equation 8.7)

$$N(\rho_{\text{realised}} - \rho_{\text{fixed}}) = 8{,}000{,}000 \times (0.50 - 0.35) = \text{EUR } 1{,}200{,}000$$

⇒ *This is what the fixed-rate* payer *(the correlation buyer, the bank's counterparty) receives.*

Step 4 – Receiver's payoff

$$\text{Payoff}_{\text{receiver}} = N(\rho_{\text{fixed}} - \rho_{\text{realised}}) = \text{EUR } -1{,}200{,}000$$

⇒ *The bank sold correlation at 35% and correlation rose to 50%, so the bank pays EUR 1.2 million.*

Interpretation

Selling correlation is effectively short protection against a sell-off; the loss is the cost of correlation rising. The stock volatilities are irrelevant: Equation 8.7 depends only on realised versus fixed correlation.

Why the other options are wrong

B is incorrect. Wrong normaliser $2/n^2 = 2/9$: $\rho = (2/9)(1.5) = 0.3333$, so the receiver payoff becomes $-8{,}000{,}000 \times (0.3333 - 0.35) = +133{,}333$.

C is incorrect. Wrong normaliser $1/(n^2 - n)$ (forgot the 2): $\rho = 1.5/6 = 0.25$, receiver payoff $-8{,}000{,}000 \times (0.25 - 0.35) = +800{,}000$.

D is incorrect. Party flip: reported the payer's payoff $+1{,}200{,}000$, although the bank receives fixed.

LO: MR-8 d: Describe the structure, uses, and payoffs of a correlation swap. · Reference: Meissner, Ch.1, 8.5 Trading and Correlation, Equations 8.6 and 8.7

Question A8.3

MR-8 d · S3 · [Map #8]

An internal auditor at a Japanese securities firm is validating a booked correlation swap whose term sheet is missing the fixed correlation. The firm was the fixed-rate *receiver* on a notional of JPY 600,000,000 on three Nikkei constituents and received JPY 90,000,000 at maturity. The realised pairwise correlations were:

	Stock 1	Stock 2	Stock 3
Stock 1	1	0.55	0.40
Stock 2	0.55	1	0.25
Stock 3	0.40	0.25	1

Which of the following is closest to the fixed correlation in the swap?

A. 15%

B. 25%

C. 42%

D. 55%

Solution

Answer: D

Step 1 – Realised correlation

$$\rho_{\text{realised}} = \frac{2}{3^2 - 3}(0.55 + 0.40 + 0.25) = \frac{1.20}{3} = 0.40$$

⇒ *Average pairwise co-movement over the swap life was 40%.*

Step 2 – Return per unit notional

$$\frac{\text{Payoff}_{\text{receiver}}}{N} = \frac{90{,}000{,}000}{600{,}000{,}000} = 0.15$$

⇒ *The firm earned 15 correlation points; a positive* receiver *payoff means realised correlation came in* below *the fixed rate.*

Step 3 – Reverse-solve the receiver's payoff

$$\frac{\text{Payoff}_{\text{receiver}}}{N} = \rho_{\text{fixed}} - \rho_{\text{realised}} \;\Rightarrow\; \rho_{\text{fixed}} = 0.40 + 0.15 = 0.55$$

$\Rightarrow$ *The firm had sold correlation at 55%.*

Interpretation

The booked receipt is consistent only with a 55% strike. A term sheet showing a lower fixed correlation alongside a positive JPY 90 million receipt would be an exception for the auditor to escalate, because a receiver with a strike below realised correlation pays rather than receives.

Why the other options are wrong

A is incorrect. Intermediate result: reported the return per unit notional, $90/600 = 0.15$, as if it were the strike.
B is incorrect. Party flip: treated the firm as the payer, $\rho_{\text{fixed}} = \rho_{\text{realised}} - 0.15 = 0.25$.
C is incorrect. Wrong normaliser $2/n^2 = 2/9$: $\rho = (2/9)(1.20) = 0.2667$, then $0.2667 + 0.15 = 0.42$.

LO: MR-8 d: Describe the structure, uses, and payoffs of a correlation swap. · Reference: Meissner, Ch.1, 8.5 Trading and Correlation, Equations 8.6 and 8.7

Question A8.4

MR-8 g · S5 · [Map #8] [Map #76]

A credit risk manager at a German commercial bank is reviewing a proposal from the corporate lending desk. Today the bank has a single EUR 20 million loan to one borrower. The desk proposes to replace it with two loans of EUR 10 million each to two unrelated borrowers, X and Y, in different sectors. The manager's inputs are:

- One-year default probability of X: 8%; of Y: 12%
- Default correlation between X and Y (binomial Pearson correlation): 0.25
- Expected recovery on either loan: 40%

Using Meissner's framework, which of the following correctly states the bank's concentration ratio after the change and the probability of the worst-case scenario in which both X and Y default?

A. Concentration ratio 0.50; joint default probability 3.16%

B. Concentration ratio 0.50; joint default probability 0.96%

C. Concentration ratio 0.50; joint default probability 2.20%

D. Concentration ratio 2.00; joint default probability 3.16%

Solution

Answer: A

Step 1 – Concentration ratio

$$\text{Concentration ratio} = \frac{1}{n} = \frac{1}{2} = 0.50$$

$\Rightarrow$ *Two loans of equal size halve the ratio from 1 (a single loan) to 0.50: the default risk is more diversified, provided the default correlation is below 1.*

Step 2 – Standard deviations of the binomial default variables (Equation 8.12)

$$\sigma_X = \sqrt{0.08 \times 0.92} = 0.2713, \qquad \sigma_Y = \sqrt{0.12 \times 0.88} = 0.3250$$

Step 3 – Joint default probability (Equation 8.11)

$$P(X \cap Y) = \rho_{XY}\,\sigma_X\sigma_Y + P_X P_Y = 0.25 \times 0.2713 \times 0.3250 + 0.08 \times 0.12 = 0.0220 + 0.0096 = 3.16\%$$

$\Rightarrow$ *There is a 3.16% chance that both borrowers default within the year, the bank's worst case.*

Interpretation

Under the old single loan the worst case (default of the one borrower) had the full PD of that borrower; after splitting, losing both loans requires a joint default, here 3.16%. The benefit comes from both levers Meissner links: a lower concentration ratio and a correlation below 1. If X and Y were in the same sector, intra-sector default correlation

would be higher and the joint probability would rise. The 40% recovery is a red herring: it changes the size of the loss, not its probability.

Why the other options are wrong

B is incorrect. Dropped term: assumed independence and kept only $P_X P_Y = 0.08 \times 0.12 = 0.96\%$, ignoring the 0.25 default correlation.

C is incorrect. Dropped term: kept only the correlation term $0.25 \times 0.2713 \times 0.3250 = 2.20\%$ and forgot to add $P_X P_Y$.

D is incorrect. Polarity flip on the ratio: reported $n/1 = 2.00$ instead of $1/n$; a higher number would wrongly signal *more* concentration after diversifying.

LO: MR-8 g: Explain how correlation risk relates to systemic and concentration risk. · Reference: Meissner, Ch.1, 8.8 Correlation Risk and Concentration Risk, Example 8.3, Equations 8.11 and 8.12

Question A8.5

MR-8 c · S4 · [Map #8]

A structurer at a Singapore bank is presenting to a group of newly hired analysts. A US client has bought a quanto call on the Nikkei 225 whose yen payoff will be converted into US dollars at today's spot rate. The exchange rate X is quoted as US dollars per one yen. Which of the following statements in the presentation is correct?

A. The more positive the correlation between the Nikkei and X, the more expensive the quanto call, because the seller must convert a larger yen payoff.

B. The seller of the quanto call is uncertain only about the exchange rate at maturity, because the yen payoff it must convert is fixed at inception.

C. If the correlation between the Nikkei and X is negative, a rising Nikkei comes with a falling yen, so the seller needs more yen for the dollar payoff and the quanto is more expensive.

D. A positive correlation between the Nikkei and X favours the buyer of the quanto call, because a rising yen converts the Nikkei gain into more US dollars.

Solution

Answer: C

Why C is correct

The quanto seller does not know two things: how deep the call will be in the money (the yen amount to convert) and the exchange rate at which it is converted. Their correlation drives the price. With a negative correlation, if the Nikkei increases the yen decreases in value, so more yen are needed to meet the dollar payment; "the lower the correlation coefficient, the more expensive is the quanto option".

Interpretation

A quanto carries correlation risk between the foreign underlying and the exchange rate. The seller is effectively short the low-correlation scenario, the same negative price-correlation relationship as the exchange option in Figure 8.4.

Why the other options are wrong

A is incorrect. Polarity flip: the more positive the correlation, the *lower* the quanto price, because a rising Nikkei comes with a rising yen and fewer yen are needed for the dollar payoff.

B is incorrect. Scope error: drops the first of the seller's two unknowns. The yen payoff is not fixed; it depends on how deep in the money the call finishes ("quanto" comes from "quantity").

D is incorrect. Role misassignment: a positive correlation favours the *seller*, not the buyer. The buyer's conversion rate is fixed by the contract, so a rising yen does not give the buyer more dollars.

LO: MR-8 c: Describe how correlation impacts the price of quanto options as well as other multi-asset exotic options. · Reference: Meissner, Ch.1, 8.5 Trading and Correlation (quanto option)

A9 Securitisation: Tranches, Waterfall, Prepayment and Performance Ratios

Readings: CR-22 a-m · Structured Credit Risk (Malz Ch.9); CR-23 a-h · An Introduction to Securitisation (Choudhry Ch.12)

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Specifics reported

Your August 2026 recall: Delinquency ratio straightforward numerical.
Map #9: 'Reported: 'securitization: calculating cash flow into equity tranche'; 'calculate shares to senior and equity tranche'; 'securitization waterfall structure (GARP Mocks)'; 'prepayment rate (theory)'; 'CPR, SMM, PSA'; 'overcollateralisation account balance'. Closed-book: CPR $= 1-(1-\text{SMM})^{12}$; DSCR = NOI / debt service; WAL $= \sum t\cdot$PoolFactor. Your formula card defines overcollateralisation as funds paid less funds received.' One recall notes a tranche question came 'exactly the same from the mocks': a template-recycling flag.
Shapes covered here: S1 delinquency ratio, plain numerical as seen in August (Q1); S2 cash flow into the equity tranche after the interim waterfall, with defaults and fees (Q4); S3 overcollateralisation account balance at a year end (Q4); S4 shares paid to the senior and equity tranches in the final-year waterfall (Q5); S5 CPR, SMM and PSA conversion, with the deal's overcollateralisation as the twist (Q6); S6 prepayment rate theory and forecasting methodology (Q7); S7 closed-book formulas CPR, DSCR and WAL (Q6, Q7).

Revision notes

- **The specific: the interim waterfall** (CR-22 e). Loan interest from survivors $L_t = (r+s)\,(N-\sum_{\tau\le t} d_\tau)\times$ par; loans that default during the year pay *no* interest that year. Excess spread is $L_t - B$ (after any fees, which take priority over coupons). The amount diverted to the overcollateralisation account is $\max[\min(L_t - B, K), 0]$; the equity receives $\max(L_t - B - OC_t, 0)$. In Malz's base case (2% default rate) the full USD 1,750,000 is diverted every year.
- **Overcollateralization account and the custodian's tests** (CR-22 f, g). Value at the end of year t: $\sum_{\tau\le t}(1+r)^{t-\tau}(OC_\tau + R_\tau)$, i.e. last year's balance grown at the money market rate, plus this year's diversion, plus this year's recoveries (40% of defaulted par in the example). Two-step test each year: is $L_t - B > 0$? If yes, is $L_t - B \ge K$? (cap K goes to the account, any surplus to equity; otherwise all of it goes to the account and the equity gets nothing). If negative, can the account plus current recoveries cover $B - L_t$? If yes, the shortfall is paid out of it; if not, the account is depleted. Funds are released to the equity *only at maturity*. Malz calls the name "a bit of a misnomer": it is soft, not hard, credit enhancement.
- **Why recoveries are escrowed:** if recoveries flowed through the waterfall, the equity would perversely benefit from defaults; in real deals recoveries go to the senior bonds, then the mezzanine.
- **The final year** (CR-22 e). Four sources: survivors' interest, redemption at par of survivors, final-year recoveries, and the account balance grown one more year. No more diversion or OC test. Sequential shortfall tests: senior first (F vs senior par plus coupon), then mezzanine on $F -$ senior due; the equity gets $\max(F -$ total bond due$, 0)$. Malz's equity IRR is 23% at a 2% default rate and -92.1% and -95.5% at 7.5% and 10%.
- **Timing of defaults:** collateral cash flows are larger early in a deal's life; accelerating defaults hurt the bond tranches; other things equal the equity benefits if defaults come *later*.
- **Product types** (CR-22 a). Covered bonds (cover pool stays on the bank's balance sheet, so not a full securitisation); mortgage pass-throughs (true securitisations, prepayment risk); CMOs (sequential-pay time tranching); structured credit (sequential distribution of credit losses). CDOs repackage ABS, MBS or CLOs; CDO-squareds go one level further. Pools are static (auto, residential mortgages), revolving (credit cards) or managed (CLOs). Synthetic deals hold CDS against a reference portfolio.
- **Tranching** (CR-22 b). Equity: no fixed coupon, residual claim, thin, first loss. Mezzanine: high coupon, thin, next in line. Senior: low coupon, bulk of liabilities; if the pool cannot be financed mainly by low-yield senior debt the deal is not viable. Attachment point of the senior tranche = detachment point of the junior one; subordination (credit enhancement) is the share of the pool that must be lost before a tranche loses. A senior bond with 20% enhancement facing 21% pool losses loses $1/80 = 1.25\%$. Default in securitisations is "material impairment".

- **Waterfall** (CR-22 c; CR-23 b). Cash is paid top down: fees, then Class A interest, A coverage tests, Class B interest, B coverage tests, and so on, with the equity taking the residual (Choudhry Figure 23.4). An independent third-party agent runs the tests; if a test fails, the vehicle pays down the notes starting with the senior notes.
- **OC, first-loss and equity piece** (CR-23 b). Over-collateralisation: the principal value of notes issued is lower than the principal value of assets (hard enhancement, present at issue). The first-loss piece is the most junior note, lowest rated or non-rated, called the equity piece "even though it is a bond", and usually held by the originator. Excess spread is soft enhancement that accrues over time.
- **Trust vs SPV and the three structures** (CR-23 a, c). An SPV is bankruptcy remote from the sponsor; a Trust (represented by a Trustee) is more common in the US, companies in the Euromarkets. Amortising (pass-through): principal and interest paid each coupon, priced on expected maturity and WAL; RMBS, CMBS, consumer ABS. Revolving: principal collections buy new receivables during the revolving period (credit cards, auto loans); then controlled amortisation or soft bullet. Master trust: frequent issuers, multiple series from one pool; MBS and credit card ABS (preferred for cards since 1991).
- **Participants and conflicts** (CR-22 d; CR-23 a). Originator (sponsor/seller); underwriter or arranger (issuer/depositor, bears warehousing risk); rating agencies (paid by issuers and involved in setting enhancement, so incentive to require less); servicer (CMBS foreclose vs extend conflict between senior and junior); managers (first-loss retention, Dodd-Frank); trustee and custodian; monoline wraps.
- **Motivations** (CR-22 m; CR-23 d). Issuers: term funding and maturity matching, cheaper funding (the notes' rating is de-linked from the originator's), diversified funding, regulatory and economic capital relief (Basel I example: 12% target, 1.5% first loss retained, frees 10.5%), higher ROE, credit risk transfer; arbitrage vs balance sheet CDOs. Investors: access to loan pools, tailored risk-return, reaching for yield. Northern Rock: over-reliance on securitised funding.
- **Credit enhancements** (CR-23 e): over-collateralisation, pool insurance, senior/junior note classes, margin step-up, excess spread held in a reserve account. The lower the asset quality, the greater the enhancement needed.
- **Simulation, PD and correlation** (CR-22 h, i). Simulate correlated normals, map to default times $t = -\log[1-\Phi(z)]/\lambda$, count defaults per year, push each thread through the waterfall. Higher default probability hurts every tranche. Higher correlation *raises* equity value (U-shaped, "lottery ticket") and *raises* senior losses; the mezzanine behaves like a senior at low default rates and like equity at high ones.
- **Default01, risk factors, implied correlation** (CR-22 j, k, l). Default01 $= -[V(\pi + 10\text{bp}) - V(\pi - 10\text{bp})]/20$, peaking near each tranche's attachment point. Risk factors: systematic risk, tranche thinness, granularity (concentration risk), prepayment, extension and refinancing risk. Implied (base) correlation is backed out of standard tranche prices (CDX: 0-3, 3-7, 7-10, 10-15, 15-30%); it varies by tranche (correlation skew) and is a market, not credit, risk concept (see D15).
- **Performance tools by asset class** (CR-23 f, g). Credit cards: delinquency ratio (overdue more than 90 days / total receivables, an early indication of quality) and default ratio (written off in the period / receivables at period end), both tied to early amortisation triggers. MPR (principal and interest collections / pool balance) for all non-amortising classes, with a minimum set as an early amortisation trigger; it is *not* a prepayment measure. CMBS: DSCR (NOI / debt payments; below 1.0 the property cannot cover debt service; crucial because CMBS is non-recourse). Mortgages: WAC, WAM, WAL. Autos: loss curves and absolute prepayment speed.
- **WAC, WAM, WAL** (CR-23 g). WAC is balance-weighted and, compared with the net investor coupon, shows the pool's ability to pay. WAM is balance-weighted remaining term in months: on stated maturity it recognises liquidity risk; on the reset date it shows the shortening effect of prepayments. WAL $= \sum(a/365)\, PF(s)$ lets investors compare an MBS with investments of similar maturity (Table 23.1 gives 2.58 years). A WAL is tied to a prepayment assumption.
- **Prepayment forecasting** (CR-23 h). For a given price, prepayments *reduce* the yield on the MBS. SMM: an SMM of 0.65% means about 0.65% of the balance at the start of the month, less scheduled principal, prepays. CPR $= 1 - (1 - \text{SMM})^{12}$, based on pool characteristics and the expected economic environment. 100% PSA: CPR starts at 0, rises 0.2% a month to 6% at month 30, then constant 6%; other speeds are multiples, reflecting slower prepayment in the first 30 months of seasoning. Applies to mortgages, home equity and student loans. Auto loans: no real prepayment culture, prepayment speed extremely stable.

Formulas

Malz three-tranche CLO (annual, year-end payments).

$$L_t = (r+s)\Big(N - \sum_{\tau=1}^{t} d_\tau\Big) \times \text{par}, \qquad OC_t = \max\big[\min(L_t - B,\, K),\, 0\big], \qquad \text{Equity}_t = \max(L_t - B - OC_t,\, 0)$$

B: coupons due to the bond tranches (plus senior fees if any); K: annual cap on diversion; $R_t = \text{RR} \times d_t \times \text{par}$ paid into the account.

$$\text{OC account}_t = (1+r)\,\text{OC account}_{t-1} + OC_t + R_t$$

$$F = (1+r)\,\text{OC account}_{T-1} + \Big(N - \sum_{\tau=1}^{T} d_\tau\Big)(1+r+s)\,\text{par} + \text{RR}\, d_T\,\text{par}$$

Senior writedown $= \max(\text{senior due} - F, 0)$; mezzanine writedown $= \max[\text{mezz due} - (F - \text{senior due}), 0]$; equity terminal flow $= \max(F - \text{total bond due}, 0)$.

Choudhry performance measures (Table 23.2).

$$\text{Delinquency ratio} = \frac{\text{receivables overdue} > 90 \text{ days}}{\text{total receivables}}, \qquad \text{Default ratio} = \frac{\text{written off in period}}{\text{receivables at period end}}$$

$$\text{MPR} = \frac{\text{collections (principal and interest)}}{\text{outstanding pool balance}}, \qquad \text{DSCR} = \frac{\text{net operating income}}{\text{debt payments}}$$

$$\text{WAC} = \frac{\sum_i B_i c_i}{\sum_i B_i}, \qquad \text{WAM} = \frac{\sum_i B_i\, m_i}{\sum_i B_i}, \qquad \text{WAL} = \sum_s \frac{a_s}{365}\, PF(s)$$

$$\text{SMM} = \frac{\text{prepayment}}{\text{start-of-month pool balance} - \text{scheduled principal}}, \qquad \text{CPR} = 1 - (1 - \text{SMM})^{12}$$

$$\text{100\% PSA: CPR}_m = \min(0.2\% \times m,\, 6\%), \qquad \text{PSA} = \frac{\text{CPR}}{0.2\% \times m} \times 100 \;\; (m \le 30)$$

Absolute prepayment speed (autos) = prepayments / outstanding pool balance.

Exam traps

- **Two kinds of OC (sibling swap):** hard over-collateralisation is collateral par in excess of note par, fixed at issue; the Malz overcollateralisation *account* is soft enhancement built from diverted excess spread and escrowed recoveries, compounding at the money market rate and released to equity only at maturity. Your card's "funds paid less funds received" is not the books' wording (outside-source): use the book definitions.
- **Dropped or mis-sized terms in the interim year:** survivors are N minus *cumulative* defaults, not this year's only; fees come before coupons; recoveries go into the account, not to the equity; last year's balance must be grown at r.
- **Final-year order (sibling swap with pass-throughs):** the senior is paid par *plus* its final coupon in full before the mezzanine, and the equity gets only the residual. Splitting terminal funds pro rata is pass-through logic, not a credit waterfall.
- **PSA ramp threshold:** up to month 30 the 100% PSA CPR is $0.2\% \times m$; after that it is a flat 6%. SMM is monthly and on the *pool* balance net of scheduled principal; in an overcollateralised deal the note balance is the wrong denominator.
- **Polarity flips:** faster prepayments *reduce* yield for a given price and *shorten* WAL; correlation *helps* the equity and *hurts* the senior; an MPR *below* the minimum trips the early amortisation trigger.
- **Asset-class sibling swaps:** delinquency (overdue > 90 days) vs default (written off); MPR is not a prepayment measure; DSCR is a CMBS credit measure; ABS speed and loss curves are for autos.

Question A9.1

CR-23 g · S1 · [Aug-2026]

A junior surveillance analyst at an Indian asset manager is preparing the monthly investor report on a credit card master trust. The month-end ageing of receivables is:

Status	INR billion
Total credit card receivables (month end)	48.0
30 to 59 days past due	2.4
60 to 89 days past due	1.5
90 to 119 days past due	1.2
120 days or more past due	0.6
Receivables written off during the month	0.36

Which of the following is closest to the delinquency ratio as defined for credit card ABS performance analysis?

A. 0.75%

B. 2.50%

C. 3.75%

D. 11.88%

Solution **Answer: C**

Step 1 – Delinquent receivables (overdue more than 90 days)

$$1.2 + 0.6 = \text{INR } 1.8 \text{ billion}$$

⇒ *Only the 90-plus buckets meet the book's definition of delinquent.*

Step 2 – Delinquency ratio

$$\text{Delinquency ratio} = \frac{1.8}{48.0} = 3.75\%$$

⇒ *3.75% of the pool is seriously past due.*

Interpretation

Because delinquencies precede write-offs, a rising delinquency ratio is the early warning that the default ratio and the early amortisation triggers may be hit in coming months.

Why the other options are wrong

A is incorrect. Sibling swap: used write-offs, which gives the default ratio $0.36/48.0 = 0.75\%$.
B is incorrect. Scope error: counted only the 90 to 119 day bucket, $1.2/48.0 = 2.50\%$, omitting receivables 120 days or more past due.
D is incorrect. Scope error: counted every past-due bucket, $(2.4 + 1.5 + 1.2 + 0.6)/48.0 = 11.88\%$.

LO: CR-23 g: Define and calculate the delinquency ratio, default ratio, monthly payment rate (MPR), debt service coverage ratio (DSCR), the weighted average coupon (WAC), the weighted average maturity (WAM), and the weighted average life (WAL) for relevant securitized structures. · Reference: Choudhry, Structured Credit Products 2e, Ch.12, Performance Analysis for Credit Card ABS

Question A9.2 CR-23 f, g · [self-gen]

A rating agency analyst in Sydney is testing a credit card ABS against its monthly payment rate trigger. The trust documents set a minimum MPR of 15.0% as an early amortisation trigger: if the month's MPR falls below it, the revolving period ends. During the month, the trust's data are:

- Outstanding pool balance: AUD 1,500 million
- Principal collections: AUD 210 million
- Finance charge (interest) collections: AUD 27 million
- Receivables written off: AUD 6 million
- New receivables purchased by the trust (revolving period): AUD 180 million

Which of the following correctly states the MPR and whether the early amortisation trigger is hit?

A. MPR 3.8%; trigger hit

B. MPR 14.0%; trigger hit

C. MPR 15.8%; trigger not hit

D. MPR 15.8%; trigger hit

Solution **Answer: C**

Step 1 – Collections (principal and interest)

$$210 + 27 = \text{AUD } 237 \text{ million}$$

$\Rightarrow$ *MPR measures the proportion of the principal and interest on the pool repaid in the period.*

Step 2 – MPR

$$\text{MPR} = \frac{\text{collections}}{\text{outstanding pool balance}} = \frac{237}{1{,}500} = 15.8\%$$

$\Rightarrow$ *Cardholders repay 15.8% of the pool in the month.*

Step 3 – Trigger test

$$15.8\% \ \geq \ 15.0\% \text{ (minimum)} \ \Rightarrow \ \text{trigger not hit}$$

$\Rightarrow$ *The revolving period continues: principal collections keep buying new receivables.*

Interpretation

A falling MPR signals slower repayment; the rating agencies require every non-amortising ABS to set a minimum MPR as an early amortisation trigger. Write-offs and new purchases do not enter the ratio (deducting write-offs would give 15.4%, still above the floor).

Why the other options are wrong

A is incorrect. Netted off the new receivables bought in the revolving period, $(237 - 180)/1{,}500 = 3.8\%$, which then falls below the floor.

B is incorrect. Dropped interest collections, $210/1{,}500 = 14.0\%$, which falls below the floor.

D is incorrect. Right MPR, polarity flip on the trigger: treated the 15.0% minimum as a ceiling, so an MPR above it was read as a breach.

LO: CR-23 g: Define and calculate the delinquency ratio, default ratio, monthly payment rate (MPR), debt service coverage ratio (DSCR), the weighted average coupon (WAC), the weighted average maturity (WAM), and the weighted average life (WAL) for relevant securitized structures. · CR-23 f: Explain the various performance analysis tools for securitized structures and identify the asset classes they are most applicable to. · Reference: Choudhry, Ch.12, Performance Analysis for Credit Card ABS; Table 23.2

Question A9.3 CR-23 g · [self-gen]

A portfolio manager at a Canadian pension fund is reviewing an RMBS whose collateral pool consists of three loan groups. The notes pay investors a net coupon of 4.10%.

Loan group	Balance (CAD million)	Mortgage rate	Remaining term (months)
1	60	4.0%	300
2	90	5.2%	240
3	150	6.0%	120

Which of the following is closest to the pool's weighted average coupon (WAC)?

A. 1.26%

B. 4.80%

C. 5.07%

D. 5.36%

Solution **Answer: D**

Step 1 – Balance times rate for each group

$$60 \times 4.0\% = 2.40, \quad 90 \times 5.2\% = 4.68, \quad 150 \times 6.0\% = 9.00 \text{ (CAD million)}$$

Step 2 – Total balance

$$60 + 90 + 150 = \text{CAD } 300 \text{ million}$$

Step 3 – WAC

$$\text{WAC} = \frac{2.40 + 4.68 + 9.00}{300} = \frac{16.08}{300} = 5.36\%$$

⇒ *The pool earns 5.36% on average, weighted by loan balance.*

Interpretation

Against the 4.10% net investor coupon, the pool has 1.26 percentage points of coupon headroom, an indication of its ability to pay. WAC drifts as loans repay (here the high-rate group 3 has the shortest term). The remaining terms are irrelevant to WAC; they would give a WAM of 192 months.

Why the other options are wrong

A is incorrect. Intermediate result: reported WAC minus the net coupon, $5.36\% - 4.10\% = 1.26\%$.
B is incorrect. Wrong weights: weighted the rates by remaining term, $(300 \times 4.0 + 240 \times 5.2 + 120 \times 6.0)/660 = 4.80\%$.
C is incorrect. Wrong weights: simple average of the three rates, $(4.0 + 5.2 + 6.0)/3 = 5.07\%$.

LO: CR-23 g: Define and calculate the delinquency ratio, default ratio, monthly payment rate (MPR), debt service coverage ratio (DSCR), the weighted average coupon (WAC), the weighted average maturity (WAM), and the weighted average life (WAL) for relevant securitized structures. · Reference: Choudhry, Ch.12, Performance Analysis for MBS

Question A9.4

CR-22 e, f, g · S2, S3 · [Map #9]

A structured credit analyst at a Shanghai insurer, which holds the equity tranche of a five-year CLO, is checking the trustee's report for the end of year 2. The deal terms are:

- Collateral: 80 identical floating-rate loans of CNY 25 million each, bought at par, paying the reference rate + 400 bps
- Liabilities: senior notes CNY 1,600 million at reference + 60 bps; mezzanine notes CNY 280 million at reference + 450 bps; equity CNY 120 million
- The reference rate is flat at 3%; all payments are annual, at year end
- Trustee and management fees of CNY 4 million a year are paid ahead of all note coupons
- Up to CNY 20 million of excess spread a year is diverted to the overcollateralisation account, which earns 3%; any remainder goes to the equity
- A loan that defaults pays no interest for that year; 40% of the defaulted par is recovered at year end and paid into the overcollateralisation account
- Defaults: 3 loans in year 1 and 5 loans in year 2; the account held CNY 50 million at the end of year 1

Which of the following correctly states the overcollateralisation account balance and the cash flow to the equity tranche at the end of year 2?

A. Account CNY 121.5 million; equity CNY 23.40 million

B. Account CNY 120.0 million; equity CNY 23.40 million

C. Account CNY 121.5 million; equity CNY 28.65 million

D. Account CNY 71.5 million; equity CNY 73.40 million

Solution

Answer: A

Step 1 – Surviving loans and loan interest

$$N - \sum d = 80 - (3 + 5) = 72, \qquad L_2 = 72 \times 25 \times (0.03 + 0.04) = \text{CNY } 126.0 \text{ million}$$

⇒ *Only loans that have not defaulted by the end of year 2 pay interest; the eight defaulted loans pay nothing.*

Step 2 – Excess spread and the custodian's test

$$B = 1{,}600 \times 3.6\% + 280 \times 7.5\% = 57.6 + 21.0 = 78.6$$

$$L_2 - \text{fees} - B = 126.0 - 4.0 - 78.6 = 43.4 > 0, \quad 43.4 \geq K = 20 \;\Rightarrow\; OC_2 = 20.0$$

$\Rightarrow$ *Excess spread is positive and exceeds the cap, so exactly CNY 20 million is diverted.*

Step 3 – Cash flow to the equity

$$\text{Equity}_2 = 43.4 - 20.0 = \text{CNY } 23.40 \text{ million}$$

Step 4 – Overcollateralisation account at the end of year 2

$$R_2 = 0.40 \times 5 \times 25 = 50.0, \qquad 50.0 \times 1.03 + 20.0 + 50.0 = \text{CNY } 121.5 \text{ million}$$

$\Rightarrow$ *Last year's balance earns 3%, then this year's diversion and recoveries are added.*

Interpretation

The equity flow falls from CNY 32.15 million in year 1 to 23.40 million as cumulative defaults erode loan interest. The account, now CNY 121.5 million, is soft enhancement: it pays bond interest if loan interest ever falls short and reaches the equity only at maturity. (Dropping the fee would give equity 27.40; ignoring the cap would divert all 43.4 and leave the equity nothing.)

Why the other options are wrong

B is incorrect. Dropped term: did not grow the year-1 balance at 3%, $50.0 + 20.0 + 50.0 = 120.0$. The equity flow is unaffected.

C is incorrect. Wrong input: counted only year-2 defaults, so 75 loans pay interest, $L_2 = 131.25$, excess spread 48.65, equity $48.65 - 20 = 28.65$. The cap still binds, so the account is unchanged.

D is incorrect. Recoveries routed through the waterfall to the equity: equity $23.4 + 50.0 = 73.40$, account $51.5 + 20.0 = 71.5$. This is exactly the perverse benefit from defaults that escrowing recoveries prevents.

LO: CR-22 e: Calculate and evaluate one or two iterations of interim cashflows in a three-tiered securitization structure. · CR-22 f: Describe the treatment of excess spread in a securitization structure and estimate the value of the overcollateralization account at the end of each year. · CR-22 g: Explain the tests on the excess spread that a custodian must go through at the end of each year to determine the cash flow to the overcollateralization account and to the equity noteholders. · Reference: Malz, Financial Risk Management, Ch.9, Tracking the Interim Cash Flows; Table 22.1

Question A9.5

CR-22 c, e; CR-23 b · S4 · [Map #9]

A risk consultant is presenting the final-year waterfall of a four-year CLO to a group of newly hired analysts at a Frankfurt bank. The deal has:

- Collateral: 120 identical floating-rate loans of EUR 1 million each, bought at par, paying the reference rate + 380 bps; surviving loans pay their final interest and redeem at par at the end of year 4
- Liabilities: senior notes EUR 96 million at reference + 70 bps; mezzanine notes EUR 15 million at reference + 600 bps; equity EUR 9 million
- The reference rate is flat at 4%; no fees
- Defaults: 13 loans in years 1 to 3 and 5 loans in year 4; 40% of defaulted par is recovered at year end
- The overcollateralisation account held EUR 7.2 million at the end of year 3 and earns 4%

Which of the following correctly states the amounts paid at the end of year 4 to the senior notes and to the equity tranche?

A. Senior EUR 100.512 million; equity EUR 0.432 million

B. Senior EUR 100.512 million; equity EUR 2.432 million

C. Senior EUR 96.000 million; equity EUR 6.944 million

D. Senior EUR 95.555 million; equity EUR 8.958 million

Solution **Answer: B**

Step 1 – Overcollateralisation account at maturity

$$7.2 \times 1.04 = \text{EUR } 7.488 \text{ million}$$

⇒ *The account earns one more year of interest and is now released into the terminal waterfall.*

Step 2 – Terminal available funds

$$\text{Survivors} = 120 - 18 = 102: \quad \text{interest } 102 \times 7.8\% = 7.956, \quad \text{redemption } 102.0, \quad \text{recovery } 0.4 \times 5 = 2.0$$

$$F = 7.488 + 7.956 + 102.0 + 2.0 = \text{EUR } 119.444 \text{ million}$$

Step 3 – Senior shortfall test

$$\text{Senior due} = 96 \times (1 + 0.047) = 100.512 \le F \Rightarrow \text{paid in full; } 119.444 - 100.512 = 18.932 \text{ left}$$

Step 4 – Mezzanine test, then the equity residual

$$\text{Mezzanine due} = 15 \times (1 + 0.10) = 16.5 \le 18.932, \qquad \text{Equity} = 18.932 - 16.5 = \text{EUR } 2.432 \text{ million}$$

⇒ *Both bonds are money good; the equity's terminal claim is only what is left.*

Interpretation

With 18 cumulative defaults (15% of the pool) the equity collects only EUR 2.432 million at maturity on a EUR 9 million tranche: it absorbed the first loss, exactly as the capital structure intends. A few more defaults would push the shortfall into the mezzanine before the senior is touched.

Why the other options are wrong

A is incorrect. Dropped term: left out the final-year recoveries of EUR 2.0 million, so $F = 117.444$ and the equity gets 0.432.

C is incorrect. Paid the senior only its principal of 96.000 and forgot its final coupon, which inflates the equity residual to $119.444 - 96 - 16.5 = 6.944$.

D is incorrect. Sibling swap with a pass-through: split F pro rata to par, $119.444 \times 96/120 = 95.555$ and $119.444 \times 9/120 = 8.958$, ignoring the sequential priority of the waterfall.

LO: CR-22 c: Describe a waterfall structure in a securitization. · CR-22 e: Calculate and evaluate one or two iterations of interim cashflows in a three-tiered securitization structure. · CR-23 b: Explain the terms over-collateralization, first-loss piece, equity piece, and cash waterfall within the securitization process. · Reference: Malz, Ch.9, Tracking the Final-Year Cash Flows, Table 22.2; Choudhry, Ch.12, Figure 23.4

Question A9.6 CR-23 h · S5, S7 · [Map #9]

A treasury analyst at a US life insurer is reviewing the monthly trustee report on a non-agency RMBS, 18 months after the pool's origination. The report shows:

- Collateral pool balance at the start of the month: USD 820.0 million
- Notes outstanding at the start of the month: USD 779.0 million (the deal is overcollateralised)
- Overcollateralisation reserve account: USD 6.5 million
- Scheduled principal received in the month: USD 1.4 million
- Unscheduled prepayments received in the month: USD 5.2 million

Which of the following is closest to the pool's CPR and its speed expressed against the PSA benchmark?

A. CPR 0.64%; 17.6% PSA

B. CPR 7.36%; 123% PSA

C. CPR 7.36%; 204% PSA

D. CPR 7.74%; 215% PSA

Solution **Answer: C**

Step 1 – SMM

$$\text{SMM} = \frac{\text{prepayment}}{\text{pool balance} - \text{scheduled principal}} = \frac{5.2}{820.0 - 1.4} = \frac{5.2}{818.6} = 0.6352\%$$

$\Rightarrow$ *0.635% of the pool's start-of-month balance, net of scheduled principal, prepaid this month.*

Step 2 – CPR

$$\text{CPR} = 1 - (1 - 0.006352)^{12} = 7.36\%$$

$\Rightarrow$ *At this pace roughly 7.36% of the pool would prepay over a year.*

Step 3 – 100% PSA benchmark at month 18

$$m = 18 \leq 30 \;\Rightarrow\; \text{CPR}_{100\%\text{PSA}} = 0.2\% \times 18 = 3.6\%$$

$\Rightarrow$ *The pool is still on the seasoning ramp, so the 6% plateau does not apply yet.*

Step 4 – PSA speed

$$\text{PSA} = \frac{7.36\%}{3.6\%} \times 100 = 204\% \text{ PSA}$$

Interpretation

The pool is prepaying about twice as fast as the benchmark. For a given price this reduces the yield on the MBS and shortens the WAL, so the insurer should re-run its cash flow projections at the faster speed. The reserve account balance is irrelevant to the prepayment measures.

Why the other options are wrong

A is incorrect. Unit error: used the monthly SMM as if it were the annual CPR, $0.6352/3.6 \times 100 = 17.6\%$ PSA.
B is incorrect. Threshold error: compared with the 6% plateau although month 18 is still on the ramp, $7.36/6 \times 100 = 123\%$ PSA.
D is incorrect. Wrong input twin (the overcollateralisation twist): divided by the note balance instead of the pool balance, SMM $= 5.2/(779.0 - 1.4) = 0.6687\%$, CPR 7.74%, 215% PSA.

LO: CR-23 h: Explain the prepayment forecasting methodologies and calculate the constant prepayment rate (CPR) and the Public Securities Association (PSA) rate. · Reference: Choudhry, Ch.12, Example 23.2 Forecasting Prepayment Levels; Table 23.2

Question A9.7 CR-23 f, h; CR-22 k · S6, S7 · [Map #9]

An internal auditor at a Singapore life insurer is reviewing the investment team's methodology note on how prepayments are forecast for the insurer's MBS holdings. Which of the following statements in the note is correct?

A. The WAL of a pass-through note is quoted under a stated prepayment assumption, and a faster assumed PSA speed shortens it.

B. For a given price, faster prepayments increase the yield on the MBS, because principal is returned to investors earlier.

C. For commercial MBS, prepayment speed is tracked with the DSCR, the ratio of net operating income to debt payments.

D. A pool at 100% PSA prepays at a constant 6% CPR from origination, and faster speeds are multiples of that rate.

Solution **Answer: A**

Why A is correct

Amortising (pass-through) structures are priced and traded on expected maturity and WAL, and "a WAL approach incorporates various pre-payment assumptions"; Malz adds that "a WAL is associated with a particular prepayment assumption". With WAL $= \sum(a/365)\, PF(s)$, faster prepayment cuts the pool factor sooner, so the WAL shortens.

Interpretation

Prepayment forecasts (SMM, CPR, PSA) drive the projected cash flows, the yield and the WAL together; an audit should check that all three are run off the same prepayment assumption.

Why the other options are wrong

B is incorrect. Polarity flip: Choudhry states that prepayments, for a given price, *reduce* the yield on the MBS.
C is incorrect. Sibling swap: DSCR (net operating income / debt payments) is the CMBS *credit* measure of the borrower's ability to repay. Prepayment is measured by SMM, CPR and PSA for mortgages (Table 23.2).
D is incorrect. Scope error: dropped the seasoning ramp. 100% PSA starts at 0, rises 0.2% a month and reaches 6% only at month 30, because mortgages prepay slower in their first 30 months.

LO: CR-23 h: Explain the prepayment forecasting methodologies and calculate the constant prepayment rate (CPR) and the Public Securities Association (PSA) rate. · CR-23 f: Explain the various performance analysis tools for securitized structures and identify the asset classes they are most applicable to. · CR-22 k: Describe risk factors that impact structured products. · Reference: Choudhry, Ch.12, Amortising Structures, Example 23.2, Table 23.2; Malz, Ch.9, Capital Structure and Credit Losses in a Securitization

A10 Altman Z-Score and Linear Discriminant Analysis

Readings: CR-9 a, b, c · Estimating Default Probabilities (Hull, RMFI 6e, Ch.17): 9.1 Credit Ratings; 9.1.1 Internal Credit Ratings; 9.1.2 Altman's Z-Score; 9.2 Historical Default Probabilities; also CR-5 d · Introduction to Credit Risk Modeling and Assessment (Doumpos et al., Ch.1, 5.6.2 Data-Driven Empirical Models)

AUG 2026 RECALL | MAP TIER B · #15 · 6/16 SOURCES | MAP TIER D · #74 · 2/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: Altman z score - direct plug value question.
Map #15: "Reported as 'Altman Z score (1)', 'Zscore', and once as 'calculate z score (LDA)', linking it to the linear discriminant analysis framing. Know the five ratios, the weights, and the cut-off zones for classifying a firm. Also examinable: the relationship between borrower rating and PD (LO 27.c) and rating-migration matrices (LO 27.d)."
Map #74: "Reported: 'LDA approach'; and 'calculate z score (LDA)', which suggests a discriminant-analysis framing rather than the loss-distribution approach: these are two different things sharing an acronym and the recall is ambiguous. Cover both: loss frequency × severity convolution for operational capital, and linear discriminant analysis for scoring."
Shapes covered here:
S1 direct plug: compute Z from a set of filings (Q1)
S2 the five ratios and their weights recalled closed-book, none given in the stem (Q1, Q2, Q3)
S3 cut-off zones used to classify a firm (Q2, Q3)
S4 'calculate z score (LDA)': Z as Altman's linear discriminant function assigning a firm to a group (Q2)
S5 the two LDAs: linear discriminant analysis for scoring versus the loss distribution approach for operational capital (Q2; the loss distribution approach itself is in D19).

Revision notes

- **Agency ratings aim for stability** (CR-9 a): agencies want to avoid ratings reversals, so ratings change only when there is reason to believe in a long-term change in creditworthiness. They "rate through the cycle": a downturn that raises the six-month PD but barely moves the three-to-five-year PD would not usually change the rating. Bond traders, often bound by rating rules, are major users. See also D7 for point-in-time versus through-the-cycle.
- **Market-based PDs** (CR-9 a): Moody's KMV and Kamakura estimates are built from equity prices and other variables, do *not* have stability as an objective, and respond more quickly to market information than ratings (see A5).
- **Internal ratings** (CR-9 a): needed because agency ratings usually exist only for companies with publicly traded debt; many small and medium-sized firms are unrated. Inputs are profitability ratios (return on assets) and balance sheet ratios (current ratio, debt to equity). Banks convert the financials to a *cash flow* statement because it is cash rather than profit that repays a loan.
- **Altman (1968)** (CR-9 b): pioneered the use of accounting ratios to predict default, using "a statistical technique known as discriminant analysis" on five ratios. Doumpos dates the roots of data-driven empirical models to these late-1960s statistical bankruptcy models (CR-5 d).
- **Linear discriminant function** (S4): Z is a fixed-weight linear combination of the five ratios, and cut-off values of Z assign a firm to a group (fail or not fail). The weights were estimated so that the score separates the failed from the non-failed firms in the estimation sample (the statistical mechanics of discriminant analysis are outside-source; Hull gives only the technique's name and the fitted equation).
- **Estimation sample** (CR-9 b): 66 publicly traded manufacturing companies, 33 that failed within one year and 33 that did not. The model "proved to be fairly accurate when tested out of sample".
- **Type I and Type II errors** (CR-9 b): Type I = predicted *not* to go bankrupt but did; Type II = predicted to go bankrupt but did not. Both were small. Type I errors are liable to be the more costly for a commercial bank's lending department.
- **Scope** (CR-9 b): the 1.2 / 1.4 / 3.3 / 0.6 / 0.999 equation is for *publicly traded manufacturing* companies. Variations exist for manufacturers that are not publicly traded and for non-manufacturers, and the methodology has been revised and extended since 1968.

- **Direction and inputs:** a higher Z means a *lower* likelihood of default. Every coefficient is positive, so negative working capital or negative retained earnings pulls Z down. X_4 is the only ratio not over total assets: *market* value of equity over *book* value of total liabilities. X_3 uses EBIT, not net income.
- **Cut-off zones** (CR-9 b, S3): above 3.0 unlikely to default; 2.7 to 3.0 "on alert"; 1.8 to 2.7 a good chance of default; below 1.8 the probability of "a financial embarrassment" very high. Example 9.1: $Z = 5.46$, not in danger of defaulting in the near future.
- **Rating and PD** (CR-9 c): Table 9.1 shows cumulative PDs rising sharply as ratings fall. For investment grade, the PD in a year tends to *increase* with time in the first few years (A: 0.05%, 0.08%, 0.09%, 0.11%); for the lowest grade it can *decrease* (CCC/C: 28.3%, 10.03%, 5.09%, 2.94%) because the next year or two are critical. Migration matrices and hazard rates: see A15.
- **Empirical models, strengths and limits** (CR-5 d): they add transparency and consistency, can be validated against actual data, support scenario analysis and stress testing, and can be customised (large firms, SMEs, sectors). Their main weakness is reliance on historical data that are backward looking and static (financial statements reflect only the current status) and not updated in real time.
- **The other LDA** (S5): in operational risk, LDA is the *loss distribution approach*: a frequency distribution and a severity distribution are combined (convolution) into an annual loss distribution for capital. It shares only the acronym with discriminant analysis. See D19.

Formulas

Altman's original Z-score (Hull eq. 9.1), publicly traded manufacturing companies:

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 0.999X_5$$

- X_1 = Working capital / Total assets (working capital = current assets − current liabilities, can be negative)
- X_2 = Retained earnings / Total assets
- X_3 = Earnings before interest and taxes (EBIT) / Total assets
- X_4 = Market value of equity / Book value of total liabilities
- X_5 = Sales / Total assets

Zones (as stated in the book):

Z-score	Interpretation
$Z > 3.0$	Unlikely to default
2.7 to 3.0	"On alert"
1.8 to 2.7	Good chance of default
$Z < 1.8$	Probability of a financial embarrassment very high

Reverse-solve for one input. Required ratio $=$ (target Z − contribution of the other four)/its coefficient; then multiply by that ratio's denominator.
Book check (Example 9.1). WC 170,000; TA 670,000; EBIT 60,000; Sales 2,200,000; MVE 380,000; TL 240,000; RE 300,000 give $X_1 = 0.254$, $X_2 = 0.448$, $X_3 = 0.0896$, $X_4 = 1.583$, $X_5 = 3.284$ and $Z = 5.46$.

Exam traps

- **Wrong input twin in X_4:** book value of equity instead of *market* value, or total assets instead of total *liabilities* as the denominator.
- **Wrong input twin in X_3:** net income instead of EBIT.
- **Coefficient swap:** 1.4 belongs to retained earnings, 3.3 to EBIT; 0.6 to the equity ratio, 0.999 to sales. A swap moves Z by enough to change the zone.
- **Working capital twin:** current assets instead of current assets minus current liabilities; dropping the minus sign when working capital is negative.
- **Zone boundaries (polarity):** high Z is safe. The boundaries are 3.0, 2.7 and 1.8; 1.8 to 2.7 is "good chance of default", 2.7 to 3.0 is "on alert".

- **Acronym swap:** "LDA" in a credit-scoring stem is linear discriminant analysis; in an operational risk capital stem it is the loss distribution approach.

Question A10.1

CR-9 b · S1, S2 · [Aug-2026] [Map #15]

A junior credit analyst at a Canadian commercial bank is preparing an internal credit memo on a listed auto-parts manufacturer. Extracts from its latest filings (CAD million):

Current assets	180	Total assets	600
Current liabilities	126	Total liabilities	390
Retained earnings	96	Book value of equity	210
EBIT	42	Market value of equity	330
Sales	690		

Which of the following is closest to the firm's Altman Z-score?

A. 2.03

B. 2.10

C. 2.22

D. 2.47

Solution

Answer: C

Step 1 – Working capital

$$\text{WC} = \text{CA} - \text{CL} = 180 - 126 = 54$$

Step 2 – The five ratios

$$X_1 = 54/600 = 0.0900, \qquad X_2 = 96/600 = 0.1600, \qquad X_3 = 42/600 = 0.0700,$$
$$X_4 = 330/390 = 0.8462, \qquad X_5 = 690/600 = 1.1500$$

$\Rightarrow$ *X_4 uses the market value of equity (330) over total liabilities (390); the book equity of 210 is not an input.*

Step 3 – Weighted sum

$$\begin{aligned} Z &= 1.2(0.0900) + 1.4(0.1600) + 3.3(0.0700) + 0.6(0.8462) + 0.999(1.1500) \\ &= 0.1080 + 0.2240 + 0.2310 + 0.5077 + 1.1489 = 2.2195 \end{aligned}$$

Interpretation

$Z = 2.22$ lies between 1.8 and 2.7: by Altman's original cut-offs there is a good chance of default. The memo should flag the name for closer review rather than standard approval.

Why the other options are wrong

A is incorrect. Used the book value of equity in X_4: $X_4 = 210/390 = 0.5385$, giving $Z = 2.03$.
B is incorrect. Swapped the coefficients of X_4 and X_5 (0.999 on the equity ratio, 0.6 on sales): $Z = 2.10$.
D is incorrect. Used current assets instead of working capital in X_1: $X_1 = 180/600 = 0.30$, giving $Z = 2.47$.

LO: CR-9 b: Describe Altman's Z-score methodology and apply it to predict a firm's default and to classify a sample of firms by credit quality. · Reference: Hull, RMFI 6e, Ch.17, 9.1.2 eq. 9.1, Example 9.1

Question A10.2

CR-9 b · S2, S3, S4, S5 · [Map #15] [Map #74]

A credit risk consultant is validating an Australian lender's scoring engine. The engine applies Altman's original linear discriminant function: the five accounting ratios are combined with Altman's fixed discriminant weights into a single score Z, and the cut-off values of Z assign each listed manufacturer to a credit-quality group. The consultant re-scores one listed packaging manufacturer (AUD million):

- Total assets 800; current assets 250; current liabilities 280
- Retained earnings 40; EBIT 40; net income 8
- Market value of equity 330; book value of equity 240; total liabilities 560; sales 1,040

Which of the following correctly states the discriminant score and the group to which the firm is assigned?

A. 1.71; probability of financial embarrassment very high

B. 1.75; probability of financial embarrassment very high

C. 1.84; on alert

D. 1.84; good chance of default

Solution **Answer: D**

Step 1 – Working capital

$$\text{WC} = 250 - 280 = -30$$

$\Rightarrow$ *Negative working capital: short-term obligations exceed short-term assets, so X_1 enters with a minus sign.*

Step 2 – The five ratios

$$X_1 = -30/800 = -0.0375, \quad X_2 = 40/800 = 0.0500, \quad X_3 = 40/800 = 0.0500,$$
$$X_4 = 330/560 = 0.5893, \quad X_5 = 1{,}040/800 = 1.3000$$

Step 3 – Discriminant score

$$\begin{aligned} Z &= 1.2(-0.0375) + 1.4(0.05) + 3.3(0.05) + 0.6(0.5893) + 0.999(1.3) \\ &= -0.0450 + 0.0700 + 0.1650 + 0.3536 + 1.2987 = 1.8423 \end{aligned}$$

Step 4 – Assign the group

$1.8 < 1.84 \leq 2.7$, so the firm falls in the zone where there is a good chance of default.

Interpretation

The engine is doing exactly what Altman's discriminant analysis does: collapse five ratios into one linear score and read the group off fixed cut-offs. The firm sits just above the 1.8 boundary, so any input error that lowers Z by more than 0.05 moves it into the worst group. "LDA" here is linear discriminant analysis, a scoring tool; it has nothing to do with the loss distribution approach used for operational risk capital.

Why the other options are wrong

A is incorrect. Used net income instead of EBIT in X_3: $X_3 = 8/800 = 0.01$, $Z = 1.71$, which wrongly drops the firm below 1.8.

B is incorrect. Used the book value of equity in X_4: $X_4 = 240/560 = 0.4286$, $Z = 1.75$.

C is incorrect. Correct score, wrong zone: "on alert" is the 2.7 to 3.0 band; 1.8 to 2.7 is "good chance of default".

LO: CR-9 b: Describe Altman's Z-score methodology and apply it to predict a firm's default and to classify a sample of firms by credit quality. · Reference: Hull, RMFI 6e, Ch.17, 9.1.2 eq. 9.1

Question A10.3 CR-9 b · S2, S3 · [Map #15]

The treasurer of a listed Indian textile manufacturer is preparing for a refinancing. The firm's current Altman Z-score places it in the zone in which a company is considered unlikely to default. Holding all other items fixed (INR crore):

- Total assets 400; working capital 60; retained earnings 48; current EBIT 170
- Market value of equity 210; total liabilities 280; sales 380

Which of the following is closest to the largest fall in EBIT the firm can absorb while remaining in that zone?

A. INR 18.1 crore

B. INR 42.7 crore

C. INR 54.5 crore

D. INR 163.6 crore

Solution **Answer: A**

Step 1 – Z contribution excluding EBIT

$$X_1 = 60/400 = 0.15, \quad X_2 = 48/400 = 0.12, \quad X_4 = 210/280 = 0.75, \quad X_5 = 380/400 = 0.95$$
$$Z_{\text{ex } X_3} = 1.2(0.15) + 1.4(0.12) + 0.6(0.75) + 0.999(0.95) = 0.180 + 0.168 + 0.450 + 0.949 = 1.7471$$

Step 2 – Current Z

$$X_3 = 170/400 = 0.425, \qquad Z = 1.7471 + 3.3(0.425) = 1.7471 + 1.4025 = 3.1496$$

$\Rightarrow$ *The firm sits about 0.15 above the 3.0 boundary of the "unlikely to default" zone.*

Step 3 – EBIT at the 3.0 boundary

$$3.3X_3 = 3.0 - 1.7471 = 1.2530 \;\Rightarrow\; X_3 = 0.3797 \;\Rightarrow\; \text{EBIT} = 0.3797 \times 400 = 151.9$$

Step 4 – Largest fall

$$170 - 151.9 = 18.1$$

Interpretation

EBIT can fall by only about INR 18 crore (11%) before the firm drops into the "on alert" band. EBIT carries the largest coefficient (3.3), so it is the most powerful single lever in both directions.

Why the other options are wrong

B is incorrect. Used 1.4 (the retained-earnings coefficient) on EBIT: $(3.1496 - 3.0)/1.4 \times 400 = 42.7$.
C is incorrect. Targeted 2.7, the lower edge of the "on alert" zone: $170 - 115.5 = 54.5$.
D is incorrect. Targeted the 1.8 cut-off: $170 - 6.4 = 163.6$.

LO: CR-9 b: Describe Altman's Z-score methodology and apply it to predict a firm's default and to classify a sample of firms by credit quality. · Reference: Hull, RMFI 6e, Ch.17, 9.1.2 eq. 9.1

A11 Liquidity-Adjusted VaR and Cost of Liquidation

Readings: LTR-1 a · Liquidity Risk (Hull, Ch.24); LTR-2 e, f · Liquidity and Leverage (Malz, Ch.12)

AUG 2026 RECALL | MAP TIER B · #11 · 6/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: LVaR CoL; liquidity-adjusted VaR theory.
Map #11: "Reported: '1 numerical on liquidity adjusted VaR'; 'liquidity var numerical'; 'CAR, LVaR'. Two constructions must be kept apart: the constant-spread approach (add $\frac{1}{2}\times$ spread $\times$ position) and the exogenous-spread approach using the spread's own mean and volatility: stressed cost of liquidation $= 0.5 \cdot V \cdot (\mu_{\text{spread}} + z \cdot \sigma_{\text{spread}})$. Picking the wrong one is the sibling swap."
Shapes covered here: S1 LVaR numerical with the normal-market (constant-spread) cost, the 'CAR, LVaR' pairing (CAR itself: see D3) (Q1); S2 stressed (exogenous-spread) cost of liquidation $0.5\,V(\mu+z\sigma)$, with the constant-spread input as the sibling-swap distractor (Q1, Q2); S3 cost of liquidation (CoL) as its own number, plain form of the August recall (Q1, Q2); S4 LVaR theory: normal vs stressed spread, perfect spread correlation, liquidation horizon (Q5); S5 Malz expected transactions cost and 99% spread risk factor (Q3); S6 Malz multi-day liquidation adjustment, with time to escape from trading volume (Q4).

Revision notes

- **The specific: two cost-of-liquidation constructions** (LTR-1 a). Hull's *normal market* cost charges half the current proportional spread on each position, $\sum s_i\alpha_i/2$ (eq. 1.1): this is the map's "constant-spread" approach. The *stressed market* cost replaces s_i with $\mu_i + \lambda\sigma_i$, the mean plus λ standard deviations of the proportional spread (eq. 1.2): the map's "exogenous-spread" approach (labels outside-source). Hull's Example 1.2: stressed cost USD 36.58 million, almost five times the normal-market USD 7.5 million.
- **The confidence parameter λ.** $\lambda = 2.326$ for a 99% spread if spreads are normal. Spreads are not normally distributed in practice, so λ can be set from the empirical distribution (e.g. 3.6 if the 99th percentile is 3.6 standard deviations above the mean).
- **Two LVaR definitions** (Hull eq. 1.3 and its stressed version, from Bangia et al.). LVaR = regular VaR + cost of unwinding in a *normal* market, or regular VaR + cost of unwinding in a *stressed* market. VaR captures mark-to-market price moves; the cost of liquidation assumes market prices do *not* change.
- **Mid-market price is the fair price.** Sells are done at the bid, buys at the offer, so each trade costs half the spread, $s\alpha/2$. Spreads range from 0.05% to 5% or even 10% of mid. Above the market maker's size limit, the bid falls and the offer rises with trade size (in retail markets the pattern can be the opposite).
- **Perfect spread correlation is not over-conservative.** Equation (1.2) assumes spreads on all instruments are perfectly correlated; "this may seem overly conservative, but it is not": when liquidity is tight, spreads widen together.
- **Diversification and position size.** Diversification reduces market risk but does *not* necessarily reduce liquidity trading risk. Because s_i rises with position size, many small positions carry less liquidity risk than a few large ones; position-size limits reduce liquidity trading risk.
- **Liquidation horizon.** Both cost measures are *decreasing* functions of the time allowed for liquidation. Fast unwind: large spreads, small risk of adverse mid-market moves. Slow unwind: smaller spreads, larger market-move risk.
- **Optimal unwinding (Almgren and Chriss).** Choose daily trades q_i to minimise a liquidity-adjusted VaR of the unwind. Over n days trade *more* than $1/n$ on day 1, since the longer any part is held the greater the risk of adverse moves. Hull's Example 1.3 at 95%: 48.9, 30.0, 14.1, 5.1, 1.9 million units; at 50% (expected cost only) equal amounts of 20 million a day.
- **Other market liquidity measures and transparency.** Trading volume per day, price impact of a trade, and the Amihud measure (average of absolute daily return / daily dollar volume); expected return rises as liquidity falls. Opaque products (ABS CDOs after 2007) lose liquidity quickly: Merrill Lynch sold USD 30.6 billion of former AAA tranches for 22 cents on the dollar. Predatory trading (London Whale, LTCM) makes large exits costlier.

- **Liquidity black holes** (LTR-1 d). Liquidity dries up because everyone wants to sell (a "crowded exit"). Negative feedback traders (buy when prices fall) stabilise; positive feedback traders destabilise. Causes of positive feedback trading: trend trading, stop-loss rules, dynamic hedging of *short* option positions, creating options synthetically (1987 portfolio insurance), margin calls, predatory trading, LTCM's forced unwind. Basel III LCR and NSFR (LTR-1 c): see A32.
- **Malz's two liquidity risks** (LTR-2 e). *Transactions liquidity risk*: moving the price adversely in the act of trading. *Funding liquidity risk* (balance sheet risk): creditors withdraw credit or change its terms. Hull's labels for the same pair: liquidity trading risk and liquidity funding risk.
- **Measuring funding liquidity risk** (LTR-2 e). Part of bank ALM: track and forecast cash and funding sources against cash needs, keep ratios of ready cash and marketable securities. Goldman Sachs' "Global Core Excess" liquidity buffer (about 20% of the balance sheet) pre-funds crisis needs. Hedge fund liquidity sources: cash (broker balances carry broker risk; money funds can break the buck), unpledged assets (limited by price volatility and collateral eligibility, and subject to transactions liquidity risk), and unused borrowing capacity (revocable through haircuts).
- **Market liquidity vocabulary** (Malz). Tightness (round-trip cost: bid-ask spread and commissions), depth (order size that moves the market), resiliency (how long a lumpy order keeps price from equilibrium). Illiquidity shows up as bid-ask spread, adverse price impact and slippage. Quantitative transactions-liquidity measures are less widely used than funding-liquidity measures, partly because they are not in the regulatory framework.
- **Transaction cost liquidity risk** (LTR-2 f). Daily changes in the relative spread are normal with *zero mean* and constant variance σ_s^2; $\bar{s}$ is set to the latest observed relative spread, P to the latest midprice. The expected transactions cost is the half spread; $\frac{1}{2}(\bar{s} + 2.33\sigma_s)$ is the 99% *spread risk factor*, the worst-case bid-ask cost of changing a position.
- **Adverse price impact: multi-day LVaR** (LTR-2 f). Estimate the days T for orderly liquidation ("time to escape", usually position size vs daily volume). Liquidating in equal parts means holding $1, \frac{T-1}{T}, \ldots, \frac{1}{T}$ of the position, so $\sqrt{T}$ overstates: the LVaR lies between the one-day VaR and one-day VaR $\times\sqrt{T}$. $T = 5$ adds 48%; $T = 10$ roughly doubles the overnight VaR.

Formulas

Proportional bid-offer spread (Hull).

$$s = \frac{\text{Offer price} - \text{Bid price}}{\text{Mid-market price}}, \qquad \text{mid} = \tfrac{1}{2}(\text{bid} + \text{offer})$$

Cost of liquidation, normal market (eq. 1.1) and stressed market (eq. 1.2).

$$\text{Cost}_{\text{normal}} = \sum_{i=1}^{n} \frac{s_i\,\alpha_i}{2}, \qquad \text{Cost}_{\text{stressed}} = \sum_{i=1}^{n} \frac{(\mu_i + \lambda\sigma_i)\,\alpha_i}{2}$$

s_i: proportional spread in normal conditions; α_i: mid-market value of position i; μ_i, σ_i: mean and standard deviation of the *proportional* spread (a dollar spread moment is divided by the mid price); λ: spread confidence parameter (2.326 at 99% if normal).

Liquidity-adjusted VaR (eq. 1.3 and stressed version).

$$\text{LVaR} = \text{VaR} + \sum_i \frac{s_i\alpha_i}{2} \qquad \text{or} \qquad \text{LVaR} = \text{VaR} + \sum_i \frac{(\mu_i + \lambda\sigma_i)\alpha_i}{2}$$

Malz: expected transactions cost and 99% transactions cost (per unit; multiply by units held).

$$\bar{s} = \frac{\text{ask} - \text{bid}}{\text{midprice}}, \qquad \text{expected cost} = \tfrac{1}{2}\,\bar{s}\,P, \qquad 99\%\text{ cost} = P \cdot \tfrac{1}{2}(\bar{s} + 2.33\,\sigma_s)$$

Malz: liquidation over T days in equal parts.

$$\text{LVaR}_T = \text{VaR}_{\text{1-day}} \times \sqrt{\frac{(1+T)(1+2T)}{6T}}, \qquad 1 < \sqrt{\frac{(1+T)(1+2T)}{6T}} < \sqrt{T} \;\; (T > 1)$$

Factor: $T = 5 \Rightarrow 1.4832$; $T = 8 \Rightarrow 1.7854$; $T = 10 \Rightarrow 1.962$ ("doubles").

Exam traps

- **Constant vs exogenous spread (sibling swap):** the normal-market cost uses the *current* quoted spread; the stressed cost uses the spread's *mean* and standard deviation. Plugging the current quote into $\mu + \lambda\sigma$, or adding $\lambda\sigma$ to a normal-market definition, is the reported trap. Read which LVaR definition the stem gives.
- **Full spread instead of half:** the cost is *half* the spread times mid value (Hull), and Malz's $\frac{1}{2}$ applies to *both* $\bar{s}$ and $2.33\sigma_s$. Halving only $\bar{s}$, or halving twice, are the twin slips.
- λ **on the wrong term:** $\mu + \lambda\sigma$, not $\lambda(\mu + \sigma)$.
- $\sqrt{T}$ **trap (scope error):** the position shrinks each day, so the multi-day factor is $\sqrt{(1+T)(1+2T)/(6T)}$, not $\sqrt{T}$; T comes from position size vs the volume the firm will actually trade per day.
- **Intermediate results:** the cost of liquidation is not the LVaR; Malz's expected cost $\frac{1}{2}\bar{s}P$ is not the 99% cost; the adjustment increment is not the adjusted VaR.
- **Polarity flips in theory:** longer liquidation horizon *lowers* spread cost but *raises* market-move risk; perfect spread correlation is *not* overly conservative; diversification does *not* necessarily reduce liquidity trading risk.

Question A11.1

LTR-1 a · S1, S2, S3 · [Aug-2026] [Map #11]

A market risk analyst at a UK asset manager is reporting liquidity-adjusted VaR, defined by the firm as regular VaR plus the cost of unwinding positions in a *normal* market. The one-day 99% VaR of the book is GBP 9.4 million. Positions and spread data are:

Position	Mid-market value	Current quote	Mean prop. spread	SD prop. spread
Equity block	GBP 200 million	bid 49.80 / offer 50.20	0.90%	0.60%
Corporate bond	GBP 150 million	bid 98.70 / offer 99.30	0.70%	0.40%

Which of the following is closest to the liquidity-adjusted VaR?

A. GBP 10.03 million

B. GBP 10.65 million

C. GBP 11.91 million

D. GBP 12.92 million

Solution

Answer: B

Step 1 – Proportional spreads from current quotes

$$s_1 = \frac{50.20 - 49.80}{50.00} = 0.008000, \qquad s_2 = \frac{99.30 - 98.70}{99.00} = 0.006061$$

⇒ *A round trip in the equity costs 0.80% of mid value; in the bond 0.61%.*

Step 2 – Normal-market cost of liquidation

$$\text{Cost}_{\text{normal}} = \tfrac{1}{2}(200)(0.008000) + \tfrac{1}{2}(150)(0.006061) = 0.8000 + 0.4545 = \text{GBP } 1.2545 \text{ million}$$

⇒ *Selling everything at the bid rather than mid would cost about GBP 1.25 million if prices did not move.*

Step 3 – Liquidity-adjusted VaR

$$\text{LVaR} = 9.4 + 1.2545 = \text{GBP } 10.65 \text{ million}$$

Interpretation

Liquidity adds about 13% to the market VaR in normal conditions. The spread mean and SD columns are irrelevant under the firm's definition; they would only enter a stressed-market LVaR (at 95%, $\lambda = 1.645$, that would be GBP 12.31 million).

Why the other options are wrong

A is incorrect. Halved twice: took the half-spread $(\text{offer} - \text{bid})/(2 \times \text{mid})$ as s and then applied the $\frac{1}{2}$ of eq. 1.1 again, $9.4 + 1.2545/2 = 10.03$.

C is incorrect. Charged the full spread instead of half: $9.4 + 2(1.2545) = 11.91$.

D is incorrect. Sibling swap: used the stressed (exogenous-spread) cost at 99% when the firm's definition is normal-market, $9.4 + \frac{1}{2}(200)(0.009 + 2.326 \times 0.006) + \frac{1}{2}(150)(0.007 + 2.326 \times 0.004) = 9.4 + 3.518 = 12.92$.

LO: LTR-1 a: Explain and calculate liquidity trading risk via cost of liquidation and liquidity-adjusted VaR (LVaR). · Reference: Hull, Risk Management and Financial Institutions 5e, Ch.24, 1.1.2 Measuring Market Liquidity, eq. (1.1); 1.1.3 Liquidity-Adjusted VaR, eq. (1.3)

Question A11.2

LTR-1 a · S2, S3 · [Aug-2026] [Map #11]

The head of market risk at a Canadian dealer computes liquidity-adjusted VaR as regular VaR plus the cost of unwinding in a *stressed* market, with spreads assumed normally distributed and a 99% spread confidence level ($\lambda = 2.326$). The one-day 99% VaR is CAD 7.5 million. The book holds:

- Provincial bonds: mid-market value CAD 120 million; proportional spread mean 1.0%, standard deviation 1.2%
- 2 million shares of a mid-cap stock, currently quoted bid CAD 39.60 / offer CAD 40.40; the dollar bid-offer spread has a mean of CAD 0.16 and a standard deviation of CAD 0.20

Which of the following is closest to the stressed liquidity-adjusted VaR?

A. CAD 10.40 million

B. CAD 11.04 million

C. CAD 11.41 million

D. CAD 13.30 million

Solution

Answer: A

Step 1 – Convert the stock's dollar spread moments to proportional

$$\text{mid} = \tfrac{1}{2}(39.60 + 40.40) = 40.00, \qquad \alpha_2 = 2 \times 40.00 = \text{CAD } 80 \text{ million}$$

$$\mu_2 = \frac{0.16}{40.00} = 0.0040, \qquad \sigma_2 = \frac{0.20}{40.00} = 0.0050$$

$\Rightarrow$ *Equation (1.2) needs proportional spreads, so dollar moments are divided by the mid price, as in Hull's Example 1.2.*

Step 2 – Stressed spread for each position

$$\mu_1 + \lambda\sigma_1 = 0.010 + 2.326(0.012) = 0.037912, \qquad \mu_2 + \lambda\sigma_2 = 0.004 + 2.326(0.005) = 0.015630$$

$\Rightarrow$ *A 1-in-100 spread for the bonds is 3.79% of mid, almost four times its average.*

Step 3 – Stressed cost of liquidation

$$\text{Cost}_{\text{stressed}} = \tfrac{1}{2}(120)(0.037912) + \tfrac{1}{2}(80)(0.015630) = 2.2747 + 0.6252 = \text{CAD } 2.8999 \text{ million}$$

Step 4 – Stressed LVaR

$$\text{LVaR} = 7.5 + 2.8999 = \text{CAD } 10.40 \text{ million}$$

Interpretation

The stressed liquidation cost is about 3.8 times the normal-market cost built from the mean spreads (CAD 0.76 million). Equation (1.2) assumes spreads move together, which Hull argues is realistic when liquidity is tight, so no diversification credit is taken.

Why the other options are wrong

B is incorrect. Sibling swap (constant-spread input in the exogenous-spread formula): used the current quoted spread $0.80/40.00 = 2.0\%$ as μ_2, so $7.5 + 2.2747 + \frac{1}{2}(80)(0.020 + 2.326 \times 0.005) = 11.04$.

C is incorrect. Applied λ to the sum of mean and SD, $\lambda(\mu + \sigma)$: $7.5 + \frac{1}{2}(120)(2.326)(0.022) + \frac{1}{2}(80)(2.326)(0.009) = 11.41$.

D is incorrect. Charged the full stressed spread instead of half: $7.5 + 2(2.8999) = 13.30$.

LO: LTR-1 a: Explain and calculate liquidity trading risk via cost of liquidation and liquidity-adjusted VaR (LVaR). · Reference: Hull, Ch.24, 1.1.2 Measuring Market Liquidity, eq. (1.2), Example 1.2; 1.1.3 Liquidity-Adjusted VaR

Question A11.3

LTR-2 f · S5 · [self-gen]

A risk analyst at a Singapore hedge fund measures transactions cost liquidity risk following Malz. The fund holds 2 million shares of a stock quoted bid SGD 25.10 / ask SGD 25.30. Daily changes in the relative spread are assumed normal with zero mean and a volatility of 0.45%. The one-day 99% market VaR of the position is SGD 2.6 million. Which of the following is closest to the 99% transactions cost of liquidating the position?

A. SGD 0.200 million

B. SGD 0.387 million

C. SGD 0.464 million

D. SGD 0.728 million

Solution

Answer: C

Step 1 – Midprice, position value and relative spread

$$P = \tfrac{1}{2}(25.10 + 25.30) = 25.20, \qquad \text{Value} = 2{,}000{,}000 \times 25.20 = \text{SGD } 50.4 \text{ million}$$

$$\bar{s} = \frac{25.30 - 25.10}{25.20} = 0.0079365$$

Step 2 – Expected transactions cost (intermediate)

$$\tfrac{1}{2}\bar{s} \times \text{Value} = \tfrac{1}{2}(0.0079365)(50.4) = \text{SGD } 0.200 \text{ million}$$

$\Rightarrow$ *This is the typical mid-to-bid cost of selling out on an ordinary day.*

Step 3 – 99% spread risk factor

$$\tfrac{1}{2}(\bar{s} + 2.33\sigma_s) = \tfrac{1}{2}(0.0079365 + 2.33 \times 0.0045) = \tfrac{1}{2}(0.0184215) = 0.0092107$$

Step 4 – 99% transactions cost

$$0.0092107 \times 50.4 = \text{SGD } 0.464 \text{ million}$$

Interpretation

At 99% confidence the bid-ask cost of changing the position is SGD 0.46 million, more than twice the expected cost. The market VaR is irrelevant to this measure; it would matter only for the multi-day adverse-price-impact adjustment.

Why the other options are wrong

A is incorrect. Intermediate result: stopped at the expected transactions cost $\frac{1}{2}\bar{s}P$, 0.200.
B is incorrect. Wrong-input twin: used the 95% multiplier 1.645 instead of 2.33, $50.4 \times \frac{1}{2}(0.0079365 + 1.645 \times 0.0045) = 0.387$.
D is incorrect. Halved only $\bar{s}$ and not the volatility term: $50.4 \times (0.0039683 + 2.33 \times 0.0045) = 0.728$.

LO: LTR-2 f: Calculate the expected transactions cost and the spread risk factor for a transaction and calculate the liquidity adjustment to VaR for a position to be liquidated over a number of trading days. · Reference: Malz, Financial Risk Management, Ch.12, 2.5 Liquidity Risk Measurement, Transaction Cost Liquidity Risk

Question A11.4

LTR-2 e, f · S6 · [self-gen]

A portfolio risk manager at an Australian superannuation fund holds 2.4 million shares of a mid-cap stock. The position's one-day 99% VaR is AUD 4.0 million. The stock trades about 1.2 million shares a day, and the fund's execution policy caps its own selling at 25% of daily volume. The position would be liquidated in equal parts at the end of each trading day. Using Malz's approach, which of the following is closest to the liquidity-adjusted VaR of the position?

A. AUD 4.47 million

B. AUD 7.14 million

C. AUD 11.31 million

D. AUD 12.75 million

Solution **Answer: B**

Step 1 – Time to escape

$$T = \frac{\text{position}}{\text{shares sold per day}} = \frac{2.4}{0.25 \times 1.2} = \frac{2.4}{0.30} = 8 \text{ days}$$

⇒ *Malz: the time to liquidate is usually based on comparing position size with daily transactions volume; here the volume the fund will actually trade is 0.30 million shares a day.*

Step 2 – Liquidity adjustment factor

$$\sqrt{\frac{(1+T)(1+2T)}{6T}} = \sqrt{\frac{(9)(17)}{48}} = \sqrt{3.1875} = 1.7854$$

⇒ *Holding a declining fraction* $1, \frac{7}{8}, \ldots, \frac{1}{8}$ *raises VaR by 78.5%, far less than* $\sqrt{8} = 2.83$.

Step 3 – Liquidity-adjusted VaR

$$\text{LVaR} = 4.0 \times 1.7854 = \text{AUD } 7.14 \text{ million}$$

Interpretation

The fund should hold risk capital or set limits on the basis of AUD 7.14 million. The figure lies between the one-day VaR (4.0) and the square-root-of-time VaR (11.31), as Malz requires; the increment alone is AUD 3.14 million.

Why the other options are wrong

A is incorrect. Ignored the 25% participation cap: $T = 2.4/1.2 = 2$, factor $\sqrt{3 \times 5/12} = 1.1180$, $4.0 \times 1.1180 = 4.47$.
C is incorrect. Scope error: scaled by $\sqrt{T}$ as if the whole position were held for 8 days, $4.0 \times \sqrt{8} = 11.31$.
D is incorrect. Dropped the square root on the adjustment factor: $4.0 \times 3.1875 = 12.75$.

LO: LTR-2 f: Calculate the expected transactions cost and the spread risk factor for a transaction and calculate the liquidity adjustment to VaR for a position to be liquidated over a number of trading days. · LTR-2 e: Describe and compare methods to measure and manage funding liquidity risk and transactions liquidity risk. · Reference: Malz, Ch.12, 2.5 Liquidity Risk Measurement, Measuring the Risk of Adverse Price Impact

Question A11.5 LTR-1 a; LTR-2 f · S4 · [Aug-2026] [Map #11]

The head of market risk at a Mumbai bank is reviewing a consultant's note on adding liquidity to the bank's VaR. The bank holds large corporate bond and mid-cap equity positions that would take several days to unwind. Which of the following statements in the note is correct?

A. The stressed cost of liquidation assumes spreads on all instruments are perfectly correlated, so it overstates liquidity risk and should be reduced to reflect diversification across positions.

B. Allowing more days to unwind the book lowers both the bid-offer spread cost and the exposure to adverse mid-market moves, so the longest feasible liquidation horizon should always be assumed.

C. The normal-market cost charges half of each position's current proportional spread on its mid-market value, while the stressed cost replaces that spread with its mean plus a confidence multiple of its standard deviation.

D. If the book is liquidated in equal parts over T days, its liquidity-adjusted VaR equals the one-day VaR scaled by the square root of T, because the whole position stays exposed for all T days.

Solution **Answer: C**

Why C is correct

Hull's eq. (1.1) charges $s_i \alpha_i / 2$ using the proportional spread in normal conditions; eq. (1.2) replaces s_i with $\mu_i + \lambda \sigma_i$, where "the parameter λ gives the required confidence level for the spread" (2.326 at 99% if normal, or an empirical multiple such as 3.6). Either cost is added to regular VaR to give an LVaR.

Interpretation

Keeping the two constructions apart is the map's reported sibling swap: the current quote belongs in the normal-market measure, the mean and volatility of the spread in the stressed one. For a book that takes days to exit, the bank should also price the horizon trade-off and use Malz's declining-position adjustment.

Why the other options are wrong

A is incorrect. Polarity flip: Hull says the perfect-correlation assumption "may seem overly conservative, but it is not", because spreads widen together when liquidity is tight; diversification does not necessarily reduce liquidity trading risk.

B is incorrect. Absolute claim: a longer horizon lowers the spread cost but *raises* the risk of adverse mid-market moves, which is why Almgren and Chriss optimise the schedule (trading more than $1/n$ on day 1) rather than always stretching it.

D is incorrect. Scope error: dropped the condition that the position shrinks each day. Holding $1, \frac{T-1}{T}, \ldots, \frac{1}{T}$ gives the factor $\sqrt{(1+T)(1+2T)/(6T)}$, so $\sqrt{T}$ overstates the LVaR.

LO: LTR-1 a: Explain and calculate liquidity trading risk via cost of liquidation and liquidity-adjusted VaR (LVaR). · LTR-2 f: Calculate the expected transactions cost and the spread risk factor for a transaction and calculate the liquidity adjustment to VaR for a position to be liquidated over a number of trading days. · Reference: Hull, Ch.24, 1.1.2 to 1.1.4; Malz, Ch.12, Measuring the Risk of Adverse Price Impact

A12 Incremental, Marginal and Component VaR

Readings: IM-5 a, b, c, d, e · Portfolio Risk: Analytical Methods (Jorion, *Value at Risk* 3e, Ch.7: 5.1 Portfolio VaR; 5.2 VaR Tools; 5.3 Summary; 5.6 From VaR to Portfolio Management)

AUG 2026 RECALL | MAP TIER B · #13 · 6/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: Incremental VaR 1Q.
Map #13: "Reported: 'calculate incremental VaR of including platinum asset in the silver asset': a two-asset incremental setup; 'allocation using MVaR'; 'MVaR, vol, rf is given: calculate Sharpe ratio', which chains marginal VaR into a performance measure. Closed-book: marginal VaR = (VaR/P)·β; component VaR = individual VaR × ρ. Shortcut to find: when ranking marginal VaRs, only beta matters. The intermediate-result trap between marginal and component VaR is explicitly in GARP's repertoire."
Shapes covered here:
S1 incremental VaR of including a platinum position in a silver book, full revaluation versus the marginal VaR shortcut (Q1)
S2 allocation using marginal VaR: where to add (Q2) and which position to grow for a better return-to-risk trade-off (Q4)
S3 marginal VaR, volatility and the risk-free rate chained into a Sharpe ratio (Q4)
S4 closed-book formulas: marginal VaR $= (\text{VaR}/W)\beta$; component VaR $=$ individual VaR $\times\ \rho$ (Q3, Q4)
S5 shortcut: when ranking marginal VaRs only beta matters (Q2, Q4)
S6 intermediate-result trap between marginal and component VaR (Q3; tangent versus curve in Q1).

Revision notes

- **Delta-normal (covariance matrix) approach** (IM-5 a): asset payoffs are linear (delta) functions of normally distributed risk factors, so the portfolio return is normal and VaR has a closed form. Jorion: the method "lends itself to a simple decomposition of the portfolio VaR". VaR treats each component as a *risk factor* and x_i as the linear exposure to it; x is the vector of currency positions, kept distinct from the total value W (and from the weights $w_i = x_i/W$).
- **Diversified, individual and undiversified VaR** (IM-5 a): diversified VaR is the portfolio VaR "taking into account diversification benefits between components"; individual VaR is "the VaR of one component taken in isolation", $\alpha\sigma_i|w_i|W$ (absolute value, because a weight can be negative but risk cannot). Undiversified VaR is the sum of individual VaRs, or the portfolio VaR when there is no short position and all correlations are unity. Example 5.1: individual VaRs USD 165,000 and 198,000 sum to USD 363,000 against a diversified VaR of USD 257,738.
- **Correlation** (IM-5 b): with $\rho = 0$, $\text{VaR}_p = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2}$; with $\rho = 1$ and both positions long, $\text{VaR}_p = \text{VaR}_1 + \text{VaR}_2$. With equal weights, risks and correlations, $\sigma_p = \sigma\sqrt{1/N + (1 - 1/N)\rho} \to \sigma\sqrt{\rho}$: at 20% asset risk and $\rho = 0.5$ risk falls to 14.8% at $N = 10$ and never below 14.1%. For a long/short position (the hedge fund long corporates, short Treasuries) the worst case is $\rho = -1$, so undiversified VaR is then the upper bound at correlation -1.
- **Marginal VaR** (IM-5 a): "the change in portfolio VaR resulting from taking an additional dollar of exposure to a given component"; the partial (linear) derivative with respect to the position, unitless. It equals $\alpha\,\text{cov}(R_i, R_p)/\sigma_p = (\text{VaR}/W)\beta_i$. Because Σx is already an intermediate step of the VaR run, β and marginal VaR come almost for free. Under normality marginal VaR is proportional to β_i, so ranking marginal VaRs needs only the betas (S5).
- **Using marginal VaR** (IM-5 c): to lower VaR by cutting every position by the same fixed amount, "rank all marginal VaR numbers and pick the asset with the largest ΔVaR because it will have the greatest hedging effect". Table 5.1: the EUR marginal VaR (0.1521) is three times the CAD's (0.0528), so cutting EUR is far more effective.
- **Incremental VaR** (IM-5 a): "the change in VaR owing to a new position", $\text{VaR}_{p+a} - \text{VaR}_p$ (a "before and after" comparison). If VaR falls the trade is a hedge; otherwise it is risk-increasing. It differs from marginal VaR because the amount added or subtracted can be large, in which case VaR changes in a nonlinear fashion.

- **Full revaluation versus the shortcut** (IM-5 a, c): full incremental VaR needs a re-run of portfolio VaR (Jorion's bank: 100,000 trades, 10 minutes per run). The shortcut $(\Delta\text{VaR})'a$ uses the marginal VaR of the *existing* portfolio and ignores second-order terms; it trades accuracy for speed, is good when the trade is small relative to the portfolio, and "allows real-time trading limits". Example 5.1: adding USD 10,000 CAD gives USD 528 by the shortcut versus USD 529 exactly.
- **Component VaR** (IM-5 a): "a partition of the portfolio VaR that indicates how much the portfolio VaR would change approximately if the given component was deleted". $\text{CVaR}_i = \Delta\text{VaR}_i\, x_i = \text{VaR}\,\beta_i w_i = \text{VaR}_i\,\rho_i$; components sum exactly to the portfolio VaR because $\sum w_i\beta_i = 1$. Negative components act as a hedge for the rest of the portfolio. Adding individual VaRs and taking percentages is *not* a valid decomposition: it ignores diversification.
- **Curve versus tangent** (IM-5 a, Figure 5.4): incremental VaR of deleting the EUR position is USD 92,738, measured along the VaR curve; component VaR is USD 152,108, measured along the tangent. The linear approximation improves when components are small, so the decomposition is most useful for large portfolios of many small positions.
- **Reports and Barings** (IM-5 a, c): Table 5.2 (global equity, tracking error 1.82%): Japan and Brazil carry over 50% of risk although not the largest deviations (Hot Spots, over 5%); cutting Japan by 4.93% lowers volatility to 1.48%. Barings: long USD 7.7 billion Nikkei, short USD 16 billion JGB with a negative correlation, so the bond leg added risk instead of hedging; component VaRs USD 147.15 and 688.01 million sum to USD 835.16 million.
- **General distributions** (IM-5 a): VaR is homogeneous of degree one, so Euler's theorem gives $\text{VaR} = \sum \Delta\text{VaR}_i\, x_i$ for any distribution. With simulations, component VaR is the expected component return conditional on the portfolio being at its VaR, averaged over a window of observations because a single data point is unreliable.
- **Risk-minimizing position** (IM-5 d): the best hedge adds $a^* = -W\beta_i\sigma_p^2/\sigma_i^2$ to one asset. Under a fully invested constraint, cut where marginal VaR is greatest and add where it is lowest until all marginal VaRs (all betas) are equal. Table 5.4: EUR from 33.33% to 14.79%, volatility from 15.62% to 13.85%, VaR from USD 257,738 to 228,462.
- **Return-to-risk maximizing position** (IM-5 d): with returns in excess of the risk-free rate, maximize $\text{SR} = E_p/\sigma_p$ (or with VaR in the denominator). At the optimum $E_i/\Delta\text{VaR}_i$ is constant, equivalently E_i/β_i is constant: a restatement of the CAPM (Roll 1977). Table 5.5: E/β is 0.1301 for CAD versus 0.0282 for EUR, so CAD is increased from 66.67% to 90.21% and the Sharpe ratio rises from 0.448 to 0.551 (both ratios end at 0.0771).
- **Risk management versus portfolio management** (IM-5 e): marginal and component VaR serve risk control (which position to alter, whether a client trade is risk-reducing, in which case the bid-offer spread can be adjusted to win it). Portfolio management balances increasing risk against increasing expected return; expected returns are additive but risk is not, which is why the VaR tools are needed. See also A18 for the Sharpe ratio as a performance measure.

Formulas

Portfolio (diversified) VaR, Eq. 5.7 and 5.12. $\text{VaR}_p = \alpha\sigma_p W = \alpha\sqrt{x'\Sigma x}$; for two assets $\text{VaR}_p = \alpha\sqrt{x_1^2\sigma_1^2 + x_2^2\sigma_2^2 + 2\rho_{12}x_1x_2\sigma_1\sigma_2}$
Individual VaR, Eq. 5.8. $\text{VaR}_i = \alpha\sigma_i|x_i|$; undiversified VaR $= \sum_i \text{VaR}_i$
Marginal VaR, Eq. 5.17 to 5.20.

$$\Delta\text{VaR}_i = \frac{\partial\text{VaR}}{\partial x_i} = \alpha\frac{\text{cov}(R_i, R_p)}{\sigma_p} = \alpha\frac{(\Sigma x)_i}{\sqrt{x'\Sigma x}} = \frac{\text{VaR}}{W}\,\beta_i, \qquad \beta_i = \frac{\text{cov}(R_i, R_p)}{\sigma_p^2} = W\frac{(\Sigma x)_i}{x'\Sigma x}$$

Incremental VaR, Eq. 5.21 and 5.23. Exact: $\text{VaR}_{p+a} - \text{VaR}_p$; shortcut: $(\Delta\text{VaR})'a$
Component VaR, Eq. 5.27 to 5.30. $\text{CVaR}_i = \Delta\text{VaR}_i\, x_i = \text{VaR}\,\beta_i w_i = \text{VaR}_i\,\rho_i$, $\sum_i \text{CVaR}_i = \text{VaR}$, $\text{CVaR}_i/\text{VaR} = w_i\beta_i$
Best hedge, Eq. 5.26. $a^* = -W\beta_i\dfrac{\sigma_p^2}{\sigma_i^2}$. **Minimum risk, Eq. 5.35:** $\Delta\text{VaR}_i = (\text{VaR}/W)\beta_i = \text{constant}$.
Optimal return to risk, Eq. 5.37 and 5.38. $\text{SR} = \dfrac{E_p}{\sigma_p}$ (excess returns); at the optimum $\dfrac{E_i}{\Delta\text{VaR}_i} = \text{constant} \Leftrightarrow \dfrac{E_i}{\beta_i} = \text{constant}$. A position whose E_i/β_i exceeds E_p raises SR when increased.

Back out σ_p from marginal VaRs (Euler). $\dfrac{\text{VaR}}{W} = \sum_i w_i \, \Delta\text{VaR}_i, \qquad \sigma_p = \dfrac{\text{VaR}/W}{\alpha}$

Confidence rescaling (normal). $\text{VaR}_{99\%} = \text{VaR}_{95\%} \times 2.326/1.645$; the same factor applies to marginal and component VaR.

Exam traps

- **Shortcut used on a large trade (scope error):** $(\Delta\text{VaR})'a$ is built from the *existing* portfolio and is accurate only for trades small relative to it. Adding a position as large as the book (S1) needs $\text{VaR}_{p+a} - \text{VaR}_p$.
- **Marginal versus component (intermediate result):** marginal VaR is unitless (VaR per unit of position); component VaR is a currency amount, marginal VaR times the position. A "CAD 0.27 million" option is usually a marginal VaR reported as money.
- **Individual VaR as a contribution (dropped term):** $\alpha\sigma_i x_i$ ignores correlation; component VaR multiplies it by ρ_i.
- **Cut versus add (polarity flip):** cut the position with the *largest* marginal VaR; add to the one with the *lowest*. For return to risk, grow the position with the *highest* $E_i/\Delta\text{VaR}_i$ using *excess* returns, not the lowest marginal VaR (that is the minimum-risk rule) and not the highest expected return.
- **Diversification benefit as incremental VaR (sibling swap):** undiversified minus diversified VaR is the diversification benefit, not the change in VaR from adding a position.
- **Confidence level:** rescale every VaR measure, including marginal and component VaR, by $2.326/1.645$ when moving from 95% to 99%.

Question A12.1

IM-5 a, c · S1, S6 · [Aug-2026] [Map #13]

A senior risk analyst at a UK bank's precious metals desk is presenting VaR tools to newly hired analysts. The desk's book currently holds only silver: a GBP 6 million position with an annual return volatility of 10%. The desk proposes including a GBP 4 million platinum position in the book. Platinum has an annual return volatility of 18% and a correlation of 0.3 with silver. The bank's real-time limit screen estimates the effect of any trade from the marginal VaRs of the current book. The analyst uses delta-normal VaR with $\alpha = 2.33$. Which of the following is closest to the incremental VaR of including the platinum position in the silver book?

A. GBP 0.503 million

B. GBP 0.590 million

C. GBP 0.786 million

D. GBP 1.087 million

Solution

Answer: D

Step 1 – VaR of the current book (silver only)

$$\text{VaR}_p = \alpha\sigma_1 x_1 = 2.33 \times 0.10 \times 6 = 1.398$$

⇒ *Before the trade the desk's VaR is GBP 1.398 million.*

Step 2 – VaR of the book including platinum, Eq. 5.12

$$\begin{aligned}\sigma_{p+a} W_{p+a} &= \sqrt{(6\times 0.10)^2 + (4\times 0.18)^2 + 2\times 0.3\times(6\times 0.10)(4\times 0.18)} \\ &= \sqrt{0.36 + 0.5184 + 0.2592} = \sqrt{1.1376} = 1.06658 \\ \text{VaR}_{p+a} &= 2.33 \times 1.06658 = 2.4851\end{aligned}$$

Step 3 – Incremental VaR, Eq. 5.21

$$\text{VaR}_{p+a} - \text{VaR}_p = 2.4851 - 1.398 = 1.087$$

⇒ *Including platinum raises VaR by GBP 1.09 million: the trade is risk-increasing, not a hedge.*

Interpretation

The platinum position is two-thirds the size of the existing book, so VaR moves along its curve, not its tangent: the screen's linear estimate (option A) misses most of the change. Measured along the tangent of the *new* book, platinum's component VaR would be $2.33 \times (0.3 \times 0.10 \times 0.18 \times 6 + 0.18^2 \times 4)/1.06658 \times 4 = 1.416$, again different from the exact 1.087, as in Jorion's Figure 5.4.

Why the other options are wrong

A is incorrect. Used the screen's shortcut $(\Delta\text{VaR})'a$ on a large trade. In the silver-only book $\Delta\text{VaR}_{\text{plat}} = \alpha\,\text{cov}(R_{\text{plat}}, R_p)/\sigma_p = 2.33 \times 0.3 \times 0.18 = 0.1258$, times 4 gives 0.503. The shortcut ignores second-order terms, valid only for small trades.

B is incorrect. Computed the diversification benefit, undiversified minus diversified VaR: $1.398 + 2.33 \times 0.18 \times 4 - 2.4851 = 1.398 + 1.6776 - 2.4851 = 0.590$.

C is incorrect. Set the correlation to zero: $\sqrt{1.398^2 + 1.6776^2} - 1.398 = 2.1837 - 1.398 = 0.786$.

LO: IM-5 a: Define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR. IM-5 c: Apply the concept of marginal VaR in making portfolio management decisions. · Reference: Jorion, *Value at Risk* 3e, Ch.7, 5.2 (Eq. 5.21 to 5.23, Figure 5.4)

Question A12.2

IM-5 a, c · S2, S5 · [Map #13]

A treasury analyst at a Japanese bank manages a two-asset portfolio: JPY 50 billion in asset 1 (annual volatility 8%) and JPY 30 billion in asset 2 (annual volatility 15%). The correlation between the two returns is 0.4. The bank must add JPY 2 billion of new exposure to one of the two assets and wants the choice that increases portfolio VaR the least. To give an answer in real time, the VaR system estimates the change with marginal VaR rather than a full revaluation, using $\alpha = 1.65$. Which of the following correctly identifies the asset to add to and the estimated incremental VaR of the trade?

A. Asset 1; JPY 0.148 billion

B. Asset 1; JPY 0.215 billion

C. Asset 1; JPY 0.264 billion

D. Asset 2; JPY 0.424 billion

Solution

Answer: B

Step 1 – Covariance vector Σx

$$(\Sigma x)_1 = 0.08^2 \times 50 + 0.4 \times 0.08 \times 0.15 \times 30 = 0.320 + 0.144 = 0.464$$
$$(\Sigma x)_2 = 0.4 \times 0.08 \times 0.15 \times 50 + 0.15^2 \times 30 = 0.240 + 0.675 = 0.915$$

$\Rightarrow$ *The shortcut: $\beta_i = W(\Sigma x)_i / x'\Sigma x$, so the marginal VaRs rank exactly as $(\Sigma x)_i$: asset 1 already has the lower marginal VaR.*

Step 2 – Portfolio volatility and VaR

$$\sigma_p W = \sqrt{50 \times 0.464 + 30 \times 0.915} = \sqrt{50.65} = 7.1169, \qquad \text{VaR}_p = 1.65 \times 7.1169 = 11.743$$

Step 3 – Marginal VaRs, Eq. 5.17

$$\Delta\text{VaR}_1 = 1.65 \times \frac{0.464}{7.1169} = 0.1076, \qquad \Delta\text{VaR}_2 = 1.65 \times \frac{0.915}{7.1169} = 0.2121$$

$\Rightarrow$ *Equivalently $\beta_1 = 80 \times 0.464/50.65 = 0.733$ and $\beta_2 = 1.445$. Each extra yen in asset 1 adds 0.11 yen of VaR; in asset 2, 0.21 yen.*

Step 4 – Incremental VaR of adding JPY 2 billion to asset 1, Eq. 5.23

$$(\Delta\text{VaR})'a = 0.1076 \times 2 = 0.215$$

$\Rightarrow$ *VaR rises by about JPY 0.215 billion. Full revaluation gives 0.216, so the shortcut is accurate for a trade this small relative to the JPY 80 billion book.*

Interpretation

Allocating by marginal VaR: new exposure goes where the marginal VaR (the beta) is lowest; a reduction would come from where it is highest. Here the lower-volatility asset is also the lower-beta asset, so both views agree.

Why the other options are wrong

A is incorrect. Dropped the covariance term from $(\Sigma x)_1$: $1.65 \times 0.320/7.1169 \times 2 = 0.148$. This ignores the asset's co-movement with the rest of the portfolio.

C is incorrect. Used the individual VaR of the new position, $1.65 \times 0.08 \times 2 = 0.264$, which ignores diversification against the existing book.

D is incorrect. Polarity flip of the marginal VaR rule: picked the asset with the *largest* marginal VaR (the one to cut when reducing risk) and added to it: $0.2121 \times 2 = 0.424$.

LO: IM-5 a: Define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR. IM-5 c: Apply the concept of marginal VaR in making portfolio management decisions. · Reference: Jorion, *Value at Risk* 3e, Ch.7, 5.2 (Eq. 5.17 to 5.23)

Question A12.3

IM-5 a · S4, S6 · [Map #13]

A portfolio manager at a Canadian pension fund reviews the risk report for a CAD 120 million equity portfolio. The VaR system is delta-normal and reports, at the 95% level:

Position	Value (CAD m)	Marginal 95% VaR (per CAD 1)	Individual 95% VaR (CAD m)
X	54	0.1200	7.9
Y	36	0.1875	7.2
Z	30	0.1590	5.3

The 1-day 95% diversified portfolio VaR is CAD 18.0 million. Use $z_{0.95} = 1.645$ and $z_{0.99} = 2.326$. Which of the following is closest to the 1-day 99% component VaR of position Y?

A. CAD 0.27 million

B. CAD 6.75 million

C. CAD 9.54 million

D. CAD 10.18 million

Solution

Answer: C

Step 1 – Check the report is internally consistent (Euler)

$$\sum_i \Delta\text{VaR}_i \, x_i = 0.1200(54) + 0.1875(36) + 0.1590(30) = 6.48 + 6.75 + 4.77 = 18.0$$

$\Rightarrow$ *The components add up to the diversified VaR, so the marginal VaRs belong to this portfolio.*

Step 2 – 95% component VaR of Y, Eq. 5.27

$$\text{CVaR}_Y = \Delta\text{VaR}_Y \times x_Y = 0.1875 \times 36 = 6.75$$

$\Rightarrow$ *Y contributes CAD 6.75 million of the CAD 18.0 million 95% VaR (37.5%) on 30% of the capital.*

Step 3 – Rescale to 99%

$$\text{CVaR}_Y^{99\%} = 6.75 \times \frac{2.326}{1.645} = 9.54$$

Step 4 – Closed-book cross-checks

$$\beta_Y = \frac{\Delta\text{VaR}_Y \, W}{\text{VaR}} = \frac{0.1875 \times 120}{18.0} = 1.25, \qquad \rho_Y = \frac{\text{CVaR}_Y}{\text{VaR}_Y} = \frac{6.75}{7.2} = 0.9375$$

$\Rightarrow$ *Component VaR = individual VaR* $\times$ *ρ:* $7.2 \times 0.9375 \times 2.326/1.645 = 9.54$. *The beta above 1 makes Y a hot spot.*

Interpretation

Deleting Y would cut the 99% VaR by roughly CAD 9.5 million, a linear approximation of the incremental VaR of deletion. The individual VaR column is only needed for the ρ cross-check.

Why the other options are wrong

A is incorrect. Intermediate result: reported the 99% *marginal* VaR of Y, $0.1875 \times 2.326/1.645 = 0.265$, as if it were a currency amount. Marginal VaR is unitless and must be multiplied by the position.
B is incorrect. Forgot to rescale from 95% to 99%: $0.1875 \times 36 = 6.75$.
D is incorrect. Used the individual VaR as the contribution, rescaled: $7.2 \times 2.326/1.645 = 10.18$. This sets $\rho_Y = 1$ and ignores diversification.

LO: IM-5 a: Define, calculate, and compare the following portfolio VaR measures: diversified and undiversified portfolio VaR, individual VaR, incremental VaR, marginal VaR, and component VaR. · Reference: Jorion, *Value at Risk* 3e, Ch.7, 5.2 (Eq. 5.27 to 5.30, Table 5.1) and 5.5 (Eq. 5.33)

Question A12.4

IM-5 c, d, e · S2, S3, S4, S5 · [Map #13]

A portfolio manager at an Australian superannuation fund runs an AUD 500 million portfolio of three assets. Returns are normal and the risk system reports 1-year 95% marginal VaRs ($\alpha = 1.645$):

Asset	Weight	Expected return	Marginal 95% VaR (per AUD 1)
A	50%	9.0%	0.12
B	30%	12.0%	0.20
C	20%	4.5%	0.05

The risk-free rate is 3%. The manager wants to move the portfolio toward the highest ratio of expected excess return to risk, starting by increasing one position funded from another. Which of the following correctly identifies the position to increase and the portfolio's current Sharpe ratio?

A. Increase C; Sharpe ratio 0.76

B. Increase B; Sharpe ratio 0.76

C. Increase A; Sharpe ratio 1.14

D. Increase A; Sharpe ratio 0.76

Solution

Answer: D

Step 1 – VaR per AUD of portfolio from the marginal VaRs (Euler)

$$\frac{\text{VaR}}{W} = \sum_i w_i \, \Delta\text{VaR}_i = 0.5(0.12) + 0.3(0.20) + 0.2(0.05) = 0.06 + 0.06 + 0.01 = 0.13$$

$\Rightarrow$ *The 95% VaR is 13% of value, AUD 65 million.*

Step 2 – Portfolio volatility

$$\sigma_p = \frac{\text{VaR}/W}{\alpha} = \frac{0.13}{1.645} = 7.90\%$$

Step 3 – Sharpe ratio (excess returns, Eq. 5.37)

$$E_p - r_f = 0.5(9\%) + 0.3(12\%) + 0.2(4.5\%) - 3\% = 9.0\% - 3\% = 6.0\%, \qquad \text{SR} = \frac{6.0\%}{7.90\%} = 0.76$$

Step 4 – Which position to grow (Eq. 5.38)

Betas: $\beta_i = \Delta\text{VaR}_i/(\text{VaR}/W)$: 0.923, 1.538, 0.385 (they sum to 1 when weighted). Excess return per unit of marginal VaR:

$$\frac{E_A}{\Delta\text{VaR}_A} = \frac{6\%}{0.12} = 0.50, \qquad \frac{E_B}{\Delta\text{VaR}_B} = \frac{9\%}{0.20} = 0.45, \qquad \frac{E_C}{\Delta\text{VaR}_C} = \frac{1.5\%}{0.05} = 0.30$$

$\Rightarrow$ *Only beta and excess return matter. A has the highest ratio and C the lowest, so A is increased, funded from C. Check: $E_A/\beta_A = 6.5\%$ exceeds $E_p = 6.0\%$, while B (5.85%) and C (3.9%) fall short.*

Interpretation

At the optimum every $E_i/\Delta\text{VaR}_i$ is equal (Jorion's restatement of the CAPM). Shifting 1% from C to A raises the Sharpe ratio from 0.759 to about 0.761. Growing C would lower risk but also the Sharpe ratio: that is risk management (minimum risk), not portfolio management (best return per unit of risk).

Why the other options are wrong

A is incorrect. Applied the risk-minimizing rule (add where marginal VaR is lowest) instead of the return-to-risk rule. (Using raw expected returns, $4.5\%/0.05 = 0.90$, would also point to C.)

B is incorrect. Chose the highest expected return without dividing by marginal VaR: B's high return comes with the highest beta, and $E_B/\Delta\text{VaR}_B = 0.45 < 0.50$.

C is incorrect. Right position, but the Sharpe ratio used the raw expected return: $9.0\%/7.90\% = 1.14$. The numerator must be in excess of the risk-free rate.

LO: IM-5 c: Apply the concept of marginal VaR in making portfolio management decisions. IM-5 d: Explain and calculate the risk-minimizing position and position that maximizes the ratio of expected return to risk. IM-5 e: Explain the difference between risk management and portfolio management and describe how to use marginal VaR in portfolio management. · Reference: Jorion, *Value at Risk* 3e, Ch.7, 5.6 (Eq. 5.35 to 5.39, Tables 5.4 and 5.5)

A13 Lognormal VaR

Readings: MR-1 a, b, c, d, e, f · Estimating Market Risk Measures: An Introduction and Overview (Dowd, Measuring Market Risk 2e, Ch.3; 1.3 Eqs. 1.14 to 1.16, Examples 1.3 to 1.5); MR-2 c for the volatility-regime decision (Dowd Ch.4, Volatility-weighted Historical Simulation)

AUG 2026 RECALL | MAP TIER B · #14 · 6/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: Log-normal VaR, straight forward formula, 2Q.
Map #14: "The complication is documented verbatim: 'a simple lognormal calculation was complicated by asking to choose between 2 volatilities, increasing from 16% to 18% six weeks ago.' That is a scope/condition trap: the candidate must decide which volatility regime governs the horizon, not just plug in. Second reported form: 'calculate lognormal VaR (direct formulae based)'. Keep the lognormal form (based on ln returns, VaR $= P(1 - e^{\mu - z\sigma})$) distinct from the normal-arithmetic form."
Shapes covered here:
S1 choose between two volatilities, 16% rising to 18% six weeks ago: which regime governs the VaR horizon (Q3)
S2 direct lognormal VaR from the formula, the plain form seen in August (Q2, Q4)
S3 lognormal form $1 - e^{\mu - z\sigma}$ kept distinct from the normal arithmetic form $-\mu + \sigma z$ (Q1, Q2, Q3, Q4)

Revision notes

- **Data forms** (MR-1 b): P/L, L/P (losses positive), arithmetic returns $r_t = (P_t + D_t - P_{t-1})/P_{t-1}$ and geometric returns $R_t = \ln[(P_t + D_t)/P_{t-1}] = \ln(1 + r_t)$. Geometric returns assume interim payments are continuously reinvested and keep the asset price non-negative; arithmetic returns "should not be used when we are concerned with long horizons". Example 1.1: $r = 0.05 \Rightarrow R = 0.0488$; $R = 0.05 \Rightarrow r = 0.0513$.
- **Historical simulation VaR** (MR-1 a): order the loss observations; with n observations and confidence level α, VaR is the $(1-\alpha)n+1$th highest loss (1000 observations at 95%: 50 in the tail, VaR is the 51st highest loss; the book's example gives 1.704). No distribution is assumed. See also A19 for weighted and bootstrapped HS.
- **Parametric approaches** (MR-1 b) require us to *explicitly specify* the statistical distribution of the data; they fit a curve through the data and read VaR off it. Both the distribution and the data type must be chosen, and in practice VaR is computed from *estimates* m and s of the parameters (eq. 1.9).
- **Normal P/L, L/P and arithmetic-return VaR** are equivalent formulations: with the same underlying assumptions they give the same VaR (1.645 for mean 0, s.d. 1 at 95%). Example 1.2: normal P/L with mean 10, s.d. 20 gives 95% VaR 22.9 and 99% VaR 36.52. Example 1.3: arithmetic returns with mean 0.1, s.d. 0.25 give 0.331 and 0.482.
- **Drawback of normal (arithmetic) VaR:** it assigns a positive probability to the asset value P_t becoming *negative*.
- **Lognormal VaR** assumes *geometric* returns are normal (dividends zero or reinvested), so P_t is lognormal: never negative, skewed with a long *right*-hand tail, and VaR is lognormally distributed. Lognormal VaR can never exceed the portfolio value. Standardised case ($\mu_R = 0$, $\sigma_R = 1$): 95% VaR $= 0.807$. Example 1.4: $\mu_R = 0.05$, $\sigma_R = 0.20$ gives 0.244 (95%) and 0.340 (99%), "quite close" to Example 1.3.
- **Normal versus lognormal** (Example 1.5): annual mean 0.10 and volatility 0.40, 1-day 95%: normal 4.12%, lognormal 4.04%. They are "much the same" for short holding periods and realistic parameters; with the same inputs the lognormal VaR is the smaller.
- **Choosing the volatility input** (S1): the VaR is a forecast for the coming holding period, so σ_R must describe the regime that governs that period. Dowd's volatility-weighting logic (MR-2 c): if current volatility is 1.5% a day and was only 1% a month ago, "data a month old understate the changes we can expect to see tomorrow"; an unusually quiet data period gives VaR estimates "that are too low for the risks we are actually facing". See also A19.
- **Expected shortfall** (MR-1 c): the probability-weighted average of tail losses. Estimate it by slicing the tail into n equal-probability slices and averaging the $n - 1$ tail VaRs: with $n = 10$ the 95% normal ES estimate is 2.0250; the estimate *rises* with n and converges to the true 2.063 (versus 95% VaR 1.645).

- **Coherent risk measures** (MR-1 d) are weighted averages of quantiles q_p with a user-chosen risk-aversion function $\phi(p)$; ES gives every tail quantile weight $1/(1-\alpha)$ and all others 0. The exponential spectral measure ($\gamma = 0.05$) converges to about 1.854 but needs a much larger n; double n until the halving error falls below a tolerance. An existing VaR engine needs only small adjustments to produce coherent measures.
- **Quantile standard error** (MR-1 e): $\text{se}(q) = \sqrt{p(1-p)/n}\,/\,f(q)$. It *falls* as n rises and *rises* as we move further into the tail. Example 1.6 (95% VaR, $n = 1000$): bin width $h = 0.1$ gives the 90% interval $[0.6081, 2.6819]$; $h = 0.2$ narrows it to $[1.1496, 2.1404]$.
- **Weaknesses of the quantile-standard-error approach:** relies on asymptotic theory and large samples, gives wide confidence intervals, depends on the arbitrary bin width, and its symmetric intervals mislead for extreme quantiles whose true intervals are asymmetric. ES versus VaR: heavy tails raise the standard errors of ES estimators (Yamai and Yoshiba), but Acerbi finds the accuracies roughly comparable in empirically realistic ranges; always estimate the standard error alongside the risk measure.
- **QQ plot** (MR-1 f): empirical quantiles plotted against a reference distribution's quantiles. Linear means the reference fits (also for a linear transformation, i.e. different location and scale; intercept and slope then hint at location and scale). Steeper slopes in the tails mean the empirical tails are *heavier* than the reference (Student-t data against a normal reference). Outliers stand out.
- **The three core issues:** which risk measure, which level (portfolio or position), which method (the "VaR trinity": non-parametric, parametric, Monte Carlo simulation).

Formulas

Normal VaR (arithmetic returns normal), eq. 1.14. $\alpha\text{VaR} = (-\mu_r + \sigma_r z_\alpha)\, P_{t-1}$, with $z_{0.95} = 1.645$, $z_{0.99} = 2.326$.

Lognormal VaR (geometric returns normal), eqs. 1.15 and 1.16.

$$R^* = \mu_R - \sigma_R z_\alpha, \qquad P^* = P_{t-1}\exp[\mu_R - \sigma_R z_\alpha]\,, \qquad \alpha\text{VaR} = P_{t-1} - P^* = P_{t-1}\,(1 - \exp[\mu_R - \sigma_R z_\alpha])$$

R^*: critical geometric return; P^*: critical asset value. Set $P_{t-1} = 1$ for VaR as a fraction of portfolio value.

Horizon scaling (Example 1.5, 250 trading days). $\mu_h = \mu_{\text{annual}} \times h/250, \qquad \sigma_h = \sigma_{\text{annual}} \times \sqrt{h/250}$

Reverse-solving for σ_R. From $V = 1 - \exp[\mu_R - \sigma_R z_\alpha]$: $\sigma_R = \dfrac{\mu_R - \ln(1-V)}{z_\alpha}$

Why lognormal is smaller. $1 - e^{-x} < x$ for $x > 0$: the gap is tiny at short horizons and material at long horizons with high volatility.

ES and coherent measures. $M_\phi = \int_0^1 \phi(p) q_p\, dp$; ES: $\phi(p) = 1/(1-\alpha)$ for $p \geq \alpha$, 0 otherwise.

Quantile standard error and interval, eqs. 1.26 and 1.27. $\text{se}(q) = \dfrac{\sqrt{p(1-p)/n}}{f(q)}$, 90% interval $q \pm 1.645\,\text{se}(q)$.

Exam traps

- **Wrong formula twin:** plugging the inputs into the normal formula $-\mu + \sigma z$ instead of $1 - \exp(\mu - \sigma z)$. It gives a slightly larger number, and GARP places it right next to the correct answer.
- **Stale volatility (scope/condition):** when the stem reports a volatility that changed recently (16% rising to 18% six weeks ago), the VaR horizon is governed by the current regime. Using the old figure, or an average across both regimes, understates VaR.
- **$\sqrt{t}$ scaling errors:** scaling σ by h/T instead of $\sqrt{h/T}$, or scaling μ by $\sqrt{h/T}$ instead of h/T.
- **Dropped mean:** using $1 - \exp(-\sigma z)$ and ignoring μ_R, which overstates VaR when $\mu_R > 0$.
- **Wrong critical value:** 1.645 for a 99% question, 2.326 for a 95% question, or the two-tailed 2.576 for a one-tailed 99% VaR.
- **Sign inside the exponent:** the critical return is $\mu_R - \sigma_R z_\alpha$, a left-tail value. Writing $\ln(1+V)$ in a reverse-solve flips the tail.

Question A13.1

MR-1 b · S3 · [self-gen]

A model validator at a Singapore insurer is reviewing an equity sleeve whose reported 1-year 99% lognormal VaR is 35% of portfolio value. The sleeve's geometric returns have an annual mean of 7%. Using $z_{0.99} = 2.326$, which of the following is closest to the annual volatility of geometric returns implied by the reported VaR?

A. 15.9%

B. 18.1%

C. 18.5%

D. 21.5%

Solution

Answer: D

Step 1 – Write the lognormal VaR as a fraction

$$V = 1 - \exp[\mu_R - \sigma_R z_\alpha] \;\Rightarrow\; \ln(1 - V) = \mu_R - \sigma_R z_\alpha$$

Step 2 – Take the log of the surviving value

$$\ln(1 - 0.35) = \ln(0.65) = -0.43078$$

$\Rightarrow$ *The critical geometric return R^* is -43.08%: a 35% price fall corresponds to a log return of -43%.*

Step 3 – Solve for σ_R

$$\sigma_R = \frac{\mu_R - \ln(1 - V)}{z_{0.99}} = \frac{0.07 + 0.43078}{2.326} = 0.2153$$

$\Rightarrow$ *The reported VaR is consistent with roughly 21.5% annual volatility of log returns.*

Interpretation

If the validator's own volatility estimate differs materially from 21.5%, the reported VaR is inconsistent with the model's stated inputs and should be challenged.

Why the other options are wrong

A is incorrect. Wrong tail: treated the 35% move as a gain and used $\ln(1 + V) = \ln 1.35 = 0.30010$ as the magnitude of the critical log return instead of $-\ln(1 - V) = 0.43078$: $(0.07 + 0.30010)/2.326 = 15.9\%$.
B is incorrect. Reverse-solved the normal formula $V = -\mu + \sigma z$: $\sigma = (0.35 + 0.07)/2.326 = 18.1\%$.
C is incorrect. Dropped the mean: $\sigma = 0.43078/2.326 = 18.5\%$.

LO: MR-1 b: Estimate VaR using a parametric approach for both normal and lognormal asset price distributions. · Reference: Dowd, Measuring Market Risk 2e, Ch.3, 1.3 Estimating Lognormal VaR, eq. 1.16

Question A13.2

MR-1 b · S2, S3 · [Aug-2026] [Map #14]

A risk consultant at an Indian mutual fund is reporting VaR as a percentage of portfolio value. The fund's geometric returns are normally distributed with an annualised mean of 12% and an annualised volatility of 30%. The fund is worth INR 8,000 million and its benchmark beta is 1.05. Assuming 250 trading days per year, which of the following is closest to the 10-day 95% lognormal VaR, expressed as a percentage of portfolio value?

A. 3.03%

B. 8.96%

C. 9.39%

D. 12.61%

Solution

Answer: B

Step 1 – Scale the mean to 10 days

$$\mu_{10} = 0.12 \times \frac{10}{250} = 0.0048$$

$\Rightarrow$ *Expected 10-day log return of 0.48%: the mean scales linearly with time.*

Step 2 – Scale the volatility to 10 days

$$\sigma_{10} = 0.30 \times \sqrt{\frac{10}{250}} = 0.30 \times 0.2 = 0.06$$

$\Rightarrow$ *10-day volatility of 6%: volatility scales with* $\sqrt{t}$.

Step 3 – Critical geometric return

$$R^* = 0.0048 - 0.06 \times 1.645 = -0.0939$$

Step 4 – Lognormal VaR as a fraction ($P_{t-1} = 1$)

$$\text{VaR} = 1 - e^{-0.0939} = 0.0896 = 8.96\%$$

$\Rightarrow$ *Over 10 days there is a 5% chance the fund loses more than 8.96% of its value (about INR 717 million).*

Interpretation

Reporting as a percentage makes the number comparable across funds of different size. The portfolio value and beta are not needed for a percentage VaR. Scaling volatility linearly (1.48%) or scaling the mean by $\sqrt{10/250}$ (7.20%) are the other classic slips on this template.

Why the other options are wrong

A is incorrect. Computed the 1-day VaR and stopped: $1 - e^{0.12/250-(0.30/\sqrt{250})\times 1.645} = 3.03\%$.
C is incorrect. Applied the normal (arithmetic) formula to the same inputs: $-0.0048 + 0.06 \times 1.645 = 9.39\%$. Close, but it is the normal VaR, not the lognormal VaR.
D is incorrect. Used the 99% critical value 2.326 instead of 1.645: $1 - e^{0.0048-0.06\times 2.326} = 12.61\%$.

LO: MR-1 b: Estimate VaR using a parametric approach for both normal and lognormal asset price distributions. · Reference: Dowd, Measuring Market Risk 2e, Ch.3, 1.3, eqs. 1.14 and 1.16, Example 1.5

Question A13.3

MR-1 b; MR-2 c · S1, S3 · [Map #14]

A market risk analyst at a US asset manager is computing the 10-day 99% VaR of a USD 50 million equity portfolio for the weekly risk report. Geometric returns are assumed normally distributed, dividends are reinvested and the desk uses 250 trading days per year. The desk's volatility model reports:

- Annualised mean of geometric returns: 8%
- Annualised volatility of geometric returns: 16% for most of the past year, increasing to 18% six weeks ago and unchanged since

The analyst must choose between the two volatilities. Which of the following is closest to the 10-day 99% lognormal VaR?

A. USD 3.44 million

B. USD 3.87 million

C. USD 4.03 million

D. USD 4.28 million

Solution

Answer: B

Step 1 – Choose the volatility

The VaR is a forecast for the next 10 trading days. The 18% regime has held for about 30 trading days, longer than the horizon, and is the current state of the market, so it governs the forecast. Dowd's volatility-weighting argument makes the same point: when volatility has risen, older data "understate the changes we can expect to see tomorrow".

Step 2 – Scale to 10 days

$$\mu_{10} = 0.08 \times \frac{10}{250} = 0.0032, \qquad \sigma_{10} = 0.18 \times \sqrt{\frac{10}{250}} = 0.036$$

Step 3 – Critical geometric return

$$R^* = \mu_{10} - \sigma_{10} z_{0.99} = 0.0032 - 0.036 \times 2.326 = -0.080536$$

⇒ *On the worst 1% of 10-day periods, the log return is at or below* -8.05%.

Step 4 – Lognormal VaR

$$\text{VaR} = 50 \times \left(1 - e^{-0.080536}\right) = 50 \times 0.077378 = \text{USD } 3.87 \text{ million}$$

⇒ *A 1% chance of losing more than about USD 3.87 million (7.74% of the portfolio) over the next 10 days.*

Interpretation

The volatility choice moves VaR by about USD 0.43 million, far more than the choice between the normal and lognormal formulas at this horizon (USD 0.16 million). Reporting the stale-regime figure would understate the risk the desk is running now.

Why the other options are wrong

A is incorrect. Used the stale 16% volatility (scope/condition error): $\sigma_{10} = 0.032$, $R^* = 0.0032 - 0.032 \times 2.326 = -0.071232$, VaR $= 50(1 - e^{-0.071232}) = 3.44$.

C is incorrect. Correct volatility, but applied the normal (arithmetic) formula: $50 \times (-0.0032 + 0.036 \times 2.326) = 4.03$.

D is incorrect. Used the two-tailed 99% critical value 2.576 for a one-tailed VaR: $50 \times (1 - e^{0.0032 - 0.036 \times 2.576}) = 4.28$.

LO: MR-1 b: Estimate VaR using a parametric approach for both normal and lognormal asset price distributions. MR-2 c: Compare and contrast the age-weighted, the volatility-weighted, the correlation-weighted, and the filtered historical simulation approaches. · Reference: Dowd, Measuring Market Risk 2e, Ch.3, 1.3 eq. 1.16 and Example 1.5; Ch.4, 2.4 Volatility-weighted Historical Simulation

Question A13.4

MR-1 b · S2, S3 · [Aug-2026]

The CRO of an Australian pension fund asks a risk analyst to compare normal and lognormal VaR for a long-horizon equity mandate worth AUD 120 million. Annual returns have a mean of 6% and a volatility of 22%, and dividends are reinvested. The CRO wants the figure under the assumption that *geometric* returns are normal. Which of the following is closest to the 1-year 99% VaR?

A. AUD 31.3 million

B. AUD 43.6 million

C. AUD 48.1 million

D. AUD 54.2 million

Solution

Answer: B

Step 1 – Critical geometric return (1-year horizon, no scaling needed)

$$R^* = \mu_R - \sigma_R z_{0.99} = 0.06 - 0.22 \times 2.326 = -0.45172$$

⇒ *The 1st percentile log return over one year is* -45.17%.

Step 2 – Critical portfolio value ratio

$$e^{R^*} = e^{-0.45172} = 0.63653$$

⇒ *In the 1% worst case the portfolio retains about 63.7% of its value.*

Step 3 – Lognormal VaR

$$\text{VaR} = 120 \times (1 - 0.63653) = \text{AUD } 43.6 \text{ million}$$

Step 4 – Contrast with the normal VaR

$$\text{Normal VaR} = 120 \times (-0.06 + 0.22 \times 2.326) = \text{AUD } 54.2 \text{ million}$$

⇒ *Over a 1-year horizon with 22% volatility the gap is about AUD 10.6 million: the lognormal caps losses because the price cannot go below zero.*

Interpretation

For long horizons the choice of distribution matters materially; the "much the same" result of Example 1.5 is a short-horizon result. Dowd's advice is to use geometric returns when dealing with longer periods.

Why the other options are wrong

A is incorrect. Used $z_{0.95} = 1.645$: $120 \times (1 - e^{0.06 - 0.22 \times 1.645}) = 31.3$.
C is incorrect. Dropped the mean: $120 \times (1 - e^{-0.22 \times 2.326}) = 48.1$.
D is incorrect. Used the normal formula, which is what the CRO did not ask for: $120 \times (0.22 \times 2.326 - 0.06) = 54.2$.

LO: MR-1 b: Estimate VaR using a parametric approach for both normal and lognormal asset price distributions. · Reference: Dowd, Measuring Market Risk 2e, Ch.3, 1.1 Geometric Return Data; 1.3 eqs. 1.14 and 1.16, Examples 1.4 and 1.5

A14 Artificial Intelligence and GenAI

Readings: CI-1 a, b, c, d · Advances in Artificial Intelligence: Implications for Capital Markets Activities (IMF, October 2024); CI-2 a, b, c · The Financial Stability Implications of Artificial Intelligence (FSB, November 2024) (source: SchweserNotes Book 5, Readings 100 and 101, one rung below a GARP text)

AUG 2026 RECALL | MAP TIER B · #20 · 6/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: AI - 2Q.
Map #20: "Reported: 'AI (2 to 3)'; 'GEN AI 2 Qs'; 'AI vs Gen AI': an explicit discrimination question between the two; 'disadvantage of GenAI'; 'application of AI in market risk'. The AI-vs-GenAI split and the financial-stability implications (herding, model monoculture, third-party concentration, explainability) are the two examinable axes."
Shapes covered here:
S1 AI vs GenAI: the IMF's two model groups, what GenAI adds, and its signature failure mode (Q1)
S2 disadvantages of GenAI: fraud, deepfakes, cyber attack vectors, the short-term edge of bad actors (Q2)
S3 application of AI in market risk: volatility and liquidity models, risk scenarios, dynamic risk sensing (Q3)
S4 financial-stability axis: herding and correlated models, third-party concentration, explainability and model risk (Q2, Q3)

Revision notes

The specifics first: AI vs GenAI, GenAI's downsides, AI in market risk

- **Two model groups (S1, CI-1 a).** The IMF groups models as (1) *AI/ML models*: established predictive analytics tools such as natural language processing (NLP), neural networks, decision trees and clustering; (2) *sophisticated AI models*: the most advanced innovations, including large language models (LLMs), deep learning and reinforcement learning.
- **What GenAI is and does (S1).** GenAI builds on existing strategies and models: investment idea generation, code development for analysts, customer-facing work; LLMs are being integrated into analytical frameworks for more accurate forecasting. The **Transformer** architecture (which uses an *attention* mechanism and processes data in parallel) laid the foundation for LLMs and GenAI. GenAI moved chatbots from rules-based systems to LLM-based ones. Future benefits: lower barriers to entry for less liquid asset classes and faster market reactions to news.
- **Human in the loop (S1).** Sophisticated AI has so far mainly generated *signals* as inputs to existing systems; human traders still make the final decisions, not fully autonomous AI trading agents. More autonomy is expected (particularly in equities), but a human-in-the-loop approach is likely to remain.
- **GenAI failure mode: hallucination (S1, S2).** A type of model inaccuracy associated with GenAI in which a model produces a highly confident but incorrect response. Because LLM outputs are typically unstructured text, outcome analysis and error-rate measurement are *difficult*.
- **GenAI disadvantages (S2, CI-2 c).** Fraud: video and voice generation enables deepfakes, fake profiles and IDs, bypassed security checks; generating fraudulent AI content may currently be easier than detecting it. Disinformation becomes more convincing and scalable, dangerous in stress. Cyber: GenAI enables more effective attacks at higher frequency and lowers barriers to entry for less sophisticated threat actors; bad actors may benefit more than legitimate users in the *short* term (legitimate users are expected to gain more over the long term). Cautious legitimate actors create an asymmetry with risk-taking malicious actors. Remedy for fraud: AI literacy and financial literacy among customers.
- **Attack vectors.** *Prompt injection* extracts confidential information; *model/data poisoning* intentionally manipulates model weights and/or training data. Best practice: a new AI risk management framework or AI-specific additions to the existing one, plus AI questions in vendor due diligence.
- **AI in market risk (S3).** CI-1: in risk management (reporting and monitoring) AI streamlines compliance monitoring, generates tailored risk insights, develops advanced risk scenarios and enables dynamic risk sensing. CI-2: operations use cases include volatility and liquidity risk management, building models that better predict how heightened volatility may affect positions; firms increasingly use AI for liquidity, market and credit risk, but representative data on that use are lacking.

IMF (CI-1): uses, market dynamics, stability, regulators

- **Uses and benefits (CI-1 a).** IMF outreach: 27 firms, buy side and sell side. Four broad risk areas: operational risk; volatility and speed under stress; cyber and manipulation; monitoring and transparency (opacity). Functions: client profiling, asset allocation (signals from alternative and unstructured data), trading (liquidity, execution, less market impact, price discovery), risk management. Value is greatest in liquid public markets (equities, listed derivatives, then government bonds and FX); strong link between market liquidity and AI adoption. Survey benefits: productivity, unstructured data processing, *narrower* bid-ask spreads, cheaper trading algorithm design. High fixed costs favour large firms; North America leads foundational models.
- **Market dynamics (CI-1 b).** AI may favour NBFIs (over half of global financial assets) over legacy-bound banks. US algorithmic share: about 70% of equities, over 50% of futures, over 40% of options, under 10% of fixed income. Normal times: better efficiency and liquidity, fewer non-informational price swings; stress: volatility, occasional large swings, auto-shutdowns create feedback loops. *Flighty liquidity under stress*: HFTs *reduce* order cancellations as they step back; institutions *increase* hidden orders. New dynamics: faster, more accurate reactions to news (FOMC minutes), higher and procyclical volumes (AI ETFs have higher turnover), possible AI manipulation or unintended collusion.
- **Stability concerns (CI-1 c).** Principal concern: *concentration* (strategies built on open-source models trained on data from few vendors; few dominant providers). Also market fragility (flash crashes, data poisoning, cyberattacks), hallucinations and lack of explainability, manipulation via deepfakes, high fixed costs. Lesser: data access, customer fraud, shock transmission to emerging and developing economies. Respondents prefer best practices and flexible guidelines to strict rules and want extreme-scenario stress testing. Challenges: volatility under stress (herding, flight to safety), opacity as activity shifts to lightly overseen NBFIs, cyber and manipulation, third-party dependence.
- **Regulators and supervisors (CI-1 d).** Institutions remain fully responsible for AI they deploy, built or bought. Focus areas: governance, transparency and explainability, outsourcing, data quality and bias, reporting, human oversight (FSB, BCBS, BIS, NIST, IOSCO). Supervisory AI: automated data quality checks, data integration, anomaly detection, real-time monitoring, misinformation detection; GenAI for legacy code, retrieval, code generation and debugging. Priorities: (1) speed and volatility: circuit breakers, algorithm testing, *initial margin in normal times*, margin call volatility, variation margin, CCP margin models; (2) third-party concentration: map links between critical IT and AI providers, cyber resilience across the AI lifecycle; (3) opacity: map model, data and infrastructure interdependencies, NBFI disclosure, liquidity buffers; (4) OTC markets: transaction data, central clearing incentives, margin for non-centrally cleared derivatives, backstops.

FSB (CI-2): drivers, use cases, vulnerabilities

- **Drivers since 2017 (CI-2 a).** Evidence says the *supply side* has been the more influential driver. Supply: technology (embeddings for unstructured data, the Transformer, GPUs and cheaper compute; human capital shortage), data (digital engagement, unstructured data, possible need for synthetic data), business models (cloud and pre-trained models; trend toward *open-source*). Demand: profitability (revenue, cost reduction, risk management), competition, regulation (compliance).
- **Use cases (CI-2 b).** Industry, four areas: customers (credit scoring, marketing, adviser matching, insurance pricing, LLM chatbots); operations (model risk management, capital optimisation, code generation, market impact, volatility and liquidity models; GenAI content, retrieval, legacy code, voice transcription); trading and portfolio management (reinforcement learning execution, sentiment in text); regulatory compliance (RegTech: fraud, AML, CFT). SupTech: information collection, statistics, payments oversight, real-time analysis, LLM-drafted inspection reports; future GenAI simulation of stress tests and bank runs.
- **Third-party concentration (S4).** Few cloud and chip suppliers, vertical integration, data aggregator consolidation. Concentration pushed up by complexity and cost, vertical integration, GPU bottlenecks, incumbent investment, multimodality; pushed down by hardware competition, open-source models, *architectural innovations*, domain- and task-specific models. Mitigant: third-party risk standards (periodic assessments, vendor inventories, ongoing monitoring and due diligence).
- **Market correlations (S4).** Similar models and data (limited model choice, herding, network externalities, opaque design) raise correlations: flash crashes, amplified volatility, liquidity crunches. Pre-trained core models share architectures and data, so outputs correlate; GenAI output re-used as training data adds to it. Risk is *higher* when LLMs inform risk management, investment strategy or new AI applications, *far lower*

for back-office uses. AI can also reduce correlations (customised strategies, new entrants). Mitigants: circuit breakers, transparency, monitoring and surveillance.

- **Model risk, data, governance (S4).** In crises models are harder to monitor, validate and correct; complexity and lower explainability raise model risk; opaque, unstructured training data complicate data quality. Regulated institutions handle this through their *existing* model risk frameworks: data quality assessment, validation, performance and outcome analysis, ongoing monitoring, plans for limited data.
- **Other and longer term.** Misalignment: profit-maximising AI that learns collusion or spreads disinformation triggering runs. Longer term: compute and data demand, uneven regulation, effects on rates, inflation and labour (greater risk for high-skilled workers), energy (about 1% of global use, rising). FSB priorities: close data gaps, test whether frameworks cover these vulnerabilities, build supervisory capability.

Exam traps

- **Sibling swap on the model groups:** NLP, neural networks, decision trees and clustering are the established AI/ML group; LLMs, deep learning and reinforcement learning are the sophisticated group. Also: prompt injection extracts data, poisoning manipulates weights or training data; cloud and pre-trained models are a supply-side (business model) driver, not demand side.
- **Polarity flips:** bad actors gain more in the *short* term; LLM text output makes outcome analysis *harder*; under stress HFT order cancellations *fall* and hidden orders *rise*; the survey cited *narrower* spreads and *high* fixed costs.
- **Scope error on correlation:** correlation risk is high when LLMs inform risk management or investment strategy, far lower for back-office uses such as retrieval, training or drafting.
- **Role misassignment:** the institution stays fully responsible for bought AI; model risk sits in the institution's existing model risk framework; humans make the final trading decisions.
- **Sibling swap on concentration:** architectural innovations, open-source and domain-specific models push concentration *down*; multimodality, vertical integration and GPU bottlenecks push it *up*.
- **Absolute claims:** circuit breakers and transparency mitigate, they do not eliminate; respondents want flexible guidelines, not strict rules; a human in the loop is expected to remain.

Question A14.1

CI-1 a; CI-2 a, c · S1 · [Aug-2026] [Map #20]

A junior analyst at a Singapore asset manager is presenting to newly hired analysts on the firm's model inventory, which contains a decision-tree credit model, a clustering tool for client segmentation, an NLP news classifier, and a recently licensed LLM-based assistant that drafts research notes and code. She wants one slide that correctly distinguishes established AI/ML from GenAI. Which of the following statements should she include?

A. Decision trees, clustering and NLP belong to the sophisticated AI group, while LLMs and reinforcement learning belong to the group of established predictive analytics tools.

B. GenAI is now used mainly as fully autonomous trading agents that execute without human intervention, taking over the final decisions that human traders made with earlier AI/ML signals.

C. Decision trees, clustering and NLP are established predictive analytics tools, while the LLM is a GenAI tool whose characteristic inaccuracy is a hallucination: a highly confident but incorrect response.

D. Because LLM outputs are structured numerical forecasts, outcome analysis and error rates are easier to establish for the GenAI assistant than for the established AI/ML models.

Solution

Answer: C

Step 1 – Place each tool in the IMF grouping

AI/ML models are established predictive analytics tools: NLP, neural networks, decision trees, clustering. Sophisticated AI models are the most advanced innovations: LLMs, deep learning, reinforcement learning.

⇒ *Three of the four inventory items are established AI/ML; the LLM assistant is the sophisticated, generative one.*

Step 2 – What makes the LLM generative

LLMs, built on the Transformer architecture (attention mechanism), generate content: research drafts, code, chatbot answers. GenAI's characteristic inaccuracy is the hallucination, a highly confident but incorrect response.

⇒ *The discriminator between the groups is not "better accuracy" but a different output (generated text) and a different failure mode.*

Step 3 – How such models are used today

Sophisticated AI has mainly generated signals that feed existing systems; human traders still make the final decisions, and a human-in-the-loop approach is expected to remain.

Interpretation

C is correct. For model governance the slide matters: the LLM's text outputs need different validation (outcome analysis is harder) and different controls (hallucination checks) from the three predictive models.

Why the other options are wrong

A is incorrect. Sibling swap: the two groups are reversed. NLP, decision trees and clustering are the established AI/ML tools; LLMs and reinforcement learning are sophisticated AI.

B is incorrect. Role misassignment: the final decision still rests with human traders; sophisticated AI has been applied mainly to generate signals for existing systems, not as autonomous agents.

D is incorrect. Polarity flip: LLM outputs are typically unstructured text, which makes outcome analysis and error rates *harder* to determine, not easier.

LO: CI-1 a: Describe current uses of artificial intelligence (AI) and machine learning (ML) in capital markets, and potential future uses of sophisticated AI models including GenAI. · CI-2 c: Assess the potential implications of AI for financial stability, with a focus on the following vulnerabilities: third-party dependencies and service provider concentration; market correlations; cyber risks; model risk, data, and governance; other vulnerabilities, such as fraud, disinformation, and misalignment. · Reference: IMF (October 2024), Advances in Artificial Intelligence; FSB (November 2024), The Financial Stability Implications of AI; SchweserNotes Book 5, Readings 100.1 and 101.1, 101.3

Question A14.2

CI-2 c; CI-1 d · S2, S4 · [Aug-2026] [Map #20]

The CRO of a German retail bank is preparing a paper for the board risk committee on a proposal to deploy a vendor's pre-trained GenAI model for customer onboarding, voice-based customer service and internal code generation. The project sponsor expects EUR 40 million of annual cost savings. The committee has asked for one paragraph on the disadvantages of GenAI. Which of the following statements is correct to include?

A. Legitimate firms are expected to benefit more than malicious actors from GenAI in the short term, so GenAI-enabled cyberattacks are mainly a long-term concern for the bank.

B. Prompt injection attacks manipulate model weights or training data, whereas model or data poisoning attacks are used to extract confidential information from the model.

C. Because the model is bought from a vendor rather than built in-house, responsibility for its model risk and outcomes passes to the vendor under its outsourcing contract.

D. GenAI's voice and video generation enables deepfakes, fake IDs and bypassed security checks, and generating fraudulent AI content may currently be easier than detecting it.

Solution — Answer: D

Step 1 – Identify the vulnerability the proposal touches

Onboarding and voice service rely on identity checks. The FSB's "other vulnerabilities" include fraud: GenAI's strong video and voice generation creates deepfakes, fake profiles and IDs, bypassed security checks, and fraud targeting customers and institutions.

⇒ *The bank's new voice channel is exactly the surface that GenAI-enabled fraud attacks.*

Step 2 – The asymmetry

Generating fraudulent AI content may currently be easier than detecting it, and cautious legitimate actors adopt more slowly than risk-taking malicious actors. Mitigant named in the reading: AI literacy and financial literacy among customers.

Step 3 – Accountability for a bought model

Institutions remain fully responsible for AI they deploy, whether developed internally or obtained from external providers; AI questions belong in vendor due diligence.

Interpretation

D is correct. The EUR 40 million saving is irrelevant to the risk paragraph: the committee should weigh it against fraud and cyber controls on the onboarding and voice channels.

Why the other options are wrong

A is incorrect. Polarity flip on timing: bad actors may benefit *more* than legitimate users in the *short* term; legitimate users are expected to gain more over the long term.

B is incorrect. Sibling swap: the two attack vectors are reversed. Prompt injection extracts confidential information; model or data poisoning manipulates weights or training data.

C is incorrect. Role misassignment: the bank remains fully responsible for the AI systems it deploys regardless of whether they are developed internally or obtained from external providers.

LO: CI-2 c: Assess the potential implications of AI for financial stability, with a focus on the following vulnerabilities: third-party dependencies and service provider concentration; market correlations; cyber risks; model risk, data, and governance; other vulnerabilities, such as fraud, disinformation, and misalignment. · CI-1 d: Describe best practices for regulators pertaining to AI, including the use of AI by supervisors and recommendations for policy on the use of AI by regulated entities. · Reference: FSB (November 2024), Section on cyber risks and other vulnerabilities; IMF (October 2024), regulatory and supervisory considerations; SchweserNotes Book 5, Readings 101.3 and 100.3

Question A14.3

CI-2 b, c; CI-1 a, c · S3, S4 · [Map #20]

The head of market risk at an Australian bank plans to use a pre-trained LLM, the same core model licensed by most of the bank's peers, in two ways: (1) to build volatility and liquidity risk models that predict how heightened volatility would affect the trading book, to generate advanced stress scenarios, and to suggest hedges and risk limits; (2) to draft internal policy documents and retrieve information for staff training. An internal auditor reviewing the plan must state the financial-stability implication. Which of the following assessments is correct?

A. Use (2) carries higher market correlation risk than use (1), because back-office drafting and retrieval are the most widespread uses of LLMs across the industry.

B. Circuit breakers and transparency requirements fully eliminate the correlation vulnerability, so the bank need not monitor how the LLM's risk outputs move with those of its peers.

C. Validation and outcome analysis of the LLM's risk models should be governed by the vendor's own model risk standards rather than by the bank's existing model risk framework.

D. Use (1) raises market correlation risk relative to use (2), because models sharing architectures and training data produce correlated outputs that can amplify volatility and liquidity crunches in stress.

Solution

Answer: D

Step 1 – Classify the two uses

Use (1) is AI in market risk: the FSB lists volatility and liquidity risk management (models predicting how heightened volatility affects positions) among operations use cases, and the IMF lists advanced risk scenarios and dynamic risk sensing under risk management. Use (2) is back-office: information retrieval, employee training, document drafting.

Step 2 – Apply the correlation channel

Pre-trained core models share architectures and training data, so their outputs are naturally correlated. If LLMs inform risk management or investment strategy, correlation risk is higher; if they address internal back-office inefficiencies, it is far lower.

⇒ *When most peers hedge and set limits from the same model, they react alike in stress: herding, flash crashes, amplified volatility and liquidity crunches.*

Step 3 – Governance and mitigants

Model risk for AI is addressed through the institution's existing framework (data quality, validation, performance and outcome analysis, ongoing monitoring). Circuit breakers, transparency, monitoring and surveillance mitigate correlation; they do not remove it.

Interpretation

D is correct. The auditor's recommendation: treat use (1) as a model under full validation, monitor how its signals co-move with peers, and keep a human decision on hedges and limits; use (2) is low correlation risk.

Why the other options are wrong

A is incorrect. Polarity flip: correlation risk is higher when LLMs inform risk management and far lower for back-office uses; how widespread a use is does not reverse that.

B is incorrect. Absolute claim: circuit breakers and transparency are mitigants, complemented by consistent monitoring and surveillance; nothing in the reading says they eliminate the vulnerability.

C is incorrect. Role misassignment: AI-related data quality challenges and model risk must be addressed by the regulated institution through its existing risk management framework, not delegated to the vendor.

LO: CI-2 c: Assess the potential implications of AI for financial stability, with a focus on the following vulnerabilities: third-party dependencies and service provider concentration; market correlations; cyber risks; model risk, data, and governance; other vulnerabilities, such as fraud, disinformation, and misalignment. · CI-2 b: Identify and analyze existing and emerging AI use cases in financial services, including industry use cases as well as regulatory and supervisory use cases. · Reference: FSB (November 2024), market correlations and model risk sections; IMF (October 2024), uses of AI in risk management; SchweserNotes Book 5, Readings 101.2, 101.3 and 100.1

A15 Hazard Rates, CDS Spreads and Conditional vs Unconditional PD

Readings: CR-9 d, e, f, g, h, i, j, k · Estimating Default Probabilities (Hull, RMFI 6e, Ch.17): 9.2 Historical Default Probabilities; 9.2.1 Hazard Rates; 9.3 Recovery Rates; 9.4 Credit Default Swaps; 9.5 Credit Spreads; 9.6 Estimating Default Probabilities from Credit Spreads; 9.7 Comparison of Default Probability Estimates; also CR-10 · Credit Value at Risk (Hull, RMFI 6e, Ch.19), 10.1 Ratings Transition Matrices

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Specifics reported

Your August 2026 recall: Hazard rate and PD calculation - 2Q; hazard rate from CDS spread.
Map #17: "Reported: 'hazard rate calculation from PD (direct formula)'; 'hazard rate & PD'. Closed-book: $\lambda = \text{spread}/(1 - \text{RR})$; survival $= e^{-\lambda t}$; cumulative PD $= 1 - e^{-\lambda t}$; marginal PD = difference of successive cumulative PDs. Correction to your formula card: it labels pd $= 1 - e^{-\lambda t}$ as 'hazard rate'. That expression is the cumulative default probability implied by a hazard rate, not the hazard rate itself: a sibling swap waiting to happen. The conditional-vs-unconditional distinction is named by GARP-facing sources as one of the most commonly confused pairs in Part II."
Shapes covered here:
S1 hazard rate and PD: from a constant hazard rate to cumulative and year-t default probabilities (Q1)
S2 hazard rate from PD by the direct formula $\bar{\lambda} = -\ln(1 - Q)/T$ (Q4, Q7)
S3 hazard rate from a CDS spread, $\bar{\lambda} = s/(1 - R)$ (Q5; term structure in Q6; reversed in Q7)
S4 marginal PD as the difference of successive cumulative PDs (Q1, Q2, Q3)
S5 sibling swap: $1 - e^{-\lambda t}$ is the cumulative PD, not the hazard rate (Q1, Q7)
S6 conditional versus unconditional default probability (Q1, Q2, Q3).

Revision notes

- **Cumulative default probability** (CR-9 d; Table 9.1, S&P 1981 to 2020): the probability of default *by* the end of year t for issuers starting in a given rating (BBB: 0.16% by year 1, 0.43% by year 2).
- **Unconditional default probability** (CR-9 e): the probability of default *during* year t "as seen at time zero": the difference of consecutive cumulative PDs (BBB year 2: $0.43 - 0.16 = 0.27\%$; CCC/C year 3: $43.42 - 38.33 = 5.09\%$). This is the marginal PD.
- **Conditional default probability** (CR-9 e): the unconditional PD divided by the probability of surviving to the *start* of the year (CCC/C year 3: $5.09/61.67 = 8.25\%$). It is always at least the unconditional value.
- **Hazard rate (default intensity)** (CR-9 e): $\lambda(t)\Delta t$ is the probability of default between t and $t + \Delta t$ *conditional on no earlier default*; it is the conditional PD for a very short period. $1 - e^{-\bar{\lambda} t}$ is the cumulative PD it implies, not the hazard rate itself.
- **Rating and PD** (CR-9 c, d): for investment grade the marginal PD tends to *increase* with time in the first few years (A: 0.05%, 0.08%, 0.09%, 0.11%); for the lowest grades it can *decrease* (CCC/C: 28.3%, 10.03%, 5.09%, 2.94%) because the next year or two are critical and survivors' health is likely to improve. See also A10.
- **Rating migration matrix** (CR-9 d; CR-10, Table 10.1): one-year probabilities of moving between ratings. Assuming independence across periods, the two-year matrix is the matrix multiplied by itself and the five-year matrix its fifth power; six months needs the square root. The independence assumption is not exactly true (ratings momentum). Schroeck makes a related point: a counterparty's PD in the second year of a loan is typically higher than in the first year, modelled with Markov migration matrices.
- **Recovery rate** (CR-9 f): the price of the bond about one month after default as a percent of principal. Table 9.2 (Moody's, 1983 to 2021): first lien 54.6%, second lien 44.3%, senior unsecured 38.0%, senior subordinated 31.0%, subordinated 32.1%, junior subordinated 22.3%.
- **Recovery depends negatively on default rates** (CR-9 f): in a low-default year recoveries might be as high as 60%, in a high-default year as low as 30%. "A bad year for defaults is doubly bad for a lender." The same link appeared in mortgages in the 2007 to 2008 crisis.
- **CDS mechanics** (CR-9 g): insurance against default by the *reference entity*; the default is a *credit event* (failure to pay, restructuring, bankruptcy; restructuring sometimes excluded in North America). The face

value of bonds that can be sold is the *notional principal*. The *buyer* pays periodic premiums (usually quarterly in arrears) until maturity or the credit event, plus a final accrual payment, and gains the right to sell bonds at face value; the *seller* receives the premiums and pays face value minus the auction value of the cheapest-to-deliver bond (cash settlement, now usual) or buys the bonds at face value (physical settlement).

- **CDS quotes and market** (CR-9 g, h): the CDS spread is the total paid per year as a percent of notional. A market maker's bid is the price at which it *buys* protection, the ask the price at which it *sells*. Banks tend to be net buyers, insurers net sellers. Fixed coupons: if the quoted spread is below the coupon, the *seller* pays the PV difference upfront; if above, the *buyer* pays. Indices (CDX NA IG, iTraxx Europe) hold 125 investment-grade names.
- **Spread-implied hazard rate** (CR-9 h, S3): the credit spread is roughly an average annual *loss* rate, so $\bar{\lambda} = s(T)/(1-R)$ (eq. 9.3): 240 bp with 40% recovery gives 4%. Spreads for several maturities bootstrap the term structure (Example 9.3). The approximation works well for CDS spreads, and for bond yield or asset swap spreads only when the bond trades near par; otherwise use the more exact bond-pricing calculation (Table 9.3, $Q = 3.03\%$).
- **CDS spread and bond yields** (CR-9 h, i): the n-year CDS spread should be close to the par yield on an n-year corporate bond minus the par yield on an n-year risk-free bond. The implied risk-free rate is *higher* than Treasury rates, close to the short-term borrowing rate of a AA company.
- **CDS-bond basis** (CR-9 i): CDS spread minus bond yield spread; in theory close to zero. Bond above par: negative; below par: positive; CDS counterparty risk: negative; cheapest-to-deliver option: positive; no accrued interest in the payoff: negative; restructuring clause: positive; a non-market risk-free benchmark: either way. Positive before 2007 (about 16 bp in 2004 to 2005), very negative at times in the crisis, usually small and negative since.
- **Historical versus spread-implied PDs** (CR-9 j, Tables 9.4 and 9.5): seven-year hazard rates implied by spreads (risk-neutral) are much *higher* than those from historical default rates (real-world), and the gap *widens* as credit quality declines (A: 0.109% versus 1.145%). Investors are compensated for more than the actuarial cost of default.
- **Why the gap** (CR-9 j): illiquidity of corporate bonds (perhaps 25 bp in normal markets); traders' subjective depression scenarios (hard to see as a large part); but "by far the most important reason" is that bonds do not default independently, so systematic risk earns a premium. Skewed, hard-to-diversify idiosyncratic risk also contributes.
- **Which PD to use** (CR-9 k): risk-neutral (spread-implied) PDs for *valuation*; real-world (historical) PDs for *scenario analysis*. Merton-based PDs are covered in A5.

Formulas

From a cumulative default table (C_t = cumulative PD by end of year t):

$$\text{Unconditional PD in year } t = C_t - C_{t-1}, \qquad \text{Conditional PD in year } t = \frac{C_t - C_{t-1}}{1 - C_{t-1}}$$

Hazard rate relationships (Hull eq. 9.2). $V(t)$: survival probability to t; $Q(t)$: cumulative PD by t; $\bar{\lambda}(t)$: average hazard rate from 0 to t.

$$V(t) = e^{-\int_0^t \lambda(\tau)d\tau}, \qquad Q(t) = 1 - e^{-\bar{\lambda}(t)\,t}, \qquad \bar{\lambda}(t) = -\frac{1}{t}\ln\big(1 - Q(t)\big)$$

With a constant hazard rate λ: unconditional PD in year $t = e^{-\lambda(t-1)} - e^{-\lambda t}$; conditional PD in any year $= 1 - e^{-\lambda}$.
Book check (Example 9.2). $\lambda = 1.5\%$: $Q(1) = 0.0149$, $Q(3) = 0.0440$, $Q(4) = 0.0582$; unconditional year-4 PD $= 0.0142$; conditional $= 0.0142/(1 - 0.0440) = 0.0149$.
Two-year cumulative PD from a migration matrix (start in rating i): $C_2 = p_{iD} + \sum_{j \neq D} p_{ij}\, p_{jD}$.

Average hazard rate from a spread (eq. 9.3). $\bar{\lambda} = \dfrac{s(T)}{1-R}$, with s in decimals (1 bp $= 0.0001$) and $1 - R$ the loss given default.

Bootstrapping (Example 9.3). $\bar{\lambda}_{T_1,T_2} = \dfrac{T_2\bar{\lambda}(T_2) - T_1\bar{\lambda}(T_1)}{T_2 - T_1}$. Book: 50, 60, 100 bp at 3, 5, 10 years, $R = 60\%$: $\bar{\lambda} = 1.25\%, 1.5\%, 2.5\%$; years 3 to 5: 1.875%; years 5 to 10: 3.5%.
Reverse: spread from a cumulative PD. $\bar{\lambda} = -\ln(1-Q)/T$, then $s = \bar{\lambda}(1-R)$.

Book check (Table 9.5). A-rated: $Q_7 = 0.76\%$ gives $\bar{\lambda} = 0.109\%$; spread 68.68 bp with $R = 40\%$ gives 1.145%.
Premium leg and basis. Quarterly payment $= 0.25 \times s\times$ notional (90 bp on USD 100 million $\approx$ USD 225,000); CDS-bond basis = CDS spread − bond yield spread.

Exam traps

- **Unconditional versus conditional (intermediate result):** the difference of cumulative PDs is the *unconditional* PD; dividing by survival to the *start* of the year gives the conditional one. Read the stem for "as seen today" versus "conditional on no earlier default".
- **Wrong survival denominator:** dividing by survival to the *end* of the year $(1-C_t)$ instead of the start $(1-C_{t-1})$.
- **Cumulative PD is not the hazard rate (sibling swap):** $1 - e^{-\lambda t}$ is the probability of default by t. Feeding a cumulative PD into $s = \bar{\lambda}(1 - R)$ as if it were $\bar{\lambda}$ overstates the spread; the hazard needs $-\ln(1 - Q)/T$.
- **Recovery versus LGD (wrong input twin):** divide the spread by $1 - R$, not by R; and divide, do not multiply. 180 bp is 0.0180.
- **Forward versus average hazard:** the hazard *between* two maturities needs the T-weighted bootstrap divided by $T_2 - T_1$, not the longer average hazard and not a simple average of the two.
- **Continuous versus discrete:** $Q = 1 - e^{-\lambda t}$, not $1 - (1 - \lambda)^t$ and not λt.

Question A15.1

CR-9 e · S1, S4, S5, S6 · [Aug-2026] [Map #17]

A quantitative analyst at an Indian bank models a corporate borrower with a constant hazard rate of 2.5% per annum. The borrower's expected recovery rate is 40% and the loan's notional is INR 500 crore. For the bank's expected-loss schedule, which of the following is closest to the probability, as seen today, that the borrower defaults during the fourth year?

A. 2.29%

B. 2.32%

C. 2.47%

D. 9.52%

Solution

Answer: A

Step 1 – Cumulative PDs from the hazard rate

$$Q(3) = 1 - e^{-0.025\times 3} = 0.07226, \qquad Q(4) = 1 - e^{-0.025\times 4} = 0.09516$$

$\Rightarrow$ *There is a 7.23% chance of default by year 3 and a 9.52% chance by year 4.*

Step 2 – Unconditional (marginal) PD in year 4

$$Q(4) - Q(3) = e^{-0.075} - e^{-0.100} = 0.09516 - 0.07226 = 0.02291$$

$\Rightarrow$ *Seen from today, there is a 2.29% chance that default happens during year 4.*

Step 3 – Check against the conditional PD

$$\frac{0.02291}{1 - 0.07226} = 0.02469 = 1 - e^{-0.025}$$

$\Rightarrow$ *The conditional PD is larger because it assumes the borrower is still alive at the start of year 4; it is not what "as seen today" asks for.*

Interpretation

An expected-loss schedule built today weights each year's loss by the unconditional PD, so year 4 gets 2.29%. Recovery and notional are red herrings for the probability; they enter only once the loss is computed.

Why the other options are wrong

B is incorrect. Used discrete compounding: $\lambda(1 - \lambda)^3 = 0.025 \times 0.975^3 = 2.32\%$.
C is incorrect. Computed the conditional PD (sibling swap with the unconditional one): $0.02291/0.92774 = 2.47\%$.

D is incorrect. Reported the cumulative four-year PD $Q(4) = 1 - e^{-4\lambda} = 9.52\%$, the expression the formula card mislabels as a hazard rate.

LO: CR-9 e: Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities. · Reference: Hull, RMFI 6e, Ch.17, 9.2.1 eq. 9.2, Example 9.2

Question A15.2

CR-9 d, e · S4, S6 · [Map #17]

A junior credit analyst at a UK pension fund is studying a rating agency's cumulative default table for B-rated issuers:

Year	1	2	3	4
Cumulative default rate (%)	3.10	7.40	11.20	14.60

Which of the following is closest to the probability that a B-rated issuer defaults during year 3, conditional on no default in years 1 and 2?

A. 3.73%

B. 3.80%

C. 4.10%

D. 4.28%

Solution

Answer: C

Step 1 – Unconditional PD in year 3

$$C_3 - C_2 = 11.20 - 7.40 = 3.80\%$$

$\Rightarrow$ *Seen from time zero, there is a 3.80% chance the issuer defaults during year 3.*

Step 2 – Survival to the start of year 3

$$1 - C_2 = 100 - 7.40 = 92.60\%$$

Step 3 – Conditional PD

$$\frac{3.80}{92.60} = 4.10\%$$

Interpretation

Given the issuer has survived two years, the chance of defaulting in year 3 is 4.10%. Conditional PDs approximate the hazard rate over the year and are the right input when assessing a surviving name.

Why the other options are wrong

A is incorrect. Divided the three-year cumulative rate by three: $11.20/3 = 3.73\%$ (an average, not a year-3 probability).

B is incorrect. Stopped at the unconditional PD $C_3 - C_2 = 3.80\%$ (intermediate result).

D is incorrect. Divided by survival to the *end* of year 3: $3.80/88.80 = 4.28\%$.

LO: CR-9 e: Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities. CR-9 d: Describe a rating migration matrix and calculate the probability of default, cumulative probability of default, and marginal probability of default. · Reference: Hull, RMFI 6e, Ch.17, 9.2 and 9.2.1, Table 9.1

Question A15.3

CR-9 d; CR-10 · S4, S6 · [Map #17]

A credit portfolio manager at a Swiss private bank (CHF book) uses a simplified one-year rating migration matrix with two non-default grades. Rating changes in successive years are assumed independent.

Initial rating	To X	To Y	Default
X	92%	6%	2%
Y	8%	82%	10%

Which of the following is closest to the unconditional (marginal) probability that an issuer rated X today defaults during year 2?

A. 2.44%

B. 2.49%

C. 4.00%

D. 4.44%

Solution **Answer: A**

Step 1 – Two-year cumulative PD (row X of the squared matrix)

$$C_2 = p_{XD} + p_{XX}p_{XD} + p_{XY}p_{YD} = 0.02 + 0.92(0.02) + 0.06(0.10) = 0.0444$$

$\Rightarrow$ *4.44% of X-rated issuers default within two years, including those that slip to Y first.*

Step 2 – Marginal PD in year 2

$$C_2 - C_1 = 0.0444 - 0.0200 = 0.0244$$

Interpretation

2.44% is the probability, seen today, of default in year 2. It exceeds the year-1 PD of 2% because migration to Y (with a 10% default rate) raises year-2 risk: the investment-grade pattern of rising marginal PDs.

Why the other options are wrong

B is incorrect. Computed the conditional year-2 PD: $0.0244/(1 - 0.02) = 2.49\%$.
C is incorrect. Ignored migration and doubled the one-year PD: $2 \times 2\% = 4.00\%$.
D is incorrect. Reported the two-year cumulative PD $C_2 = 4.44\%$ (intermediate result).

LO: CR-9 d: Describe a rating migration matrix and calculate the probability of default, cumulative probability of default, and marginal probability of default. · Reference: Hull, RMFI 6e, Ch.17, 9.2; Ch.19, 10.1 Ratings Transition Matrices

Question A15.4 CR-9 e, j, k · S2 · [Map #17]

A risk consultant at an Australian fund manager is explaining the gap between market-implied and historical default estimates to the investment committee for a BBB-type issuer class:

- Seven-year cumulative historical default probability: 2.80%
- Average seven-year credit spread: 150 basis points
- Expected recovery rate: 40%

Which of the following is closest to the excess of the spread-implied (risk-neutral) average hazard rate over the historical (real-world) average hazard rate, per annum?

A. -0.30%

B. 1.09%

C. 2.09%

D. 3.34%

Solution **Answer: C**

Step 1 – Real-world average hazard rate (direct formula)

$$\bar{\lambda}_{\text{hist}} = -\tfrac{1}{7}\ln(1 - 0.028) = 0.00406 = 0.406\%$$

$\Rightarrow$ *Historical data imply defaults arrive at about 0.41% per year.*

Step 2 – Risk-neutral average hazard rate

$$\bar{\lambda}_{\text{spread}} = \frac{s}{1 - R} = \frac{0.0150}{1 - 0.40} = 0.0250 = 2.50\%$$

Step 3 – Difference

$$2.50\% - 0.406\% = 2.09\%$$

Interpretation

Bond investors are compensated for about 2.09% per year more default intensity than history justifies. Hull attributes most of this to systematic (non-diversifiable) default risk, with a small liquidity component. For valuation use the 2.50%; for scenario analysis use the 0.41%.

Why the other options are wrong

A is incorrect. Treated the seven-year cumulative 2.80% as an annual hazard rate: $2.50 - 2.80 = -0.30\%$.
B is incorrect. Did not divide the spread by LGD: $1.50 - 0.41 = 1.09\%$.
D is incorrect. Divided the spread by the recovery rate instead of $1 - R$: $1.50/0.40 - 0.41 = 3.34\%$.

LO: CR-9 j: Compare default probabilities calculated from historical data with those calculated from credit yield spreads. CR-9 k: Describe the difference between real-world and risk-neutral default probabilities and determine which one to use in the analysis of credit risk. · Reference: Hull, RMFI 6e, Ch.17, 9.7, Tables 9.4 and 9.5, 9.7.1 and 9.7.2

Question A15.5

CR-9 h · S3 · [Aug-2026] [Map #17]

A treasury analyst at a Canadian bank hedging a loan to a mining company observes:

- Five-year CDS spread on the company: 180 basis points
- Expected recovery rate: 35%
- Loan notional: CAD 75 million; five-year risk-free rate: 3.2%

Which of the following is closest to the average hazard rate implied by the CDS spread over the five years?

A. 0.28%

B. 1.17%

C. 2.77%

D. 5.14%

Solution

Answer: C

Step 1 – Convert the spread

$$s = 180 \text{ bp} = 180 \times 0.0001 = 0.0180$$

Step 2 – Loss given default

$$1 - R = 1 - 0.35 = 0.65$$

Step 3 – Average hazard rate

$$\bar{\lambda} = \frac{0.0180}{0.65} = 0.0277$$

$\Rightarrow$ *The market prices a 2.77% per year (risk-neutral) default intensity, conditional on survival.*

Interpretation

Protection buyers pay 1.80% a year because the market expects to lose 65% of notional on a default arriving at about 2.77% a year. The notional and risk-free rate are not needed.

Why the other options are wrong

A is incorrect. Converted 180 bp as 0.0018: $0.0018/0.65 = 0.28\%$.
B is incorrect. Multiplied by LGD instead of dividing: $0.018 \times 0.65 = 1.17\%$.
D is incorrect. Divided by the recovery rate instead of LGD: $0.018/0.35 = 5.14\%$.

LO: CR-9 h: Describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates. · Reference: Hull, RMFI 6e, Ch.17, 9.6.1 eq. 9.3

Question A15.6

CR-9 h · S3 · [Map #17]

A credit derivatives trader at a Hong Kong dealer (USD contracts) observes the following CDS spreads on a reference entity, with an expected recovery rate of 40%:

Maturity	3 years	5 years
CDS spread (bp)	400	350

Which of the following is closest to the average hazard rate between year 3 and year 5?

A. 1.83%

B. 2.75%

C. 4.58%

D. 5.83%

Solution

Answer: C

Step 1 – Three-year average hazard

$$\bar{\lambda}(3) = \frac{0.0400}{1 - 0.40} = 0.0667$$

Step 2 – Five-year average hazard

$$\bar{\lambda}(5) = \frac{0.0350}{0.60} = 0.0583$$

Step 3 – Bootstrap the forward hazard

$$\bar{\lambda}_{3,5} = \frac{5(0.0583) - 3(0.0667)}{5 - 3} = \frac{0.2917 - 0.2000}{2} = 0.0458$$

$\Rightarrow$ *An issuer that survives to year 3 is expected to default at about 4.58% a year over years 4 and 5.*

Interpretation

An inverted CDS curve (shorter spread higher) means the market sees near-term distress: if the firm survives the next three years, its default intensity falls. This mirrors the book's observation that poorly rated issuers can have decreasing marginal PDs.

Why the other options are wrong

A is incorrect. Divided the bootstrapped difference by $T_2 = 5$ instead of $T_2 - T_1 = 2$: $0.0917/5 = 1.83\%$.
B is incorrect. Bootstrapped the raw spreads without dividing by LGD: $(5 \times 0.035 - 3 \times 0.040)/2 = 2.75\%$, a forward loss rate, not a hazard rate.
D is incorrect. Reported the five-year average hazard $\bar{\lambda}(5) = 5.83\%$ (intermediate result).

LO: CR-9 h: Describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates. · Reference: Hull, RMFI 6e, Ch.17, 9.6.1, Example 9.3

Question A15.7

CR-9 e, h · S2, S3, S5 · [Map #17]

A credit risk manager at a German insurer is checking whether a quoted five-year CDS spread on a counterparty is consistent with the firm's internal view. The internal model gives a five-year cumulative default probability of 8% and an expected recovery rate of 45%. Assume a constant hazard rate and the approximation in Hull eq. 9.3. Which of the following is closest to the five-year CDS spread consistent with these inputs?

A. 75 basis points

B. 92 basis points

C. 167 basis points

D. 440 basis points

Solution **Answer: B**

Step 1 – Average hazard from the cumulative PD (direct formula)

$$\bar{\lambda} = -\tfrac{1}{5}\ln(1-0.08) = \tfrac{1}{5}(0.08338) = 0.01668$$

$\Rightarrow$ *An 8% five-year cumulative PD corresponds to a 1.67% per year default intensity.*

Step 2 – Loss given default

$$1 - R = 1 - 0.45 = 0.55$$

Step 3 – Spread

$$s = \bar{\lambda}(1-R) = 0.01668 \times 0.55 = 0.00917 = 91.7 \text{ bp}$$

Interpretation

If the market quotes materially more than about 92 bp, either the market's risk-neutral PD exceeds the internal (closer to real-world) estimate, which Hull shows is normal, or the market sees deterioration the internal model has not captured.

Why the other options are wrong

A is incorrect. Multiplied by the recovery rate instead of LGD: $0.01668 \times 0.45 = 75.0$ bp.
C is incorrect. Stopped at the hazard rate, omitting LGD: 166.8 bp.
D is incorrect. Sibling swap: treated the five-year cumulative PD $1 - e^{-5\lambda} = 8\%$ as if it were the hazard rate, $0.08 \times 0.55 = 4.40\% = 440$ bp.

LO: CR-9 h: Describe CDS spreads and explain how CDS spreads can be used to estimate hazard rates. CR-9 e: Define the hazard rate and use it to define probability functions for default time as well as to calculate conditional and unconditional default probabilities. · Reference: Hull, RMFI 6e, Ch.17, 9.2.1 eq. 9.2; 9.6.1 eq. 9.3

A16 Duration Gap and Interest-Sensitive Gap

Readings: LTR-17 a, b, c, d · Risk Management for Changing Interest Rates: Asset-Liability Management and Duration Techniques (Rose and Hudgins, Ch.7)

AUG 2026 RECALL | MAP TIER C · #24 · 5/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: Duration gap (recalled twice).
Map #24: "Reported: 'duration gap'; 'impact of asset and liability values on net worth given duration of asset and liability'; 'duration of asset liability (2)'; 'IS gap' and 'ISR gap LR'. Closed-book: $\Delta NW = -[D_A \cdot A - D_L \cdot L] \cdot \Delta r/(1+r)$. The polarity of the leveraged-adjusted duration gap against a rate move is the single most corruptible fact here."
Shapes covered here: S1 duration gap, plain form of the August recall (Q2); S2 impact on net worth given asset and liability durations, with direction (Q1); S3 duration of assets and liabilities: the liability duration that immunises net worth (Q4); S4 IS gap and ISR: cumulative gap and the change in net interest income (Q3); S5 polarity of the leverage-adjusted gap against a rate move, including a sign reversal from the leverage adjustment (Q1, Q2).

Revision notes

- **The specific: change in net worth** (LTR-17 c). $\Delta NW = [-D_A \frac{\Delta i}{1+i} A] - [-D_L \frac{\Delta i}{1+i} L]$, where i is the *original* discount rate. Book: $D_A = 3.047$, $D_L = 2.669$, $A = 300$, $L = 275$, rates 8% to 10%: $-16.93 + 13.59 = -$USD 3.34 million; rates 8% to 6%: $+3.34$ million. Three size factors drive it: the size of the duration gap, the size of the institution, the size of the rate change.
- **Leverage-adjusted duration gap** (eq. 18.21). $D_A - D_L \times L/A$. Because assets exceed liabilities, liability values must change by slightly *more* than asset values to immunise. Book: $3.047 - 2.669 \times 275/300 = +0.60$ year.
- **Polarity of the gap.** Positive: liabilities change by less than assets, so rates up $\Rightarrow$ net worth down, rates down $\Rightarrow$ net worth up. Negative: rates up $\Rightarrow$ net worth up, rates down $\Rightarrow$ net worth down. Zero: immunised. Management expecting rising rates moves toward a negative gap (reduce D_A, increase D_L); expecting falling rates, toward a positive gap.
- **Immunisation target.** Zero leverage-adjusted gap $\Rightarrow D_L = D_A \times A/L$ (book: $3.047 \times 300/275 = 3.324$ years; footnote: $3 \times 100/92 = 3.261$ years). Close the gap by changing D_A, D_L, or the relative amounts of A and L.
- **Duration** (LTR-17 c). A value- and time-weighted measure of maturity: the average time needed to recover funds committed. $\Delta P/P \approx -D\,\Delta i/(1+i)$; price risk is directly proportional to duration (book: $D = 4$, 10% to 11% gives -3.64%). Portfolio duration is the market-value-weighted average (assets 3.047 years in Table 18.4). Convexity: price moves more at low rates than at high rates, rises with duration, falls as the coupon rises.
- **ALM strategies** (LTR-17 a). Asset management (sources of funds taken as given by customers), liability management (price, the offer rate, as the control lever), and today's funds management: coordinate control of volume, mix and return or cost on both sides. The ALCO meets regularly, sets funding and limits, and reports to the board. Two interest rate risks: price risk (rates rise, asset values fall) and reinvestment risk (rates fall, incoming funds earn less).
- **Two targets, two tools.** Interest-sensitive gap management protects the *net interest margin* (NIM = net interest income / total earning assets; book: $(4 - 2.6)/40 = 3.50\%$). Duration gap management protects *net worth*. Protecting NIM does not mean net worth is protected.
- **Repriceable items** (LTR-17 b). Interest-sensitive assets: short-term securities and loans about to mature, variable-rate loans and securities. Interest-sensitive liabilities: money market borrowings (fed funds, RPs), short-term savings, money market deposits, floating-rate deposits. Nonrepriceable: cash and reserves, long-term fixed-rate loans and securities, buildings, demand deposits, long-term savings, equity.
- **Asset vs liability sensitive.** Asset sensitive: positive dollar ISGAP, positive relative ISGAP, ISR > 1; NIM *rises* when rates rise. Liability sensitive: negative ISGAP, ISR < 1; NIM *falls* when rates rise and rises when rates fall. Book: ISA 150, ISL 200 gives ISGAP -50, relative ISGAP -0.33, ISR 0.75.

- **Cumulative gap** (eq. 18.13). $\Delta\text{NII} \approx \Delta i \times$ cumulative gap. Book: -600 million and $+1$ point gives -6 million. The gap measured over 24 hours can have the opposite sign to the gap over a week, so the planning horizon is crucial.
- **Aggressive vs defensive GAP management.** Aggressive: expect rising rates, go positive (increase ISA, decrease ISL); expect falling rates, go negative. Defensive: keep the gap as close to zero as possible to reduce the volatility of net interest income. Consistently correct rate forecasting is impossible.
- **Weighted gap and basis risk.** A zero gap still leaves risk because asset and liability rates are not perfectly correlated (deposit rates lag loan rates): basis risk. The weighted interest-sensitive gap multiplies each item by its rate-sensitivity weight; the book's gap flips from -23 to $+75$, and a 2 point rise changes NII by $+1.50$ instead of -0.46.
- **ISGAP limitations** (LTR-17 d). Repricing points are hard to identify (checkable and savings deposits); the choice of planning periods is arbitrary and items fall between the cracks; ISGAP ignores the effect of rate changes on net worth, and aggressive gapping raises earnings volatility.
- **Duration gap limitations** (LTR-17 d). Hard to find assets and liabilities of matching duration; duration is less than maturity except for zero-coupon instruments, single-payment loans and T-bills; deposit cash flows are ill-defined; prepayments and defaults distort cash flows; assumes a linear price-rate relationship (convexity) and small, parallel yield-curve shifts; duration itself drifts as rates change. Research suggests duration balancing still works under moderate violations.
- **Fannie Mae** reported a negative duration gap of about 14 months at the end of 2002 (target band ± 6 months): with longer liabilities than assets, falling rates would have kept eroding net worth.

Formulas

Net interest margin and gap measures (eq. 18.5, 18.7).

$$\text{NIM} = \frac{\text{Interest income} - \text{Interest expense}}{\text{Total earning assets}}, \qquad \text{ISGAP} = \text{ISA} - \text{ISL}$$

$$\text{Relative ISGAP} = \frac{\text{ISGAP}}{\text{Size of institution}}, \qquad \text{ISR} = \frac{\text{ISA}}{\text{ISL}}$$

Cumulative gap and change in net interest income (eq. 18.13).

$$\Delta\text{NII} \approx \Delta i \text{ (decimal)} \times \text{Cumulative gap}, \qquad \text{Cumulative gap} = \sum_{\text{buckets in horizon}} (\text{ISA} - \text{ISL})$$

Price sensitivity (eq. 18.18).

$$\frac{\Delta P}{P} \approx -D \times \frac{\Delta i}{1+i}$$

Leverage-adjusted duration gap (eq. 18.21) and change in net worth (eq. 18.22, 18.23).

$$\text{DGAP} = D_A - D_L \times \frac{L}{A}, \qquad \Delta\text{NW} = \left[-D_A \frac{\Delta i}{1+i} A\right] - \left[-D_L \frac{\Delta i}{1+i} L\right] = -\text{DGAP} \times \frac{\Delta i}{1+i} \times A$$

D_A, D_L: dollar-weighted average durations; A, L: total assets and liabilities; i: original discount rate.
Immunisation target. $\text{DGAP} = 0 \Rightarrow D_L = D_A \times A/L$.

Exam traps

- **Polarity of the leverage-adjusted gap (the map's flag):** positive DGAP loses net worth when rates *rise*; negative DGAP loses net worth when rates *fall*. Read the sign only after the leverage adjustment: $D_A < D_L$ can still give a positive DGAP when L/A is low enough.
- **Dropped or misplaced leverage:** $D_A - D_L$ instead of $D_A - D_L(L/A)$; L/A applied to D_A; multiplying instead of dividing by L/A in the immunisation reverse-solve; setting $D_L = D_A$.
- **Dropped $(1+i)$:** the rate change is divided by one plus the *original* rate.
- **ISGAP vs DGAP polarity (sibling swap):** a positive ISGAP *gains* NII when rates rise; a positive DGAP *loses* net worth when rates rise.

- **Bucket vs cumulative, and the horizon:** ΔNII uses the cumulative gap over the whole planning horizon, not one bucket's gap and not a shorter horizon.

Question A16.1 LTR-17 c · S2, S5 · [Aug-2026] [Map #24]

An ALM analyst at a eurozone savings bank is preparing a net-worth sensitivity for the asset-liability committee. Total assets are EUR 500 million with a dollar-weighted average duration of 4.2 years; total liabilities are EUR 450 million with an average duration of 1.8 years. All rates are currently 6% and are expected to rise by 150 basis points. The bank's interest-sensitivity ratio for the next 90 days is 0.92. Which of the following is closest to the change in the market value of the bank's net worth?

A. Decrease of EUR 29.72 million

B. Decrease of EUR 19.35 million

C. Decrease of EUR 18.25 million

D. Increase of EUR 18.25 million

Solution **Answer: C**

Step 1 – Relative rate change

$$\frac{\Delta i}{1+i} = \frac{0.015}{1.06} = 0.014151$$

Step 2 – Change in asset value

$$-D_A \times \frac{\Delta i}{1+i} \times A = -4.2 \times 0.014151 \times 500 = -\text{EUR } 29.717 \text{ million}$$

$\Rightarrow$ *Long-duration assets lose almost EUR 30 million of market value.*

Step 3 – Change in liability value

$$-D_L \times \frac{\Delta i}{1+i} \times L = -1.8 \times 0.014151 \times 450 = -\text{EUR } 11.462 \text{ million}$$

$\Rightarrow$ *Liabilities also fall in value, which is a gain to the owners.*

Step 4 – Change in net worth

$$\Delta\text{NW} = -29.717 - (-11.462) = -\text{EUR } 18.25 \text{ million}$$

Step 5 – Check with the leverage-adjusted gap

$$\text{DGAP} = 4.2 - 1.8 \times \frac{450}{500} = 2.58 \text{ years}, \qquad -2.58 \times 0.014151 \times 500 = -18.25$$

Interpretation

A positive leverage-adjusted duration gap of 2.58 years means rising rates destroy net worth. ALCO should shorten asset duration, lengthen liability duration, or hedge. The 90-day ISR concerns NIM, not net worth, and is irrelevant here.

Why the other options are wrong

A is incorrect. Dropped the liability side (asset value change only): 29.72.
B is incorrect. Omitted the $(1+i)$ divisor: $(4.2 \times 500 - 1.8 \times 450) \times 0.015 = 19.35$.
D is incorrect. Polarity flip: right size, wrong direction; a positive DGAP loses net worth when rates rise (an ISGAP sign rule carried over to net worth).

LO: LTR-17 c: Describe duration gap management and apply this strategy to protect a bank's net worth. · Reference: Rose and Hudgins, Bank Management and Financial Services 9e, Ch.7, 18.6 Using Duration to Hedge Against Interest Rate Risk, eq. (18.22), (18.23)

Question A16.2

LTR-17 c · S1, S5 · [Aug-2026] [Map #24]

A treasury analyst at an Indian private-sector bank is reporting the duration gap to the board risk committee. The bank's assets have a dollar-weighted duration of 3.6 years and its liabilities 3.8 years. Total assets are INR 2,000 billion and total liabilities INR 1,760 billion. A director remarks that because liabilities are longer than assets, the bank gains if rates rise. Which of the following correctly states the leverage-adjusted duration gap and the effect of a rise in rates on net worth?

A. -0.20 years; net worth rises when rates rise

B. $+0.26$ years; net worth falls when rates rise

C. $+0.26$ years; net worth rises when rates rise

D. -0.63 years; net worth rises when rates rise

Solution

Answer: B

Step 1 – Leverage ratio

$$\frac{L}{A} = \frac{1{,}760}{2{,}000} = 0.88$$

Step 2 – Leverage-adjusted liability duration

$$D_L \times \frac{L}{A} = 3.8 \times 0.88 = 3.344 \text{ years}$$

$\Rightarrow$ *Liabilities are a smaller dollar base, so their duration counts for only 88% against assets.*

Step 3 – Leverage-adjusted duration gap

$$\text{DGAP} = 3.6 - 3.344 = +0.256 \approx +0.26 \text{ years}$$

Step 4 – Direction

Positive DGAP: a parallel rise in rates lowers asset values by more than liability values, so net worth falls.

Interpretation

The director's reading uses the unadjusted gap $3.6 - 3.8 = -0.20$. After the leverage adjustment the sign reverses: the bank is exposed to *rising* rates. To immunise it would need $D_L = 3.6 \times 2{,}000/1{,}760 = 4.09$ years.

Why the other options are wrong

A is incorrect. Dropped the leverage adjustment: $3.6 - 3.8 = -0.20$, and read the direction off that wrong sign.
C is incorrect. Right gap, polarity flip on the direction: a positive leverage-adjusted gap loses net worth when rates rise.
D is incorrect. Applied the leverage ratio to asset duration instead of liability duration: $3.6 \times 0.88 - 3.8 = -0.63$.

LO: LTR-17 c: Describe duration gap management and apply this strategy to protect a bank's net worth. · Reference: Rose and Hudgins, Ch.7, 18.6, eq. (18.21) and the table of net worth changes by gap sign

Question A16.3

LTR-17 b · S4 · [Map #24]

The ALCO secretary at a US community bank presents the following repricing schedule (USD million) for the one-year planning horizon:

Bucket	Next 7 days	8 to 30 days	31 to 90 days	91 to 360 days
Interest-sensitive assets	60	140	95	210
Interest-sensitive liabilities	45	190	120	225

Market rates on all interest-sensitive items are expected to rise by 125 basis points. Which of the following is closest to the approximate change in the bank's annual net interest income over the one-year horizon?

A. $-$USD 0.94 million

B. $-$USD 0.75 million

C. $-$USD 0.63 million

D. $+$USD 0.94 million

Solution **Answer: A**

Step 1 – Gap per bucket (ISA − ISL)

$$+15, \quad -50, \quad -25, \quad -15$$

Step 2 – Cumulative gap over the one-year horizon

$$15 - 50 - 25 - 15 = -\text{USD } 75 \text{ million}, \qquad \text{ISR} = \frac{505}{580} = 0.87 < 1$$

⇒ *The bank is liability sensitive over the year: more liabilities than assets reprice. (It is asset sensitive over the next 7 days only.)*

Step 3 – Change in net interest income

$$\Delta\text{NII} \approx \Delta i \times \text{Cumulative gap} = 0.0125 \times (-75) = -\text{USD } 0.94 \text{ million}$$

Interpretation

Rising rates squeeze NII because funding costs reprice faster than asset yields. A defensive manager would shorten asset maturities or lengthen liability maturities (or increase ISA, decrease ISL) to move the cumulative gap toward zero.

Why the other options are wrong

B is incorrect. Wrong horizon: used the cumulative gap only to 90 days, $0.0125 \times (15 - 50 - 25) = -0.75$.
C is incorrect. Used the single largest bucket gap instead of the cumulative gap: $0.0125 \times (-50) = -0.63$.
D is incorrect. Polarity flip: treated a negative cumulative gap as benefiting from rising rates.

LO: LTR-17 b: Describe interest-sensitive gap management and apply this strategy to maximize a bank's net interest margin. · Reference: Rose and Hudgins, Ch.7, 18.4 Interest-Sensitive Gap Management, eq. (18.11), (18.13)

Question A16.4 LTR-17 c · S3 · [Map #24]

The CFO of a Japanese regional bank wants to immunise net worth against parallel rate changes. Total assets are JPY 600 billion with an average duration of 2.8 years; total liabilities are JPY 540 billion. The current discount rate is 7%. Which of the following is closest to the average liability duration the bank must target?

A. 2.52 years

B. 2.80 years

C. 3.11 years

D. 3.33 years

Solution **Answer: C**

Step 1 – Zero leverage-adjusted gap condition

$$0 = D_A - D_L \times \frac{L}{A} \;\Rightarrow\; D_L = D_A \times \frac{A}{L}$$

Step 2 – Solve

$$D_L = 2.8 \times \frac{600}{540} = 2.8 \times 1.1111 = 3.11 \text{ years}$$

⇒ *Liabilities must be longer than assets because their smaller dollar base must produce the same dollar value change.*

Step 3 – Check

$$2.8 - 3.11 \times \frac{540}{600} = 2.8 - 2.8 = 0$$

Interpretation

With $D_L = 3.11$ years the bank's net worth is immunised: asset and liability value changes offset for any small parallel shift. The 7% discount rate does not affect the target.

Why the other options are wrong

A is incorrect. Multiplied by L/A instead of dividing: $2.8 \times 0.9 = 2.52$.
B is incorrect. Dropped the leverage adjustment, setting $D_L = D_A = 2.80$.
D is incorrect. Scaled by one plus the discount rate, an irrelevant input: $3.11 \times 1.07 = 3.33$.

LO: LTR-17 c: Describe duration gap management and apply this strategy to protect a bank's net worth. · Reference: Rose and Hudgins, Ch.7, 18.6, eq. (18.21) and the zero-gap example (3.324 years)

A17 Extreme Value Theory: GEV vs POT

Readings: MR-3 a, b, c, d, e, f · Parametric Approaches (II): Extreme Value (Dowd, Measuring Market Risk 2e, Ch.7; 3.1 Generalised Extreme-Value Theory, 3.2 The Peaks-over-Threshold Approach, GEV vs POT, 3.3 Refinements, 3.4 Conclusions); ORR-4 f · Risk Measurement and Assessment (GARP, Operational Risk and Resilience, Ch.4, Extreme Value Theory and EVT: A Word of Caution)

AUG 2026 RECALL | MAP TIER C · #25 · 5/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: GEV theory question.
Map #25: "Reported: 'GEV vs POT distribution'; 'peaks over threshold'; 'POT threshold low consideration': i.e. the consequence of setting the threshold too low (more data, more bias, less 'extreme'). GEV uses block maxima and needs no threshold but wastes data; POT retains more tail observations but requires a subjective threshold. The shape parameter $\xi > 0$ means fat tails. Also examinable: the operational-risk framing of EVT vs. block maxima, which one sheet flags as a comparison."
Shapes covered here:
S1 GEV theory: Fisher-Tippett, three parameters, the Frechet, Gumbel and Weibull cases (Q1)
S2 GEV versus POT: block maxima versus exceedances, data use, threshold (Q1)
S3 peaks over threshold: the GPBdH theorem and the GP distribution of excess losses (Q2)
S4 POT threshold set too low: more data, more bias, less extreme (Q2)
S5 shape parameter $\xi > 0$ means fat tails (Q1)
S6 operational-risk framing of EVT and block maxima (Q1)

Revision notes

Why extremes are hard (MR-3 a)

- **The core problem:** extreme events are rare by definition, so there are few extreme observations; estimates are very uncertain, especially for events *beyond* the range of the historical data (an unprecedented market fall).
- **Why not fit one distribution to everything:** a distribution fitted to the whole data set accommodates the many central observations, not the sparse extremes: good for the centre, ill-suited to the tails. Extremes are governed by extreme-value theorems, not central limit theorems; do *not* use distributions justified by central limit theory (especially the normal) for extremes. The standard analogy is hydrology: how high to build dikes and sea walls.

GEV: the distribution of block maxima (MR-3 b)

- **Fisher-Tippett theorem**: for iid losses, as sample size n grows, the distribution of the sample maximum M_n converges to the generalised extreme value (GEV) distribution. Three parameters: location μ (central tendency of M_n), scale σ (dispersion of M_n), and the **tail index** ξ (shape, heaviness of the tail). μ and σ are related to, but are *not*, the mean and standard deviation.
- **Three cases of** ξ: $\xi > 0$ **Frechet** (power-law, heavy tails: Levy, Student t, Pareto parents; "particularly useful for financial returns", with estimates typically positive but below 0.35). $\xi = 0$ **Gumbel** (exponential, relatively light tails: normal and lognormal parents). $\xi < 0$ **Weibull** (lighter than normal tails; not useful for financial returns). The parent's *domain of attraction* decides the case.
- **Frechet versus Gumbel:** both right-skewed; Frechet is more skewed with a longer right tail. Frechet quantiles *rise* with ξ and tend to the Gumbel values as $\xi \to 0$. Example 3.5 ($\mu = 2\%$, $\sigma = 0.7\%$, $\xi = 0.3$, $n = 100$): 99.5% VaR 2.537% Frechet versus 2.483% Gumbel; 99.9%: 4.322% versus 3.612%. The GEV probabilities refer to the extreme-loss distribution: its 5th percentile is a small extreme loss but sits in the right tail of the parent.
- **Choosing Frechet or Gumbel:** pick the EV distribution whose domain of attraction contains the parent; or test the significance of ξ (danger: wrongly accepting $\xi = 0$ underestimates risk); safest given model risk: *always* choose the Frechet.
- **Parameter estimation:** maximum likelihood (well grounded, consistent and asymptotically normal if $\xi > -1/2$, but needs numerical methods, may not be robust, small-sample issues); Gumbel's regression method;

moment-based (easy but unreliable); the semi-parametric **Hill estimator** of ξ. Choosing the Hill threshold k is a bias/variance trade-off: higher k means more data, lower variance, *higher* bias. Hill plots can turn into "Hill horror plots" that never settle; Danielsson and de Vries minimise MSE but need at least 1500 observations.

- **Horizon scaling:** the mean scales with h, the standard deviation with $\sqrt{h}$, the tail index not at all; empirical VaR scaling exponent is about 0.45, so the square-root rule is usually inappropriate for EV distributions.

POT and the generalised Pareto (MR-3 c, e)

- **Peaks over threshold**: models the distribution of *excess losses* $X - u$ over a high threshold u. **GPBdH theorem** (Gnedenko-Pickands-Balkema-deHaan): as u gets large, the excess-loss distribution converges to the **generalised Pareto** (GP) distribution, "pretty much regardless" of the parent distribution. Two parameters only: positive scale β and shape ξ (the same tail index as GEV).
- **POT risk measures:** $F(u)$ is estimated by the proportion of observations below the threshold, $(n - N_u)/n$; VaR follows from inverting eq. 3.21, and ES = VaR plus the mean excess loss over VaR, valid for $\xi < 1$. Example 3.6 ($\beta = 0.8$, $\xi = 0.15$, $u = 2\%$, $N_u/n = 4\%$): 99.5% VaR 3.952%, ES 5.238%; at 99.9%: 5.942 and 17.578.
- **Threshold choice, the weak spot of POT**: u must be high enough for GPBdH to apply reasonably closely, but if too high there are too few exceedances for reliable estimates. It is "inevitably arbitrary and therefore judgmental", "very much akin to the problem of choosing k": a lower u gives more data and lower variance but more bias. Aid: choose u where the mean-excess function becomes horizontal. ML estimators are asymptotically normal for $\xi > -0.5$.

GEV versus POT (MR-3 d)

- **Same theory, different targets:** GEV models the distribution of extremes (maxima); POT models exceedances over a high threshold. GEV disadvantages: typically one *additional* parameter, and the block maxima approach can lose useful data because some blocks contain more than one extreme. POT disadvantage: the threshold must be chosen, a problem that does not arise with GEV. "Either approach is usually reasonable"; choose what suits the problem (e.g. limited data).
- **Operational-risk framing** (ORR-4 f): the LDA is complemented by EVT or scenario data for the tail. Block maxima (Fisher-Tippett) examines "the maxima equally spaced in time" (the maximum loss per period and per unit of measure); POT focuses on observations above a high threshold u, whose excess converges to the GPD. Word of caution: EVT is valid only "if a single mechanism is responsible for all observed losses"; with losses from different institutions, business lines and risk types this is unlikely. Alternatives: scenarios, causal modelling, fixed multipliers, macro determinants.

Refinements and multivariate EVT (MR-3 f)

- **Conditional EV:** unconditional EVT suits *long* horizons; for short horizons with dynamic structure, fit a GARCH-type model and apply EVT to its iid residuals (McNeil and Frey two-step).
- **Dependent (clustered) data:** violates iid, can produce very poor estimator performance and changes how often a VaR is breached. Remedies: GEV on block maxima (maxima are less clustered; costs efficiency and creates a block-length bandwidth problem), or the tail of the conditional (GARCH-standardised) distribution; advanced: the extremal index.
- **Multivariate EVT**: beyond elliptical distributions, correlation *no longer suffices*; the limiting multivariate extreme distribution is a member of the family of **EV copulas**, and **tail dependence** is the central focus. Curse of dimensionality: with independent variables and 1-in-100 univariate extremes, a joint extreme is 1 in 10,000 for two variables and 1 in 1,000,000 for three. Extremes are not independent in practice (earthquake, tsunami, market crash).
- **Limitations of EVT:** few extreme observations and wide confidence intervals (not a fault of EVT), considerable model risk and subjective judgment (thresholds), and sensitivity to small-sample effects, biases and non-linearities. Estimate confidence intervals and stress test. "We shouldn't ask more of the theory than it can deliver" (Diebold et al.).

Formulas

GEV, eq. 3.1. $H_{\xi,\mu,\sigma}(x) = \exp\left[-\left(1+\xi\frac{x-\mu}{\sigma}\right)^{-1/\xi}\right]$ for $\xi \neq 0$; $\exp\left[-\exp\left(-\frac{x-\mu}{\sigma}\right)\right]$ for $\xi = 0$

GEV VaR, eq. 3.5. Frechet: $\text{VaR} = \mu_n - \frac{\sigma_n}{\xi_n}\left[1-(-n\ln\alpha)^{-\xi_n}\right]$; Gumbel: $\text{VaR} = \mu_n - \sigma_n \ln[-n\ln\alpha]$

GP distribution, eq. 3.19. $G_{\xi,\beta}(x) = 1-(1+\xi x/\beta)^{-1/\xi}$ for $\xi \neq 0$; $1-\exp(-x/\beta)$ for $\xi = 0$

POT VaR and ES, eqs. 3.22 and 3.23.

$$\text{VaR} = u + \frac{\beta}{\xi}\left\{\left[\frac{n}{N_u}(1-\alpha)\right]^{-\xi} - 1\right\}, \qquad \text{ES} = \frac{\text{VaR}}{1-\xi} + \frac{\beta-\xi u}{1-\xi} \quad (\xi < 1)$$

Hill estimator, eq. 3.17. $\hat{\xi}^{(H)}_{n,k} = \frac{1}{k}\sum_{i=1}^{k} \ln X_i - \ln X_{k+1}$ (losses ordered from highest)

EV horizon scaling. $\text{VaR}(h) = \text{VaR}(1)\, h^{\kappa}$, $\kappa \approx 0.45$ empirically

Exam traps

- **Polarity flip on ξ:** $\xi > 0$ is Frechet, heavy tails, the relevant case for financial and loss data; $\xi < 0$ is Weibull, lighter than normal tails. Statements that attach "lighter" or "Weibull" to a positive tail index are wrong.
- **Sibling swap GEV/POT:** "three parameters" and "wastes data in blocks" belong to GEV (block maxima); "two parameters" and "needs a threshold" belong to POT (GP). Swapped attributes are the standard distractor.
- **Threshold trade-off polarity:** a lower u (or a higher Hill k) means more data, *lower* variance and *higher* bias; a higher u means fewer exceedances and higher variance.
- **Scope on GPBdH:** the GP limit holds as the threshold gets *large*; it is not a licence to lower u to collect data, and "regardless of the parent" is a limiting result, not an exact one at any threshold.
- **Absolute claims:** EVT is not "free of judgment" (threshold choice), and in operational risk it is not valid for pooled losses unless a single mechanism generated them.
- **Wrong distribution twin:** GEV percentiles refer to the extreme-loss distribution, not the parent; and normal distributions justified by central limit theory are exactly what EVT says not to use for extremes.

Question A17.1

MR-3 b, d; ORR-4 f · S1, S2, S5, S6 · [Aug-2026] [Map #25]

The head of operational risk modelling at a European insurer is presenting to newly hired analysts on how extreme value theory can complement the loss distribution approach in the tail of the severity distribution. She compares the block maxima (GEV) approach, applied to the largest loss per quarter in each unit of measure, with the peaks-over-threshold approach. Which of the following statements should she include in the presentation?

A. The block maxima approach fits a two-parameter generalised Pareto distribution to the largest loss in each quarter, so it needs fewer parameters than the peaks-over-threshold approach.

B. The block maxima approach needs no threshold choice, but it can waste useful data because a quarter may contain more than one extreme loss; the peaks-over-threshold approach models all losses above a high threshold.

C. A positive tail index places the fitted GEV in the Weibull case with lighter than normal tails, so a positive estimate signals that extreme losses are less likely than under a normal parent.

D. Because both approaches rest on extreme-value theorems, they remain valid for losses pooled across business lines and risk types, whatever mechanism produced each loss.

Solution

Answer: B

Step 1 – What block maxima is (GEV theory)

By the Fisher-Tippett theorem, the distribution of the maximum of a large iid sample converges to the GEV, with location μ, scale σ and tail index ξ. The operational-risk reading frames it as examining "the maxima equally spaced in time (e.g., the maximum operational loss per period of time and per UoM)".

Step 2 – What POT is

POT models excess losses over a high threshold u; by the GPBdH theorem they converge to the two-parameter generalised Pareto distribution (β, ξ) as u gets large.

Step 3 – Compare them

Dowd: the GEV "typically involves an additional parameter relative to the POT", and block maxima "can involve some loss of useful data relative to the POT approach, because some blocks might have more than one extreme in them"; on the other hand, POT "requires us to grapple with the problem of choosing the threshold, and this problem does not arise with the GEV". Statement B carries exactly these properties.

Interpretation

Neither approach dominates: "either approach is usually reasonable". With a short internal loss history, POT keeps more of the scarce tail observations; block maxima avoids the subjective threshold and handles clustered losses better, at the cost of efficiency.

Why the other options are wrong

A is incorrect. Sibling swap. The two-parameter GP distribution belongs to POT; the GEV fitted to block maxima has three parameters, typically one *more* than POT.

C is incorrect. Polarity flip. $\xi > 0$ is the Frechet case: power-law, heavy tails, more extreme losses than a normal parent. The Weibull case with lighter than normal tails is $\xi < 0$.

D is incorrect. Absolute claim. The ORR reading cautions that EVT "would only be valid if a single mechanism is responsible for all observed losses"; with losses from different business lines and risk types this is unlikely, which "raises concerns about the applicability of EVT in operational risk".

LO: MR-3 b: Describe extreme value theory (EVT) and its use in risk management. MR-3 d: Compare and contrast the generalized extreme value (GEV) and POT approaches to estimating extreme risks. ORR-4 f: Describe approaches used to determine the level of operational risk capital for economic capital purposes, including their application and limitations. · Reference: Dowd, Measuring Market Risk 2e, Ch.7, 3.1 and GEV vs POT; GARP, Operational Risk and Resilience, Ch.4, Extreme Value Theory and EVT: A Word of Caution

Question A17.2

MR-3 c, d, e · S3, S4 · [Map #25]

A market risk analyst at a Canadian bank fits a generalised Pareto distribution to 1,000 daily trading losses using the peaks-over-threshold approach. At the current threshold $u = 2.0\%$, there are 40 exceedances. To obtain more data for the estimation, a colleague proposes lowering the threshold to $u = 1.2\%$, which would give 150 exceedances. Which conclusion should the head of market risk reach about the proposal?

A. Lowering u to 1.2% raises the number of exceedances to 150 and also makes the GPBdH approximation more accurate, because the theorem holds for any threshold that leaves many exceedances.

B. Lowering u to 1.2% leaves the estimated tail index unchanged, because the generalised Pareto distribution describes excess losses whatever the parent distribution of the losses.

C. Lowering u to 1.2% raises the number of exceedances to 150 and lowers the variance of the estimates, but adds bias, because more exceedances come from closer to the centre of the distribution.

D. Lowering u to 1.2% reduces the bias of the estimates but raises their variance, because the additional exceedances come from closer to the centre of the distribution.

Solution

Answer: C

Step 1 – What the proposal changes

$$\frac{N_u}{n} : \frac{40}{1{,}000} = 4\% \longrightarrow \frac{150}{1{,}000} = 15\%$$

$\Rightarrow$ *The threshold moves from roughly the 96th percentile of losses to roughly the 85th: the sample of "extremes" becomes larger and less extreme.*

Step 2 – The trade-off in the reading

Dowd: "we want a threshold u to be sufficiently high for the GPBdH theorem to apply reasonably closely; but if u is too high, we won't have enough excess-threshold observations". The threshold problem "is very much akin to the problem of choosing k" for the Hill estimator, where more data "increases the precision of our estimator (and therefore reduces its variance), but also increases the bias of the tail estimator by placing relatively more weight on observations closer to the centre".

Interpretation

The proposal buys precision with bias: the GP limit is a large-threshold result, so the extra exceedances near the body of the distribution pull the fitted tail toward central behaviour. A defensible process documents the choice, checks where the mean-excess function becomes horizontal, and reports confidence intervals; the choice remains "inevitably arbitrary and therefore judgmental".

Why the other options are wrong

A is incorrect. Scope/condition error. GPBdH holds "as u gets large"; lowering u weakens the approximation. The number of exceedances is not the condition.

B is incorrect. Absolute claim. "Pretty much regardless" of the parent is a limiting result for a high threshold; the estimated ξ is sensitive to the threshold, which is why its choice is called the weak spot of POT.

D is incorrect. Polarity flip on the trade-off. More data near the centre *lowers* variance and *raises* bias, not the reverse.

LO: MR-3 c: Describe the peaks-over-threshold (POT) approach. MR-3 d: Compare and contrast the generalized extreme value (GEV) and POT approaches to estimating extreme risks. MR-3 e: Discuss the application of the generalized Pareto (GP) distribution in the POT approach. · Reference: Dowd, Measuring Market Risk 2e, Ch.7, 3.1 Semi-Parametric Estimation Methods; 3.2 Theory and Estimation, eqs. 3.18 to 3.24

A18 Performance Measures and Attribution

Readings: IM-7 a-i · Portfolio Performance Evaluation (Bodie, Kane and Marcus, *Investments* 12e, Ch.24); IM-3 b, c, d · Alpha (and the Low-Risk Anomaly) (Ang, *Asset Management*, Ch.10: Definition of Alpha, Benchmarks Matter, Ideal Benchmarks, Creating Alpha)

AUG 2026 RECALL | MAP TIER C · #36 · 5/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: Jensen's alpha, Sharpe and other metrics theory.
Map #36: "Reported: 'Sharpe ratio (this was bit hard)'; 'Mvar, vol, rf is given, calculate Sharpe ratio'; 'IR, alpha'; 'portfolio manager asset allocation and selection attributes', i.e. performance attribution split into allocation and selection effects. Grinold's fundamental law and the maximum attainable information ratio (LO 85.d) sit alongside."
Shapes covered here:
S1 Sharpe ratio, the harder form: Sharpe paired with M^2 (Q4); Sharpe as the wrong lens for one of many subportfolios (Q3)
S2 marginal VaR, volatility and the risk-free rate chained into a Sharpe ratio (A12.4, see also)
S3 Jensen's alpha and the information ratio (Q1 plain Jensen's alpha; Q2)
S4 attribution split into asset allocation and security selection effects (Q5)
S5 Grinold's fundamental law and the maximum attainable information ratio (Q6)
S6 which measure suits which use: Sharpe, Treynor, information ratio (Q2, Q3, Q6).

Revision notes

- **Time-weighted versus dollar-weighted return** (IM-7 a): the geometric average is a *time-weighted* average (each period's return has equal weight). The dollar-weighted return is the IRR of all cash flows, so periods with more money invested weigh more. Book example: buy one share at 50, a second at 53, sell both at 54: dollar-weighted 7.117% versus time-weighted 7.81%, lower because the weaker second year had more money in it. Time-weighting judges the manager, who does not control the client's cash flows.
- **Comparison universes** (IM-7 b): percentile rankings against funds of similar style are a useful first step, but subgroups differ in risk (high-beta growth, different durations) and the median fund is not an investable benchmark known in advance.
- **Sharpe ratio** (IM-7 b; IM-3 b): $(\bar{r}_P - \bar{r}_f)/\sigma_P$, reward to *total* volatility, the slope of the capital allocation line. Relevant "when choosing among portfolios competing for the overall risky portfolio"; the benchmark is the market index's Sharpe ratio. Ang: when the benchmark is the risk-free rate, the information ratio coincides with the Sharpe ratio (Martingale: 8.59/12.22 = 0.70 versus 0.45 for the Russell 1000).
- M^2 (IM-7 c): mix P with T-bills (or borrow) so that the adjusted portfolio P* has the index's standard deviation; $M^2 = r_{P^*} - r_M = (S_P - S_M)\sigma_M$, the vertical distance between M and P*. Same ranking as Sharpe, read as a return. Example 8.1: weights 0.714 and 0.286, $r_{P^*} = 26.7\%$, $M^2 = -1.3\%$.
- **Treynor and T^2** (IM-7 b, c): excess return per unit of beta, the slope of the T-line; the SML slope is the market's Treynor. Relevant "when ranking many portfolios that will be mixed to form the overall risky portfolio" (CalPERS' funds of funds), because each manager's nonsystematic risk diversifies away. T^2 mixes with T-bills to a beta of 1: Q ($w = 0.8$) gives $T^2 = 2.8\%$, U ($\beta = 0.8$, $w = 1.25$, borrowing) gives 3.75%, so U wins despite a lower alpha (3.0% versus 3.5%).
- **Jensen's alpha** (IM-7 b; IM-3 b): $\alpha_P = \bar{r}_P - [\bar{r}_f + \beta_P(\bar{r}_M - \bar{r}_f)]$; multifactor versions subtract SMB and HML loadings. Positive alpha is *necessary* to beat the index under every measure (so it is the most widely used), but not sufficient: departing from diversification can raise volatility until the Sharpe ratio falls. No situation uses alpha alone to choose between funds.
- **Information ratio** (IM-7 b; IM-3 b): BKM: $\alpha_P/\sigma(e_P)$, alpha per unit of nonsystematic risk ("tracking error"); some call this the appraisal ratio and define the IR with excess return, but BKM uses alpha consistently. Relevant "when evaluating a portfolio to be mixed with the benchmark portfolio" (a university adding a hedge fund to an indexed core): the improvement in the Sharpe ratio is driven by the IR (Eq. 8.2). Ang: alpha is the average excess (active) return over a benchmark, tracking error its standard deviation; IRs above one are not common.

- **Benchmark choice** (IM-3 c): alpha is "first and foremost a statement about a benchmark". Martingale against a naive Russell 1000 benchmark (beta assumed 1): alpha 1.50%, IR 0.24; against the risk-adjusted benchmark 0.27 T-bills plus 0.73 Russell 1000: alpha 3.44%, IR 0.78. A sound benchmark is well defined, tradeable, replicable and adjusted for risk; most industry benchmarks are not risk adjusted.
- **Grinold's fundamental law** (IM-3 d): the *maximum* attainable IR $=$ IC $\times \sqrt{\text{BR}}$, ignoring transaction costs and trading restrictions. IC is the correlation of forecasts with actual returns; breadth is the number of independent bets. A market timer making 4 bets a year needs IC 0.25 for IR 0.50; a stock selector making 400 bets needs IC 0.025. "Play often and play well."
- **Fundamental law: assumptions and limitations** (IM-3 d): derived under mean-variance utility, so it ignores downside and higher-moment risk and assumes information is used optimally; forecasts must be independent. Limitation 1: IC is assumed constant across BR, yet ICs fall as assets under management rise (decreasing returns to scale in mutual funds, hedge funds and private equity). Limitation 2: bets are correlated: 1,000 value overweights against 1,000 growth underweights is one bet on a value-growth factor; one hundred managers reaching for yield is one bet on illiquidity.
- **Statistical significance** (IM-7 d): $t(\alpha) = \alpha\sqrt{N}/\sigma(e)$. Joe Dart: monthly alpha 0.2%, $\sigma(e) = 2\%$; $t = 1.96$ needs $N = 384$ months (32 years), while typical manager tenure is under 5 years.
- **Selection and survivorship bias** (IM-7 f): with thousands of funds a few "winners" appear by luck; restricting a sample to funds alive for the whole period drops closed funds and creates survivorship bias, which can be substantial even when few funds fail. See also A24.
- **Style analysis** (IM-7 e): Sharpe regresses fund returns on asset-class indexes with coefficients constrained to be non-negative and to sum to 100%; R^2 is the share of return variability attributable to style choice, the intercept the average return from security selection. Magellan (Lynch era): 61% large cap, 35% medium cap, 5% growth, $R^2 = 97.5\%$, intercept 32 bp a month (19.19% over 5 years). A 100% R^2 can coexist with a positive intercept.
- **Changing portfolio composition** (IM-7 f): conventional measures assume constant risk. Example 8.3: Sharpe 0.5 in each year (1%/2%, then 9%/18%) but 0.37 on the eight quarters pooled (mean 5%, SD 13.42%), below the market's 0.4: shifting means look like volatility and bias the Sharpe ratio downward.
- **Performance manipulation** (IM-7 g): the Sharpe ratio is invariant to a constant T-bill mix, but a manager can raise leverage after poor early returns and deleverage after good ones, reweighting the subperiods; investors lose on average. Only the Morningstar Risk-Adjusted Rating (MRAR, $\gamma = 2$ for funds) is manipulation-proof; stars by percentile: 0 to 10, 10 to 32.5, 32.5 to 67.5, 67.5 to 90, 90 to 100.
- **Market timing** (IM-7 h): Treynor-Mazuy adds $c(r_M - r_f)^2$; Henriksson-Merton adds $c(r_M - r_f)D$ with beta b in bear and $b + c$ in bull markets; $c > 0$ indicates timing. In Table 8.2, Q's alpha falls from 5.26% to -2.29% once timing ($c = 0.10$) is allowed. Perfect timing is a free call: $C = 2N(\frac{1}{2}\sigma_M\sqrt{T}) - 1 = 7.98\%$ at $\sigma_M = 20.04\%$; imperfect timing is worth $(P_1 + P_2 - 1) \times C$; a coin flip scores zero.
- **Performance attribution** (IM-7 i): the bogey is the passive benchmark (neutral weights, index within each class). Asset allocation contribution $= \sum(w_P - w_B) \times r_{\text{index}}$; selection contribution $= \sum w_P \times (r_P - r_{\text{index}})$; together they equal the excess return. BKM example: 31 bp allocation plus 106 bp selection = 137 bp; within equity, 147 bp excess splits into 129 bp sector allocation and 18 bp security selection, weighted by the 70% equity weight (102.9 bp) before adding to the portfolio total.

Formulas

First question: which risk is in the denominator? Total σ_P, systematic β_P, or residual $\sigma(e_P)$.

$$S_P = \frac{\bar{r}_P - \bar{r}_f}{\sigma_P}, \qquad T_P = \frac{\bar{r}_P - \bar{r}_f}{\beta_P}, \qquad \alpha_P = \bar{r}_P - [\bar{r}_f + \beta_P(\bar{r}_M - \bar{r}_f)], \qquad IR_P = \frac{\alpha_P}{\sigma(e_P)}$$

M^2 (Eq. 8.1). $w = \dfrac{\sigma_M}{\sigma_P}$, $\quad r_{P^*} = w\bar{r}_P + (1-w)\bar{r}_f$, $\quad M^2 = r_{P^*} - \bar{r}_M = (S_P - S_M)\sigma_M$

T^2. $w = 1/\beta_P$, $\quad T^2 = w\bar{r}_P + (1-w)\bar{r}_f - \bar{r}_M$. **Beta from correlation:** $\beta_P = \rho_{PM}\sigma_P/\sigma_M$.

Active portfolio added to the index (Eq. 8.2). $S_P^2 = S_M^2 + [\alpha_H/\sigma(e_H)]^2$ (equation body not legible in the source text; standard BKM form, outside-source).

Fundamental law (Ang Eq. 3.5). $\text{IR}_{\max} = \text{IC} \times \sqrt{\text{BR}}$. **Ang IR:** IR $= \bar{r}^x/\text{stdev}(r^x)$, excess over the benchmark.

Significance (Eq. 8.3). $t(\alpha) = \alpha\sqrt{N}/\sigma(e)$

Timing. $C = 2N(\frac{1}{2}\sigma_M\sqrt{T}) - 1$ per dollar; imperfect timer $(P_1 + P_2 - 1)\,C$

Attribution. Allocation $= \sum_i (w_{P,i} - w_{B,i})\, r_{B,i}$; Selection $= \sum_i w_{P,i}\,(r_{P,i} - r_{B,i})$; within a class: class excess return = sector allocation + security selection, each multiplied by the class's portfolio weight.

Measure	Risk in denominator	Use when (BKM)
Sharpe / M^2	total, σ_P	the portfolio is the entire risky investment
Treynor / T^2	systematic, β_P	one subportfolio of many
Information ratio	residual, $\sigma(e_P)$	active portfolio mixed with the passive benchmark
Jensen's alpha	none	positive alpha needed under all three

Exam traps

- **Wrong lens (sibling swap):** Sharpe for one of many managers (should be Treynor), Treynor for an active add-on to an index (should be the information ratio). Name the decision context before computing anything.
- **Raw instead of excess return (dropped term):** dividing $\bar{r}_P$ by σ_P or β_P, or forgetting the T-bill leg of M^2. This can flip a ranking.
- **Naive benchmark (wrong-input twin):** alpha as $\bar{r}_P - \bar{r}_M$ assumes beta of 1 (Martingale: 1.50% versus 3.44%); correlation used as beta; σ_P/σ_M used as beta without ρ.
- **Highest alpha wins (absolute claim):** alpha is necessary, not sufficient; Table 8.1's U beats the higher-alpha Q.
- **Attribution weights:** allocation uses *index* returns, selection uses *portfolio* weights; a selection effect inside equity must be multiplied by the equity weight before it enters the portfolio total.
- **Fundamental law scope:** it gives a *maximum* IR before costs, assumes independent bets and an IC that does not decay with size.

Question A18.1

IM-7 b; IM-3 b, c · S3 · [Aug-2026] [Map #36]

A junior risk analyst at an Indian asset manager is preparing a report on an INR large-cap equity fund. Over the period the fund returned 16% with a standard deviation of 20%. Its correlation with the NIFTY index was 0.9. The index returned 11% with a standard deviation of 15%, and the average T-bill rate was 4%. Which of the following is closest to the fund's Jensen's alpha?

A. 2.7%

B. 3.6%

C. 5.0%

D. 5.7%

Solution

Answer: B

Step 1 – Beta from correlation

$$\beta_P = \frac{\operatorname{cov}(r_P, r_M)}{\operatorname{var}(r_M)} = \rho \frac{\sigma_P}{\sigma_M} = 0.9 \times \frac{20\%}{15\%} = 1.20$$

$\Rightarrow$ *The fund moves 1.2% for each 1% move in the index: more systematic risk than the market.*

Step 2 – Market risk premium

$$\bar{r}_M - \bar{r}_f = 11\% - 4\% = 7\%$$

Step 3 – CAPM required return

$$4\% + 1.20 \times 7\% = 12.4\%$$

Step 4 – Jensen's alpha

$$\alpha_P = 16\% - 12.4\% = 3.6\%$$

⇒ *After paying for a beta of 1.2, the fund still earned 3.6% a year above the SML.*

Interpretation

Positive alpha is necessary for the fund to beat the index on any risk-adjusted basis, but not sufficient: if its non-systematic risk is large, its Sharpe ratio can still lag the market's.

Why the other options are wrong

A is incorrect. Used $\sigma_P/\sigma_M = 1.333$ as beta, dropping ρ: $16\% - (4\% + 1.333 \times 7\%) = 2.7\%$.
C is incorrect. Naive benchmark: subtracted the raw index return, $16\% - 11\% = 5.0\%$, implicitly assuming $\beta = 1$ (Ang's benchmark-choice trap).
D is incorrect. Used the correlation as beta: $16\% - (4\% + 0.9 \times 7\%) = 5.7\%$.

LO: IM-7 b: Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant. IM-3 b: Define and calculate alpha, tracking error, the information ratio, and the Sharpe ratio. · Reference: Bodie, Kane and Marcus, Ch.24, Risk-Adjusted Performance Measures; Ang, Ch.10, Benchmarks Matter

Question A18.2

IM-7 b; IM-3 b, c · S3, S6 · [Map #36]

The CIO of a Singapore university endowment, whose core holding is a passive SGD-hedged global equity index fund, is evaluating an active manager to add on top of that index position. The manager's portfolio data are:

- Average return: 15%; beta relative to the index: 1.2
- Standard deviation of returns: 22%
- Standard deviation of residual returns (tracking error): 6%
- Index average return: 11%; average risk-free rate: 3%

Which of the following is closest to the value of the measure the CIO should use to evaluate this manager?

A. 0.40

B. 0.55

C. 0.67

D. 2.00

Solution

Answer: A

Step 1 – Choose the measure

An active portfolio is being mixed with a passive benchmark position, so the relevant measure is the information ratio: alpha per unit of residual risk.

Step 2 – CAPM required return

$$\bar{r}_f + \beta_P(\bar{r}_M - \bar{r}_f) = 3\% + 1.2 \times (11\% - 3\%) = 12.6\%$$

Step 3 – Jensen's alpha

$$\alpha_P = 15\% - 12.6\% = 2.4\%$$

⇒ *The manager adds 2.4% a year beyond what beta exposure explains.*

Step 4 – Information ratio

$$IR = \frac{\alpha_P}{\sigma(e_P)} = \frac{2.4\%}{6\%} = 0.40$$

⇒ *Each percentage point of diversifiable risk the endowment takes on by tilting away from the index earns 0.40 percentage points of alpha.*

Interpretation

By Eq. 8.2 the gain in the endowment's Sharpe ratio from optimally adding this manager is driven by the IR. An IR of 0.40 is respectable; Ang notes that IRs above one are not common.

Why the other options are wrong

B is incorrect. Computed the Sharpe ratio, $(15\% - 3\%)/22\% = 0.55$: the total-risk measure, relevant only if the manager were the endowment's entire risky portfolio.
C is incorrect. Naive benchmark: return in excess of the index, $(15\% - 11\%)/6\% = 0.67$, implicitly assuming a beta of 1.
D is incorrect. Put the excess return over the risk-free rate in the numerator, $(15\% - 3\%)/6\% = 2.00$: the Sharpe numerator over the IR denominator.

LO: IM-7 b: Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant. IM-3 c: Explain the impact of benchmark choice on alpha and describe characteristics of an effective benchmark to measure alpha. · Reference: Bodie, Kane and Marcus, Ch.24, The Information Ratio; Ang, Ch.10, Definition of Alpha

Question A18.3

IM-7 b, c · S1, S6 · [Map #36]

A portfolio manager at a large EUR-denominated pension fund allocates assets across more than 20 external equity managers using a funds-of-funds approach, so each manager's nonsystematic risk is largely diversified away. She must choose between two candidate managers. Annual data for the evaluation period are:

	Fund X	Fund Y	Market index
Average return	17%	13%	10%
Standard deviation	18%	24%	15%
Beta	1.4	0.8	1.0

The average risk-free rate was 4%. Which of the following correctly identifies the preferred fund and the supporting measure?

A. Fund X, because its Sharpe ratio is 0.72 versus 0.38 for Fund Y

B. Fund Y, because its Treynor measure is 11.25 versus 9.29 for Fund X

C. Fund Y, because its Treynor measure is 16.25 versus 12.14 for Fund X

D. Fund X, because its Jensen's alpha is 4.6% versus 4.2% for Fund Y

Solution

Answer: B

Step 1 – Choose the measure

Each manager is one of many components of the overall risky portfolio and nonsystematic risk diversifies away, so only beta matters: the Treynor measure.

Step 2 – Excess returns

$$\bar{r}_X - \bar{r}_f = 17\% - 4\% = 13\%, \qquad \bar{r}_Y - \bar{r}_f = 13\% - 4\% = 9\%$$

Step 3 – Treynor measures

$$T_X = \frac{13\%}{1.4} = 9.29\%, \qquad T_Y = \frac{9\%}{0.8} = 11.25\%$$

$\Rightarrow$ *Per unit of beta, Y delivers 11.25% of excess return and X 9.29%; the market's Treynor is 6%, so both beat the SML, Y by more.*

Step 4 – Sharpe ranking (to see the flip)

$$S_X = \frac{13\%}{18\%} = 0.72, \qquad S_Y = \frac{9\%}{24\%} = 0.375, \qquad S_M = \frac{6\%}{15\%} = 0.40$$

$\Rightarrow$ *On total risk X is far better: Y carries a lot of nonsystematic risk, which washes out in a multi-manager structure.*

Interpretation

The rankings flip because Y's volatility is mostly nonsystematic (high σ, low β). Sharpe ranks a standalone overall portfolio; Treynor ranks one of many.

Why the other options are wrong

A is incorrect. Right numbers, wrong lens (sibling swap): the Sharpe ratio uses total risk, relevant only if the fund were the investor's entire risky portfolio.

C is incorrect. Used raw returns instead of excess returns: $13\%/0.8 = 16.25$ and $17\%/1.4 = 12.14$. Right fund, wrong values.
D is incorrect. The alphas are right ($17 - (4 + 1.4 \times 6) = 4.6\%$; $13 - (4 + 0.8 \times 6) = 4.2\%$), but highest alpha is not a selection rule: BKM's Table 8.1 shows the lower-alpha portfolio winning on Treynor.

LO: IM-7 b: Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant. IM-7 c: Describe the uses for the Modigliani-squared and Treynor's measure in comparing two portfolios and the graphical representation of these measures. · Reference: Bodie, Kane and Marcus, Ch.24, The Treynor Ratio (Table 8.1, Figure 8.3)

Question A18.4

IM-7 b, c · S1 · [Map #36]

An investment consultant at a USD family office is presenting to newly hired analysts how to express the risk-adjusted performance of a concentrated portfolio P that the family holds as its entire risky portfolio. Over the period, P returned an average of 22% with a standard deviation of 28% and a beta of 1.3. The market index returned 12% with a standard deviation of 16%. The average T-bill rate was 4%. Which of the following correctly states the Sharpe ratio of P and its M^2?

A. Sharpe ratio 0.64; M^2 0.14%

B. Sharpe ratio 0.64; M^2 2.29%

C. Sharpe ratio 0.64; M^2 4.00%

D. Sharpe ratio 0.79; M^2 0.57%

Solution

Answer: B

Step 1 – Sharpe ratios

$$S_P = \frac{22\% - 4\%}{28\%} = 0.643, \qquad S_M = \frac{12\% - 4\%}{16\%} = 0.500$$

$\Rightarrow$ P earns 0.64% of excess return per 1% of total volatility, against 0.50% for the index.

Step 2 – Weight in P that matches the index's volatility

$$w_P = \frac{\sigma_M}{\sigma_P} = \frac{16\%}{28\%} = 0.5714$$

$\Rightarrow$ Hold 57.14% in P and 42.86% in T-bills so P has a 16% standard deviation.*

Step 3 – M^2

$$r_{P^*} = 0.5714 \times 22\% + 0.4286 \times 4\% = 14.286\%, \qquad M^2 = 14.286\% - 12\% = 2.29\%$$

Check

$$(S_P - S_M)\,\sigma_M = 0.143 \times 16\% = 2.29\%$$

Interpretation

At the index's volatility P would have beaten the index by 2.29 points a year. Because P is the family's whole risky portfolio, total risk is the right measure; the beta of 1.3 is a red herring (it would feed T^2).

Why the other options are wrong

A is incorrect. Reported $S_P - S_M = 0.143$ as a return of 0.14%; the Sharpe difference must be scaled by σ_M.
C is incorrect. Scaled the Sharpe difference by σ_P instead of σ_M: $0.143 \times 28\% = 4.00\%$, which matches P's volatility rather than the index's.
D is incorrect. Used raw returns throughout: $S_P = 22/28 = 0.79$, $S_M = 12/16 = 0.75$, $M^2 = (0.786 - 0.75) \times 16\% = 0.57\%$, equivalent to dropping the T-bill leg $(1 - w)\bar{r}_f$.

LO: IM-7 c: Describe the uses for the Modigliani-squared and Treynor's measure in comparing two portfolios and the graphical representation of these measures. IM-7 b: Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant. · Reference: Bodie, Kane and Marcus, Ch.24, The M^2 Measure and the Sharpe Ratio (Eq. 8.1, Example 8.1)

Question A18.5 IM-7 i · S4 · [Map #36]

A fund-of-funds analyst at a Canadian pension plan is preparing a monthly attribution report for a balanced manager. The neutral (bogey) weights, the manager's weights, and the returns are:

Market	Bogey weight	Manager weight	Index return	Manager's return in market
Equity	55%	65%	4.0%	5.2%
Fixed income	35%	20%	1.2%	1.6%
Cash	10%	15%	0.4%	0.4%

A separate sector analysis of the equity component shows that the manager's sector weights within equity contributed 0.80% of the equity component's excess return over the equity index. Which of the following correctly states the contribution of the asset allocation decision to the portfolio's excess return, and the contribution of security selection within equity sectors to the portfolio's excess return?

A. Asset allocation 0.24%; security selection within equity 0.40%

B. Asset allocation 0.24%; security selection within equity 0.78%

C. Asset allocation 0.30%; security selection within equity 0.26%

D. Asset allocation 0.24%; security selection within equity 0.26%

Solution Answer: D

Step 1 – Bogey, portfolio and excess return

$$r_B = 0.55(4.0\%) + 0.35(1.2\%) + 0.10(0.4\%) = 2.66\%$$
$$r_P = 0.65(5.2\%) + 0.20(1.6\%) + 0.15(0.4\%) = 3.76\%, \qquad r_P - r_B = 1.10\%$$

Step 2 – Asset allocation contribution (active weight times index return)

$$(0.10)(4.0\%) + (-0.15)(1.2\%) + (0.05)(0.4\%) = 0.40\% - 0.18\% + 0.02\% = 0.24\%$$

$\Rightarrow$ *Overweighting equity in a month when the equity index returned 4% added 24 bp.*

Step 3 – Equity component: total selection, then split

$$\text{equity excess} = 5.2\% - 4.0\% = 1.20\%, \qquad \text{security selection within sectors} = 1.20\% - 0.80\% = 0.40\%$$

Step 4 – Contribution to the portfolio (multiply by the equity weight)

$$0.40\% \times 0.65 = 0.26\%$$

$\Rightarrow$ *Stock picking inside equity sectors added 26 bp to the whole portfolio.*

Check: the full decomposition

Sector allocation in equity $0.80\% \times 0.65 = 0.52\%$; bond selection $0.20 \times (1.6\% - 1.2\%) = 0.08\%$; total $0.24 + 0.52 + 0.26 + 0.08 = 1.10\%$.

Interpretation

All decisions added value this month, with most of it coming from sector bets inside equity rather than from the asset mix or stock picking.

Why the other options are wrong

A is incorrect. Stopped at the equity-level figure $1.20\% - 0.80\% = 0.40\%$ without multiplying by the 65% equity weight (BKM multiplies 147 bp by 0.70).

B is incorrect. Intermediate result: took the whole equity selection effect, $1.20\% \times 0.65 = 0.78\%$, which still contains the sector allocation decision.

C is incorrect. Asset allocation computed with the manager's own returns instead of index returns: $0.10(5.2\%) - 0.15(1.6\%) + 0.05(0.4\%) = 0.30\%$, which mixes selection into allocation.

LO: IM-7 i: Describe and apply performance attribution procedures, including the asset allocation decision, sector and security selection decision, and the aggregate contribution. · Reference: Bodie, Kane and Marcus, Ch.24, 8.5 Performance Attribution Procedures (Tables 8.6 to 8.9)

Question A18.6 IM-3 d; IM-7 b · S5, S6 · [Aug-2026] [Map #36]

The CIO of a European pension fund is presenting to the investment committee on two candidate active equity managers. Manager P is a market timer who makes 4 independent bets a year with an information coefficient of 0.20. Manager Q is a stock selector who makes 225 independent bets a year with an information coefficient of 0.03. Which of the following statements should the CIO include in the presentation?

A. Q's maximum attainable information ratio is 0.45 versus 0.40 for P, so Q will deliver the higher realized information ratio after transaction costs and trading restrictions.

B. Q's maximum attainable information ratio is 0.45 versus 0.40 for P; if Q is added as an active position on top of the fund's passive index holding, BKM would rank it by its Treynor measure.

C. Q's maximum attainable information ratio is 0.45 versus 0.40 for P; if Q's 225 positions are overweights in value stocks against underweights in growth stocks, its true breadth and attainable ratio are far lower.

D. Q's maximum attainable information ratio is 0.45 versus 0.40 for P; doubling Q's assets under management would leave that ratio unchanged, because the information coefficient is constant across breadth.

Solution **Answer: C**

Step 1 – Fundamental law (Ang Eq. 3.5)

$$\mathrm{IR}_P = 0.20 \times \sqrt{4} = 0.40, \qquad \mathrm{IR}_Q = 0.03 \times \sqrt{225} = 0.03 \times 15 = 0.45$$

$\Rightarrow$ *Q has a tiny edge per bet but plays often; P plays well but rarely. The shared first clause is true in every option.*

Step 2 – Test the second clause

Breadth counts *independent* bets. Ang: an equity manager overweight 1,000 value stocks against 1,000 growth stocks "has not placed 1,000 different bets; he's placed just one bet on a value-growth factor". Correlated bets reduce BR, so Q's attainable ratio falls well below 0.45. C is correct.

Interpretation

The fundamental law is a ceiling and a framing tool, not a forecast: it assumes independent forecasts, optimal use of information and mean-variance preferences, and says nothing about where alpha comes from.

Why the other options are wrong

A is incorrect. Scope error: the law gives the *maximum* information ratio "since it ignores transactions costs, restrictions on trading, and other real-world considerations"; it does not promise the realized ranking after costs.

B is incorrect. Sibling swap of performance measures: an active portfolio mixed with a passive benchmark is evaluated by its information ratio (BKM Eq. 8.2); Treynor applies to one of many subportfolios forming the risky portfolio.

D is incorrect. Absolute claim: constant IC across breadth is an *assumption* Ang lists as a limitation; ICs tend to fall as assets under management rise (decreasing returns to scale), so doubling AUM would likely lower the ratio.

LO: IM-3 d: Describe Grinold's fundamental law of active management, including its assumptions and limitations, and calculate the maximum attainable information ratio using this law. IM-7 b: Describe risk-adjusted performance measures, such as Sharpe's measure, Treynor's measure, Jensen's measure (Jensen's alpha), and the information ratio, and identify the circumstances under which the use of each measure is most relevant. · Reference: Ang, Ch.10, Creating Alpha; Bodie, Kane and Marcus, Ch.24, The Information Ratio

A19 Non-Parametric VaR: Weighted HS, Bootstrap and FHS

Readings: MR-2 a, b, c, d · Non-Parametric Approaches (Dowd, Measuring Market Risk 2e, Ch.4; 2.2 Bootstrapped HS and non-parametric density estimation, 2.3 confidence intervals, 2.4 Weighted HS eqs. 2.2 to 2.6, 2.5 Advantages and Disadvantages, Appendix 2 The Bootstrap)

AUG 2026 RECALL | MAP TIER C · #28 · 4/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: Age-weighted and volatility-weighted HS.
Map #28: "Reported: 'bootstrap HS property question'; 'difference in non-parametric approaches'; 'expected shortfall in filtered historical simulation'. The polarity trap is documented in GARP's own materials: volatility-weighted HS adjusts historical returns upward when current volatility is above the historical level: the reversed version is the distractor."
Shapes covered here:
S1 age-weighted HS (BRW): single weights, cumulative weights and the weighted VaR read-off (Q1, Q2, Q4)
S2 volatility-weighted HS (HW), with the polarity trap: returns scaled up when current volatility is above the historical level (Q3, Q5)
S3 bootstrap HS property question (Q5)
S4 differences among the non-parametric approaches (Q5)
S5 expected shortfall in filtered historical simulation (Q5)

Revision notes

- **Non-parametric approaches** estimate risk measures without strong assumptions about the P/L distribution: they "let the P/L data speak for themselves". Core assumption: the near future will be sufficiently like the recent past.
- **HS P/L** (eq. 2.1) is the P/L the *current* portfolio would have earned had it been held through the sample period, not the P/L actually earned. Basic HS VaR is the highest loss observation outside the tail (1000 obs at 95%: the 51st highest loss, 1.704 in the book's example; 100 obs: the 6th); ES is the average of the losses in the tail (2.196 against a true 2.063). See also A13 for MR-1 HS VaR.
- **Bootstrapped HS** (MR-2 a): resample from the existing data set *with replacement*; each resample gives a VaR, and each resample ES is "the average of the losses in each resample exceeding the resample VaR". The best estimates are the *means* of the resample estimates (1000 resamples: about 1.669 and 2.114, closer to the true 1.645 and 2.063). Often more accurate than the raw sample estimate and useful for gauging precision.
- **Bootstrap confidence intervals** (Appendix 2): the percentile interval reads the bounds off the quantiles of the bootstrapped estimates (book: 95% VaR [1.554, 1.797], ES [1.986, 2.271]); the bias-corrected and accelerated (BC_a) method corrects the end-points and is transformation-respecting. The order-statistics approach gives very similar intervals (VaR [1.552, 1.797]). Bootstrap estimates have little bias but often substantial variance.
- **Bootstrap limitation:** standard bootstrap procedures presuppose that observations are independent over time. Remedies: bootstrap GARCH residuals (exposes us to model and parameter risk), a block approach (can "whiten" the data), or selection probabilities that depend on recently chosen observations.
- **Non-parametric density estimation** (MR-2 b): basic HS only gives VaR at discrete confidence levels set by sample size (100 obs: 95% yes, 95.1% no). Drawing lines through the tops of histogram bars, or using kernels, gives a surrogate pdf, so VaR can be estimated at *any* confidence level. Kernels are theoretically superior, but four kernels all gave 1.735 versus basic HS 1.704: "they do not necessarily produce much better estimates in practice".
- **Weighted HS is semi-parametric**: it adds parametric features (GARCH volatility forecasts) to the HS framework. Problems with equal weights: the $1/n$ weight is hard to justify, risk estimates are unresponsive to major events, the iid assumption is unrealistic, and weights drop suddenly to zero at age n, creating ghost effects.
- **Age-weighted HS** (Boudoukh, Richardson and Whitelaw, BRW): weights decay exponentially with age; λ close to 1 means *slow* decay. Equal-weight HS is the special case $\lambda \to 1$. Rear-view mirror analogy: equal-weight HS is safe only if the road is straight; age-weighting is safe if it bends gently.

- **Age-weighting attractions**: (1) generalises traditional HS; (2) more responsive to large recent losses and better with loss clusters; (3) reduces distortions from unlikely-to-recur events and reduces ghost effects (an observation leaves with weight $\lambda^n w(1) < 1/n$); (4) lets the sample size grow over time, improving efficiency and effectively eliminating ghost effects.
- **Age-weighting drawbacks**: reduces the *effective* sample size; a sequence of major profits or losses can distort the risk profile; VaR can still be insufficiently responsive (Pritsker); the approach is *ad hoc*: except for $\lambda = 1$ no asset-return process makes BRW theoretically correct.
- **Volatility-weighted HS** (Hull and White, HW): replace each historical return by $r^*_{t,i} = (\sigma_{T,i}/\sigma_{t,i})\, r_{t,i}$. Returns are *increased* when the current forecast is greater than the forecast for day t, and decreased when it is less; then estimate VaR or ES in the traditional equal-weight way.
- **HW advantages**: handles volatility changes in a natural, direct way (age-weighting treats them "in a rather arbitrary and restrictive way"); sensitive to current GARCH information; VaR and ES *can exceed the maximum historical loss*; HW evidence shows superior VaR to BRW. Can be combined with age-weighting and with order-statistics or bootstrap confidence intervals.
- **Correlation-weighted HS** (Duffie and Pan suggestion): adjust returns to the *current* correlation matrix using Choleski matrices, $\tilde{R} = \tilde{A}A^{-1}R$. A major generalisation of HW (correlations as well as volatilities). Example 2.1: historical $\rho = 0.3$ to current 0.9 gives a second row $[0.7629,\ 0.4569]$.
- **Filtered HS** (Barone-Adesi et al.): a *semi-parametric bootstrap*. Fit an asymmetric GARCH (leverage effect), divide realised returns by their volatility forecasts to get iid standardised returns, bootstrap these and multiply by tomorrow's volatility forecast. For a multi-asset portfolio, bootstrap whole standardised vectors, which keeps the correlation structure without a variance-covariance matrix.
- **FHS attractions and ES**: combines HS with a sophisticated volatility treatment; fast; VaR *and ES* can exceed the maximum historical loss; can be combined with order-statistics or bootstrap intervals. FHS "can also be extended easily" to ES (Giannopoulos and Tunaru). Evidence is mixed: Pritsker finds FHS VaR may not pay enough attention to extreme observations or time-varying correlations; "a partial response to the first point would be to use ES instead of VaR".
- **Advantages of non-parametric methods** (MR-2 d): intuitive and simple; accommodate fat tails, skewness and other non-normal features; any position type including derivatives; easy on a spreadsheet; no covariance matrices or curse of dimensionality; readily available data; easy to report; confidence intervals easy to produce; can be refined with semi-parametric add-ons.
- **Disadvantages**: results are very (often completely) dependent on the historical data set; a quiet sample gives VaR too low, a volatile sample too high; slow to reflect shifts and major events; unlikely-to-recur extremes can dominate; ghost (shadow) effects; no allowance for plausible events that did not occur; constrained by the largest historical loss, so poorly suited to extremes with small or medium samples.
- **Window length**: most experts want at least 500 daily observations; a longer window worsens aged data, ghost effects, drowning-out of current news and data-collection problems. Never rely on non-parametric methods alone: complement them with stress testing.

Formulas

Age weights (BRW), eq. 2.2. $w(1) = \dfrac{1-\lambda}{1-\lambda^n}, \qquad w(i) = \lambda^{i-1}w(1) = \dfrac{\lambda^{i-1}(1-\lambda)}{1-\lambda^n}$

i: age in days ($i = 1$ is yesterday); $\lambda \in (0,1)$: decay factor; n: observations in the window. Equal-weight HS gives each observation $1/n$.

Cumulative weight on the k most recent observations. $\displaystyle\sum_{i=1}^{k} w(i) = \frac{1-\lambda^k}{1-\lambda^n}$

Age-weighted VaR. Rank P/L from the worst loss, attach each $w(i)$, accumulate, and stop where the cumulative weight reaches the tail probability $1-\alpha$ (interpolation allowed).

Equal-weight HS VaR. The $[n(1-\alpha)+1]$th largest loss: the highest loss outside the tail.

Volatility-adjusted return (HW), eq. 2.3. $r^*_{t,i} = \dfrac{\sigma_{T,i}}{\sigma_{t,i}}\, r_{t,i}$ (current forecast over the forecast for day t)

Correlation adjustment, eqs. 2.4 and 2.5. $R = A\varepsilon,\ \tilde{R} = \tilde{A}\varepsilon \ \Rightarrow\ \tilde{R} = \tilde{A}A^{-1}R$

FHS (AGARCH), eq. 2.6. $r_t = \mu + \varepsilon_t, \quad \sigma_t^2 = \omega + \alpha(\varepsilon_{t-1}+\gamma)^2 + \beta\sigma_{t-1}^2$; standardised return ε_t/σ_t, bootstrapped and multiplied by tomorrow's forecast.

Bootstrap point estimate and percentile interval, eqs. 2.14 and 2.11. $\hat{\theta}_B = \frac{1}{B}\sum_i \hat{\theta}_i^B$, interval $[\hat{\theta}_\alpha^B, \hat{\theta}_{1-\alpha}^B]$

Exam traps

- **Volatility polarity (the documented trap):** HW scales returns *up* when current volatility is above the volatility on the day they were observed. The reversed statement, and the inverted ratio σ_t/σ_T, are the distractors.
- **Dropped normaliser:** using $w(i) = \lambda^{i-1}(1-\lambda)$ without dividing by $1-\lambda^n$ (or using raw λ^{i-1} as probabilities). Weights then do not sum to 1.
- **Exponent off by one:** the weight on an i-day-old observation uses λ^{i-1}; yesterday's observation carries the full $w(1)$.
- **Tail-counting convention:** VaR is the first observation *outside* the tail (100 obs at 95%: the 6th largest loss, not the 5th). In age-weighted HS, keep accumulating until the cumulative weight first exceeds $1-\alpha$.
- **Sibling swap on the maximum loss:** basic HS and the bootstrap cannot produce a VaR or ES above the largest historical loss; HW and FHS can, because returns are rescaled to current volatility.
- **Bootstrap scope:** the standard bootstrap assumes independence over time; "resampling fixes clustering" is false.

Question A19.1

MR-2 c · S1 · [Aug-2026]

A junior risk analyst at a Singapore asset manager is implementing the Boudoukh, Richardson and Whitelaw age-weighted historical simulation approach on a window of the most recent 100 daily returns of an SGD 300 million fixed income portfolio. The decay parameter is $\lambda = 0.98$. Which of the following is closest to the probability weight assigned to the return observed 12 days ago?

A. 0.0160

B. 0.0185

C. 0.0231

D. 0.2482

Solution

Answer: B

Step 1 – Weight on the most recent observation, $w(1)$

$$w(1) = \frac{1-\lambda}{1-\lambda^n} = \frac{1-0.98}{1-0.98^{100}} = \frac{0.02}{1-0.13262} = \frac{0.02}{0.86738} = 0.023058$$

$\Rightarrow$ *Yesterday's return carries a probability weight of about 2.31%, versus 1% under equal weighting.*

Step 2 – Decay to age 12

$$\lambda^{i-1} = 0.98^{11} = 0.80073$$

$\Rightarrow$ *An observation 12 days old keeps about 80% of the weight of yesterday's observation.*

Step 3 – Age weight, eq. 2.2

$$w(12) = \lambda^{11}\, w(1) = 0.80073 \times 0.023058 = 0.018463$$

$\Rightarrow$ *The 12-day-old return counts as a 1.85% probability mass, almost twice its equal-weight value of 1%.*

Interpretation

With $\lambda = 0.98$ recent observations dominate the tail estimate, so a large recent loss raises VaR more than under basic HS. The portfolio value is irrelevant: weights depend only on λ, n and age.

Why the other options are wrong

A is incorrect. Dropped the normaliser $1-\lambda^n$: $0.98^{11} \times 0.02 = 0.0160$. These weights would not sum to 1 over a finite window.

C is incorrect. Reported $w(1) = 0.0231$, the weight on yesterday's observation (an intermediate result), rather than decaying it to age 12.

D is incorrect. Reported the cumulative weight on the 12 most recent observations, $(1-0.98^{12})/(1-0.98^{100}) = 0.2482$, instead of the weight on the single 12-day-old observation.

LO: MR-2 c: Compare and contrast the age-weighted, the volatility-weighted, the correlation-weighted, and the filtered historical simulation approaches. · Reference: Dowd, Measuring Market Risk 2e, Ch.4, 2.4 Age-weighted Historical Simulation, eq. 2.2

Question A19.2

MR-2 c · S1 · [Aug-2026]

A market risk manager at a UK bank estimates 1-day 95% VaR for a GBP trading book using age-weighted historical simulation with $n = 100$ daily observations and $\lambda = 0.96$. The nine largest historical losses, ranked from worst, and their ages in days are:

Rank	1	2	3	4	5	6	7	8	9
Loss (GBP million)	7.4	6.9	6.1	5.6	5.2	4.8	4.5	4.3	4.0
Age (days)	85	3	58	10	47	72	6	30	21

The bank takes VaR to be the first ranked loss at which the cumulative probability weight exceeds 5% (no interpolation). Which of the following is closest to the age-weighted 95% VaR?

A. GBP 4.8 million

B. GBP 5.6 million

C. GBP 6.1 million

D. GBP 6.9 million

Solution

Answer: B

Step 1 – Weight on the most recent observation

$$w(1) = \frac{1 - 0.96}{1 - 0.96^{100}} = \frac{0.04}{1 - 0.016870} = \frac{0.04}{0.98313} = 0.040686$$

Step 2 – Age weights of the largest losses, $w(i) = 0.96^{\,i-1} \times 0.040686$

$$\begin{aligned} w(85) &= 0.96^{84} \times 0.040686 = 0.00132 \\ w(3) &= 0.96^{2} \times 0.040686 = 0.03750 \\ w(58) &= 0.96^{57} \times 0.040686 = 0.00397 \\ w(10) &= 0.96^{9} \times 0.040686 = 0.02818 \end{aligned}$$

$\Rightarrow$ *The worst loss is 85 days old and counts for only 0.13%; the 3-day-old loss counts for 3.75%.*

Step 3 – Cumulate down the ranking

$$\begin{aligned} &\text{after rank } 1 : 0.00132 \\ &\text{after rank } 2 : 0.00132 + 0.03750 = 0.03882 \\ &\text{after rank } 3 : 0.03882 + 0.00397 = 0.04279 \\ &\text{after rank } 4 : 0.04279 + 0.02818 = 0.07096 > 0.05 \end{aligned}$$

$\Rightarrow$ *The 5% tail is exhausted at rank 4: the first three losses sit inside the tail and the 10-day-old loss of GBP 5.6 million is where the cumulative weight first exceeds 5%.*

Step 4 – Age-weighted VaR

$$\text{VaR}_{95\%} = \text{GBP } 5.6 \text{ million}$$

Interpretation

Two recent large losses (ages 3 and 10) carry heavy weights, so the age-weighted VaR (5.6) is above the equal-weight HS VaR (4.8). This is the BRW attraction: VaR is more responsive to recent large losses and clusters. Reversing the age ordering (heaviest weights on the oldest data) would give 4.3.

Why the other options are wrong

A is incorrect. Used equal-weight HS: each loss has weight $1/100$, the tail holds 5 losses, and VaR is the 6th largest loss, 4.8.
C is incorrect. Stopped one rank early, at the last loss inside the tail: after rank 3 the cumulative weight is only 4.28%, so GBP 6.1 million has not yet reached the 5% tail probability.
D is incorrect. Used the raw decay factors λ^{i-1} as probabilities, dropping the $(1-\lambda)/(1-\lambda^n)$ scaling: $0.96^{84} = 0.0323$, then $+0.96^2 = 0.9216$ exceeds 5% at rank 2, giving 6.9.

LO: MR-2 c: Compare and contrast the age-weighted, the volatility-weighted, the correlation-weighted, and the filtered historical simulation approaches. · Reference: Dowd, Measuring Market Risk 2e, Ch.4, 2.4 Age-weighted Historical Simulation, eq. 2.2

Question A19.3

MR-2 c · S2 · [Aug-2026] [Map #28]

A risk analyst at a US hedge fund applies the Hull and White volatility-weighted historical simulation approach to a USD 25 million single-asset position. The asset's beta to the S&P 500 is 1.3. The most recent EWMA forecast of daily volatility is 1.2%. The ten daily returns in the window and the EWMA volatility forecast that applied on each day are:

Day	1	2	3	4	5	6	7	8	9	10
Return (%)	−3.1	1.2	−2.6	−0.9	1.8	−2.2	0.7	−1.7	0.4	−1.2
Vol forecast (%)	2.4	1.5	2.0	1.1	1.6	1.3	1.0	0.9	1.2	1.4

Using the convention that VaR is the highest loss outside the tail, which of the following is closest to the 1-day 90% volatility-weighted HS VaR?

A. USD 0.51 million

B. USD 0.57 million

C. USD 0.65 million

D. USD 1.08 million

Solution

Answer: A

Step 1 – Volatility-adjusted returns, eq. 2.3

$$r_t^* = \frac{\sigma_T}{\sigma_t} r_t = \frac{1.2\%}{\sigma_t} r_t$$

$$\text{Day 1}: \tfrac{1.2}{2.4} \times (-3.1\%) = -1.550\% \qquad \text{Day 3}: \tfrac{1.2}{2.0} \times (-2.6\%) = -1.560\%$$
$$\text{Day 4}: \tfrac{1.2}{1.1} \times (-0.9\%) = -0.982\% \qquad \text{Day 6}: \tfrac{1.2}{1.3} \times (-2.2\%) = -2.031\%$$
$$\text{Day 8}: \tfrac{1.2}{0.9} \times (-1.7\%) = -2.267\% \qquad \text{Day 10}: \tfrac{1.2}{1.4} \times (-1.2\%) = -1.029\%$$

(The four positive returns stay positive after scaling and do not enter the left tail.)
$\Rightarrow$ *Day 1's -3.1% happened when volatility was 2.4%, twice today's level, so it is halved; Day 8's -1.7% happened in a calm 0.9% regime and is scaled up to -2.27%.*

Step 2 – Locate the VaR observation

With 10 observations and a 10% tail, one observation sits in the tail, so VaR is the 2nd largest adjusted loss:

$$\text{ranked adjusted losses: } 2.267\%,\ 2.031\%,\ 1.560\%,\ 1.550\%,\ \ldots$$

$\Rightarrow$ *The 90% VaR return is -2.031% (Day 6).*

Step 3 – Convert to currency

$$\text{VaR}_{90\%} = 0.02031 \times 25 = \text{USD } 0.508 \text{ million}$$

Interpretation

Because current volatility (1.2%) is below the levels prevailing on the worst raw days, volatility-weighting lowers VaR relative to basic HS (USD 0.65 million). The HW approach makes the estimate appropriately sensitive to current volatility. The beta is irrelevant.

Why the other options are wrong

B is incorrect. Took the largest adjusted loss (inside the tail) instead of the highest loss outside it: $0.02267 \times 25 = 0.57$.

C is incorrect. Ignored the volatility adjustment and used basic HS: 2nd largest raw loss $2.6\% \times 25 = 0.65$.

D is incorrect. Inverted the ratio, scaling by σ_t/σ_T (the polarity trap): Day 3 becomes $-2.6\% \times 2.0/1.2 = -4.333\%$ (2nd largest after Day 1's -6.2%), giving $0.04333 \times 25 = 1.08$.

LO: MR-2 c: Compare and contrast the age-weighted, the volatility-weighted, the correlation-weighted, and the filtered historical simulation approaches. · Reference: Dowd, Measuring Market Risk 2e, Ch.4, 2.4 Volatility-weighted Historical Simulation, eq. 2.3

Question A19.4

MR-2 c, d · S1 · [Aug-2026]

A model validator at an Indian insurer is reviewing the age-weighted historical simulation engine used for an INR 12,000 million equity portfolio. The engine uses $n = 250$ daily observations and $\lambda = 0.99$. To judge how concentrated the model is on recent data, the validator wants the total probability weight carried by the 50 most recent observations. Which of the following is closest?

A. 0.200

B. 0.395

C. 0.430

D. 0.544

Solution

Answer: C

Step 1 – Normalising constant

$$1 - \lambda^n = 1 - 0.99^{250} = 1 - 0.081059 = 0.918941$$

⇒ *An infinite EWMA would leave about 8.1% of weight beyond day 250; the BRW normaliser rescales so the 250 in-sample weights sum to 1.*

Step 2 – Un-normalised weight on the 50 most recent observations

$$\sum_{i=1}^{50} \lambda^{i-1}(1-\lambda) = 1 - \lambda^{50} = 1 - 0.99^{50} = 1 - 0.605006 = 0.394994$$

Step 3 – Normalise

$$\sum_{i=1}^{50} w(i) = \frac{1-\lambda^{50}}{1-\lambda^{250}} = \frac{0.394994}{0.918941} = 0.4298$$

⇒ *The most recent 20% of the window carries about 43% of the probability mass.*

Interpretation

Age-weighting concentrates information in recent data, which raises responsiveness but reduces the effective sample size (a BRW drawback). It also explains the reduced ghost effect: the oldest observation leaves with weight $\lambda^{249}w(1) = 0.00089$, far below $1/250 = 0.004$.

Why the other options are wrong

A is incorrect. Used equal weights: $50/250 = 0.200$.

B is incorrect. Dropped the normaliser and reported the intermediate result $1 - \lambda^{50} = 0.395$ (Step 2), so the 250 weights would sum to only 0.919.

D is incorrect. Multiplied $w(1)$ by 50, ignoring decay: $50 \times 0.0108821 = 0.544$.

LO: MR-2 c: Compare and contrast the age-weighted, the volatility-weighted, the correlation-weighted, and the filtered historical simulation approaches. MR-2 d: Identify advantages and disadvantages of non-parametric estimation methods. · Reference: Dowd, Measuring Market Risk 2e, Ch.4, 2.4 Age-weighted Historical Simulation, eq. 2.2

Question A19.5 MR-2 a, c, d · S2, S3, S4, S5 · [Map #28]

A model validator at a Canadian bank is reviewing a proposal to replace the bank's basic HS engine (500 daily observations) with a refined non-parametric method. The current EWMA volatility forecast is 1.6% a day. One observation in the window, 20 days old, is a return of -3.0% recorded on a day whose volatility forecast was 1.2%. Which of the following statements should the validator include in the report?

A. Under volatility-weighted HS, the -3.0% return enters the adjusted series as -2.25%, because historical returns are scaled down when the current volatility forecast is above the forecast for the day they were observed.

B. Under bootstrapped HS, each resample is drawn from the 500 observations with replacement, the best VaR estimate is the mean of the resample VaRs, and each resample ES is the average of the losses beyond that resample's VaR.

C. Under filtered HS, the standardised return of -2.5 is bootstrapped and rescaled by the 1.6% forecast to -4.0%, but the resulting VaR and ES still cannot exceed the largest loss in the historical sample.

D. Under bootstrapped HS, the standard resampling procedure remains reliable when returns are dependent over time, because each observation is drawn at random from the full window of 500 observations.

Solution **Answer: B**

Step 1 – The correct statement

Dowd: a bootstrap "involves resampling from our existing data set with replacement"; each resample gives a VaR, "our 'best' estimate [is] the mean of these resample-based estimates", and each resample ES is "the average of the losses in each resample exceeding the resample VaR". B states all three properties.

Step 2 – The numbers behind A and C

$$\text{HW: } r^* = \frac{\sigma_T}{\sigma_t} r = \frac{1.6}{1.2} \times (-3.0\%) = -4.0\%; \qquad \text{FHS: } \frac{-3.0\%}{1.2\%} = -2.5, \quad -2.5 \times 1.6\% = -4.0\%$$

$\Rightarrow$ *Both semi-parametric methods inflate the old return to -4.0% because volatility has risen since it was observed; that rescaling is exactly what lets them produce losses beyond the historical maximum.*

Interpretation

The bootstrap improves the precision of the estimate and gives confidence intervals, but it stays inside the historical data: it cannot exceed the largest historical loss and, in standard form, it assumes independence. HW and FHS are the refinements that respond to the change in volatility; FHS also extends "easily" to ES, and using ES is Dowd's partial response to the criticism that FHS VaR pays too little attention to extremes.

Why the other options are wrong

A is incorrect. Polarity flip (the documented GARP trap). Returns are *increased* when the current forecast is above the forecast for day t: the return enters as -4.0%, not -2.25% (which is the inverted ratio $1.2/1.6$).

C is incorrect. Sibling swap. "Cannot exceed the largest historical loss" is the constraint of basic HS (and of the plain bootstrap). FHS, like HW, "allows us to get VaR and ES estimates that can exceed the maximum historical loss".

D is incorrect. Scope/condition error. "Perhaps the main limitation of the bootstrap is that standard bootstrap procedures presuppose that observations are independent over time"; remedies are GARCH residuals, blocks, or time-dependent selection probabilities.

LO: MR-2 a: Apply the bootstrap historical simulation approach to estimate coherent risk measures. MR-2 c: Compare and contrast the age-weighted, the volatility-weighted, the correlation-weighted, and the filtered historical simulation approaches. MR-2 d: Identify advantages and disadvantages of non-parametric estimation methods. · Reference: Dowd, Measuring Market Risk 2e, Ch.4, 2.2 Bootstrapped Historical Simulation; 2.4 Volatility-weighted and Filtered Historical Simulation; Appendix 2, Time Dependency and the Bootstrap

A20 Wrong-Way and Right-Way Risk

Readings: CR-20 k, l, m, n, o, p · CVA (Gregory, The xVA Challenge 4e, Ch.17: 20.6 Wrong-Way Risk; 20.6.1 Overview; 20.6.2 Quantification of WWR in CVA; 20.6.3 Wrong-Way Risk Models; 20.6.4 Jump Approaches; 20.6.5 Credit Derivatives; 20.6.6 Collateralisation and WWR; 20.6.7 Central Clearing and WWR); CR-17 e, j · Margin (Collateral) and Settlement (Gregory, Ch.7: 17.3.5, 17.5.3)

AUG 2026 RECALL | MAP TIER C · #26 · 4/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: Wrong way risk.
Map #26: "Reported verbatim: 'RWR & WWR (2 theory questions): confusing, study…'. Your formula card frames it correctly as 'identify if there is any correlation'. WWR = exposure rises as counterparty credit quality falls; RWR = the favourable case. Examinable sub-items: wrong-way collateral, the effect on CCPs, and the modelling approaches (hazard-rate, structural, parametric, jump)."
Shapes covered here:
S1 classify trades as wrong-way or right-way, plain form as seen in August (Q1)
S2 the test: is the exposure linked to the counterparty's credit quality, and in which direction (Q1)
S3 wrong-way collateral (Q2)
S4 the effect of WWR on CCPs (Q2)
S5 modelling approaches: hazard rate (intensity), structural, parametric, jump (Q2).

Revision notes

- **The specific: the test** (CR-20 k; S2). *Wrong-way risk* (WWR) is an *unfavourable* dependence between exposure (EPE) and counterparty credit quality: the exposure is high when the counterparty is more likely to default. *Right-way risk* (RWR) is the favourable dependence and *reduces* CVA. Standard CVA assumes EPE, PD and LGD are independent. To classify a trade, ask two questions: what makes my exposure to this counterparty rise, and does the same event weaken the counterparty?
- **Character of WWR:** hard to identify, model and hedge because of subtle macro-economic and structural effects; often reasonable to ignore, but potentially dramatic. Losses: Asian crisis 1997/98 (sovereign and corporate defaults with collapsing local currencies); from 2007, banks buying protection from monoline insurers. Mortgage lending is a natural example: recession brings falling property prices and higher default rates together.
- **General vs specific WWR** (Table 20.7): *general* WWR is macro-economic, may be detectable in historical data, can be incorporated into pricing models, and should be priced and managed. *Specific* WWR comes from structural or causal links, is hard to detect without knowledge of the market, the counterparty and the rationale of the trade, is dangerous to model with naive correlation (address it qualitatively, e.g. stress testing), and should in general be *avoided*. Example: the 2010 European sovereign crisis, which historical data did not bear out.
- **Put options** (CR-20 l; S1): buying a put on a stock (or index) whose fortunes are highly correlated with the counterparty's is WWR (a put on one bank's stock bought from another bank). Equity *calls* should be right-way.
- **FX forwards and cross-currency swaps:** paying a sovereign its local currency (or hedging that trade with a bank in the same region) is WWR: currency weakening and default go together, either way round. A cross-currency swap is like a loan collateralised by the opposite currency. Levy and Levin: residual currency values on sovereign default average 17% (AAA) to 62% (CCC).
- **Interest rate swaps:** a bank *receiving fixed* from a corporate that pays fixed while the economy is strong has WWR (rates are cut in a recession, so the swap moves in the bank's favour as the corporate weakens); paying fixed during a recovery may be right-way. Duffee (1998): corporate defaults cluster when rates are falling. Uncollateralised receiver swaps with weakening sovereigns or corporates show the "cross-gamma" of spreads and rates even without defaults.
- **Commodity swaps:** a producer (e.g. a mining company) hedging its output owes money only when the commodity price is high and it is profitable: *right-way*, arising from *hedging* as opposed to speculation.

- **Credit default swaps** (CR-20 l, 20.6.5): *buying* protection where the reference entity and the counterparty are strongly linked is extreme WWR (a bank selling protection on its own sovereign); *selling* protection in that case is right-way. For credit derivatives WWR is unavoidable and may be specific, and margin may help little. Book example (reference spread 250 bps, counterparty 500 bps, LGDs 60%): at 60% correlation the fair premium for buying protection is about 200 bps (CVA 50 bps, one-fifth of the premium); at 100% it is just above 100 bps $= 250 \times 40\%$, the recovery value alone.
- **Credit quality and WWR:** WWR *increases* as the counterparty's credit quality *increases*: the default of a strong name is a less probable, bigger shock. Levy and Levin implied FX jumps: 83% AAA vs 27% BBB sovereigns; euro jumps for Greece, Italy, Spain, Germany of 9%, 17%, 20%, 25%; Chung and Gregory (Japan): sovereign 38.4%, financials 13.5%, non-financials 8%.
- **Collateral and WWR** (CR-20 m): timing decides. If exposure rises gradually before default, margin can be received; a *jump* in exposure at default makes margin useless. Jump approaches show collateral as near useless against WWR; continuous approaches (intensity, structural) show it as effective; the truth is in between and depends on the trade and the counterparty. Pykhtin and Sokol: allow for jumps and higher volatility during the MPoR; WWR reduces the benefit of collateralisation and is greater for systemic parties (banks), which are the ones that post margin, while corporates often do not post at all.
- **Wrong-way collateral** (CR-20 n; S3): (i) a *payer* swap collateralised by a high-quality government bond: rates up raise the swap value and lower the bond value (general WWR, minor because rate volatility is low; for a *receiver* swap it is right-way margin); (ii) a cross-currency swap collateralised by cash in the currency being paid: one FX move raises the exposure and cuts the margin value (more significant); (iii) an entity posting its *own* bonds: specific WWR, a very weak mitigant for exposure and CVA, though it may mitigate FVA; (iv) a bank posting bonds of its own sovereign. CR-17 adds euros received from European sovereigns or banks, and LTCM's Russian bonds; the bilateral margin rules forbid "wrong-way" margin (own bonds or equity).
- **CCPs are particularly prone to WWR** (CR-20 o; S4): they rely on variation and initial margin as their primary protection, especially when clearing CDSs. The danger is that the defaulter-pays resources are insufficient, losses are imposed on surviving members, and the CCP itself may become insolvent.
- **CCPs disassociate credit quality and exposure:** membership needs a minimum credit quality (set by the CCP, not by external ratings), but initial margin and default fund contributions are driven mainly by the *market risk* of each portfolio, so CCPs risk implicitly ignoring WWR. Because WWR rises with credit quality, the logic perversely implies *more* initial margin and default fund from better-quality, more systemically important members (Pykhtin and Sokol: a large dealer is more WWR than a smaller or weaker one).
- **The CCP waterfall as a CDO:** defaulter's initial margin, default fund and CCP equity form the first loss; surviving members' default fund contributions and other loss allocation are a rather *senior* second-loss tranche, heavily concentrated in systemic risk. CCPs also face WWR and adverse selection on margin received: members post the riskiest eligible assets relative to the haircut (a European bank posting European sovereign debt).
- **Quantifying WWR** (CR-20 p; S5): replace EPE in eq. 20.3 with EPE *conditional* on default at that date. The regulatory alpha factor plus stressed data partially capitalise general WWR. With a single (abstract) correlation parameter, $+50\%$ roughly doubles EPE and -50% at least halves it. Pitfalls: uninformative historical data; misspecification (causality rather than correlation: zero correlation does not prove independence, e.g. $Y = X^2$); unclear direction (high-rate credit stress is possible).
- **Hazard rate (intensity) approach:** a stochastic credit spread correlated with the exposure drivers; tractable (spread paths first, exposures only on default paths); correlations observable from historical spread data. It generates only *weak* dependence even near $\pm 100\%$ correlation, so it may *underestimate* WWR. For a payer swap, positive correlation of spreads with rates raises the positive exposure (WWR).
- **Structural approach:** maps the exposure distribution and the default time separately onto a bivariate distribution (e.g. a Gaussian copula); no re-simulation, WWR added on top of pre-computed exposures; a much *stronger* effect than the intensity model. Drawbacks: the correlation is *opaque* and hard to calibrate, and the unconditional exposures may not hold all the relevant information.
- **Parametric approach** (Hull and White 2011): links the default probability to the exposure through a simple functional relationship, calibrated by a what-if scenario or by history (past portfolio values against the counterparty's spread); it needs today's portfolio to resemble the historical one.

- **Jump approach** (Levy and Levin 1999): the FX rate jumps at the counterparty's default; more realistic for *specific* WWR. Supported by quanto CDS data: euro CDS on Italy about 30% cheaper than dollar CDS at all maturities; Ehlers and Schönbucher show intensity models cannot match these data even at $\pm 100\%$ correlation.
- **See also** A6 for the CVA formulas WWR breaks, A21 for margin and the MPoR, and D24 for the CCP loss waterfall.

Exam traps

- **Polarity flip (credit quality):** "WWR is greatest for weak, low-rated counterparties." It *increases* with credit quality.
- **Role misassignment (CDS):** *buying* protection on a reference entity linked to the counterparty is WWR; *selling* it is right-way. State the buyer and the seller before classifying.
- **Scope error (commodity hedge):** the producer's right-way risk arises from *hedging*; a speculative position does not inherit it.
- **Absolute claim (collateral):** "margin eliminates WWR". Jump approaches show it near useless; WWR reduces the benefit of collateral.
- **Sibling swap (models):** intensity = weak dependence, may underestimate; structural = strong effect, opaque correlation; parametric = functional link (Hull and White); jump = specific WWR, quanto CDS evidence.
- **Polarity flip (wrong-way collateral):** a government bond securing a *payer* swap is wrong-way (the bond falls as the swap gains); securing a *receiver* swap it is right-way.

Question A20.1 CR-20 k, l · S1, S2 · [Aug-2026] [Map #26]

A counterparty risk officer at a Singapore bank is classifying four proposed uncollateralised trades before a credit committee. The bank's rule is to tag a trade wrong-way if its exposure to the counterparty is expected to rise when the counterparty's credit quality falls, and right-way if the reverse holds.

- Trade 1: the bank receives fixed from a domestic manufacturer on a 7-year interest rate swap, agreed while the economy is strong.
- Trade 2: a copper producer that is hedging its future output pays the floating copper price to the bank and receives a fixed price on a commodity swap.
- Trade 3: in an FX forward with an emerging-market sovereign, the bank pays the sovereign's local currency and receives US dollars.
- Trade 4: the bank buys CDS protection on its domestic sovereign from a domestic bank.

Which of the following classifications should the officer record?

A. Trade 1 is right-way, because a recession that weakens the manufacturer would also push rates up and move the swap against the bank.

B. Trade 2 is right-way whatever the producer's motive, because a producer's commodity swap always pays the bank when the producer is profitable.

C. Trade 3 is wrong-way, because a weakening of the local currency would raise the bank's exposure as the sovereign's credit quality deteriorates.

D. Trade 4 is right-way, because buying protection from a seller linked to the reference entity is the right-way side of a CDS.

Solution **Answer: C**

Step 1 – Apply the test to Trade 3

What raises the bank's exposure? The bank pays local currency and receives dollars, so a fall in the local currency makes the forward more valuable to the bank. Does the same event weaken the counterparty? Yes: a currency collapse and a sovereign default go together, either way round (Gregory: "paying its local currency in an FX forward or cross-currency swap"). Unfavourable dependence, so wrong-way.

⇒ *Levy and Levin put residual currency values at sovereign default at 17% (AAA) to 62% (CCC): the exposure jumps exactly when the counterparty fails.*

Step 2 – Check the others with the same two questions

Trade 1: rates fall in a recession, the receive-fixed swap gains as the manufacturer weakens: wrong-way. Trade 2: the producer owes the bank only when copper is high and it is profitable: right-way, *because* it is a hedge. Trade 4: the bank's exposure is the protection's value, which rises as the sovereign weakens, and the domestic bank weakens with its sovereign: extreme wrong-way.

Interpretation

The committee should price or limit Trades 1 and 3 (general WWR) and decline Trade 4, a case of specific WWR that Gregory says should in general be avoided; Trade 2 reduces CVA.

Why the other options are wrong

A is incorrect. Polarity flip: Gregory says rates are likely to be *cut* in a recession, so the receive-fixed swap moves *in* the bank's favour as the manufacturer weakens. That is wrong-way, not right-way.

B is incorrect. Scope/condition error (absolute claim): Gregory ties the right-way character to *hedging* ("as opposed to speculation"). Dropping the condition "whatever the motive" turns a true statement false.

D is incorrect. Role misassignment: Gregory gives the right-way side to the protection *seller*. The bank is the *buyer*: its exposure rises as the reference sovereign weakens, which drags the domestic bank (the seller) down with it, so it is extreme wrong-way.

LO: CR-20 k: Describe wrong-way risk and contrast it with right-way risk. CR-20 l: Identify examples of wrong-way risk and examples of right-way risk. · Reference: Gregory, The xVA Challenge 4e, Ch.17, 20.6.1 and Table 20.7

Question A20.2

CR-20 m, n, o, p · S3, S4, S5 · [Map #26]

A risk consultant is reviewing how a European derivatives clearing house and its largest clearing members measure and mitigate wrong-way risk, and is drafting recommendations. Which of the following statements is correct to include in the report?

A. Variation and initial margin are effective mitigants of wrong-way risk in all cases, because margin received as the exposure rises offsets the higher exposure before default.

B. The hazard rate (intensity) approach suits specific wrong-way risk, because correlating credit spreads with exposure drivers produces a strong dependence that matches quanto CDS data.

C. Setting initial margin and default fund contributions mainly from the market risk of each portfolio may ignore wrong-way risk, which argues for more from higher-quality members.

D. A payer interest rate swap collateralised by a high-quality government bond is right-way collateral, because a rise in rates raises both the swap value and the bond value.

Solution

Answer: C

Step 1 – The CCP statement

Gregory: CCPs require a minimum credit quality for membership but then charge initial margins and default fund contributions "driven primarily by the market risk of their portfolio"; in doing this "CCPs are in danger of implicitly ignoring WWR". Because WWR *increases* with credit quality, this "perversely" suggests greater initial margin and default fund from better-credit-quality, more systemically important members.

⇒ *The surviving members' default fund sits in a senior, systemic second-loss position of a CDO-like waterfall, so under-margined WWR lands on them.*

Interpretation

The consultant should recommend that the clearing house quantify WWR in initial margin and default fund setting (hard for CDSs), and review eligible margin for adverse selection, since members will post the riskiest assets relative to the haircut.

Why the other options are wrong

A is incorrect. Absolute claim: margin helps only if exposure rises gradually; a jump at default makes it useless. Jump approaches show collateral as near useless, and Pykhtin and Sokol find WWR reduces the benefit of collateralisation. "In all cases" is false.

B is incorrect. Sibling swap: the intensity approach generates only *weak* dependence and may underestimate WWR; Ehlers and Schönbucher show it cannot match quanto CDS data even at $\pm 100\%$ correlation. The *jump* approach is the one suited to specific WWR and supported by quanto data.

D is incorrect. Polarity flip: a rate rise raises the payer swap's value but *lowers* the bond's value, so the margin shrinks as the exposure grows: general wrong-way collateral (minor, given low rate volatility). The receiver swap is the right-way case.

LO: CR-20 m: Discuss the impact of collateral on wrong-way risk. CR-20 n: Identify examples of wrong-way collateral. CR-20 o: Discuss the impact of wrong-way risk on central counterparties (CCPs). CR-20 p: Describe the various wrong-way modeling methods including hazard rate approaches, structural approaches, parametric approaches, and jump approaches, and explain the implications of central clearing on wrong-way risk. · Reference: Gregory, The xVA Challenge 4e, Ch.17, 20.6.3 to 20.6.7

A21 Collateral, Two-Way CSA, Margin Period of Risk and Netting

Readings: CR-16 a, b, c, d, e, f · Netting, Close-out and Related Aspects (Gregory, The xVA Challenge 4e, Ch.6); CR-17 a, b, c, d, e, f, g, h, i, j, k · Margin (Collateral) and Settlement (Gregory, Ch.7: 17.1 Termination and Reset Features; 17.2 Basics; 17.3 Margin Terms; 17.4 Bilateral Margin Requirements; 17.5 Impact of Margin; 17.6 Margin and Funding); CR-20 j · CVA (Gregory, Ch.17: 20.5 Impact of Margin)

AUG 2026 RECALL | MAP TIER C · #27 · 4/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: Two-way CSA; margin period of risk.
Map #27: "Reported: 'calculate collateral required given the values of MtM exposure, threshold, etc.'; 'overcollateralisation account balance'. The sheet singles out LO 35.g (know the terms very well), LO 35.f (disputes and reconciliation) and LO 35.j (how market, operational and liquidity risk arise through collateralisation, a role-misassignment favourite). From Reading 34, termination features and trade compression are flagged as crucial."
Shapes covered here:
S1 collateral required from the MtM value, threshold, MTA, rounding and margin already held (Q1, Q2)
S2 the threshold-plus-MTA test that fails, so no transfer is made (Q2)
S3 overcollateralisation: initial margin and haircuts (Q5); the securitisation overcollateralisation account is drilled in A9 (Q4)
S4 features and terms of a collateral agreement, CR-17 g (LO 35.g) (Q4)
S5 disputes and reconciliation, CR-17 f (LO 35.f) (Q4, Q6)
S6 market, operational and liquidity risk through collateralisation, with role misassignment, CR-17 j (LO 35.j) (Q6)
S7 termination features and trade compression, CR-16 d (Reading 34) (Q6)
S8 two-way vs one-way CSA and rating-linked terms, as seen in August (Q1, Q2, Q4)
S9 margin period of risk: components and effect on collateralised exposure and CVA, as seen in August (Q3, Q5).

Revision notes

The CSA and its terms (CR-17 b, c, g; S1, S4)

- **CSA**: an annex to the ISDA Master Agreement in which the parties agree contractual margin-posting; the market-standard bilateral margin agreement. It covers valuation method and timing, the margin amount, transfer mechanics, eligible margin, substitutions, dispute resolution, remuneration, initial margin, haircuts, rehypothecation and triggers (e.g. downgrades). Terms are fixed at signing and change only by mutual amendment (a CCP can change terms *unilaterally*). There can be two CSAs (legacy and new) under one Master Agreement.
- **Threshold**: the amount below which margin is not required, producing *under*collateralisation. Zero threshold: margin always posted (above the MTA); infinite threshold: the party never posts (a one-way CSA). Thresholds are increasingly zero and can be non-zero only for parties exempt from the bilateral margin rules.
- **Initial margin (independent amount)**: the *opposite* of a threshold (a negative threshold), extra margin posted independently of portfolio value, giving *over*collateralisation against delays and close-out costs (S3). It relates to the *variability* of value, is increasingly dynamic, is not netted against variation margin, and two-way initial margins are paid gross. Threshold and initial margin are not used together.
- **Minimum transfer amount (MTA)** and rounding: the MTA is the smallest amount that can be transferred. Threshold and MTA are additive for *when* a call can be made, but the MTA is *not* deducted from the amount due (S2). MTAs create *path dependency* (Table 17.7). Rounding has a small effect; MTA and rounding are generally zero for cash-only margin (variation margin rules, CCPs).
- **Haircuts**: a haircut of $x\%$ gives a valuation percentage of $(1-x)\%$; the margin *giver* accounts for it, so haircuts also overcollateralise (S3). They reflect liquidation time, volatility, default risk, maturity, liquidity and wrong-way linkage. Cash may carry none except for FX mismatch; assets with significant credit or liquidity risk are avoided because haircuts cannot be large enough. Example (eq. 17.1): 99%, two days, 1% rate volatility, 10-year bond: about 2%.
- **Credit support amount** (CR-17 c): the amount of margin that may be requested at a point in time; because of thresholds, MTAs and initial margins it is *not* the portfolio value. Table 17.6: value 350, threshold 100, balance 0, call 250; then value 325, balance 250, amount -25 (return), though 75 is uncollateralised.

One-way vs two-way CSA, credit-quality links, valuation agent, disputes (CR-17 d, f, h; S5, S8)

- **Types of CSA**: *no CSA* is typical bank-corporate (corporates cannot manage the liquidity and operational demands). A *two-way* CSA, where both post, is typical between financial counterparties; two-way CSAs with *zero thresholds* are standard interbank and aim to benefit both parties. A *one-way* CSA, where only one party receives, is *additional risk for the margin giver*, leaving it *worse off than no CSA*: triple-A sovereigns and supranationals receive but do not post; banks have demanded one-way CSAs in their favour from some hedge funds. Funding and capital costs have made one-way CSAs problematic.
- **Rating-linked parameters** (Table 17.4): AAA threshold ∞; AA+ USD 100m; AA USD 50m; AA− USD 25m; A+ zero; A and A− initial margin 1%; BBB+ 2% of notional. Aim: low workload when quality is high, tighter terms as it falls. Problems: ratings react too slowly, and triggers create liquidity problems and cliff-edge effects (AIG: USD 20bn extra margin after its downgrade; Fed facility up to USD 85bn). Basel gives *no* capital benefit for rating triggers; the LCR requires pre-funding (an A+ bank funds 2% initial margin for a three-notch downgrade to BBB+).
- **Valuation agent, disputes and reconciliation**: the valuation agent is normally the party *calling* for delivery or return of margin; it calculates the current value including netting, the haircut-adjusted value of margin held, the uncollateralised exposure, and the credit support amount. Where both parties act as valuation agent, the potential for disputes is significant. Disputes arise from trade population, trade valuation methodology, application of CSA rules (thresholds, eligible collateral), market data and close time, and valuation of posted collateral. Reconciliations agree valuations and margin requirements across portfolios in advance; most large portfolios are reconciled daily; Dodd-Frank and EMIR require more frequent reconciliation; disputes can raise capital. Not relevant for centrally cleared trades (the CCP is valuation agent).

Mechanics, reuse and regulation (CR-17 a, e, i, k)

- **Rationale and transfer**: margin mitigates future exposure; on exchanges it is daily settlement, in OTC markets security ("collateralised to market"). Variation margin uses the base value; it cannot cover the delay to default and close-out costs, hence initial margin. *Security interest* (New York law, 46.8%) keeps ownership with the giver; *title transfer* (English law, 30.1%) passes it, which favours the receiver. Cash is remunerated at OIS; substitutions follow notice, consent where required, delivery, then return.
- **Rehypothecation and segregation**: reuse is natural for variation margin (quasi-settlement, reduces demand for high-quality collateral); not for initial margin, which should be segregated (Lehman, MF Global: USD 1.6bn of segregated customer margin moved illegally). Legal segregation (all three holding methods) vs operational segregation (third party or tri-party only). Ideal: pledged, unaffiliated custodian, no reuse. Segregation raises funding costs.
- **Bilateral margin rules (UMR)**: variation margin daily, zero threshold, MTA $\leq$ EUR 500,000, reusable; initial margin from both parties, gross, 99%, 10 days with stressed data, segregated, no reuse, threshold up to EUR 50m, phased in to EUR 8bn (Phase 6). FX swaps and forwards exempt from initial margin. Standard schedule net IM $= (0.4 + 0.6 \times \text{NGR}) \times$ gross IM; ISDA SIMM; no cherry-picking between model and schedule.

Risks created by margin (CR-17 j; S6, S9)

- **Margin period of risk**: the effective time between a counterparty ceasing to post margin and all trades being closed out and replaced (or hedged); market risk over it is uncollateralised. It is counterparty-risk specific and *not relevant for funding*, and is *not* the margin call frequency (daily calls, 10-day MPoR). *Pre-default*: valuation and margin call, receiving collateral (including disputes), settlement, grace (cure) period. *Post-default*: macro-hedging, close-out, rehedging and replacement (one-to-one or portfolio), liquidation of margin securities.
- **MPoR values**: Basel II minimum 10 business days (daily calls; add days if calls are less frequent); Basel III 20 days or more in some cases (over 5,000 trades, illiquid margin or exotics, a history of disputes); CCPs about 5 days; the ISDA Resolution Stay Protocol (24 or 48 hours) may lengthen it. It is a simple effective catch-all, not the literal close-out time: risk declines post-default (10 days constant equals 31 days linearly declining), while higher post-default volatility argues for longer (20% more volatility means about 44% more time). It is the primary driver of the need for initial margin.
- **Other risks**: *liquidity* risk falls on the holder that must sell margin after a default (bid-offer, price falls, large positions; concentration limits of 5% to 10%); *FX* risk from currency mismatch; *wrong-way* margin (own sovereign's bonds, LTCM's Russian bonds); *operational* risk (missed calls, failed deliveries, human

error), which Basel III answers with a longer MPoR; *legal* risk (Citigroup and Lehman, MF Global); *funding liquidity* risk for the party that must post: margin converts counterparty risk into funding liquidity risk (AIG, Ashanti: USD 280m of calls on a USD 450m loss). Margin, like netting, *redistributes* risk to other creditors (no margin: 67% recovery each; with VM: 100% vs 50%; with IM used: OC 25%).

Collateralised CVA (CR-20 j; S9)

- **Effect on CVA**: margin only changes EPE, so eq. 20.3 applies with collateralised EPE; 10 days has become standard even for accounting. Best case with variation margin only: risk falls by about $\sqrt{\text{maturity}/\text{MPoR}}$ (CR-17: 11 times for 5 years at 10 days); CR-20 quotes about 8.4 times for its 10-year swap. Reality is worse, so only initial margin makes risk negligible. Table 20.6 pay-fixed: CVA -28.9 to -3.0 bps, BCVA -21.5 to -1.5. High thresholds take CVA to the uncollateralised value, high initial margin to zero. Collateralised DVA is controversial (stopping posting 10 days before own default). Segregated initial margin posted does not change CVA but creates MVA.

Netting, close-out, compression and termination (CR-16; S7)

- **ISDA Master Agreement and close-out netting** (CR-16 a, b): the market-standard contract for OTC derivatives; it gives the contractual basis for close-out (right to terminate and stop payments) and netting (offset values into one amount, owed or claimed in bankruptcy); legal opinions in 54 jurisdictions. Payment netting reduces settlement risk; close-out netting reduces pre-settlement (counterparty) risk. 1992 ISDA: market quotation (at least three quotes) or loss; 2002 ISDA: a single close-out amount, commercially reasonable (2009 protocol). Walkaway clauses (Drexel, Enron/TXU, Lehman) create moral hazard. Set-off can extend offset to other agreements (loans).
- **Effectiveness and side effects** (CR-16 c): close-out netting is the single biggest counterparty risk mitigant and now cuts exposure by close to 90%; netted positions are more volatile, encourage concentration with one counterparty, and reduce panic when MTM is negative. Central clearing gives multilateral netting (Table 16.3: 360 to 120 bilateral to 60) but partial clearing can increase exposure (210). Netting redistributes value from other creditors (derivatives recovery 77.5% vs 60%; others 25%).
- **Trade compression** (CR-16 d, e): multilateral tear-up of redundant trades that reduces total gross notional while keeping each participant's market risk (value and risk neutral); it lowers operational, legal and regulatory (leverage ratio) costs, but does *not* necessarily cut any individual party's notional, has diminishing returns, needs fungible standard contracts and interchangeable counterparties. Example (Table 16.1): long 150 and 25 with A, short 75 with B and 50 with C, so a net long of 50 ($175 - 125$) held with Counterparty A. Trilateral netting in Figure 16.7 cuts 1,250 to 890 (final 130 or 110). EMIR: firms with over 500 uncleared contracts consider compression twice a year.
- **Termination features** (CR-16 d, f; Gregory 17.1): break clauses are mandatory, optional, or contingent on an event; the most common Additional Termination Event is a rating downgrade (for unrated parties NAV, market cap or key person). Termination is at a base MTM, often via a panel of banks. Table 17.1: *mandatory* breaks give counterparty risk reduction, CVA reduction, capital relief and LCR relief; *optional* breaks give partial counterparty risk reduction and capital relief (if rigorously exercised); *rating-based* breaks give partial counterparty risk reduction only (Basel rules out capital relief; the LCR forces pre-funding). Dangers: systematic deterioration, slow ratings, cliff-edge effects (AIG), modelling difficulty, and moral hazard from sales staff. Resets settle the MTM periodically and are easier to model.

Formulas

Receiving only, threshold and initial margin (eq. 17.2). required margin $= \max(\text{value} - K_C, 0) + \text{IM}_C$
Two-way credit support amount for variation margin (eq. 17.3).

$$\text{CSA amount} = \max(\text{value} - K_C, 0) - \max(-\text{value} - K_P, 0) - C$$

value: current base value of the netted portfolio to the calculating party; K_C: counterparty's threshold; K_P: calculating party's own threshold; C: credit support balance (negative if the party has posted). Positive: call (or request return of) margin; negative: post (or return). A transfer happens only if $|\text{amount}| \geq$ MTA; rounding is then applied (direction is a CSA term). Initial margin is computed and transferred separately.
Haircut. Value credited $= (1 - h)\times$ market value; to deliver credit X post $X/(1 - h)$. Normal-theory haircut (eq. 17.1): $h = \Phi^{-1}(\alpha)\,\sigma_C\sqrt{\tau}$, τ the liquidation time.

MPoR and collateralised risk (square-root-of-time rule).

$$\text{reduction factor (VM only, best case)} \approx \sqrt{\frac{\text{maturity}}{\text{MPoR}}}, \qquad \text{CVA}_{\text{coll}} \approx \frac{\text{CVA}_{\text{uncoll}}}{\text{reduction factor}}$$

Maturity and MPoR in the same units (250 business days per year).
Standardised initial margin. net IM $= (0.4 + 0.6 \times \text{NGR}) \times$ gross IM, NGR $=$ net over gross replacement cost.
Netting benefit (BIS). $1 - \dfrac{\text{gross credit exposure}}{\text{gross market value}}$

Exam traps

- **MTA deducted (dropped condition):** the MTA only decides *whether* a transfer occurs; once passed, the full amount moves. If the amount is below the MTA, nothing moves, even with every input supplied.
- **Credit support balance ignored (intermediate result):** $\max(\text{value} - K, 0)$ is the *total* margin required, not the call; subtract (or add back, with the sign) margin already held.
- **Whose threshold:** when the party is out of the money, *its own* threshold K_P applies; deducting both thresholds is wrong. After a downgrade use the *new* rating-linked threshold.
- **Role misassignment (CR-17 j):** liquidation risk sits with the margin *holder*; funding liquidity risk with the party that must *post*; the valuation agent is the party *calling*. A one-way CSA hurts the margin *giver*.
- **MPoR vs margin call frequency (sibling swap):** daily calls do not mean a one-day MPoR; and the MPoR is not used for funding.
- **MPoR units:** square-root-of-time with *business* days (250) and the MPoR stated; the reduction is $\sqrt{\cdot}$, not linear.

Question A21.1

CR-17 b, c, h · S1, S8 · [Aug-2026] [Map #27]

A collateral manager at a US bank is computing today's margin call under a two-way CSA with a European insurer. Thresholds are rating-linked (AA: USD 50 million; AA−: USD 25 million; A+: zero). The data are:

- Portfolio value to the bank: +USD 58.6 million
- Insurer's rating: downgraded overnight from AA to AA−
- Credit support balance already held by the bank: USD 8.0 million (cash)
- Minimum transfer amount: USD 0.5 million; rounding: amounts rounded *up* to the nearest USD 0.25 million

Which of the following is closest to the margin the bank can call?

A. USD 0.75 million

B. USD 25.25 million

C. USD 25.50 million

D. USD 25.75 million

Solution

Answer: D

Step 1 – Applicable threshold

After the downgrade to AA−, $K_C =$ USD 25 million (the AA threshold of USD 50 million no longer applies).
⇒ *The rating link tightens collateral terms as the insurer's credit quality falls.*

Step 2 – Required margin

$$\max(\text{value} - K_C, 0) = \max(58.6 - 25, 0) = \text{USD } 33.6 \text{ million}$$

Step 3 – Credit support amount before MTA and rounding

$$33.6 - \max(-58.6 - K_P, 0) - 8.0 = 33.6 - 0 - 8.0 = \text{USD } 25.6 \text{ million}$$

⇒ *The bank's own threshold term is zero because the portfolio has positive value to the bank.*

Step 4 – MTA test and rounding

$25.6 > 0.5$, so a call is valid and the full amount is called. Rounded up to the nearest 0.25: USD 25.75 million.

Interpretation

The downgrade alone triggers a USD 25.75 million call on the insurer: the cliff-edge liquidity effect Gregory warns about with rating triggers (the AIG pattern).

Why the other options are wrong

A is incorrect. Used the pre-downgrade AA threshold: $58.6 - 50 - 8.0 = 0.6$, which passes the MTA and rounds up to 0.75.

B is incorrect. Deducted the MTA from the amount: $25.6 - 0.5 = 25.1$, rounded up to 25.25. The MTA only gates the transfer.

C is incorrect. Rounded down instead of up as the CSA specifies: 25.50.

LO: CR-17 b: Describe the terms of a collateral agreement and features of a credit support annex (CSA) within the ISDA Master Agreement including threshold, initial margin, minimum transfer amount and rounding, haircuts, credit quality, and credit support amount. CR-17 c: Calculate the credit support amount (margin) under various scenarios. CR-17 h: Differentiate between a two-way and one-way CSA agreement and describe how collateral parameters can be linked to credit quality. · Reference: Gregory, The xVA Challenge 4e, Ch.7, 17.3.4 Table 17.4; 17.3.6 eq. 17.3

Question A21.2

CR-17 c, g, h · S1, S2, S8 · [Aug-2026] [Map #27]

A treasury operations analyst at a UK bank is processing today's margin under a two-way CSA with another bank. The data are:

- Portfolio value to the UK bank: −GBP 24.3 million
- Threshold for each party: GBP 5.0 million
- Margin already posted by the UK bank (credit support balance from its view): −GBP 18.9 million
- Minimum transfer amount: GBP 0.5 million; no rounding; independent amount: none

Which of the following is the correct action for the UK bank today?

A. No transfer

B. Post GBP 0.4 million

C. Post GBP 5.4 million

D. Request the return of GBP 4.6 million

Solution

Answer: A

Step 1 – Counterparty term

$$\max(\text{value} - K_C, 0) = \max(-24.3 - 5, 0) = 0$$

⇒ *The portfolio is negative to the UK bank, so it cannot call margin.*

Step 2 – Own term

$$\max(-\text{value} - K_P, 0) = \max(24.3 - 5, 0) = 19.3$$

⇒ *The UK bank's own GBP 5.0 million threshold is the uncollateralised amount it is allowed to owe.*

Step 3 – Credit support amount

$$0 - 19.3 - (-18.9) = -\text{GBP } 0.4 \text{ million}$$

⇒ *A negative amount would mean the UK bank posts.*

Step 4 – MTA test

$|-0.4| = 0.4 < 0.5$: the amount is below the minimum transfer amount, so no margin moves today.

Interpretation

The counterparty holds GBP 18.9 million against a liability of GBP 24.3 million: GBP 5.4 million uncollateralised, of which 5.0 is the agreed threshold and 0.4 the MTA gap. Full data are supplied, yet the threshold-plus-MTA test fails, which is the red-herring pattern the map flags. The 0.4 will be called once the shortfall reaches the MTA, which is why MTAs make collateralised exposure path dependent.

Why the other options are wrong

B is incorrect. Ignored the MTA test and posted the raw credit support amount of 0.4.
C is incorrect. Ignored the UK bank's own threshold: $24.3 - 18.9 = 5.4$.
D is incorrect. Deducted both parties' thresholds: required $24.3 - 5 - 5 = 14.3$ against 18.9 posted, so it wrongly asks for 4.6 back. Only the out-of-the-money party's own threshold applies.

LO: CR-17 c: Calculate the credit support amount (margin) under various scenarios. CR-17 g: Explain the features of a collateralization agreement. CR-17 h: Differentiate between a two-way and one-way CSA agreement and describe how collateral parameters can be linked to credit quality. · Reference: Gregory, The xVA Challenge 4e, Ch.7, 17.3.4; 17.3.6 eq. 17.3 and Table 17.7

Question A21.3

CR-20 j; CR-17 h, j · S9 · [Aug-2026]

A CVA desk analyst at a Swiss bank is estimating how far a new two-way, zero-threshold CSA with daily variation margin calls will reduce the CVA on an 8-year portfolio with a hedge fund. The data are:

- Uncollateralised CVA: −EUR 4.2 million
- The netting set contains over 5,000 trades, so the bank applies a 20-business-day MPoR
- 250 business days per year; no initial margin

Using the square-root-of-time approximation in the reading, which of the following is closest to the best-case collateralised CVA?

A. −EUR 0.420 million

B. −EUR 0.348 million

C. −EUR 0.297 million

D. −EUR 0.042 million

Solution

Answer: A

Step 1 – Maturity in business days

$$8 \times 250 = 2{,}000 \text{ business days}$$

Step 2 – Reduction factor

$$\sqrt{\frac{\text{maturity}}{\text{MPoR}}} = \sqrt{\frac{2{,}000}{20}} = \sqrt{100} = 10$$

⇒ With only variation margin, exposure is limited to the market move over the 20-day MPoR instead of up to 8 years, so risk falls by about 10 times at best.

Step 3 – Collateralised CVA

$$\text{CVA}_{\text{coll}} \approx \frac{-4.2}{10} = -\text{EUR } 0.420 \text{ million}$$

Interpretation

Even a zero-threshold daily CSA leaves about 10% of the CVA, and Gregory notes reality is worse than this best case; only initial margin can push CVA towards zero. The longer MPoR for large netting sets cuts the benefit compared with the 10-day standard. Margin changes only the EPE in the CVA formula, so the PD and LGD are untouched.

Why the other options are wrong

B is incorrect. Used 365 calendar days with a business-day MPoR: $4.2/\sqrt{2{,}920/20} = 0.348$.
C is incorrect. Used the standard 10-day MPoR instead of the 20 days applied to netting sets over 5,000 trades: $4.2/\sqrt{200} = 0.297$.
D is incorrect. Scaled linearly instead of by the square root: $4.2 \times 20/2{,}000 = 0.042$.

LO: CR-20 j: Explain the impact of incorporating collateralization into the CVA calculation, including the impact of margin period of risk, thresholds, and initial margins. CR-17 j: Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization. · Reference: Gregory, The xVA Challenge 4e, Ch.7, 17.5.2 and 17.5.4; Ch.17, 20.5.1 and 20.5.2

Question A21.4 CR-17 d, f, g, h · S4, S5, S8 · [Aug-2026] [Map #27]

A collateral manager at a Canadian bank is recommending CSA terms for three new relationships: a triple-A supranational development bank, an unrated hedge fund, and a dealer bank. Which of the following recommendations is consistent with the reading?

A. Accept the supranational's one-way CSA in its favour, since a one-way CSA adds risk only for the party that receives the margin.

B. Sign a two-way CSA with zero thresholds with the dealer bank, the interbank standard, which aims to benefit both parties on counterparty risk.

C. Link the hedge fund's threshold to its net asset value so that the threshold rises as the net asset value falls, easing its workload under stress.

D. Let both parties act as valuation agent on every CSA, since two valuation agents remove the potential for disputes over margin calls.

Solution **Answer: B**

Step 1 – The dealer relationship

Gregory: "Two-way CSAs with zero thresholds are standard in the interbank market and aim to be beneficial (from a counterparty risk point of view, at least) to both parties." Both parties post against a negative value, so both exposures are mitigated.

⇒ *The plain two-way CSA from the August recall: each party's own threshold (here zero) governs when it posts.*

Step 2 – The other two relationships

The supranational will want to receive but not post (an infinite threshold for itself): a one-way CSA that is extra risk for the bank as margin giver, worse than no CSA, and costly in funding and capital. For the hedge fund, an unrated party, terms can be linked to other metrics such as net asset value, but they must *tighten* as quality falls (lower threshold, then initial margin).

Interpretation

The bank should price the supranational's one-way CSA through xVA rather than treat it as risk-free, and keep the hedge fund's trigger mechanics aware of cliff-edge and liquidity effects.

Why the other options are wrong

A is incorrect. Role misassignment: a one-way CSA "represents additional risk for the margin giver", putting it in a worse situation than no CSA. The receiver (the supranational) is the party that benefits.

C is incorrect. Polarity flip: credit-quality links minimise workload when quality is high and *tighten* terms (lower threshold, initial margin) as it deteriorates; a threshold that rises as net asset value falls loosens collateralisation exactly when risk is highest.

D is incorrect. Absolute claim: where both counterparties are valuation agents, each calling for margin when it has an exposure, "the potential for margin disputes is significant". Reconciliation reduces disputes; nothing removes them.

LO: CR-17 h: Differentiate between a two-way and one-way CSA agreement and describe how collateral parameters can be linked to credit quality. CR-17 g: Explain the features of a collateralization agreement. CR-17 d: Describe the role of a valuation agent. CR-17 f: Explain the process for the reconciliation of collateral disputes. · Reference: Gregory, The xVA Challenge 4e, Ch.7, 17.2.7, 17.3.2 and 17.3.4 Table 17.4

Question A21.5 CR-17 j; CR-20 j · S3, S9 · [Aug-2026]

An xVA quantitative analyst at a UK bank is documenting the assumptions of the collateralised CVA model for portfolios under two-way, zero-threshold CSAs with daily variation margin calls. Which of the following statements should the documentation include?

A. Because margin is called daily, the model's MPoR should be one business day, matching the contractual period between margin calls.

B. A grace (cure) period after a missed call is a pre-default component of the MPoR, while rehedging and liquidating margin securities are post-default.

C. Because the MPoR measures the delay in receiving margin, the same MPoR should be used to assess the funding cost of the collateralised portfolio.

D. Because the MPoR is the primary driver of the need for initial margin, a longer MPoR reduces the initial margin the bank should require.

Solution **Answer: B**

Step 1 – Components

Gregory splits the MPoR into *pre-default* (valuation and margin call; receiving collateral, including any dispute; settlement; grace or cure period) and *post-default* (macro-hedging; close-out of transactions; rehedging and replacement; liquidation of margin securities).

⇒ *The grace period runs before the counterparty is legally in default; liquidation of the margin it posted can only happen after.*

Step 2 – Effect on collateralised exposure and CVA

Market risk over the MPoR is uncollateralised. With variation margin only, the best-case reduction is about $\sqrt{\text{maturity}/\text{MPoR}}$; reality is worse. Margin changes only EPE in the CVA formula: in Table 20.6 the 10-year pay-fixed swap's CVA falls from -28.9 to -3.0 bps with a 10-day MPoR. Initial margin (overcollateralisation) pushes CVA towards zero; a threshold acts as a negative initial margin.

Interpretation

The model should use a 10-business-day MPoR at least (more for large or disputed netting sets) and size initial margin to the MPoR, since the choice of initial margin is closely tied to the assumed MPoR.

Why the other options are wrong

A is incorrect. Sibling swap: the margin call frequency is only one component of the MPoR; "it could, therefore, be that the margin call frequency is daily, but the MPoR is assumed to be 10 days".

C is incorrect. Scope error: the MPoR is defined assuming the counterparty's default and is "not relevant when assessing funding costs"; funding uses the normal, much shorter posting delay.

D is incorrect. Polarity flip: the MPoR is indeed the primary driver of the need for initial margin, but a *longer* MPoR leaves more uncollateralised market risk and so *raises* the initial margin needed.

LO: CR-17 j: Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization. CR-20 j: Explain the impact of incorporating collateralization into the CVA calculation, including the impact of margin period of risk, thresholds, and initial margins. · Reference: Gregory, The xVA Challenge 4e, Ch.7, 17.3.3 and 17.5.2; Ch.17, 20.5.2 and 20.5.3

Question A21.6 CR-17 f, j; CR-16 d · S5, S6, S7 · [Map #27]

An internal auditor at an Australian bank is reviewing the unit that manages collateral and other counterparty risk mitigants, and is drafting findings for the audit committee. Which of the following statements is correct to include?

A. A rating-based break clause offers the same benefits as a mandatory break: counterparty risk reduction, CVA reduction, capital relief and LCR relief.

B. Liquidity risk from holding non-cash margin falls on the margin giver, which bears the bid-offer costs and price falls when the securities are sold after a default.

C. Because a compression cycle lowers the total notional of all participants, it guarantees that each participant ends with a lower notional to every counterparty.

D. Daily portfolio reconciliation reduces margin disputes, and a history of disputes is one reason Basel III requires a longer MPoR, raising capital and rewarding lower operational risk.

Solution **Answer: D**

Step 1 – Disputes, reconciliation and operational risk

Reconciliations "aim to minimise the chance of a dispute by agreeing on valuations and margin requirements across portfolios"; most large portfolios are reconciled daily. Basel III recognises the operational risk of collateralisation by requiring a larger MPoR where a netting set exceeds 5,000 trades or holds illiquid margin or exotics, and "a history of margin call disputes also requires a larger MPoR". Larger MPoRs mean higher capital and so an incentive to reduce operational risk.

Interpretation

The auditor should test reconciliation frequency and dispute logs alongside capital: an operational weakness in margining shows up directly as a longer MPoR and a higher capital charge.

Why the other options are wrong

A is incorrect. Sibling swap: Table 17.1 gives all four benefits only to *mandatory* breaks; a rating-based break gives at most some counterparty risk reduction, Basel explicitly rules out capital relief from rating triggers, and the LCR forces banks to pre-fund them.

B is incorrect. Role misassignment: liquidation risk sits with the non-defaulting party that *holds* the margin and must sell it for cash to rehedge; haircuts are set to cover it. The giver's risk is funding liquidity risk when it must post.

C is incorrect. Absolute claim: compression aims to reduce the total notional of all parties, and "it does not necessarily follow that an individual party will experience a lower notional"; positions between pairs can even rise or change sign, which is why participants set constraints.

LO: CR-17 f: Explain the process for the reconciliation of collateral disputes. CR-17 j: Explain how market risk, operational risk, and liquidity risk (including funding liquidity risk) can arise through collateralization. CR-16 d: Describe the mechanics of termination provisions and trade compressions and explain their advantages and disadvantages. · Reference: Gregory, The xVA Challenge 4e, Ch.6, 16.2.4; Ch.7, 17.1.1 Table 17.1, 17.2.7, 17.5.3 and 17.5.4

A22 Three Lines of Defence and Operational Risk Governance

Readings: ORR-2 a, b, c, d · Risk Governance (GARP ORR Ch.2); CR-2 a · Governance (Bouteille and Coogan-Pushner Ch.2); ORR-1 b · Introduction to Operational Risk and Resilience (GARP ORR Ch.1); ORR-3 c, d · Risk Identification (GARP ORR Ch.3, Section 3.3)

AUG 2026 RECALL | MAP TIER C · #29 · 4/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: Three lines of defence - 2Q.
Map #29: "Reported: 'three lines of Operational Risk Management'; 'Ops Risk Level 3 categorisation': the Basel event-type taxonomy at its third level, a separate recall item. Business lines own risk (first line); the risk function sets framework and challenges (second); internal audit provides independent assurance (third). Any statement moving an activity across those boundaries is the defect."
Shapes covered here:
S1 three lines of operational risk management: who does what, with every distractor moving an activity across a line (Q1)
S2 Basel event-type taxonomy, Level 1, Level 2 and the Level 3 examples (Q3)
S3 committees versus the board, and who owns risk appetite and limits (Q2).

Revision notes

The specific: three lines, who does what (ORR-2 c; CR-2 a)

- **Line 1, risk owners.** The front line of the business; BCBS (RPSMOR) calls it "business unit management". "Risk is managed where it is generated": risk is owned by the business, not by the risk function. Better to say *risk owners* than "the business": the head of IT owns IT risk. Duties: identify and assess the materiality of inherent operational risks; establish controls; assess control design and effectiveness; monitor and report the unit's risk profile; report residual risks (loss events, control deficiencies, process inadequacies, breaches of tolerance) and report a lack of resources, tools or training to the second line.
- **Line 1.5, risk champions.** Risk specialists, "stewards" or "risk correspondents" inside each business department (line "1.5" or "1.6"): main correspondent for risk issues, collect and record events and losses, map risks and controls to group definitions, follow up control rules, action plans and audit tracking. Ownership of operational risk must *not* be assumed by, or passed to, them.
- **Line 2, the CORF.** An "independent corporate operational risk management function": develops and maintains policies, standards and training; forms an independent view of the business units' risk identification, key controls and respect of appetite; *challenges* the business's use of the tools and provides evidence of that challenge; reviews and contributes to monitoring and reporting. It is a sounding board for decisions that change the risk profile and needs delegated authority to **veto** decisions that breach regulation or board risk appetite. It must not be directly involved in line 1 risk management. In larger banks the CORF reports independently of the risk-generating units; at small banks independence may come from separation of duties.
- **Guidance versus challenge.** First role of the risk function in young or small units is to educate (what is operational risk, how incidents are recognised and reported, benefits, pitfalls). A second-line risk manager may run an RCSA workshop yet stay independent by not steering the answers. "The second line owns the methodology; the first line owns the risks": the first line gives the final sign-off on its risk and control assessment.
- **Line 3, internal audit.** "Independent assurance"; the clearest boundaries. Independent of the risk process; assesses the control system and compliance, including of the risk function; builds its own audit universe. IIA: internal audit is neither responsible for nor part of risk, compliance or finance; it must *in no circumstances rely exclusively* on their work and always examines its own sample; less intense scrutiny only after evaluating that function. Reports in solid line to a non-executive (audit committee chair, senior independent director or non-executive chairman). External audit, also independent, gives an opinion on the financial statements.
- **Hybrid functions.** In small and medium firms finance, HR, legal, ICT and security functions act in both lines. BCBS: document and distinguish the first and second line responsibilities, emphasising second line independence. Table 2.4: legal drafts contracts (line 1) vs oversees legal activity in the business (line 2); HR administers pay (1) vs oversees people management (2); IT security encrypts data, partitions and monitors

the network (1) vs verifies ICT controls in business operations and enforces security policies (2); finance manages treasury and accounting (1) vs verifies exposures remain under authorised limits (2). What defines a line is the role, not the department's name.

- **IIA Three Lines Model (2021) and "three lines of accountability".** The 2021 model reinforces the *expertise and support* role of the second line, with challenge limited to risk-related matters; the second line is part of, and reports to, management. "Three lines of accountability" is a renaming, not a departure from the model.
- **CR-2 a version (IIA 2013 paper).** First line: business owners, who primarily own and manage risk. Second line: ERM, compliance and legal, who establish policies and procedures and serve as management oversight. Third line: internal audit, external auditors and special audit committees, giving independent assurance over lines 1 and 2. Credit guidelines are owned and maintained by the CRO's office, sponsored by a senior executive (CRO or CFO) and **approved by the board**; for listed firms the board audit committee reviews risk policies, and the board risk committee oversees and approves them.

Committees, board, Basel expectations (ORR-2 a, b)

- **Committee pyramid.** Lowest: business-type or regional risk committees, which escalate significant issues and limit breaches to a firm-wide or second-line committee. The group operational risk committee reports the consolidated picture to the executive risk committee, the management committee and the board risk committee. The **board risk committee** oversees all operational risk and makes recommendations to the **full board**; members need recent and relevant experience. A small firm may have a single operational risk committee reporting to the management committee and board risk committee. Each committee has Terms of Reference (charter); minutes record discussions, decisions and their motivation.
- **The board.** Ultimately responsible for the firm, including risk management; determines the nature and extent of significant risks it will take (sets risk appetite and keeps the firm within it); validates and periodically reviews the ORMF; gets assurance that senior management implements it; establishes risk culture; must be trained. For resilience (BCBS resilience Principle 1): uses existing governance, oversees senior management's implementation, ensures financial and technical resources.
- **Basel II and RPSMOR.** Pillar 1 for operational risk includes mandatory management principles (the only risk type where this is so); Pillar 2 add-ons for concentration, compliance, governance gaps, aggressive growth (ICAAP/SREP, ORSA, CCAR); Pillar 3 disclosure. RPSMOR (March 2021) has 12 principles: culture led by the board and implemented by senior management (1), sound ORMF (2), board approval and review of ORMF (3), board-approved risk appetite and tolerance statement (4), senior management develops and implements policies and systems (5), identification (6), change management (7), monitoring (8), control environment (9), ICT (10), business continuity (11), disclosure (12). From January 2023 Pillar 1 capital uses the SA: ORC $=$ BIC $\times$ ILM.
- **Use test.** ORM must be embedded in decisions, not rubber-stamped. "Absence of evidence is evidence of absence": committees and management should record issues, discussions and decisions in approved minutes.

Risk appetite and culture (ORR-2 d)

- **Risk appetite.** Statements must be easy to communicate, justify taking, limiting or avoiding risks, align with strategy, be monitored by indicators on exposure, control environment and losses, and be forward-looking (scenario and stress tested). A claim such as "minimal tolerance for data breaches" must be corroborated with strong controls and monitoring. Structure: exposure limits, key controls, tolerated incidents, monitoring metrics ("appetite KRIs"). The board **ultimately owns the risk limits** and validates them periodically; in practice it often delegates this to the board risk committee, which uses the risk function to implement and report on them. Governance: risk owners (line 1), control owners, metrics owners.
- **Risk culture.** Led by the board and implemented by senior management ("tone at the top"); code of conduct approved by the board and attested by staff; skin in the game; training; reinforcement without unnecessary blame ("no unnecessary blame"); "if you see something, say something"; whistleblowing. Toxic culture: overambitious targets, undue pressure, resistance to bad news (Wells Fargo). Most telling sign: talking about internal incidents without undue blame.

The Basel taxonomy (ORR-1 b; ORR-3 c, d)

- **Seven Level 1 event types** (ORX 2014 to 2019, frequency / severity): IF internal fraud 2% / 2%; EF external fraud 30% / 9%; EPWS employment practices and workplace safety 15% / 5%; CPBP clients, products and business practices 22% / 52%; DPA damage to physical assets 1% / 1%; BDSF business disruption and

system failures 2% / 5%; EDPM execution, delivery and process management 28% / 27%. CPBP's severity is driven by regulatory fines; legal losses cluster in EPWS and EDPM.

- **Levels.** Level 1 is the highest category, Level 2 a detailed version of Level 1, and Level 3 provides *examples*. BCBS recognises only Levels 1 and 2 as regulatory categories; regulated firms report using this categorisation.
- **Level 2 by Level 1.** IF: Unauthorised Activity; Theft and Fraud ("at least one internal party"). EF: Theft and Fraud; Systems Security (third party). EPWS: Employee Relations; Safe Environment; Diversity and Discrimination. CPBP: Suitability, Disclosure and Fiduciary; Improper Business or Market Practices; Product Flaws; Selection, Sponsorship and Exposure; Advisory Activities. DPA: Disasters and other events. BDSF: Systems. EDPM: Transaction Capture, Execution and Maintenance; Monitoring and Reporting; Customer Intake and Documentation; Customer/Client Account Management; Trade Counterparties; Vendors and Suppliers.
- **Level 3 examples that discriminate.** Mismarking of position (intentional): IF / Unauthorised Activity. Insider trading *not on firm's account*: IF / Theft and Fraud; insider trading *on firm's account*: CPBP / Improper Business or Market Practices (with antitrust, market manipulation, unlicensed activity, **money laundering**). Check kiting: under both IF and EF Theft and Fraud, decided by whether an internal party is involved. Hacking damage: EF / Systems Security. Account churning, breach of privacy, misuse of confidential information: CPBP / Suitability, Disclosure and Fiduciary. Model errors: CPBP / Product Flaws, whereas model or system *misoperation*: EDPM / Transaction Capture. Utility outage: BDSF / Systems. Workers compensation: EPWS / Safe Environment. Outsourcing and vendor disputes: EDPM / Vendors and Suppliers. Failed mandatory reporting: EDPM / Monitoring and Reporting.
- **Other taxonomies.** ORX reference taxonomy (2019): 14 Level 1 risks versus Basel's 7, with third party, statutory reporting and tax, business continuity, data management, information security (including cyber) and model risk elevated to Level 1; ORX causes taxonomy uses PPSE (people, process, systems, external events). Impact categories: financial loss, reputational damage, regulatory non-compliance, customer detriment (plus service disruption). Control categories: preventive, detective, corrective, directive. Firms place cyber in EF, IT risk or a separate Level 1. See also D22 for ORR-3 a, b (identification tools, scenarios).

Exam traps

- **Role misassignment (the map's defect):** the second line facilitates the RCSA and owns the methodology, but the *first* line signs off; the risk function never "owns" operational risk; risk champions never take ownership; internal audit never reports to the CRO.
- **Absolute claim on audit reliance:** internal audit may take account of others' work after evaluating it, but never relies *exclusively* on risk, compliance or finance testing.
- **Sibling swap in the committee pyramid:** the *board risk committee* recommends to the full board; the group operational risk committee reports upward to it. The board approves the ORMF and the risk appetite statement; senior management implements.
- **Scope error on hybrids and appetite:** hybrid functions must document and distinguish their line 1 and line 2 roles; an appetite statement without key controls and monitoring metrics is not credible.
- **Taxonomy siblings:** insider trading on vs not on the firm's account (CPBP vs IF); model errors (CPBP Product Flaws) vs model misoperation (EDPM); internal vs external party for check kiting; utility outage is BDSF, not DPA. Level 3 is examples only, not a regulatory category.

Question A22.1

ORR-2 c; CR-2 a · S1 · [Aug-2026] [Map #29]

The head of operational risk at a Canadian bank is reviewing a draft responsibilities matrix ahead of a supervisory visit. The bank uses the three lines of defence model, and its IT security, legal and finance departments are hybrid functions. Which of the following entries in the draft is consistent with the three lines of defence model for operational risk governance?

A. A second-line risk manager facilitates each business unit's risk and control self-assessment and gives the final sign-off on its results, as the second line owns the methodology.

B. The payments department's risk champions take over ownership of the department's operational risks, so that the head of payments can concentrate on running the business.

C. After evaluating the corporate operational risk function as effective, internal audit relies exclusively on that function's control testing when auditing the payments department.

D. In IT security, encrypting data and partitioning the network are documented as first-line roles, while verifying ICT controls and enforcing security policies are documented as second-line roles.

Solution **Answer: D**

Step 1 – Who owns the risk

The first line ("business unit management") owns the risks it generates and must give the final sign-off on its risk and control assessment. The second line may run the RCSA workshop, then collect and challenge the answers, while staying independent of the content.

⇒ *"The second line owns the methodology; the first line owns the risks." Sign-off is a first-line act.*

Step 2 – Where risk champions sit

Risk champions (line 1.5) record events, map risks and controls and follow up action plans inside the business. Firms must ensure ownership of operational risk is not assumed by, or passed to, them.

Step 3 – How far internal audit may rely on others

IIA guidance: internal audit may take account of others' work after evaluating that function, but in no circumstances relies exclusively on risk, compliance or finance; it always examines its own sample.

Step 4 – Hybrid functions

BCBS requires a hybrid function to document and distinguish its first and second line responsibilities, emphasising the independence of the second line. Table 2.4 gives exactly this split for IT security: encrypting data, partitioning and monitoring the network (line 1) versus verifying the implementation of ICT controls in business operations and enforcing security policies (line 2).

Interpretation

D is correct. The matrix entry places each IT security activity in the line its role defines, which is what the supervisor will test: the line is set by the role, not by the department's name. The fix for the other entries is to return sign-off and ownership to the first line and restore internal audit's own sampling.

Why the other options are wrong

A is incorrect. Role misassignment: the final sign-off on the risk and control assessment belongs to the first line; the second line owns only the methodology and the challenge.

B is incorrect. Sibling swap: substitutes the risk champions (line 1.5) for the risk owners (line 1); ownership must stay with the business managers.

C is incorrect. Absolute claim: a positive evaluation can justify less intense scrutiny, never exclusive reliance; internal audit must always test its own sample.

LO: ORR-2 c: Describe the "three lines of defense" model for operational risk governance and compare roles and responsibilities for each line of defense. · CR-2 a: Define risk management responsibilities in an organization and explain the three lines of defense framework for effective risk management and control. · Reference: GARP ORR Ch.2, Section 2.3 d) and Table 2.4; Bouteille and Coogan-Pushner Ch.2

Question A22.2

ORR-2 b, d · S3 · [Map #29]

A newly appointed non-executive director on the board risk committee of an Australian bank asks the company secretary for a briefing note on operational risk governance. The bank's large committee structure includes regional risk committees, a group operational risk committee, an executive risk committee and the board risk committee, and its risk appetite statement declares "minimal tolerance for data breaches". Which of the following statements is correct to include in the note?

A. The board ultimately owns the operational risk limits and validates them periodically, often delegating this to the board risk committee, which relies on the risk function to implement and report on them.

B. The group operational risk committee sits at the top of the structure and makes recommendations directly to the full board on risk-based decisions, risk exposure and risk management.

C. A statement of minimal tolerance for data breaches is credible on its own, since an appetite statement need not be tied to specific key controls or to monitoring metrics.

D. Senior management approves and periodically reviews the operational risk appetite and tolerance statement, and the board receives the approved statement for information.

Solution **Answer: A**

Step 1 – Place each committee in the pyramid

Regional and business committees escalate to a firm-wide committee; the group operational risk committee reports the consolidated picture to the executive risk committee, the management committee and the board risk committee. The board risk committee oversees all operational risk and makes recommendations to the full board.

⇒ *The top of the committee pyramid, and the link to the full board, is the board risk committee.*

Step 2 – Who approves appetite and owns limits

RPSMOR Principle 4: the risk appetite and tolerance statement is approved and periodically reviewed by the *board*. ORR-2: "It is the role of the board of directors to ultimately own the risk limits and to validate them periodically. In practice, the board often delegates this responsibility to the board risk committee, which in turn requests the support of the risk function."

Step 3 – What makes the statement credible

A firm declaring minimal tolerance for data breaches must corroborate it with strong controls and monitoring measures; an actionable appetite combines exposure limits, key controls, tolerated incidents and monitoring metrics (appetite KRIs).

Interpretation

A is correct. The director should expect to see limits owned by the board, overseen in practice by the board risk committee, and backed by key controls and KRIs on data security, with the risk function reporting against them.

Why the other options are wrong

B is incorrect. Sibling swap: attributes the board risk committee's role (recommendations to the full board) to the group operational risk committee, which reports upward to it.

C is incorrect. Scope/condition error: drops the requirement that appetite statements be supported by key controls and monitored through indicators on exposure, controls and losses.

D is incorrect. Role misassignment: the board approves and periodically reviews the risk appetite and tolerance statement; senior management develops and implements policies and systems.

LO: ORR-2 b: Describe and compare the roles of different committees and the board of directors in operational risk governance. · ORR-2 d: Explain best practices and regulatory expectations for the development of a risk appetite for operational risk and for a strong risk culture. · Reference: GARP ORR Ch.2, Sections 2.2 b), 2.3 a) and c), 2.4 a) and b)

Question A22.3 ORR-3 d; ORR-1 b · S2 · [Map #29]

An operational risk analyst at a Brazilian bank is classifying the quarter's loss events for regulatory reporting, which uses the Basel Level 1 and Level 2 categories, with Level 3 activities used as examples. The events are:

- an employee traded shares on inside information for his personal account, and the bank paid a settlement;
- an error in the pricing model embedded in a structured deposit product led to client compensation;
- a customer, acting alone, kited cheques between accounts at the bank and at another bank;
- a utility power outage halted the bank's main data centre for six hours, with no damage to equipment.

Which of the following classifications is correct?

A. The insider trading event: Level 1 Clients, Products and Business Practices; Level 2 Improper Business or Market Practices.

B. The pricing model error: Level 1 Clients, Products and Business Practices; Level 2 Product Flaws.

C. The cheque kiting event: Level 1 Internal Fraud; Level 2 Theft and Fraud.

D. The utility outage: Level 1 Damage to Physical Assets; Level 2 Disasters and Other Events.

Solution **Answer: B**

Step 1 – Insider trading: whose account?

Table 3.1 lists "Insider trading (not on firm's account)" under Internal Fraud / Theft and Fraud, and "Insider trading (on firm's account)" under CPBP / Improper Business or Market Practices. Trading for a personal account is the first.

⇒ *The Level 3 qualifier decides the Level 1 category.*

Step 2 – Model error in a product

"Model errors" is a Level 3 example under CPBP / Product Flaws (losses from the nature or design of a product).

By contrast, "Model/system misoperation" sits under EDPM / Transaction Capture, Execution and Maintenance. ⇒ *A flaw built into the product's design is CPBP; a mistake in operating a model during processing would be EDPM.*

Step 3 – Cheque kiting: who acted?

Internal fraud requires "at least one internal party"; external fraud is "by a third party". A customer acting alone is a third party, so the event is External Fraud / Theft and Fraud (Level 3: theft, robbery, forgery, check kiting).

Step 4 – Outage without damage

DPA covers loss or damage to physical assets from natural disaster or other events. A utility outage without damage is BDSF / Systems (Level 3: utility outage/disruptions).

Interpretation

B is correct. Only Levels 1 and 2 are regulatory categories, so each event must land in the right Level 1 and Level 2 cell; the Level 3 examples (on or off the firm's account, model error vs misoperation, internal vs third party, damage vs disruption) are the tie-breakers GARP uses.

Why the other options are wrong

A is incorrect. Sibling swap: uses the "on firm's account" entry; insider trading for a personal account is Internal Fraud / Theft and Fraud.

C is incorrect. Role misassignment: assigns the act to an internal party; a customer acting alone makes it External Fraud / Theft and Fraud.

D is incorrect. Scope/condition error: DPA needs loss or damage to physical assets; a disruption with no damage is Business Disruption and System Failures / Systems.

LO: ORR-3 d: Describe and apply the Level 1, 2, and 3 categories in the Basel operational risk taxonomy. · ORR-1 b: Describe the seven Basel II event risk categories and identify examples of operational risk events in each category. · Reference: GARP ORR Ch.3, Section 3.3 and Table 3.1; Ch.1, Table 1.1

A23 Cyber Risk and Operational Resilience

Readings: ORR-8 a, b, c, d, e · Cyber-Resilience: Range of Practices (BCBS, December 2018; GARP ORR Ch.8); ORR-9 a, b · Case Study: Cyberthreats and Information Security Risks (GARP ORR Ch.9); ORR-1 a, c, d · Introduction to Operational Risk and Resilience (GARP ORR Ch.1)

AUG 2026 RECALL | MAP TIER C · #31 · 4/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: Cyber risk.
Map #31: "Reported: 'cyber risk framework'; 'operational resiliency'; 'one cyber risk question'; and the detailed one, 'which bank was more vulnerable during COVID: payment bank or community bank?', i.e. an applied vulnerability-comparison rather than a definition. Operational resilience as a regulatory concept (UK first, then BCBS, ECB, MAS) is the adjacent theory item."
Shapes covered here:
S1 cyber risk framework, the one cyber risk question: supervisory practice, frameworks and the Equifax lessons (Q1)
S2 operational resiliency as a regulatory concept: UK first, then BCBS, ECB, MAS (Q2)
S3 applied vulnerability comparison: payment bank versus community bank during COVID (Q2).

Revision notes

The specific: cyber-resilience and its supervision (ORR-8 a, b)

- **Definition.** BCBS uses the FSB Cyber Lexicon: cyber-resilience is "the ability of an organisation to continue to carry out its mission by anticipating and adapting to cyber threats and other relevant changes in the environment and by withstanding, containing and rapidly recovering from cyber incidents". Cyber-risk management (vulnerabilities and threats) is narrower than IT risk management.
- **Initiatives.** G20 (March 2017) warning on malicious use of ICT; G7 Fundamental Elements of Cyber-security; CPMI-IOSCO guidance for FMIs (June 2016; resume critical operations within two hours); EU Fintech Action Plan; BCBS Operational Resilience Working Group. Banking is the only financial sector in which all FSB jurisdictions have issued regulation, guidance or supervisory practice. Most jurisdictions treat cyber through the lens of IT and general operational risk, drawing on NIST, ISO 27000 and COBIT. Examples (Box 8.1): APRA CPS 234 (board ultimately responsible for information security; notify APRA of material incidents), BaFin BAIT, US proposed rules, NYDFS (requires a CISO).
- **Strategy.** Most regulators do *not* require a standalone cyber-security strategy, but *all* expect a board-approved information security strategy, policy and procedures; cyber risk is expected to sit in the organisation-wide risk or information security framework. Three enforcement approaches: regulator-set strategy (emerging markets), bank-set strategy reviewed by the regulator, IT strategy review (Europe).
- **3LD and the CISO.** Most regulators use the 3LD model to *assess* cyber risk, but most do not *require* it or prescribe how responsibilities are split; supervisory focus is greater on lines 1 and 2 than on line 3, which could hamper checks and balances; few specify an independent reporting line from the chief audit executive to the board audit committee. The CISO usually reports to the CRO or CIO with no independent line to the CEO or board; placing the CISO under the CRO is not ideal (the CRO may prioritise compliance over innovation), so some banks have the CISO report to the CEO or board. SAMA requires a cyber-security function independent from IT.
- **Culture and workforce.** Cyber-risk awareness and a common risk culture are prerequisites; screening at hiring and periodic re-verification against insider threats. Only Hong Kong, Singapore and the UK have dedicated frameworks to certify cyber-workforce skills (HKMA PDP, MAS board training, BoE CBEST and CREST).

Testing, response, metrics, sharing, third parties (ORR-8 c, d, e)

- **Supervision and testing.** Off- and on-site reviews, risk specialists, triage tools for front-line supervisors, powers to appoint a third-party auditor. Mapping and classifying business services underpins testing. Regulator-led penetration tests are typically voluntary, bank-funded and aimed at larger institutions, mostly testing protection and detection. **TIBER-EU** (ECB, May 2018): first Europe-wide threat-intelligence-based red-teaming framework; tailor-made; *no pass or fail*.

- **Response and recovery.** Most regulators require a prevention, detection, response and recovery framework, but few issue cyber-specific continuity rules. Exercise Resilient Shield (UK/US, 2015) rehearsed communication and coordination; it was not a test of firms.
- **Metrics are not mature.** No quantitative metrics comparable to financial risk; reliance on backward-looking indicators, while adaptive adversaries (DDoS) call for forward-looking ones. **Patch ageing** is a widespread, comparable metric; no single metric suffices and no standard benchmark exists.
- **Information sharing, five types.** Among banks 75%; bank to regulator 75%; with security agencies 68%; regulator to banks 32%; among regulators 29% (least observed, often ad hoc via supervisory colleges). Threat intelligence is most shared among banks; incident information flows to regulators and security agencies. Incident reporting is fragmented (taxonomy, time frames from immediately to 72 hours, templates, thresholds) and one-directional. FS-ISAC: anonymised, member-driven. HKMA-MAS (2018): voluntary, timely, effective, confidential, Traffic Light Protocol.
- **Third parties.** Broad: outsourcing including cloud; non-outsourcing services (power, telecoms, hardware, software); interconnected counterparties and FMIs. Board-approved outsourcing framework; contracts covering strategic, compliance, security, continuity, vendor lock-in, counterparty, country, contractual, access and concentration risk; inventory of outsourced functions; rights to inspect and audit (some jurisdictions grant them to supervisors); pooled audits. If an institution outsources, the **ultimate responsibility remains with the institution**. A supervisory college model for large cloud providers could cover blind spots.

Threats, frameworks and Equifax (ORR-9 a, b)

- **Information security is wider than cyber.** Four quadrants: internal vs external causes, theft vs loss or involuntary disclosure, including paper and archives. McAfee 2017: 43% of data leaks initiated by insiders, half unintentionally. COVID changed the threat landscape (remote work, digitalisation, QR codes, social media).
- **Frameworks.** NIST CSF: voluntary; identify, protect, detect, respond, recover. CIS: 18 critical security controls (version 8). ISO/IEC 27001: certification, ISMS, PDCA cycle. CIA: confidentiality and integrity (security), availability (resilience). Behavioural controls (awareness, conduct, training) and technical controls (preventive: architecture, access, firewalls, encryption, patching; detective: DLPD, honeypot). Technical monitoring sits mostly in IT (first line); the information security department is most often second line.
- **Equifax (2017).** Apache Struts flaw exploited via the online dispute portal on 10 March; the patch was released 7 March and an alert sent on 8 March to 400 staff, but it was never applied; hackers undetected for about three months. Five weaknesses: no comprehensive IT asset inventory; failure to enforce patch management (8,500 unpatched vulnerabilities found by internal audit in 2015); inconsistent internal communication; an expired SSL certificate that blinded monitoring of encrypted traffic until 29 July; poor external communication (announced 7 September, 143 million customers, later 147 million). About USD 5bn of USD 15.5bn market capitalisation lost; CEO, CSO, CIO resigned; up to USD 700 million in fines and compensation. Lesson: large events have multiple causes in weak control environments. (Ch.1 Box 1.1 quotes 145 million and USD 3.5bn; use the Ch.9 figures.)

ORMF, characteristics and operational resilience (ORR-1 a, c, d)

- **ORMF and scope.** Operational risk: "the risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events"; includes legal risk, excludes strategic and reputational risk, though RPSMOR 2021 says they should be considered "where appropriate". Frameworks reduce to identification, assessment, mitigation, monitoring, plus governance, culture and appetite. See also A22 for the seven event types (ORR-1 b).
- **Characteristics.** Heterogeneous, idiosyncratic and diffuse ("everyone is a risk manager"; cannot be eliminated), heavy-tailed, interconnected (boundary events with credit and market risk), dynamic (cyber and resilience followed digitalisation; TSB 2018).
- **Operational resilience.** UK first (BoE, PRA, FCA discussion paper, July 2018): the ability to prevent, adapt, respond to, recover from and learn from disruptions; components: continuity of business services, **important business services** (a service-based rather than process-based view: services whose disruption causes intolerable harm to consumers or market integrity), **impact tolerances**, management of disruption, lessons learned. Mindset: severe disruptions are occasionally inevitable, so prevent, detect and correct. Fed (2020) Sound Practices. BCBS Principles (March 2021), seven: governance, operational risk management, business continuity planning and testing, mapping interconnections and interdependencies, third-party dependency management, incident management, ICT including cybersecurity; "operational resilience is an

outcome that benefits from the effective management of operational risk". ECB DORA draft (September 2020); MAS and ABS remote working paper (March 2021). COVID: remote working and critical payments under conditions beyond tested scenarios; fraud attempts tripled and phishing rose six- to seven-fold (see also A29).

Exam traps

- **Absolute claim on 3LD:** most regulators *use* 3LD to assess cyber risk; they do not require it or prescribe the split of responsibilities.
- **Polarity flip on information sharing:** sharing among regulators is the *least* observed type (29%); among banks and bank to regulator are the most common (75%).
- **Scope error on strategy and testing:** a standalone cyber-security strategy is expected but not required; TIBER-EU gives no pass or fail; cyber metrics are not mature and have no standard benchmark.
- **Case cross-mapping on Equifax:** not a zero-day attack; the patch existed for three days and the failures were asset inventory, patch enforcement, communication and an expired SSL certificate.
- **Sibling swap on resilience:** impact tolerances attach to important business *services*, not to internal processes; resilience sits alongside existing operational risk requirements rather than replacing them.
- **Role misassignment on third parties:** outsourcing never transfers ultimate responsibility to the provider or to the supervisor.

Question A23.1 ORR-8 b, d; ORR-9 b · S1 · [Aug-2026] [Map #31]

The chief information security officer of a German bank is drafting a board paper on the bank's cyber risk framework. The paper summarises supervisory practice as described by the Basel Committee's cyber-resilience range of practices report and the lessons the bank has drawn from the Equifax breach. Which of the following statements is correct to include in the paper?

A. Most regulators require banks to implement the three lines of defence model for cyber risk and prescribe how cyber responsibilities are distributed across the three lines.

B. Of the five types of cyber-security information sharing, sharing among regulators is the most widely observed, mainly through supervisory colleges and memoranda of understanding.

C. The Equifax breach exploited a vulnerability for which no patch had yet been released, so the main lesson is to invest in threat intelligence rather than in patch management.

D. Most regulators do not require a standalone cyber-security strategy, but all of them expect a board-approved information security strategy, policy and procedures.

Solution **Answer: D**

Step 1 – Strategy expectations (Section 8.3)
"Although most regulators do not require regulated entities to develop a cyber-security strategy, all expect regulated institutions to have a board-approved information security strategy, policy and procedures under the broad remit of effective oversight of technology."
⇒ *The board approves an information security strategy; cyber risk lives inside that and the firm-wide risk framework.*

Step 2 – Three lines of defence in cyber governance
The majority of regulators have adopted 3LD to *assess* cyber-security risk and controls, but most do not require its implementation and do not prescribe how responsibilities are distributed; banks are expected to define them and leave no gaps.

Step 3 – Information sharing (Section 8.5)
Among banks and from banks to regulators: 75% of jurisdictions each; with security agencies 68%; regulator to banks 32%; among regulators 29%, the least observed type.

Step 4 – Equifax (ORR-9)
The Apache Struts patch was released on 7 March 2017, an alert reached 400 employees on 8 March, and the attack came on 10 March. The patch was never applied. The weaknesses were IT asset inventory, enforcement of patch

management, internal communication, an expired SSL certificate and external crisis communication.

Interpretation

D is correct. The board paper should present the information security strategy as the board's document, then describe how cyber roles are allocated across the lines (the bank's own choice), and cite patch enforcement and asset inventory as the core Equifax lessons.

Why the other options are wrong

A is incorrect. Absolute claim: regulators use 3LD to assess cyber risk; most neither require it nor prescribe the split of responsibilities.

B is incorrect. Polarity flip: sharing among regulators is the least observed type (29%); supervisory colleges and memoranda are the ad hoc channels that explain why.

C is incorrect. Sibling swap (case cross-mapping): describes a zero-day attack; at Equifax the patch had been released three days earlier and the failures were patch enforcement and asset inventory.

LO: ORR-8 b: Describe current practices by banks and supervisors in the governance of a cyber-risk management framework, including roles and responsibilities. · ORR-8 d: Explain and assess current practices for the sharing of cybersecurity information between different types of institutions. · ORR-9 b: Describe lessons learned from the Equifax case study. · Reference: BCBS (2018), GARP ORR Ch.8, Sections 8.3 and 8.5; GARP ORR Ch.9, Equifax case study

Question A23.2

ORR-1 d; ORR-8 e · S2, S3 · [Map #31]

A risk consultant is presenting to the risk committee of a UK banking group a review of how two subsidiaries came through the COVID-19 pandemic:

- a digital payment bank whose customers transact only through its app, with payment processing running on a third-party cloud provider;
- a community bank with a branch network, a local deposit base and mainly face-to-face relationship lending.

The committee asks which subsidiary was more vulnerable from an operational resilience perspective, and what follows for its resilience framework. Which of the following responses is most appropriate?

A. Neither can be ranked, because resilience is assessed process by process, so each bank should set recovery time objectives for its internal processes rather than for its services.

B. The payment bank was less vulnerable, because sufficiently strong preventive IT controls can stop severe operational disruptions to a digital service from ever occurring.

C. The payment bank was more vulnerable, because its important business service rests on IT systems and a third party, the channels COVID-era remote fraud targeted; it should set an impact tolerance for that service.

D. The payment bank was more vulnerable, but outsourcing payment processing to the cloud provider transfers responsibility for the disruption, so the bank need not set its own impact tolerance.

Solution

Answer: C

Step 1 – The regulatory lens (UK first, then BCBS, ECB, MAS)

UK regulators (BoE, PRA, FCA, July 2018) moved from a process-based to a service-based view: firms identify *important business services* whose disruption would cause intolerable harm to consumers or market integrity, and set *impact tolerances* for them. BCBS Principles (March 2021) add mapping of interconnections and interdependencies, third-party dependency management and resilient ICT including cybersecurity. ECB (DORA draft, 2020) is ICT-focused; MAS (2021) addressed remote working risks.

⇒ *The question to ask of each subsidiary is: which important service could be disrupted, and through which dependencies?*

Step 2 – Map each bank's exposure to those principles

Payment bank: one digital service, critical payments, dependence on IT and a critical third-party (cloud) provider; attention to resilience "emanated from the financial industry's critical dependence on IT systems and infrastructure" and on cloud and payment infrastructure providers. Process-based firms such as payment processors may even organise their risk appetite around key processes (ORR-2), and a payment company's taxonomy centres on operations and IT risk (ORR-3 Box 3.4). During COVID, critical payments ran "in the most extreme conditions, much beyond any stress scenario they had tested"; fraud attempts tripled, phishing rose six- to seven-fold and the rise in remote transactions made anomalies harder to detect (ORR-11). Community bank: its service is spread across branches and people rather than concentrated in one IT channel and one provider (applied reasoning from

the scenario; the source does not rank these two bank types).
⇒ *The payment bank concentrates its important business service in exactly the dependencies (ICT, cyber, third party) that the BCBS principles single out and that COVID stressed.*

Step 3 – Who carries the responsibility

If an institution outsources, the ultimate responsibility remains with the institution (ORR-8 e; CEBS guidelines). The payment bank must map the cloud dependency, align the provider's continuity plans with its own and set its own impact tolerance.

Step 4 – Prevention is not enough

After high-profile cyberattacks the industry accepted that severe operational disruptions are inevitable, at least occasionally, and moved from purely preventive management to preventing, detecting and correcting incidents.

Interpretation

C is correct. The recommendation: treat the payment service as an important business service, set an impact tolerance for it, map its ICT and cloud dependencies, and test response and recovery against severe but plausible scenarios, including pandemic-style remote operation and fraud surges.

Why the other options are wrong

A is incorrect. Sibling swap: replaces the service-based view (important business services, impact tolerances) with the process-based view of classic business continuity that the UK approach moved beyond.
B is incorrect. Absolute claim: severe disruptions are accepted as occasionally inevitable; resilience requires detection, response and recovery, not prevention alone.
D is incorrect. Role misassignment: outsourcing does not transfer responsibility; the ultimate responsibility remains with the institution, which must manage the third-party dependency.

LO: ORR-1 d: Describe operational resilience, identify the elements of an operational resilience framework, and summarize regulatory expectations for operational resilience. · ORR-8 e: Describe practices for the governance of risks of interconnected third-party service providers. · Reference: GARP ORR Ch.1, Section 1.6 and Table 1.3; BCBS (2018), GARP ORR Ch.8, Section 8.6; GARP ORR Ch.11 (COVID fraud)

A24 Hedge Funds: Strategies, Survivorship and Other Biases, and Madoff

Readings: IM-8 a, b, c, d · Hedge Fund Investment Strategies (Stowell and Stowell, Ch.12); IM-9 a, b, c, d · Risk, Regulation and Organizational Structure (Stowell and Stowell, Ch.14); IM-14 a, b · Madoff: A Riot of Red Flags (Gregoriou and Lhabitant, 2009) (IM-8, IM-9, IM-14 source: SchweserNotes Book 5, one rung below a GARP text); IM-12 a-g · Performing Due Diligence on Specific Managers and Funds (Mirabile, Ch.12); IM-17 c · Illiquid Assets (Ang, *Asset Management*, Ch.13, Illiquid Asset Reported Returns Are Not Returns)

AUG 2026 RECALL | MAP TIER C · #34 · 3/16 SOURCES | VARDEEZ

Specifics reported

Your August 2026 recall: Hedge fund 2Q; Madoff 1Q.
Map #34: "Reported: 'survivorship bias questions'. The bias sibling set, survivorship, selection/self-selection, back-fill, reporting, is the discrimination GARP tests. Madoff (Reading 96) is rated Very Important and functions as the red-flags case study; treat it as a case↔lesson matching item against the other case studies."
Shapes covered here:
S1 hedge fund strategies: equity, macro, arbitrage, event-driven, distressed (Q1)
S2 hedge fund risks, systemic risk, regulation and structure (Q2)
S3 survivorship bias against its siblings: selection, reporting, backfill (Q3)
S4 Madoff red flags as a case↔lesson matching item against other cases (Q4).

Revision notes

- **The hedge fund edge** (IM-8 a): the ability to short sell, use leverage, use derivatives and take concentrated positions when opportunities arise, where long-only mutual funds are restricted.
- **Equity-based and macro strategies** (IM-8 a): *equity long/short* uses fundamental analysis, long and short related securities, an investment horizon per position, no single short above 2% to 3% of capital; returns are alpha on both legs, the interest rebate on short-sale proceeds and interest on the liquidity buffer; costs are margin financing, borrow fees, dividends on borrowed shares. *Dedicated short* specialises in overvaluation (aggressive accounting, legal issues) against the headwind of rising markets; risks are margin calls, squeezes, forced covering. *Nonhedged equity* is long only with full market exposure, using leverage or a bull call spread. *Global macro* makes large, leveraged, directional bets across equities, rates, FX and commodities and tends to trade *more liquid* instruments; *emerging markets* adds country-specific risk.
- **Arbitrage strategies** (IM-8 b): *fixed-income arbitrage* bets on convergence (short the on-the-run 30-year Treasury, long the cheaper off-the-run $29\frac{1}{2}$-year). *Convertible arbitrage* buys the convertible and shorts the stock, delta hedging continuously; value = investment value + conversion value + option value; returns from coupon, interest on short proceeds, *monetizing volatility* and any undervalued embedded call (example: hedged position gains USD 1.37 on a 5% fall against a USD 19.47 unhedged loss). *Relative value*: pairs trading, statistical arbitrage (including liquidity provision to large sellers), index futures against constituents.
- **Event-driven and merger arbitrage** (IM-8 c): opportunities are strategic (hard catalysts, spin-offs, proxy contests, stub trades), financial, operational, legal/regulatory and technical; activists push boards for buy-backs, dividends, spin-offs, leverage. Merger arbitrage buys the *target* and, in a share-for-share deal, shorts the *acquirer* at the exchange ratio; it earns the spread, target dividends, short interest and any higher bid, but losses when a deal fails "can far exceed the gains". The hedge fund edge is legal and antitrust expertise.
- **Distressed securities** (IM-8 d): positions in firms entering, in, or emerging from bankruptcy: discounted junk debt many institutions cannot hold, bank loan sales, trade claims, capital-structure trades (long senior, short junior). *Active* investors sit on creditor committees, supervise the workout and litigate; *passive* investors stay out and choose simpler situations. Core risk: the restructuring takes longer than expected; also valuation, illiquidity, courts, DIP financing, new senior debt.
- **Risks to investors** (IM-9 a): portfolio-level risks are benchmarking, survivorship bias and UBTI (debt-financed income taxable to tax-exempt investors; an offshore fund mitigates it); investment-level risks are counterparty, conflicts of interest, key person, strategy failure, style drift, regulatory change and correlation spikes. Five incremental risks: leverage (mostly collateralized), lighter regulation (investors must do their

own due diligence), short selling (unlimited loss, squeezes such as GameStop 2021), limited transparency (secrecy, gates), and higher, fee-driven risk tolerance.

- **Systemic risk** (IM-9 b): two channels: a failed strategy unwinding at fire-sale prices (contagion), and losses imposed on lenders and counterparties. LTCM 1998: monitor liquidity and leverage; unrelated assets become correlated in stress. Amaranth 2006: the Bank of England judged key intermediaries far more systemic than hedge funds. Leverage spiral: USD 100 of stock on a USD 75 loan is 3x; a 10% fall makes it 5x; selling USD 30 restores 3x. Banks bear revenue, credit (margin loan) and counterparty risk; Archegos 2021 (a family office using total return swaps) cost Credit Suisse about USD 5 billion and other banks another 5 billion. Mitigants: conservative bank lending, lower leverage and more diversification, and *reasonable* regulation, since overly strict rules reduce the liquidity hedge funds supply as investors of last resort.
- **Regulation** (IM-9 c): '40 Act exemptions 3(c)(1) (fewer than 100 investors) and 3(c)(7) (qualified purchasers, over USD 5 million; '34 Act registration at 500 investors); Reg D accredited investors (USD 2.5 million net worth or USD 250,000 income; Rule 506 allows 35 nonaccredited); ERISA avoided below 25% benefit-plan assets. Dodd-Frank Title IV removed the private adviser exemption: SEC registration, reports to the SEC and FSOC, Form PF (annual from USD 150 million, quarterly above USD 1.5 billion). EU AIFMD (2010; ESMA 2011 interprets, member states enforce; EUR 100 million reporting threshold; passport); Singapore MAS above USD 250 million; Hong Kong SFO licensing; China CSRC bans leverage and limits shorting to 50 CSI 300 names. Regulators' three concerns: investor protection, market integrity, financial stability.
- **Structure and tax** (IM-9 d): domicile for tax and light oversight (Delaware; Cayman 55%, BVI 15%, Bermuda 10%). Open-ended limited partnership: the manager is the *general partner*, investors are *limited partners*. Master-feeder: trading in the master; the *offshore* feeder serves non-US and US tax-exempt investors. Service providers: administrators, auditors, independent valuers (NAV), prime broker. 2 and 20: the management fee is ordinary income (Section 457 deferral ended 2008); the performance fee (carried interest) can be taxed as capital gains if assets are held at least a year.
- **Why funds fail** (IM-12 a): bad or concentrated investment decisions; accounting, valuation or misappropriation frauds; excessive leverage, tail events; a flood of withdrawals at the worst time; squeezes; weak insider-trading supervision; failure of a prime broker such as Lehman or MF Global; liquidity drying up. "If it looks too good to be true, it probably is", but "it's still about returns": do not overweight infrastructure at the expense of return.
- **The due diligence process** (IM-12 b, c): covers the investment process, risk management, the operational environment and the business model; quantitative (checklist) and qualitative (open-ended, culture). Themes: strategy and edge, concentration, gross and net exposure, stop losses, use of shorts and derivatives, capacity, equity ownership of the team, a track record that is comparable, audited, long enough and produced by the team still in place (Greenwich Roundtable), and background and reference checks (a basic report costs about USD 750; read Form ADV). Meet the actual decision makers.
- **Risk management criteria** (IM-12 d): an independent risk function (dedicated risk managers or an independent risk service provider); the *same* risk system for traders and investor reporting; stress testing; valuation policy (who can override prices); leverage and liquidity against peers; tail risk (skewness, kurtosis); and whether the terms suit the strategy (a 2 and 20 long-only fund or a one-year lock-up on liquid long/short equity are red flags).
- **Operational, business model and fraud risk** (IM-12 e, f, g): internal controls, compliance, counterparty diversification (Lehman, MF Global and Refco left funds recovering perhaps 30 to 40 percent of cash at the dealer); documents consistent across the OM, LPA and Form ADV; an *unqualified* audit opinion, no incentive fees in loss years, and general partner withdrawals as a red flag. Business model: red, yellow and green zones; break-even AUM and months of cash. FBI fraud indicators: affiliated broker-dealer execution, liquidity complaints, litigation, unusually strong performance, manager-marked illiquid assets, related-party valuation. The DDQ covers manager information, trading, compliance, AML, disaster recovery, key man, fund terms, service providers, risk controls, financial statements.
- **Madoff: history and claimed strategy** (IM-14 a): BMIS began in 1960 as a *brokerage*; Rule 390; the Cincinnati Stock Exchange; payment for order flow; Madoff chaired the NASDAQ board. The advisory arm (late 1980s, same building) ran 23 discretionary accounts and about USD 17 billion, mostly through feeder funds, so end investors could only diligence the feeders. Split-strike conversion: long a basket correlated with the S&P 100, *sell* OTM S&P 100 calls, *buy* OTM S&P 100 puts; claimed to sit in cash most of the time, positions under 30 days. Reported 8% to 12% a year for about 17 years, no down year, over 95% of months positive, returns similar to the S&P 100 with far lower volatility. Collapse: on 10 December 2008 he confessed to his family because redemptions outran new money; his sons reported him.

- **Madoff: red flags** (IM-14 b): operational: no independent service providers (execution, custody, administration in-house), broker-dealer and market maker for its own strategies, a three-person auditor (Friehling and Horowitz) that told the AICPA it had no audit clients, family in key roles, only brokerage commissions (no management or performance fee), secrecy (feeders barred from naming BMIS), late SEC registration (2006), paper trade confirmations. Investment: a black box with no negative months, style analysis showing exposures *opposite* to the stated strategy, illogical Form 13F filings, an impossible volume of S&P 100 options and no identified OTC counterparties. Markopolos warned the SEC in 2005.
- **Survivorship and other biases** (IM-17 c; IM-7): Ang's three biases that overstate returns and understate risk of illiquid assets: *survivorship bias* (poorly performing funds stop reporting and fail; only observing the entire population removes it; at least 1% to 2% for mutual funds, above 4% comparing defunct and live funds), *infrequent sampling* (volatility, correlation and beta too low), and *sample selection bias* (returns observed only when asset values are high: alpha overstated, the fitted SML flatter, beta and volatility understated; Cochrane: venture capital alpha over 90% falls to -7% once corrected). *Reporting bias*: reporting is voluntary, so funds stop when returns are low or never start if the track record is unattractive. BKM: limiting a sample to funds alive over the whole period creates survivorship bias; most mutual fund databases now include failed funds. *Backfill (instant history) bias*, where a fund joining a database adds its earlier good returns, is the remaining sibling in the map's list (outside-source).

Formulas

Leverage spiral (Stowell, Figure 91.3). Debt-to-equity target k; after a price fall, equity $E' = V' - L$; sale needed to restore k: $s = L - kE'$. Book: $L = 75$, $V' = 90$, $E' = 15$, $k = 3 \Rightarrow s = 75 - 45 = 30$.
Distressed capital-structure trade (Book 5 example). Buy a bond at 30 per 100, two 5% coupons and 35 recovery: gain $100 + 350 - 300 = 150$ per USD 1,000 face; short 100 shares at 3 that go to zero: gain 300; total 450.
Merger arbitrage, share-for-share (Book 5 example). Offer 2 acquirer shares per target share; target 95, acquirer 50: buy 100 target, short 200 acquirer; profit at close $2 \times 50 - 95 = 5$ per target share.

Exam traps

- **Legs reversed (polarity flip):** convertible arbitrage is long the convertible, short the stock; merger arbitrage is long the target, short the acquirer; Madoff *sold* calls and *bought* puts.
- **Strategy label swapped (sibling swap):** long convertible, short stock is convertible arbitrage, not fixed-income arbitrage; global macro trades *more* liquid instruments; activists and active distressed investors engage, passive ones do not.
- **Bias siblings:** dropping dead funds is survivorship bias; recording returns only when values are high is sample selection bias; choosing whether to report at all is reporting bias. Selection bias makes beta and volatility look *lower*, not higher.
- **Role misassignment:** the manager is the general partner; investors are limited partners; the offshore feeder houses US tax-exempt investors; Madoff confessed and his sons reported him (not the SEC).
- **Absolute claims:** "strict regulation always lowers systemic risk"; "reporting bias only comes from funds that stop reporting". Both are false.
- **Case cross-mapping:** correlated assets in stress is LTCM; total return swaps and bank losses is Archegos; cash stuck at a failed prime broker is Lehman, MF Global, Refco; in-house custody, a tiny auditor and impossibly smooth returns is Madoff.

Question A24.1

IM-8 a, b, c, d · S1 · [Aug-2026] [Map #34]

A senior analyst in the alternatives team of a Dutch pension fund is presenting four hedge fund managers to the investment committee. Which of the following descriptions is correct?

A. Manager W runs fixed-income arbitrage: it buys a convertible bond and sells short the issuer's stock, rebalancing the hedge as the bond's delta changes in order to monetize volatility.

B. Manager X runs merger arbitrage on a share-for-share deal: it buys the target and sells short the acquirer, accepting that losses if the deal fails can far exceed gains if it closes.

C. Manager Y runs global macro: it makes large, leveraged directional bets across rates, currencies and com-

modities, and it tends to trade less liquid instruments than other strategies.

D. Manager Z runs passive distressed investing: it sits on creditor committees and litigates to steer the workout, accepting that the restructuring may take longer than expected.

Solution **Answer: B**

Step 1 – Test each mechanic against the reading

Merger (risk) arbitrage: in a share-for-share merger the arbitrageur "may short the acquirer's stock while simultaneously purchasing the target's stock"; if the deal fails the target falls back to (or below) its pre-announcement price and the acquirer may rise, and "the losses from a failed deal can far exceed the gains earned when a deal is successfully completed". B is correct.

Interpretation

Each strategy is identified by its legs and its payoff shape: merger arbitrage resembles a short put on deal completion (small, likely gain; large, less likely loss), which is why hedge funds' legal and antitrust expertise in judging deal probability is their edge.

Why the other options are wrong

A is incorrect. Sibling swap: the mechanic described (long convertible, short stock, delta hedging, monetizing volatility) is convertible arbitrage. Fixed-income arbitrage trades yield divergences between similar bonds, such as on-the-run versus off-the-run Treasuries.

C is incorrect. Polarity flip: global macro funds "tend to trade in more liquid instruments".

D is incorrect. Role misassignment: sitting on creditor committees, supervising the workout and litigating describe *active* distressed investors; passive investors prefer less complex situations and stay uninvolved.

LO: IM-8 a: Describe and compare different types of equity-based and macro hedge fund strategies and explain their return and risk profiles. IM-8 b: Compare different types of arbitrage strategies and describe the mechanics of convertible arbitrage strategies. IM-8 c: Describe the mechanics and risks of event-driven strategies such as merger arbitrage. IM-8 d: Explain the mechanics of a distressed securities strategy, and describe challenges, risks and considerations that may arise when implementing this strategy. · Reference: Stowell and Stowell, Ch.12 (source: SchweserNotes Book 5, Reading 90, one rung below a GARP text)

Question A24.2 IM-9 a, b, c, d · S2 · [Aug-2026] [Map #34]

The CRO of a US university endowment is preparing a report for the board on the risks of investing in hedge funds and on how hedge funds are regulated and structured. Which of the following statements is correct to include in the report?

A. A fund holding USD 100 of stock on a USD 75 loan that sees a 10% price fall must sell USD 30 of stock to return to its target leverage, the forced-selling channel through which hedge funds can spread losses.

B. In a master-feeder structure, US tax-exempt investors such as pension funds invest through the onshore feeder, which keeps debt-financed income from creating unrelated business taxable income for them.

C. Because hedge funds can transmit losses to banks and markets, the reading recommends the strictest feasible regulation of hedge funds as the most effective way to reduce their systemic risk.

D. In the limited partnership, investors act as the general partners and the manager as a limited partner, and the performance fee can be taxed at capital gains rates if assets are held at least a year.

Solution **Answer: A**

Step 1 – Verify the leverage spiral

Before: stock 100, loan 75, equity 25, debt to equity 3.0x. After a 10% fall: stock 90, loan 75, equity 15, so 5.0x. To get back to 3.0x with equity of 15 the loan must fall to 45: sell $75 - 45 = 30$ of stock, leaving stock 60, loan 45, equity 15.

$\Rightarrow$ *A 10% price fall forces the sale of 30% of the original position, with no fundamental reason to sell: this is how forced selling becomes contagion.*

Interpretation

Stowell's two systemic channels are fire-sale contagion from a failing strategy and losses imposed on lenders and counterparties (revenue, credit and counterparty risk for banks, as in Archegos). The spiral in A is the first channel.

Why the other options are wrong

B is incorrect. Sibling swap: the *offshore* feeder is used for non-US investors and US tax-exempt investors such as pension funds; investing through an offshore fund is the stated mitigant for UBTI.
C is incorrect. Absolute claim: the mitigants are conservative bank lending, lower leverage with more diversification, and *reasonable* regulation; excessively strict regulation can reduce the liquidity hedge funds supply as investors of last resort and may raise systemic risk.
D is incorrect. Role misassignment: the manager is the general partner and the investors are the limited partners. The tax clause is true (carried interest).

LO: IM-9 a: Describe different risks faced by investors in hedge funds. IM-9 b: Explain how the activities of hedge funds and exposure to hedge funds can increase systemic risk. IM-9 c: Summarize different approaches taken by regulators to mitigate risks related to hedge funds. IM-9 d: Describe a typical organizational structure for a hedge fund and explain motivations for adopting this structure. · Reference: Stowell and Stowell, Ch.14 (source: SchweserNotes Book 5, Reading 91, one rung below a GARP text)

Question A24.3

IM-17 c; IM-9 a · S3 · [Map #34]

An investment consultant at a Canadian endowment is studying a commercial database of hedge fund and private equity returns. Funds report to the database voluntarily. Over the last ten years, the average annual return of funds still reporting today is 9.8%; including funds that stopped reporting or closed during the period, it is 7.1%. Which of the following statements should the consultant include in the report?

A. The 2.7-point gap reflects sample selection bias: returns are recorded only when underlying asset values are high, as when buyout companies are taken public in strong markets.

B. The 2.7-point gap reflects survivorship bias: poorly performing funds stop reporting and drop out, and only observing the entire population of funds removes the bias completely.

C. Sample selection bias in private equity data makes the fitted security market line steeper than the true line, so beta and volatility estimated from reported deals are overstated.

D. Reporting bias arises only when a fund that has been reporting stops after poor returns, so funds that never begin reporting cannot bias the database's average return.

Solution

Answer: B

Step 1 – Name the bias that the data isolate

The two averages differ only in whether dead and non-reporting funds are kept. Ang: "Survivorship bias results from the tendency of poorly performing funds to stop reporting", and "the only way to completely remove the effect of survivorship bias is to observe the entire population of funds". The 2.7-point gap is of the size Ang reports (at least 1% to 2% for mutual funds, above 4% when comparing defunct and live funds, both lower bounds for illiquid asset managers). B is correct.

Interpretation

Using the live-fund average would overstate expected returns by about 2.7 points a year. BKM gives the same warning for mutual funds: limiting a sample to funds alive over the whole period creates survivorship bias, which is why most mutual fund databases now include failed funds. Stowell lists survivorship bias among the portfolio-level risks of hedge fund investing.

Why the other options are wrong

A is incorrect. Sibling swap: recording returns only when values are high is sample selection bias, a different mechanism; the gap here comes from dropping dead funds.
C is incorrect. Polarity flip: with selection bias the fitted SML is *flatter* than the true SML, so beta and volatility are *underestimated* while alpha and expected return are overestimated.
D is incorrect. Scope error: Ang states that reporting bias "also occurs when you don't start reporting your returns in the first place because your fund never achieves a sufficiently attractive track record".

LO: IM-17 c: Assess the impact of biases on reported returns for illiquid assets. IM-9 a: Describe different risks faced by investors in hedge funds. · Reference: Ang, *Asset Management*, Ch.13, Survivorship Bias and Selection Bias; Bodie, Kane and Marcus, Ch.24, Selection Bias and Portfolio Evaluation; Stowell and Stowell, Ch.14 (SchweserNotes Book 5, Reading 91)

Question A24.4 IM-14 a, b; IM-12 a, e; IM-9 b · S4 · [Aug-2026] [Map #34]

A risk consultant is presenting failed hedge fund cases to the due diligence committee of a fund of hedge funds. Which of the following lessons is best illustrated by the case of Bernard L. Madoff Investment Securities (BMIS)?

A. Assets that normally behave independently can become highly correlated in extreme stress, so a fund's leverage and liquidity must be monitored closely.

B. Unusually high management and performance fees gave the manager an incentive to report smooth returns, which the feeder funds' due diligence failed to question.

C. Keeping execution, custody and administration in-house with a three-person auditor allowed returns with over 95% positive months to be fabricated undetected.

D. A fund can fail through no fault of its own when its prime broker goes bankrupt, because cash held as margin may be recovered only in part after litigation.

Solution **Answer: C**

Step 1 – Match the case to its lesson

BMIS performed execution, custody, administration and management itself ("no independent service providers"), was audited by Friehling and Horowitz, a three-person firm in a 550 square foot office that had told the AICPA it had no audit clients, and reported 8% to 12% a year with no down year and over 95% of months positive. In-house functions made it easy to fabricate performance; independent verification of service providers would have exposed it. C is correct.

Interpretation

Mirabile's operational due diligence lesson in one line: independent administration, audits and top-tier service providers are "generally not negotiable", and "if it looks too good to be true, it probably is". Madoff was not found by investigators: he confessed to his family on 10 December 2008.

Why the other options are wrong

A is incorrect. Case cross-mapping: this is the lesson of LTCM (1998): monitor liquidity and leverage, because assets that usually behave independently move together in stress.

B is incorrect. Polarity flip: BMIS charged *no* management or performance fee, only brokerage commissions. Forgoing fees was the red flag, and it left room for feeder funds to charge high fees of their own.

D is incorrect. Case cross-mapping: this is the lesson of Lehman, MF Global and Refco in Mirabile's counterparty discussion, where funds recovered perhaps 30 to 40 percent of cash held at the failed dealer.

LO: IM-14 b: Identify characteristics of Bernie Madoff's hedge fund that can serve as warning signs or potential "red flags" and explain how they can indicate potential fraud risk. IM-14 a: Explain the history of Bernie Madoff's hedge fund and describe the fund's reported strategy and its reported performance. IM-12 e: Explain how due diligence can be performed on a hedge fund's operational environment. · Reference: Gregoriou and Lhabitant (2009) (source: SchweserNotes Book 5, Reading 96, one rung below a GARP text); Mirabile, Ch.12; Stowell and Stowell, Ch.14 (Book 5, Reading 91)

A25 Gauss+ and Vasicek

Readings: MR-16 a, b, c, d · The Vasicek and Gauss+ Models (Tuckman & Serrat 4e, Ch.9; 16.1 to 16.4, eqs. 16.1 to 16.15, Table 16.1); MR-14 f, g, h · The Art of Term Structure Models: Drift (Tuckman & Serrat 3e, Ch.9; 14.5 The Vasicek Model: Mean Reversion)

AUG 2026 RECALL | MAP TIER D · #40 · 2/16 SOURCES | VARDEEZ

Specifics reported

Your August 2026 recall: Gauss+
Map #40: "Closed-book: $E[r_T] = r_0 e^{-kT} + \theta(1 - e^{-kT})$; half-life $= \ln 2/k$. The sheet says 'Vasicek model calculation can come' and separately flags LO 14.g (expected rate in T years). The Gauss+ multi-factor structure is the genuinely new material."
Shapes covered here:
S1 closed-book $E[r_T] = r_0 e^{-kT} + \theta(1 - e^{-kT})$, with θ built from r_∞ and the risk premium (Q4)
S2 closed-book half-life $\ln 2/k$ (Q5; Gauss+ half-lives in Q3)
S3 Vasicek model calculation: rate change and standard deviation (Q4 one-month drift; Q5 terminal standard deviation)
S4 LO 14.g expected rate in T years (Q4)
S5 Gauss+ multi-factor structure: no shock on r, the $r \to m \to l \to \mu$ cascade, two sources of risk (Q1, Q2, Q3)
S6 August plain form: Gauss+ factor changes (Q1 short rate, Q2 medium-term factor, Q3 long-term factor).

Revision notes

- **The specific: Vasicek expectation and half-life** (MR-14 g). $dr = k(\theta - r)dt + \sigma dw$. The distance between the current rate and its goal decays exponentially at the mean-reverting rate, so $E[r_T] = r_0 e^{-kT} + \theta(1 - e^{-kT})$: a weighted average of today's rate and θ, with the weight on today's rate decaying at speed k. Book example ($r_0 = 5.121\%$, $\theta = 15.339\%$, $k = 0.025$): distance $10.218\% \times e^{-0.25} = 7.9578\%$, so $E[r_{10}] = 7.3812\%$. Half-life (time to cover half the distance) $= \ln 2/k = 27.73$ years; larger k, shorter half-life.
- **Risk-neutral drift and risk premium** (MR-14 f; MR-16 b). If the true process reverts to r_∞ and a constant risk premium λ enters as a drift, $dr = k([r_\infty + \lambda/k] - r)dt + \sigma dw$, so $\theta = r_\infty + \lambda/k$. Very many (an infinite number of) combinations of r_∞ and λ give the same θ and the same prices: expectations cannot be distinguished from risk premium by observing security prices. Book: $r_\infty = 6.179\%$, $\lambda = 0.229\%$, $k = 0.025$ give $\theta = 15.339\%$.
- **Rate change and its standard deviation** (MR-14 g). One-month expected change $= k(\theta - r)dt$: $0.025 \times (15.339\% - 5.121\%)/12 = 2.13$ bps; volatility $126 \times \sqrt{1/12} = 36.4$ bps. MR-16 example ($r = 2\%$, $k = 0.0165$, $\theta = 11\%$, $\sigma = 95$ bps): drift 14.85 bps a year or about 1.2 bps a month; shock 27.4 bps a month; half-life over 42 years.
- **Terminal standard deviation** (MR-14 g). $\sqrt{\sigma^2(1 - e^{-2kT})/(2k)}$: 353 bps after 10 years in the book example versus $\sigma\sqrt{T} = 398$ bps without mean reversion. "Per year" volatility does not mean the terminal standard deviation grows with $\sqrt{T}$: the pull toward θ dampens it, particularly at long horizons.
- **The Vasicek tree** (MR-14 f). The obvious tree does not recombine because the drift grows with the distance from θ: from 5.5060% the down move gives 5.1628%, from 4.7786% the up move gives 5.1643%. Recombining method: fix the date-2 centre node at the expected rate after two steps (5.1635%), then solve for r_{uu} and p (5.8909%, 0.4990) and r_{dd} and q (4.4361%, 0.5011) so each branch matches the process's expected rate and standard deviation (36.37 bps); probabilities are no longer $\frac{1}{2}$ away from the centre.
- **Effectiveness of Vasicek** (MR-14 h). Par-rate volatilities decline with term (downward sloping everywhere, Figure 14.7), better than Model 1's flat structure but overstating short-term volatility. A 10 bp rise in the short rate lifts the 30-year spot rate only about 7 bps: not a parallel-shift model. The 30-year convexity effect is about 75 bps versus 135 bps without mean reversion. Mean reversion measures the length of economic news: low k (long half-life) means long-lived news. Calibrated to the 2- and 10-year par rates, the model is probably not flexible enough for the range of term structures observed.
- **Vasicek uses** (MR-16 b). Simple, with analytic formulae for expectations, variance, forwards and spots; one factor explains a large fraction of term structure variability, particularly at longer maturities. Reasonable to

price, compare and hedge long-term bonds first callable after some intermediate number of years (match 10- to 30-year noncallables and the 10-year rate volatility), and for trading long bonds against a model 10-year rate. Forward rate volatility is σe^{-kt}: the volatility term structure is downward sloping.

- **Vasicek weaknesses** (MR-16 b). Not widely used because too simple: (1) not flexible enough to approximate the wide variety of observed term structures; (2) a second factor captures significantly more variability, and more than one factor is needed to hedge intermediate- and shorter-term bonds; (3) it cannot capture the term structure of volatility that rises quickly with term and then flattens or declines, so it cannot handle volatility products spread across the curve.
- **Gauss+ structure** (MR-16 a). Cascade of three factors: the short rate r reverts to the medium-term factor m (business cycle and monetary policy); m reverts to the long-term factor l (long-term inflation and real rate expectations, driven by demographics and production technology); l reverts to the constant μ (long-term expectation of the short rate plus a risk premium). Expected speeds: $\alpha_r > \alpha_m > \alpha_l$.
- **No shock on** r (MR-16 a). Equation 16.9 has no volatility term, mirroring a central bank that pegs the policy rate and moves the target gradually. This keeps short-rate volatility low, so Gauss+ can match the empirically observed *hump-shaped* term structure of volatility. "Gauss" means rates are normally distributed; "+" means three factors (state variables) but only two sources of risk, dW^1 and dW^2.
- **Correlation and risk premium** (MR-16 a, c). The random terms in dm give it volatility $\sigma_m\sqrt{dt}$ and correlation ρ with dl (eqs. 16.13 to 16.15). Equations 16.9 to 16.12 are risk-neutral dynamics; the application assumes only the long-term factor earns a risk premium.
- **Factor effects** (MR-16 a). Larger α_r: changes in m and l reach the term structure faster. Larger α_m: m converges to l faster and affects longer yields less. Smaller α_l: the effect of l on all longer-term yields is more parallel. The short factor affects only the very short end; the medium factor drives the 2- to 3-year region (a "forward shadow rate" that can go deeply negative near the zero bound); l peaks at six to eight years and beyond 10 years is the sole driver of forward changes.
- **Estimation in stages** (MR-16 d). (1) r = fed funds target each day (repo rates jump idiosyncratically). (2) Choose $\alpha_r, \alpha_m, \alpha_l$ to match the coefficients of regressing yield changes on 2-year and 10-year yield changes (these depend only on mean reversion parameters). (3) Choose σ_m, σ_l, ρ to match the term structure of volatility. (4) Choose μ to minimise squared yield errors over the sample. Daily, the factors fit the fed funds rate and the one-year rates two and 10 years forward. Much easier than maximum likelihood. Table 16.1 half-lives: about 8 months, 13 months and 42 years.
- **Trading with Gauss+** (MR-16 b). Popular for proprietary relative value and macro-style trading. Relative value: positions hedged against factor changes, e.g. receive a forward that is high to the model and converging (signal = 5-day minus 40-day moving average of fitting errors), pool trades to neutralise factor exposure, and pair nine- and five-year signals. Macro: views on the factors; with the assumption that expectations stop changing past some future date, long forwards split into expectations, risk premium and convexity (the 10-year risk premium often moves opposite to rates).
- **See also** A7 for Models 1 to 3 and Ho-Lee (Model 3 with matching volatility, decay and drift reproduces the Vasicek terminal distribution but stays parallel-shift).

Formulas

Vasicek (14.8, 16.1). $dr = k(\theta - r)\,dt + \sigma\,dw$; $\theta = r_\infty + \lambda/k$ (14.10, 16.3).
Rate change. $E[dr] = k(\theta - r)\,dt$; SD of the change $= \sigma\sqrt{dt}$.
Expected rate in T years (14.23, 16.4). $E[r_T] = r_0 e^{-kT} + \theta(1 - e^{-kT})$
Variance in T years (14.26, 16.5). $V[r_T] = \sigma^2 \dfrac{1 - e^{-2kT}}{2k}$; no mean reversion: SD $= \sigma\sqrt{T}$.
Half-life (14.25, 16.8). $h = \ln 2/k$; forward-rate volatility σe^{-kt}.
Recombining tree (14.16, 14.17). $p\,r_{uu} + (1-p)\,r_{ud} = E[r \mid r_u]$; $\sqrt{p(r_{uu} - E)^2 + (1-p)(r_{ud} - E)^2} = \sigma\sqrt{dt}$.
Gauss+ cascade, risk-neutral (16.9 to 16.12).

$$
\begin{aligned}
dr_t &= -\alpha_r(r_t - m_t)\,dt && \text{no random shock} \\
dm_t &= -\alpha_m(m_t - l_t)\,dt + \sigma_m\big(\rho\,dW_t^1 + \sqrt{1-\rho^2}\,dW_t^2\big) \\
dl_t &= -\alpha_l(l_t - \mu)\,dt + \sigma_l\,dW_t^1, \qquad E\big[dW_t^1 dW_t^2\big] = 0
\end{aligned}
$$

Implied moments (16.13 to 16.15). SD$(dm) = \sigma_m\sqrt{dt}$; Cov$(dm, dl) = \rho\sigma_m\sigma_l\,dt$; Corr$(dm, dl) = \rho$.

Table 16.1 (US Treasury zeros, Jan 2014 to Jan 2022). $\alpha_r = 1.0547$ (half-life 0.66 yr); $\alpha_m = 0.6358$ (1.09 yr); $\alpha_l = 0.0165$ (42.01 yr); $\sigma_m = 109.2$ bps; $\sigma_l = 96.4$ bps; $\rho = 0.212$; $\mu = 10.555\%$.

Exam traps

- θ **versus** r_∞ **(wrong-input twin):** the risk-neutral process reverts to $\theta = r_\infty + \lambda/k$. Using the real-world long-run rate, or adding λ without dividing by k, understates the pricing expectation.
- **Weights in** $E[r_T]$**:** e^{-kT} multiplies r_0, $(1 - e^{-kT})$ multiplies θ. Swapping them leaves the rate too close to θ for short horizons and too far for long ones.
- **Half-life and SD:** half-life is $\ln 2/k$, not $1/k$. The terminal SD uses e^{-2kT} and $2k$; $\sigma\sqrt{T}$ ignores mean reversion and overstates it.
- **Shock on** r **in Gauss+:** equation 16.9 has no dW term. Adding a Vasicek-style σdw to the short rate is the defining error (and the reason for the "+").
- **Wrong speed or target (sibling swap):** r uses α_r and reverts to m; m uses α_m and reverts to l; l uses α_l and reverts to μ. A factor above its target drifts down: the sign is $-\alpha(\text{factor} - \text{target})$.
- **Shock scaling:** drifts are annual and need dt; a realised dW already contains $\sqrt{dt}$. The medium factor loads ρ on dW^1 and $\sqrt{1-\rho^2}$ on dW^2.

Question A25.1

MR-16 a, c · S5, S6 · [Aug-2026] [Map #40]

A junior risk analyst at a US asset manager is presenting the Gauss+ model to a group of newly hired analysts, using the book's estimated parameters: $\alpha_r = 1.0547$, $\alpha_m = 0.6358$, $\alpha_l = 0.0165$, $\sigma_m = 109.2$ bps, $\sigma_l = 96.4$ bps. Today the short-term rate is $r = 5.25\%$ and the medium-term factor is $m = 4.10\%$. The desk's previous one-factor Vasicek implementation used a short-rate volatility of 95 bps, and this month's realised increment is $dw = +0.20$ (already scaled by $\sqrt{dt}$). Which of the following is closest to the short-term rate one month from now ($dt = 1/12$) under Gauss+?

A. 5.149%

B. 5.189%

C. 5.339%

D. 5.351%

Solution

Answer: A

Step 1 – State the equation

$$dr_t = -\alpha_r(r_t - m_t)\,dt \qquad \text{(Equation 16.9, no random term)}$$

⇒ *The short-term rate is pushed only toward the medium-term factor, the way a central bank moves its policy target gradually.*

Step 2 – Gap to the target

$$r - m = 5.25\% - 4.10\% = 1.15\%$$

⇒ *The short rate sits 115 bps above where the medium-term factor says it is heading.*

Step 3 – Annualised drift

$$-\alpha_r(r - m) = -1.0547 \times 1.15\% = -1.2129\%$$

⇒ *At the current gap, the short rate would fall about 121 bps over a full year.*

Step 4 – One-month change and new level

$$dr = -1.2129\% \times \tfrac{1}{12} = -0.1011\%, \qquad r_{1/12} = 5.25\% - 0.1011\% = 5.149\%$$

⇒ *With no shock in Equation 16.9, the short rate's path over dt is deterministic: the expected and actual moves coincide.*

Interpretation

The short rate falls about 10 bps toward the medium-term factor, reflecting a market that expects easing. Neither σ_m, σ_l nor the Vasicek volatility and dw enter: the absence of a shock on r is what keeps short-rate volatility low and lets Gauss+ match the hump-shaped volatility term structure.

Why the other options are wrong

B is incorrect. Wrong speed (sibling swap): used $\alpha_m = 0.6358$: $5.25\% - 0.6358 \times 1.15\%/12 = 5.189\%$.
C is incorrect. Added a Vasicek-style shock to the short rate: $5.149\% + 0.95\% \times 0.20 = 5.339\%$. Equation 16.9 has no random term.
D is incorrect. Sign flip: wrote the drift as $+\alpha_r(r - m)dt$, so the rate moves away from its target: $5.25\% + 0.1011\% = 5.351\%$.

LO: MR-16 c: Calculate changes in the short-term, medium-term, and long-term interest rate factors under the Gauss+ model. MR-16 a: Describe the structure of the Gauss+ model and discuss the implications of this structure for the model's ability to replicate empirically observed interest rate dynamics. · Reference: Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.2 eq. 16.9, Table 16.1

Question A25.2

MR-16 a, c · S5, S6 · [Aug-2026] [Map #40]

A rates strategist at a hedge fund is simulating one monthly step ($dt = 1/12$) of the Gauss+ medium-term factor with the book's parameters: $\alpha_m = 0.6358$, $\sigma_m = 109.2$ bps (1.092% per year), $\rho = 0.212$. Current factor values are $r = 1.75\%$, $m = 3.00\%$ and $l = 4.20\%$. The realised increments for the month are $dW^1 = -0.2309$ and $dW^2 = -0.3464$. Which of the following is closest to the change in the medium-term factor?

A. -56.7 bps

B. -48.9 bps

C. -36.0 bps

D. $+6.4$ bps

Solution

Answer: C

Step 1 – State the equation

$$dm_t = -\alpha_m(m_t - l_t)\,dt + \sigma_m\left(\rho\,dW^1 + \sqrt{1-\rho^2}\,dW^2\right) \qquad \text{(Equation 16.10)}$$

$\Rightarrow$ *m reverts to the long-term factor l, not to r and not to μ; the short rate $r = 1.75\%$ is irrelevant here.*

Step 2 – Drift over the month

$$-\alpha_m(m - l)\,dt = -0.6358 \times (3.00\% - 4.20\%) \times \tfrac{1}{12} = +0.0636\%$$

$\Rightarrow$ *m is 120 bps below l, so the trend pulls it up by about 6.4 bps this month.*

Step 3 – Loading on the second shock

$$\sqrt{1-\rho^2} = \sqrt{1 - 0.212^2} = 0.97727$$

$\Rightarrow$ *This weight gives the medium-term fluctuation volatility $\sigma_m\sqrt{dt}$ and correlation exactly ρ with the long-term factor (Equations 16.13 to 16.15).*

Step 4 – Combined standardised shock

$$\rho\,dW^1 + \sqrt{1-\rho^2}\,dW^2 = 0.212 \times (-0.2309) + 0.97727 \times (-0.3464) = -0.04895 - 0.33853 = -0.38748$$

$\Rightarrow$ *Most of the medium-term move comes from the factor's own independent shock dW^2.*

Step 5 – Scale by volatility

$$\sigma_m \times (-0.38748) = 1.092\% \times (-0.38748) = -0.4231\%$$

Step 6 – Total change

$$dm = 0.0636\% - 0.4231\% = -0.3595\% \approx -36.0 \text{ bps}$$

$\Rightarrow$ *The negative shock overwhelms the upward drift toward l.*

Interpretation

The medium-term factor, the market's "forward shadow rate" for where the short rate will be in two to three years, fell about 36 bps. Since r reverts quickly to m ($\alpha_r = 1.0547$), this lowers the expected path of the short rate over the next year or so.

Why the other options are wrong

A is incorrect. Dropped the correlation loadings and added the raw increments: $0.0636\% + 1.092\% \times (-0.2309 - 0.3464) = -56.7$ bps.

B is incorrect. Wrong target (sibling swap): reverted m to r instead of l: $-0.6358 \times (3.00\% - 1.75\%)/12 - 0.4231\% = -48.9$ bps.

D is incorrect. Intermediate result: the drift alone, $+6.4$ bps, treating m like the shock-free short rate.

LO: MR-16 c: Calculate changes in the short-term, medium-term, and long-term interest rate factors under the Gauss+ model. · Reference: Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.2 eqs. 16.10 and 16.13 to 16.15, Table 16.1

Question A25.3

MR-16 a, c · S2, S5, S6 · [Aug-2026] [Map #40]

A model validator at a European bank is checking a monthly ($dt = 1/12$) simulation step of the long-term factor in a EUR implementation of the Gauss+ model that reuses the book's parameters: $\alpha_l = 0.0165$, $\alpha_m = 0.6358$, $\sigma_l = 96.4$ bps (0.964% per year), $\mu = 10.555\%$ and $\rho = 0.212$. The long-term factor is $l = 3.50\%$ and the medium-term factor is $m = 2.80\%$. The realised increment for the month is $dW^1 = +0.15$ (already scaled, so it includes $\sqrt{dt}$). Which of the following is closest to the change in the long-term factor?

A. $+5.1$ bps

B. $+15.4$ bps

C. $+26.1$ bps

D. $+51.8$ bps

Solution

Answer: B

Step 1 – State the equation

$$dl_t = -\alpha_l(l_t - \mu)\,dt + \sigma_l\,dW^1 \qquad \text{(Equation 16.11)}$$

$\Rightarrow$ *l reverts to the constant μ at speed α_l and loads only on dW^1; m, α_m and ρ do not enter.*

Step 2 – Gap to the target

$$l - \mu = 3.50\% - 10.555\% = -7.055\%$$

Step 3 – Drift over the month

$$-\alpha_l(l - \mu)\,dt = -0.0165 \times (-7.055\%) \times \tfrac{1}{12} = +0.0097\%$$

$\Rightarrow$ *Despite the 7-point gap, the slowest speed in the cascade (half-life $\ln 2/0.0165 = 42.0$ years) moves l up by only about 1 bp a month.*

Step 4 – Random fluctuation

$$\sigma_l\,dW^1 = 0.964\% \times 0.15 = 0.1446\%$$

$\Rightarrow$ *The shock adds about 14.5 bps; the monthly standard deviation of this term is $0.964\% \times \sqrt{1/12} = 27.8$ bps.*

Step 5 – Total change

$$dl = 0.0097\% + 0.1446\% = 0.1543\% \approx +15.4 \text{ bps}$$

Interpretation

Because α_l is so small, l behaves almost like a random walk: its effect on longer-term forwards is nearly parallel and, beyond 10 years, it is the sole factor driving forward-rate changes. A 15 bp rise in l lifts the long end almost one-for-one.

Why the other options are wrong

A is incorrect. Double $\sqrt{dt}$: multiplied the already-scaled increment by $\sqrt{1/12}$ again: $0.97 + 14.46 \times 0.28868 = 5.1$ bps.
C is incorrect. Dropped dt on the drift: $0.0165 \times 7.055\% = 11.64$ bps plus 14.46 bps $= 26.1$ bps.
D is incorrect. Wrong speed (sibling swap): used α_m: $0.6358 \times 7.055\%/12 = 37.38$ bps plus 14.46 bps $= 51.8$ bps.

LO: MR-16 c: Calculate changes in the short-term, medium-term, and long-term interest rate factors under the Gauss+ model. · Reference: Tuckman & Serrat, Fixed Income Securities 4e, Ch.9, 16.2 eq. 16.11, Table 16.1

Question A25.4

MR-14 g; MR-16 b · S1, S3, S4 · [Map #40]

A treasury analyst at a Brazilian bank prices BRL rate products with a Vasicek model. The economist's real-world view and the desk's risk premium are:

- Current short rate r_0: 2.00%; speed of mean reversion k: 0.15
- Real-world long-run value of the short rate r_∞: 4.00%
- Risk premium entering as a constant drift λ: 0.30% per year

For pricing, which of the following is closest to the expected short rate in 5 years under the risk-neutral process, and to the expected change in the short rate over the next month?

A. 3.06%; +2.5 bps

B. 3.21%; +2.9 bps

C. 3.89%; +5.0 bps

D. 4.11%; +5.0 bps

Solution

Answer: D

Step 1 – Risk-neutral long-run value

$$\theta = r_\infty + \frac{\lambda}{k} = 4.00\% + \frac{0.30\%}{0.15} = 6.00\%$$

⇒ *Prices reflect expectations plus risk premium; the risk-neutral process reverts to 6.00%, not to the economist's 4.00%.*

Step 2 – Decay factor

$$e^{-kT} = e^{-0.15\times 5} = e^{-0.75} = 0.47237$$

⇒ *After 5 years, 47.2% of today's 4-point gap to θ remains.*

Step 3 – Expected rate in 5 years

$$E[r_5] = r_0 e^{-kT} + \theta(1 - e^{-kT}) = 2.00\%(0.47237) + 6.00\%(0.52763) = 0.9447\% + 3.1658\% = 4.11\%$$

⇒ *Equivalently, θ minus the remaining gap: $6.00\% - 4.00\% \times 0.47237 = 4.11\%$.*

Step 4 – One-month expected change

$$E[dr] = k(\theta - r_0)\,dt = 0.15 \times (6.00\% - 2.00\%) \times \tfrac{1}{12} = 0.050\% = +5.0 \text{ bps}$$

Interpretation

The half-life is $\ln 2/0.15 = 4.6$ years, so after 5 years the expected rate has covered a little more than half the distance to θ. Because the drift shrinks as the gap closes, the 5-year move (211 bps) is far less than 60 months of today's 5 bp drift.

Why the other options are wrong

A is incorrect. Wrong-input twin: reverted to the real-world $r_\infty = 4.00\%$ instead of θ: $2.00\%(0.47237) + 4.00\%(0.52763) = 3.06\%$; drift $0.15 \times 2.00\%/12 = 2.5$ bps.
B is incorrect. Dropped the division by k: $\theta = 4.00\% + 0.30\% = 4.30\%$, giving 3.21% and $0.15 \times 2.30\%/12 = 2.9$ bps.

C is incorrect. Swapped the weights: $2.00\%(0.52763) + 6.00\%(0.47237) = 3.89\%$; the one-month drift is unaffected.

LO: MR-14 g: Calculate the Vasicek Model rate change, standard deviation of the rate change, expected rate in T years, and half-life. MR-16 b: Compare and contrast the dynamics, features, and applications of the Vasicek model and the Gauss+ model. · Reference: Tuckman & Serrat, Fixed Income Securities 3e, Ch.9, 14.5 eqs. 14.9, 14.10, 14.23; 4e, Ch.9, 16.1 eqs. 16.2 to 16.4

Question A25.5

MR-14 g, h · S2, S3 · [Map #40]

A risk consultant at an Australian bank is explaining to the board risk committee how long rate shocks persist in the bank's Vasicek model. Parameters: $r_0 = 3.50\%$, $\theta = 5.00\%$, $k = 0.08$, $\sigma = 110$ bps per year. Which of the following is closest to the half-life of the short rate, and to the standard deviation of the short rate 10 years from now?

A. Half-life 8.66 years; standard deviation 2.04%

B. Half-life 8.66 years; standard deviation 2.46%

C. Half-life 8.66 years; standard deviation 3.48%

D. Half-life 12.50 years; standard deviation 2.46%

Solution

Answer: B

Step 1 – Half-life

$$h = \frac{\ln 2}{k} = \frac{0.69315}{0.08} = 8.66 \text{ years}$$

$\Rightarrow$ *A shock to the short rate takes about 8.7 years to decay halfway toward θ. r_0 and θ do not enter.*

Step 2 – Variance factor

$$\frac{1 - e^{-2kT}}{2k} = \frac{1 - e^{-1.6}}{0.16} = \frac{1 - 0.20190}{0.16} = 4.9881$$

Step 3 – Terminal standard deviation

$$\text{SD} = \sigma\sqrt{4.9881} = 1.10\% \times 2.2334 = 2.46\%$$

$\Rightarrow$ *The short rate in 10 years is normal with a standard deviation of about 246 bps around its expectation.*

Interpretation

Without mean reversion the standard deviation would be $1.10\%\sqrt{10} = 3.48\%$: the pull toward θ dampens dispersion, particularly at long horizons, and the same effect makes the volatility term structure slope downward. For the committee, an 8.7-year half-life means rate news is fairly long-lived in this calibration.

Why the other options are wrong

A is incorrect. Dropped the 2 in the exponent: $1.10\%\sqrt{(1 - e^{-0.8})/0.16} = 2.04\%$.
C is incorrect. Ignored mean reversion: $\sigma\sqrt{T} = 1.10\%\sqrt{10} = 3.48\%$.
D is incorrect. Used $1/k = 12.5$ years (the exponential time constant) instead of $\ln 2/k$.

LO: MR-14 g: Calculate the Vasicek Model rate change, standard deviation of the rate change, expected rate in T years, and half-life. MR-14 h: Describe the effectiveness of the Vasicek Model. · Reference: Tuckman & Serrat, Fixed Income Securities 3e, Ch.9, 14.5 eqs. 14.25 to 14.27, Figure 14.5

A26 Monitoring Liquidity: TSECF, TSAA and Liquidity Stress Testing

Readings: LTR-7 a, b, c, d, e · Monitoring Liquidity (Castagna and Fede, Ch.6); LTR-9 a, b, c · Liquidity Stress Testing (Venkat and Baird, Ch.3); LTR-10 a, b, c · Liquidity Risk Reporting and Stress Testing (Choudhry, Ch.14); LTR-5 c · Liquidity and Reserves Management (Rose and Hudgins, Ch.11)

AUG 2026 RECALL | MAP TIER D · #64 · 2/16 SOURCES | VARDEEZ

Specifics reported

Your August 2026 recall: TSEF and TSAA numerical.
Map #64: 'Reported: 'liquidity risk trend, seasonal component question': i.e. decomposing a liquidity requirement into trend, seasonal and cyclical components, which sits in the liquidity-planning material. LCR and NSFR mechanics sit in the monitoring reading.'
Shapes covered here:
S1 TSECF and TSAA numerical, plain form of the August recall: which term structures move and by how much (Q3)
S2 liquidity requirement built from trend, seasonal and cyclical components, then the liquidity deficit or surplus (Q4)
S3 repo of an available asset: cash into the TSCLGC, haircut and accrued interest (Q1)
S4 TSECCF with initial cash and the asset sale that keeps the TSL_e non-negative (Q2).

Revision notes

- **The specific: TSECF and TSAA** (LTR-7 d, e). The book's abbreviation is TSECF (term structure of expected cash flows); "TSEF" in recall is the same object. The TSECF collects expected positive and negative cash flows by date; the TSECCF cumulates them; the TSAA records, for each date, the quantity of an asset in the bank's *possession*, regardless of ownership. A purchase records the dirty-price outflow, coupons and redemption in the TSECF and raises the TSAA until expiry; the TSLGC is unaffected.
- **The specific: trend, seasonal and cyclical components** (LTR-5 c, sources and uses of funds approach). Forecast deposits (sources) and loans (uses) as the sum of a *trend* element (constant-growth line from at least 10 years of data), a *seasonal* element (how the week or month typically compares with the most recent year-end level) and a *cyclical* element (last year's actual minus last year's trend-plus-seasonal estimate for that week, assumed to recur). Estimated liquidity deficit $(-)$ or surplus $(+)$ = change in deposits − change in loans. Book Table 5.2: deficits of 150, 200, 100, a surplus of 550, then a deficit of 200. See also A30.
- **Cash-flow taxonomy** (LTR-7 a). Time and amount, each deterministic or stochastic. Deterministic amount and time: coupons and amortisation of risk-free fixed-rate bonds and mortgages. Deterministic amount, stochastic time: one-touch option payout, the bank's withdrawals from credit lines it has received. Stochastic amount: credit related (missing flows and recoveries after default), indexed/contingent (Libor coupons, European and American option payouts), behavioural (deposit withdrawals, credit-line drawdowns, prepayments), new business (new loans and deposits, rollover, new debt issuance).
- **Liquidity options** (LTR-7 b). The right of a holder to receive cash from, or give cash to, the bank at predefined times and terms; exercised because of the holder's need for (or surplus of) liquidity, not for profit. Examples: committed credit lines sold to clients, sight and savings deposits, prepayment of fixed-rate mortgages or loans. Two impacts: a liquidity impact (amount withdrawn or repaid) and a financial impact (contract vs market rate and spread). The financial effect can be partly hedged; the liquidity impact only by cash reserves, liquid assets or credit lines.
- **Liquidity risk definitions** (LTR-7 c). Definition 7.3.1: receiving smaller than expected cash flows to meet payment obligations (covers funding and market liquidity risk). Definition 7.3.2, funding cost risk: paying a greater than expected spread over risk-free to raise funds. Definition 7.3.3: economic losses because cash flows plus existing cash differ from the desired level.
- **Causes vs sources** (LTR-7 c). Causes: flows of existing and forecast contracts from ordinary business; they enter the TSECF. Sources: factors that generate positive cash flows to manage liquidity risk and can be disposed of promptly; they form the LGC and do *not* enter the TSECF. The maturity ladder is the TSECF up to one year (short-term liquidity); beyond is structural liquidity.
- **TSECCF** (LTR-7 d). The perfect condition is a TSECCF positive at all times; it cannot be guaranteed because flows are only expected values and maturity mismatches create negative periods. A negative expected TSECCF means the bank may become insolvent.

- **Liquidity generation capacity** (LTR-7 c). The ability to generate positive cash flows *beyond contractual ones*: balance sheet expansion (unsecured funding, credit-line withdrawals, new bonds) or shrinkage (asset sales); repo is "balance sheet neutral". Three sources: asset sales (AS), secured funding and repo (RP), unsecured funding (USF). AS and RP give balance sheet liquidity (BSL). Only unencumbered ("available for sale") assets count, and a bond pledged for a period must be excluded from the TSLGC for that period.
- **Transaction effects** (LTR-7 e, Table 7.4). Repo: TSAA down, cash in the TSCLGC (haircut applied), TSECF unchanged because coupons stay with the owner bank; the repo interest enters the TSECF at the end. Reverse repo: TSAA up, cash out in the TSECF (haircut applied). Sell/buyback: TSAA down, cash in the TSCLGC. Buy/sellback: TSAA up, purchase and sellback in the TSECF, TSCLGC unchanged, coupons belong to the bank. Security lending: TSAA down, fee in the TSECF; the only contract that *decreases* LGC. Security borrowing: TSAA up, only the fee in the TSECF.
- **Term structure of expected liquidity** (LTR-7 c). $\text{TSL}_e = \text{TSECCF} + \text{TSCLGC}$, with $\text{TSECCF}(t_0, t_0) = \text{Cash}(t_0)$; it must always be positive for the bank to stay solvent. Example 7.5.1: the TSECCF is -3.2 in year 7, so 4 notional of a bond is sold at 99.00 for 3.96, and the lost coupons and redemption are fed back into the TSECF.
- **Cash flow at risk** (LTR-7 c). Positive cfaR = cash flow at confidence α minus the expected cash flow; negative cfaR at $1 - \alpha$; driven by market, credit and behavioural risk factors and usually simulated (the book uses CIR with 10,000 scenarios). Cumulating the unexpected term structures makes little sense; a term structure of liquidity at risk combines cash flows and LGC.
- **Types of liquidity** (LTR-9 a). Operational: cash to fund the business daily and clear payments; unavailable under stress. Restricted: liquid assets usable only for defined purposes (e.g. collateral for wholesale borrowings); applied only to the outflows they support. Contingent: available for general obligations under stress, held as the liquid asset buffer. Strategic: held for future business needs (acquisitions, capital expenditure); its high-quality part can be redirected to contingent needs.
- **Contingent liquidity and the buffer** (LTR-9 b). Stressed liquid asset buffer = liquid asset buffer − stressed outflows + stressed inflows (inflows may be capped, e.g. 75% of outflows under the LCR). Buffer assets need low credit and market risk, ease and certainty of valuation, an active sizable market, low concentration of buyers and sellers, and central treasury control.
- **Stress test design** (LTR-9 c). Scope: the organisational level where liquidity is commingled and accountable; transfer restrictions, currency, jurisdiction. Horizon at least 12 months, daily early on then weekly or monthly. Techniques: historical CFaR, deterministic scenarios (preferred), Monte Carlo. Scenarios: systemic, idiosyncratic and combined, graduated severity, clearly defined; historical (WaMu, Northern Rock) or hypothetical; reverse stress tests. Assumptions: deposit runoff (most important), unsecured wholesale funding, haircuts, collateral, contingent liabilities, business dial back.
- **Outputs, governance, integration** (LTR-9 c). Available liquidity relative to net outflows, survival horizon, prospective metrics; at least quarterly. ALCO owns the framework and approves scenarios; Treasury (first line) owns the modelling; independent risk management (second line) challenges and monitors limits; internal audit (third line) reviews; model risk management validates. Link to capital stress testing, ALM (consistent deposit behaviour) and FTP.
- **Liquidity reports** (LTR-10 a, b). Deposit tracker (LTD ratio, board limit e.g. 85%); daily liquidity report (liquid assets, liabilities by tenor, cumulative liquidity, counterbalancing capacity, liquidity risk factor); funding maturity gap (mismatch) and cash flow survival horizon; funding concentration (large depositor: a deposit of 5% of total liabilities; the book's top depositor at 16.5% breaches a 10% limit); undrawn commitments; liability profile; firm-specific funding yield curve; a 1-page monthly summary for senior management; qualitative reports from branches and subsidiaries.
- **Stress test reporting** (LTR-10 c). The primary output is the cash flow survival report, before and after mitigating actions (sale or repo of marketable securities, contingency funding), with behavioural stickiness (OBF) assumptions stated; a 1-slide summary for senior management; a quarterly line-by-line report of individual and combined shocks with their probability and impact. Conservative treatment of derivative collateral and undrawn commitments is best practice. LCR and NSFR: see A32.

Formulas

Cumulated cash flows (eq. 7.3) and expected liquidity (eq. 7.10).

$$CF(t_0, t_a, t_b) = \sum_{i=a}^{b} \left(cf_e^+(t_0, t_i) + cf_e^-(t_0, t_i) \right)$$

$$\text{TSL}_e(t_i) = \text{TSECCF}(t_0, t_i) + \text{TSCLGC}(t_0, t_i), \quad \text{TSECCF}(t_0, t_0) = \text{Cash}(t_0)$$

Cash amounts in the book's examples (accrued interest included).

Sale, buy/sellback, sell/buyback: $N \times (\text{Price}\% + c \times \tau)$, Repo or reverse repo: $N \times (\text{Price}\% + c \times \tau) \times (1 - h)$

Repo repayment: cash $\times (1 + r_{\text{repo}} \times \text{term})$, Lending or borrowing fee: $N \times$ fee rate $\times$ term

N: notional; c: annual coupon rate; τ: year fraction since the last coupon; h: haircut. Book: $500{,}000 \times (99.85\% + 10\% \times 0.25) \times 0.85 = 434{,}988$, repaid $434{,}988 \times (1 + 9\% \times 0.5) = 454{,}562$.

Covering a negative TSECCF by selling a bond. Minimum notional $= \dfrac{|\text{most negative TSECCF}|}{\text{expected price}}$.

Cash flow at risk. $\text{cfaR}_\alpha^+ = cf_\alpha^+ - cf_e$, $\text{cfaR}_{1-\alpha}^- = cf_{1-\alpha}^- - cf_e$.

Sources and uses of funds (Rose and Hudgins, eq. 5.4).

$$\text{Estimated total} = \text{trend} + \text{seasonal} + \text{cyclical}, \qquad \text{cyclical}_w = \text{actual}_{w,\text{last year}} - (\text{trend} + \text{seasonal})_{w,\text{last year}}$$

$$\text{Liquidity deficit } (-) \text{ or surplus } (+) = \Delta\text{deposits} - \Delta\text{loans}$$

Contingent liquidity (Baird). Stressed liquid asset buffer = liquid asset buffer − stressed outflows + stressed inflows.

Exam traps

- **Wrong term structure (sibling swap):** repo and sell/buyback cash goes to the TSCLGC; reverse repo and buy/sellback cash goes to the TSECF; the TSAA follows possession, not ownership; security lending is the only contract that decreases LGC.
- **Haircut and accrual:** haircuts apply to repo and reverse repo cash, not to sales, buy/sellbacks or sell/buybacks; accrued interest is $c \times \tau$ since the last coupon, not the full semiannual coupon.
- **Single flow vs cumulative, and initial cash:** solvency is read from the cumulated TSECCF starting from $\text{Cash}(t_0)$, and the asset sale must cover the *most negative* point, not the final one or one year's flow.
- **Intermediate result:** the repo repayment at maturity is not the cash raised at the start.
- **Sources and uses direction (polarity flip):** rising loans and falling deposits *drain* liquidity, so the gap is Δdeposits $-$ Δloans; the cyclical element is actual *minus* expected; dropping the cyclical element misstates the requirement.
- **Liquidity types (sibling swap):** restricted is tied to a specific exposure such as pledged collateral; strategic is for M&A or capital projects; operational is needed daily; only contingent liquidity meets general obligations under stress.

Question A26.1

LTR-7 c, e · S3 · [self-gen]

A liquidity analyst in the treasury of a German bank holds EUR 2,000,000 notional of an unencumbered bond paying an 8% annual coupon semiannually. Three months after a coupon date the bank repos EUR 800,000 notional for six months to generate balance sheet liquidity. The expected clean price is 99.60, the haircut is 12% and the repo rate is 7%. Which of the following is closest to the cash the bank records in the TSCLGC at the start of the repo?

A. EUR 715,264

B. EUR 729,344

C. EUR 740,298

D. EUR 812,800

Solution **Answer: A**

Step 1 – Accrued interest fraction

$$c \times \tau = 8\% \times 0.25 = 0.02$$

Step 2 – Dirty value of the repoed notional

$$800{,}000 \times (0.9960 + 0.02) = \text{EUR } 812{,}800$$

⇒ *This is what the bond would raise if sold outright.*

Step 3 – Apply the haircut

$$812{,}800 \times (1 - 0.12) = \text{EUR } 715{,}264$$

⇒ *The counterparty lends only 88% of collateral value; this cash enters the TSCLGC.*

Step 4 – Other term structures

TSAA falls from 2,000,000 to 1,200,000 for six months (possession passes); the TSECF is unaffected at the start because the bank still owns the bond and keeps the coupons. At the end the bank repays $715{,}264 \times (1 + 7\% \times 0.5) = 740{,}298$, and the interest of 25,034 enters the TSECF.

Interpretation

Repo extracts less than a sale (haircut) but keeps the bond on the bank's books; the treasurer must also track the reduced TSAA so the same bond is not counted twice in the LGC.

Why the other options are wrong

B is incorrect. Used the half-year coupon as accrued interest ($8\% \times 0.5$): $800{,}000 \times 1.036 \times 0.88 = 729{,}344$.
C is incorrect. Intermediate result: reported the repurchase amount at the end of the repo, 740,298.
D is incorrect. Dropped the haircut (treated as an outright sale): 812,800.

LO: LTR-7 e: Discuss the impact of available asset transactions on cash flows and liquidity generation capacity. · LTR-7 c: Describe and apply the concepts of liquidity risk, funding cost risk, liquidity generation capacity, expected liquidity, and cash flow at risk. · Reference: Castagna and Fede, Measuring and Managing Liquidity Risk, Ch.6, 7.4.3 The Term Structure of Available Assets, Example 7.4.2, Table 7.7

Question A26.2 LTR-7 c, d · S4 · [self-gen]

A treasurer at a French bank has built the TSECF below (EUR million, expected values). Initial cash is EUR 1.5 million. The bank holds 20 notional of an unencumbered zero-coupon government bond whose expected price is 98.40 throughout. Which of the following is closest to the minimum notional of the bond that must be sold to keep the term structure of expected liquidity non-negative at every date?

Year	1	2	3	4	5	6
TSECF	15.0	−6.0	2.0	−16.5	1.2	−0.9

A. EUR 3.76 million
B. EUR 4.00 million
C. EUR 4.07 million
D. EUR 5.59 million

Solution **Answer: C**

Step 1 – Build the TSECCF, starting from initial cash

$$\text{Yr 1: } 1.5 + 15.0 = 16.5, \quad \text{Yr 2: } 16.5 - 6.0 = 10.5, \quad \text{Yr 3: } 10.5 + 2.0 = 12.5$$
$$\text{Yr 4: } 12.5 - 16.5 = -4.0, \quad \text{Yr 5: } -4.0 + 1.2 = -2.8, \quad \text{Yr 6: } -2.8 - 0.9 = -3.7$$

⇒ *Expected cumulated cash turns negative in year 4 and stays negative; the worst point is* -4.0.

Step 2 – Liquidity needed from the LGC

The TSCLGC must cover the most negative TSECCF: EUR 4.0 million (years 5 and 6 need less).

Step 3 – Notional to sell

$$\frac{4.0}{0.9840} = \text{EUR } 4.07 \text{ million notional}$$

Step 4 – Check the TSL$_e$

$$\text{TSCLGC} = 4.065 \times 0.984 = 4.0; \qquad \text{TSL}_e:\ -4.0 + 4.0 = 0,\ \ -2.8 + 4.0 = 1.2,\ \ -3.7 + 4.0 = 0.3$$

Interpretation

Selling 4.07 notional keeps the bank in expected liquidity equilibrium; the TSAA of the bond drops from 20 to 15.93 from year 4. Being zero-coupon, the sale removes no later coupon inflows; for a coupon bond the lost flows must be fed back into the TSECF. (Covering the single-year TSECF of year 4 instead of the cumulated position would sell $16.5/0.984 = 16.77$.)

Why the other options are wrong

A is incorrect. Covered the final cumulated position instead of the most negative one: $3.7/0.984 = 3.76$, which leaves TSL$_e$ at -0.3 in year 4.

B is incorrect. Dropped the price conversion: treated notional as cash, selling 4.00 notional, which raises only $4.00 \times 0.984 = 3.94$.

D is incorrect. Ignored the initial cash in TSECCF(t_0, t_0): worst point -5.5, $5.5/0.984 = 5.59$.

LO: LTR-7 d: Interpret the term structure of expected cash flows and cumulative cash flows. · LTR-7 c: Describe and apply the concepts of liquidity risk, funding cost risk, liquidity generation capacity, expected liquidity, and cash flow at risk. · Reference: Castagna and Fede, Ch.6, 7.5 The Term Structure of Expected Liquidity, eq. (7.10), Example 7.5.1, Table 7.10

Question A26.3

LTR-7 e · S1 · [Aug-2026]

A liquidity risk officer at a Swiss bank (reporting in USD) records a buy/sellback: three months after a coupon date the bank buys 600,000 notional of a bond paying a 6% annual coupon semiannually, and will sell it back after six months at the forward price. The expected clean price is 99.40 and the applicable repo haircut on this bond is 12%. Which of the following correctly describes the effect at the start of the contract?

A. TSECF outflow of USD 605,400; TSCLGC rises by USD 605,400

B. TSECF outflow of USD 605,400; TSAA rises by 600,000

C. TSECF outflow of USD 532,752; TSAA rises by 600,000

D. TSECF outflow of USD 614,400; TSAA rises by 600,000

Solution

Answer: B

Step 1 – Accrued interest fraction

$$c \times \tau = 6\% \times 0.25 = 0.015$$

Step 2 – Cash paid at the start

$$600{,}000 \times (0.9940 + 0.015) = \text{USD } 605{,}400$$

⇒ *An outright purchase price with accrued interest; no haircut applies to a buy/sellback.*

Step 3 – Which term structures move

Ownership and possession pass to the bank, so the cash paid is a contract flow in the TSECF (and TSECCF), the TSAA rises by the notional of 600,000, and coupons received during the contract belong to the bank. The TSCLGC is not modified by the operation (the asset can, however, be repoed to raise liquidity until the sellback).

Interpretation

A buy/sellback consumes cash now but adds available assets; it does not by itself generate liquidity. At the sellback the forward price plus accrued interest returns to the TSECF and the TSAA falls back.

Why the other options are wrong

A is incorrect. Sibling swap: recorded the cash in the TSCLGC as if the operation generated liquidity; that is the sell/buyback treatment.
C is incorrect. Applied the repo haircut as if it were a reverse repo: $605{,}400 \times 0.88 = 532{,}752$.
D is incorrect. Used the full semiannual coupon as accrued interest: $600{,}000 \times (0.994 + 0.03) = 614{,}400$.

LO: LTR-7 e: Discuss the impact of available asset transactions on cash flows and liquidity generation capacity. · Reference: Castagna and Fede, Ch.6, 7.4.3 The Term Structure of Available Assets, Table 7.4, Example 7.4.2, Table 7.8

Question A26.4

LTR-5 c · S2 · [Map #64]

The liquidity manager of a Brazilian commercial bank forecasts deposits and loans with the sources and uses of funds approach, splitting each forecast into trend, seasonal and cyclical elements (BRL million). The cyclical element for a week is last year's actual level for that week minus last year's trend-plus-seasonal estimate, assumed to recur this year.

	Deposits	Loans
Week 2: estimated total (all three elements)	2,350	1,610
Week 3: trend element	2,416	1,605
Week 3: seasonal element	-96	$+95$
Week 3 last year: actual level	2,170	1,560
Week 3 last year: trend-plus-seasonal estimate	2,210	1,540

Which of the following is closest to the estimated liquidity position for week 3?

A. Deficit of BRL 180 million

B. Deficit of BRL 120 million

C. Deficit of BRL 60 million

D. Surplus of BRL 180 million

Solution

Answer: A

Step 1 – Cyclical elements for week 3

$$\text{Deposits: } 2{,}170 - 2{,}210 = -40, \qquad \text{Loans: } 1{,}560 - 1{,}540 = +20$$

$\Rightarrow$ *Last year deposits ran 40 below their trend-plus-seasonal path and loans 20 above it; the same cyclical pressure is assumed to apply this year.*

Step 2 – Estimated totals for week 3

$$\text{Deposits} = 2{,}416 - 96 - 40 = 2{,}280, \qquad \text{Loans} = 1{,}605 + 95 + 20 = 1{,}720$$

Step 3 – Changes from week 2

$$\Delta\text{deposits} = 2{,}280 - 2{,}350 = -70, \qquad \Delta\text{loans} = 1{,}720 - 1{,}610 = +110$$

$\Rightarrow$ *Sources shrink by 70 while uses grow by 110.*

Step 4 – Liquidity deficit or surplus

$$\Delta\text{deposits} - \Delta\text{loans} = -70 - 110 = -\text{BRL } 180 \text{ million}$$

Interpretation

With deposits falling and loans rising, the bank faces a BRL 180 million liquidity deficit in week 3. The manager must raise funds from the cheapest and most reliable sources, first checking which liquid assets are available and whether credit lines with correspondents are large enough.

Why the other options are wrong

B is incorrect. Dropped the cyclical element in week 3: deposits 2,320 (-30), loans 1,700 ($+90$), gap $-30-90 = -120$.
C is incorrect. Sign error on the cyclical element (estimate minus actual): deposits 2,360 ($+10$), loans 1,680 ($+70$), gap $10 - 70 = -60$.

D is incorrect. Polarity flip: treated loan growth as a source and deposit change as a use, $110 - (-70) = +180$.

LO: LTR-5 c: Estimate a bank's liquidity needs through three methods (sources and uses of funds, structure of funds, and liquidity indicators). · Reference: Rose and Hudgins, Bank Management and Financial Services, Ch.11, 5.5 Estimating Liquidity Needs, The Sources and Uses of Funds Approach, eq. (5.4), Tables 5.1 and 5.2

A27 Non-Deposit Liabilities, Pooled Cost of Funds and Liquidity Transfer Pricing

Readings: LTR-12 a, b, c, d · Managing Non-Deposit Liabilities (Rose & Hudgins, Bank Management & Financial Services 9e, Ch.13); LTR-14 a, b, c, d · Liquidity Transfer Pricing: A Guide to Better Practice (Grant, FSI Occasional Paper No.10)

AUG 2026 RECALL | MAP TIER D · #47 · 2/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: Non-deposit liability theory question.
Map #47: 'Reported: 'pooled cost approach, etc.' The examinable contrast is the pooled-funds-cost approach against the marginal-cost and historical-average-cost approaches for pricing deposits and non-deposit liabilities. LTP's role in charging liquidity cost back to business lines is the governance angle.'
Shapes covered here: S1 pooled-funds approach (forward looking hurdle rate) against the historical average cost approach, numerical (Q1); S2 sources of non-deposit liabilities and the factors in choosing among them, theory as seen in August (Q2); S3 pooled average cost of funds against matched-maturity marginal cost in LTP (Q3); S4 LTP's role in charging liquidity costs back to business lines, including the liquidity cushion (Q3).

Revision notes

The specifics first: overall cost of funds (LTR-12 d)

- **Historical average cost approach.** Looks at the *past*: what funds has the firm raised to date and what did they cost? Weighted average interest expense = all interest paid / total funds raised (book: 65/1,000 = 6.5%). Break-even cost rate on borrowed funds invested in earning assets = (interest + other operating costs) / all earning assets (book: $(65 + 10)/750 = 10\%$): the rate the firm must earn just to meet the total operating costs of raising funds. Overall cost of capital = break-even + [after-tax cost of stockholders' investment / $(1 - t)$] × (stockholders' investment / earning assets) (book: $10\% + 12\%/0.65 \times 100/750 = 12.5\%$).
- **Pooled-funds approach.** Looks at the *future*: what minimum rate of return must be earned on any future loans and investments just to cover the cost of all *new* funds raised? Pooled deposit and nondeposit funds expense = all expected operating expenses / all new funds (book: 30/300 = 10%). Hurdle rate of return over all earning assets = all expected operating costs / dollars available to place in earning assets (book: 30/200 = 15%, before taxes). New stockholders' investment is one of the pooled sources. Rose's end-of-chapter problem calls it the "pooled-funds marginal cost".
- **Effective cost rate of a single source** (LTR-12 d). (Current interest cost on amounts borrowed + noninterest costs to access the funds) / net investable funds raised. Net investable funds = total borrowed less legal reserves, deposit insurance assessments and funds placed in nonearning assets. Book: Fed funds $(1.5 + 0.0625)/24 = 6.51\%$; CDs $(1.75 + 0.1875)/24 = 8.07\%$, or 8.10% after deducting the USD 0.0675 million insurance fee (25×0.0027).

The sources (LTR-12 a)

- **Customer relationship doctrine and liability management.** The first priority is to make every loan expected to earn positive net earnings: lending decisions precede funding decisions; if deposits fall short, seek the lowest-cost borrowed funds (limitation: poor-quality loans, subprime 2007 to 2009). Liability management: buying funds, mainly from other financial institutions, with the offer rate as the control lever. Flexible (the borrower chooses amount and term, unlike deposits) but interest sensitive: suppliers are highly rate elastic. Nondeposit borrowings support about 16% of assets at the largest banks and about 4% at the smallest. See also A30.
- **Federal funds** (most popular domestic source of borrowed reserves). Short-term borrowings of immediately available money; overnight (unwritten), term, or continuing contracts (small respondents with large correspondents); no legal reserve requirement on Fed funds borrowings; usually unsecured. Key advantage: ready availability; key disadvantage: a volatile rate.
- **Repurchase agreements (RPs).** Often viewed as collateralised Fed funds: temporary sale of high-quality securities (starting leg) with an agreement to buy back at a set price and date (closing leg); reduces the lender's credit risk but may take longer to arrange. Interest cost = amount × RP rate × days/360. GCF

RPs (1998) allow collateral substitution and netting at the FICC. Bear Stearns and Lehman could not find adequate RP support. See also D6.

- **Discount window and FHLB advances.** Fed loans need acceptable collateral. Primary credit: sound institutions, usually overnight, slightly above the target Fed funds rate, no need to exhaust other sources first. Secondary credit: about 50 bp above primary, monitored, not for asset expansion. Seasonal credit: small and medium institutions with intrayear swings. Secondary carries the *highest* rate, seasonal the *lowest*. An above-market Lombard rate acts as a ceiling on overnight rates. Undercapitalised institutions: no more than 60 days in each 120. FHLB advances (12 regional banks, 1932): mortgage-collateralised, stable, below-market, overnight to 20+ years; the FHLB is first in line, ahead of the FDIC, in a failure.
- **Negotiable CDs, Eurocurrency deposits, commercial paper, long-term funds.** Negotiable CDs: legally a deposit, practically an IOU; USD 100,000+, multiples of USD 1 million, 7 days to 1 to 2 years (mainly 1 to 6 months), bearer form; domestic, Euro CDs, Yankee CDs, thrift CDs; 360-day interest-bearing basis; zero reserve requirement. Eurocurrency: the largest unregulated market, centred in London, priced off LIBOR; tap vs tranche CDs; FRCDs and FRNs of 1 to 20 years. Commercial paper via a holding-company affiliate: high volume, moderate cost, volatile capacity, credit risk. Capital notes and debentures (5 to 12 years) supplement equity and act as a barometer of perceived default risk.

How much and which source (LTR-12 b, c)

- **Available funds gap** = current and projected loans and investments desired − current and expected deposit inflows and other available funds. Book: $(150+75+135)-(185+100) =$ USD 75 million; most institutions add a small cushion.
- **Five factors in choosing a source** (LTR-12 c): (1) relative costs; (2) risk, i.e., volatility and dependability; (3) length of time funds are needed; (4) size of the borrowing institution; (5) regulations. Fed funds are among the cheapest (the shortest T-bills can be lower); CDs and Eurocurrency slightly above; commercial paper slightly above both; primary credit among the highest. Two risks: *interest rate risk* (the shorter the term, the more volatile the rate, so overnight Fed funds are the most volatile) and *credit availability risk* (CD, Eurodollar and commercial paper markets are especially sensitive). Size: the trading unit is USD 1 million; Eurodollars usually only for money-centre firms with the highest ratings; the discount window and Fed funds serve small institutions. Term: funds needed this afternoon point to Fed funds or the discount window. Nondeposit funds cost more than checkable deposits, less than thrift deposits, and are more volatile year to year.

Liquidity transfer pricing (LTR-14 a, b, c, d)

- **What LTP is** (LTR-14 a). A process that attributes the costs, benefits and risks of liquidity to business units, transferring them to a centrally managed pool: it *charges* users of funds (assets, loans), *credits* providers of funds (liabilities, deposits) and recoups the cost of the liquidity cushion by charging contingent commitments on their predicted use. Before the GFC many banks treated liquidity as a free good. BCBS Principle 4: incorporate liquidity costs, benefits and risks in internal pricing, performance measurement and new product approval.
- **Better-practice governance** (LTR-14 a). An LTP policy that defines LTP and its purpose; LTP managed centrally in group or subsidiary treasury with wholesale funding confined to it; trading book funding policies at the most granular level (desk, not book); independent oversight by risk and financial control, with regular ALCO meetings; LMIS able to attribute costs at a granular level; remuneration and profit pools adjusted for the cost of liquidity; charges attributed to the business activities, not borne by treasury.
- **Challenges found in the survey** (LTR-14 b; 38 banks, nine countries). Most lacked an LTP policy; decentralised funding centres led to inconsistent regimes, manual off-line updates and internal arbitrage; trading book policies assumed holding periods of 180 days or less, so long-term charges applied only when assets rolled into the banking book (highly rated but illiquid CDO tranches funded overnight to 90 days); weak oversight; simplistic, inflexible LMIS; bonus pools taken as a share of accrued revenues; cushions sized on idiosyncratic scenarios only and funded short term.
- **Three approaches** (LTR-14 c). *Zero cost* (liquidity as a free good, rate off the swap curve only): the worst practice, which maximises maturity transformation. *Pooled average cost*: one average rate on interest expense across existing funding sources; simple, easy to understand, workable with basic LMIS, less net interest income volatility; but it ignores the higher liquidity risk of longer-term assets and lags the market cost of funds, so it promotes maturity transformation and distorts profit assessment (separate average rates for costs and benefits are better but still flat). *Matched-maturity marginal cost*: current best practice; swap fixed-rate

wholesale costs to floating, the spread over the swap-curve reference rate is the term liquidity premium, charged to assets and credited to liabilities by maturity.

- **Worked charges** (LTR-14 c). Panel B premiums 5, 10, 18, 28, 40 bp for 1 to 5 years vs an 8 bp average: a five-year bullet is undercharged 32 bp (USD 3,200 per USD 1 million). Five-year linearly amortising loan: blended premium $(1\times5+2\times10+3\times18+4\times28+5\times40)/15=26.1$ bp (5.9 bp pre-GFC). Mortgages: charge on the behavioural maturity (WAL), e.g., four years gives 28 bp. Deposits: sticky (term) money gets larger credits than hot money; core parts of demand deposits credited on modelled behavioural maturity.
- **Contingent liquidity risk** (LTR-14 d). Cushion size from stress tests covering idiosyncratic, systemic and combined scenarios; cushion assets charged at longer-term rates, not overnight; the cost of carry (funding cost of cushion assets minus their return) is a cost of doing business recouped from the businesses that create the need, by predicted usage, not averaged across all assets (FTP = base rate + term liquidity premium + liquidity premium). Line of credit: limit USD 10 million, drawn 4 million, drawdown likelihood 60%, cushion cost 18 bp: $0.6\times0.6\times0.0018=6.48$ bp, i.e., USD 6,480; at a 0.5 bp overnight rate the book states USD 1,800, but the arithmetic gives $0.6\times0.6\times0.00005\times10\text{m}=$ USD 180 (source slip).

Formulas

Rose & Hudgins (LTR-12).

$$\begin{aligned}\text{AFG} = &\,(\text{current and projected loans and investments desired})\\ &-(\text{current and expected deposit inflows and other available funds})\end{aligned}$$

$$\text{Effective cost rate}=\frac{\text{current interest cost on amounts borrowed}+\text{noninterest costs to access funds}}{\text{net investable funds raised}}$$

$$\text{Historical break-even}=\frac{\text{interest}+\text{other operating costs}}{\text{all earning assets}}$$

$$\text{Overall cost of capital}=\text{break-even}+\frac{\text{after-tax equity return}}{1-t}\times\frac{\text{equity}}{\text{earning assets}}$$

$$\text{Pooled funds expense}=\frac{\text{all expected operating expenses}}{\text{all new funds}},\qquad \text{Hurdle rate}=\frac{\text{all expected operating costs}}{\text{dollars placed in earning assets}}$$

$$\text{RP interest}=\text{amount}\times\text{RP rate}\times\frac{\text{days}}{360},\qquad \text{CD amount due}=P+P\times\frac{\text{days}}{360}\times\text{rate}$$

Grant (LTR-14).

$$\text{FTP}=\text{base rate}+\text{term liquidity premium}+\text{liquidity premium},\qquad \text{Blended TLP}=\frac{\sum_i t_i\,\text{TLP}_i}{\sum_i t_i}\ \text{(linear amortiser)}$$

$$\text{Contingent liquidity charge}=\frac{\text{limit}-\text{drawn}}{\text{limit}}\times\text{likelihood of drawdown}\times\text{cost of funding liquidity cushion}$$

Exam traps

- **Two "pooled" approaches (sibling swap):** Rose & Hudgins' *pooled-funds approach* is forward looking and costs all *new* funds; Grant's *pooled average cost of funds* LTP approach averages the interest expense of *existing* sources and so lags the market. The backward-looking measure in Rose is the *historical average cost* approach.
- **Intermediate result vs hurdle:** expected costs over *all new funds* is the pooled expense (10% in the book); the hurdle divides by the dollars *placed in earning assets* (15%). Effective cost rates divide by *net investable* funds, not gross borrowed.
- **Polarity flips:** the shorter the term, the *more* volatile the rate; a pooled average charge *undercharges* long-term assets and *under-credits* long-term deposits, so it *encourages* maturity transformation; the lending decision comes *before* the funding decision.
- **Rank and size swaps:** secondary credit is the highest discount window rate, seasonal the lowest; Eurodollars and large CDs need money-centre size and top ratings, while Fed funds and the discount window serve small banks.
- **Role misassignment in LTP:** cushion costs are attributed to the business activities that create the need, not

absorbed by treasury; wholesale funding is centralised in treasury, not raised by business units; oversight is by independent risk and financial control.

- **Scope errors:** RPs are collateralised, Fed funds usually not; CDs, Eurodollars and commercial paper are *especially* exposed to credit availability risk; cushion assets must be charged at longer-term rates, not overnight rates.

Question A27.1 LTR-12 d · S1 · [Map #47]

The treasurer of a UK retail bank is presenting next year's funding plan to the asset-liability committee. The bank expects to raise the following new funds:

New source	Amount raised (GBP million)	Portion placed in earning assets	Interest and operating expenses (% of amount raised)
Interest-bearing transaction deposits	120	50%	7.5%
Time deposits	200	80%	6.0%
Interbank (money market) borrowings	100	95%	5.0%
New stockholders' investment	80	90%	15.0%

For reference, last year's accounts show total funds raised of GBP 2,000 million, all interest paid of GBP 100 million, other operating costs of GBP 40 million and earning assets of GBP 1,600 million. Which of the following is closest to the minimum average before-tax rate of return the bank must earn on the new loans and investments to cover the expected cost of all the new funds?

A. 7.60%

B. 8.75%

C. 9.82%

D. 12.06%

Solution **Answer: C**

Step 1 – Choose the approach

The question asks for the minimum return on *future* loans and investments to cover the cost of *new* funds: the pooled-funds approach. Last year's figures belong to the backward-looking historical average cost approach and are not needed.

Step 2 – All expected operating expenses on the new funds

$$120 \times 7.5\% + 200 \times 6.0\% + 100 \times 5.0\% + 80 \times 15.0\% = 9.0 + 12.0 + 5.0 + 12.0 = \text{GBP } 38.0 \text{ million}$$

$\Rightarrow$ *The equity's required return counts as a cost of the new funds, exactly as the new stockholders' investment does in Rose's example.*

Step 3 – Dollars available to place in earning assets

$$120 \times 0.50 + 200 \times 0.80 + 100 \times 0.95 + 80 \times 0.90 = 60 + 160 + 95 + 72 = \text{GBP } 387 \text{ million}$$

$\Rightarrow$ *Only 387 of the 500 million raised can be invested; the rest goes to reserves and other nonearning uses.*

Step 4 – Hurdle rate

$$\text{Hurdle rate} = \frac{\text{all expected operating costs}}{\text{dollars placed in earning assets}} = \frac{38.0}{387} = 9.82\%$$

$\Rightarrow$ *The bank must earn at least 9.82%, before taxes, on average, on the new earning assets to cover all expected fund-raising costs.*

Interpretation

Loans priced below 9.82% before tax would not cover the cost of the funds that finance them. The pooled expense on all new funds, $38.0/500 = 7.60\%$, understates the hurdle because about 23% of the money raised never reaches earning assets. Last year's break-even rate (8.75%) describes the cost of funds already raised, which is why it should not set the price of new business.

Why the other options are wrong

A is incorrect. Intermediate result: pooled deposit and nondeposit funds expense on all new funds, $38.0/500 = 7.60\%$, without adjusting for the portion not placed in earning assets.
B is incorrect. Sibling swap: used the backward-looking historical average cost approach, break-even $= (100 + 40)/1{,}600 = 8.75\%$, which prices funds already raised, not new funds.
D is incorrect. Wrong denominator: kept the equity's cost in the numerator but dropped its GBP 72 million from the funds placed in earning assets, $38.0/(387 - 72) = 38.0/315 = 12.06\%$.

LO: LTR-12 d: Calculate overall cost of funds using both the historical average cost approach and the pooled-funds approach. · Reference: Rose & Hudgins, Ch.13, Figuring the Overall Cost of Funds: The Historical Average Cost Approach; The Pooled-Funds Approach

Question A27.2

LTR-12 a, c · S2 · [Aug-2026]

The treasurer of a USD 400 million community bank in Kansas must cover a USD 12 million reserve deficit and a new loan commitment by the end of today, after deposits came in below forecast. She asks the funding desk which nondeposit source to tap. Which of the following statements made by the desk is correct?

A. Overnight Fed funds offer the most predictable cost, because the shorter the term of a money market borrowing, the less volatile its interest rate tends to be.

B. Seasonal credit carries the highest of the three discount window rates, so a sound bank needing funds today should request primary credit instead.

C. Eurodollar borrowings trade in multiples of USD 1 million and are usually available only to money-centre banks with top ratings, while Fed funds and the discount window can make small loans.

D. Negotiable CDs are legally deposits, so unlike Eurodollar borrowings and commercial paper they carry no credit availability risk for the issuing bank.

Solution

Answer: C

Why C is correct

Size of the borrowing institution is one of Rose & Hudgins' five factors. "The standard trading unit for most money market loans is USD 1 million", Eurodollar borrowings "usually are available only to money-center financial firms with the highest credit ratings", whereas "the central bank's discount window and the Fed funds market can make relatively small denomination loans that are suitable for smaller depository institutions". With funds needed this afternoon, the term factor also points to Fed funds or the discount window.

Interpretation

For a small bank the realistic menu today is Fed funds (cheapest, but a volatile rate) or primary credit at the window (collateral needed, priced above the Fed funds target). Size, term, cost, risk and regulation must be weighed together.

Why the other options are wrong

A is incorrect. Polarity flip: "the shorter the term of the loan, the *more* volatile the prevailing market interest rate tends to be"; overnight Fed funds have the most volatile rate of all, which is their key disadvantage.
B is incorrect. Sibling swap (rank of the three window facilities): secondary credit carries the highest rate and seasonal credit the lowest. The conclusion about primary credit is reasonable, but the stated reason is false.
D is incorrect. Absolute claim: experience shows that "the negotiable CD, Eurodollar, and commercial paper markets are especially sensitive to credit availability risks"; the legal status of a CD as a deposit does not protect the issuer from rationing.

LO: LTR-12 a: Distinguish between the various sources of non-deposit liabilities at a bank. · LTR-12 c: Discuss factors affecting the choice of non-deposit funding sources. · Reference: Rose & Hudgins, Ch.13, 13.3 Alternative Nondeposit Sources of Funds; 13.4 Choosing among Alternative Nondeposit Sources

Question A27.3

LTR-14 a, c, d · S3, S4 · [Map #47]

An internal auditor at a Hong Kong bank is reviewing its liquidity transfer pricing. Treasury charges every loan and credits every deposit a single pooled average cost of funds of 8 bps, whatever the maturity, and absorbs the cost of carrying the liquidity cushion centrally. The bank's current term liquidity premiums are 5, 10, 18, 28 and

40 bps for one to five years. Which of the following recommendations in the audit report is consistent with better LTP practice?

A. Adopt matched-maturity marginal cost, so a five-year bullet loan is charged the 40 bp five-year term liquidity premium and a five-year term deposit is credited the same premium.

B. Retain the pooled average cost, because the flat 8 bp rate overcharges long-term loans and so discourages business units from excessive maturity transformation.

C. Keep the cost of carrying the liquidity cushion in central treasury as an opportunity cost, rather than charging it to the business units that use contingent liquidity.

D. Price the funding of assets held in the liquidity cushion at overnight rates, because those assets can be sold immediately in any market disruption.

Solution **Answer: A**

Why A is correct

Grant calls the matched-maturity marginal cost of funds approach current best practice. Charges for the use of funds and credits for their benefit are based on the term liquidity premium matching the transaction's maturity: a five-year bullet needs funding for its whole life and is charged 40 bp; a five-year term deposit provides five years of funding and is credited 40 bp. Under the 8 bp average, the loan is undercharged 32 bp (3,200 per 1 million of principal) and the deposit under-credited by the same amount.

Interpretation

LTP's job is to send the cost of liquidity back to the business lines that use it, and the benefit to those that supply it, at market rates. The average approach lags the actual market cost of funds and pushes the bank toward long illiquid assets funded by short liabilities, the pattern the SSG linked to the worst funding problems in the GFC.

Why the other options are wrong

B is incorrect. Polarity flip: a single average rate *undercharges* long-term assets (8 bp against 40 bp) and so *promotes* maturity transformation; it ignores the heightened liquidity risk of longer-term assets.

C is incorrect. Role misassignment: the cost of carry is "a cost of doing business" that "should not be borne by central management (ie treasury)"; it should be recouped from the business activities that create the need, according to their predicted use of contingent liquidity.

D is incorrect. Scope error: charges on cushion assets "should not be based on short-term overnight rates but should instead be derived from longer term rates to account for the possibility of longer than expected market disruptions".

LO: LTR-14 c: Compare the various approaches to liquidity transfer pricing (zero cost, average cost, and matched-maturity marginal cost). · LTR-14 a: Discuss the process of liquidity transfer pricing (LTP) and identify best practices for the governance and implementation of an LTP process. · LTR-14 d: Describe the contingent liquidity risk pricing process and calculate the cost of contingent liquidity risk. · Reference: Grant, FSI Occasional Paper No.10, 15.3.4 to 15.3.6 (Tables 15.3, 15.4), 15.4.5 to 15.4.6, 15.5 Conclusion

A28 Private Credit, Private Markets and Their Drivers

Readings: IM-10 a, b, c, d · The Rise and Risks of Private Credit (IMF GFSR, April 2024, Ch.2); IM-11 a, b, c, d, e, f · Private Markets Investing (Demaria et al., Ch.3); CI-3 a, b, c · The Global Drivers of Private Credit (BIS, February 2025); see also IM-13 · Distress Symptoms and Remedies (Ippolito, Ch.20) (source: SchweserNotes Book 5, Readings 92, 93, 102 and 95, one rung below a GARP text)

AUG 2026 RECALL | MAP TIER D · #49 · 2/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: Private.
Map #49: "Reported: 'private credit', listed twice in the same consolidated sheet. The examinable material is the growth drivers, the bank-disintermediation story, the valuation opacity and the financial-stability concerns. Cheap marks if read; zero if skipped."
Shapes covered here:
S1 growth drivers of private credit and the BIS regression results (Q3)
S2 bank disintermediation: private credit vs bank loans, cost of equity, cost of debt and leverage (Q3)
S3 valuation opacity: stale marks, deferred losses, NAV-based metrics (Q1, Q2)
S4 financial-stability concerns and the IMF's policy fixes (Q2)
S5 private equity performance measures: IRR vs TVPI and DPI vs PME (Q1)
S6 private markets structures, co-investing and intermediated vehicles (Q4)

Revision notes

The specifics first: drivers, disintermediation, opacity, stability (CI-3, IM-10)

- **Growth drivers (S1, CI-3 b).** Four: (1) low policy rates squeeze bank margins (in the reading, floating-rate loan values fall faster than fixed-rate deposits), while PCFs are floating on both sides and avoid the squeeze; (2) low banking-sector efficiency, measured by the IMF's financial institutions (FI) index (banks reluctant to lend to SMEs lacking collateral; PCFs use targeted covenants instead); (3) higher bank regulatory stringency; (4) rising corporate leverage.
- **BIS regression (S1).** 45 countries, 2010 to 2019, dependent variable new private credit origination. FI index: coefficient -1.679 to -1.906, a one-standard-deviation *decline* gives about $+33\%$ (largest). Policy rate: -0.03, one s.d. decline gives about $+12\%$ (second). Regulatory stringency: $+0.463$ to $+0.667$, one s.d. increase gives about $+2\%$ to 4%. Corporate leverage: $+0.001$, under 1% (weakest). Coefficient size is not the ranking: rank by economic effect.
- **Bank disintermediation (S2, IM-10 b).** Growth after 2007 to 2009: low rates and stricter rules for banks and insurers (insurers entered as their capital charges were lighter than banks'); banks abandoned loans to riskier mid-sized firms. About USD 2.1 trillion in loans and undeployed capital in 2023, not much less than global high yield; mostly North America. Lenders now reach larger borrowers that used syndicated loans or bonds (tailoring, less disclosure, lower cost than a public issue).
- **Private credit vs bank loans (S2, CI-3 c).** PCF loans: flexible terms, tailored covenants, faster execution, for borrowers with limited tangible collateral, high leverage or low transparency. Cost of capital (US banks vs BDCs): BDC cost of equity has stayed *below* banks' since the GFC; BDC cost of debt has been *marginally above* banks' (gap widens in tightening). Banks' funding advantage narrowed by about **250 bp** in 2010 to 2019: bank CoE rose (market risk, stricter regulation) and BDCs levered up, replacing equity with cheaper debt (only a modest rise in their CoE). The two contributed roughly equally; CoD played almost no role. Rising policy rates could reverse it.
- **Valuation opacity (S3, IM-10 c).** Few comparables, infrequent valuations, no secondary market; no accounting standard or regulation mandates a method. Prices are less sensitive to shocks than high-yield bonds and leveraged loans but carry extra valuation discounts (larger in downturns); NAV can differ from price for several quarters. Stale marks can overstate value and returns, inflate manager fees, defer and then bunch losses. More frequent valuation is not automatically better (estimation error, short-termism).
- **Financial-stability concerns (S4, IM-10 c).** Judged *moderate* today (reasonable duration matching, relatively low leverage, low liquidity and interconnectedness risk) but untested by a serious downturn. Credit: floating-rate, leveraged borrowers; relief by adding unpaid interest to debt; yet losses historically *lower* than

high-yield bonds and *similar* to leveraged loans (PE sponsors, collateral). Liquidity: semiliquid evergreen funds, gates and suspensions, credit-line drawdowns pushed to investors via capital calls. Leverage: layers at investor, fund (SPVs, collateralized fund obligations) and borrower level; BDCs regulated with caps but rising leverage; most closed-end funds unlevered. Interconnectedness: PE, banks (under 1% of assets, secured), pensions and insurers (about 3% of AUM); PE-influenced life insurers have weaker solvency ratios. Conduct: unsophisticated investors. Underwriting: competition weakens standards and covenants.

- **Policy fixes (S4, IM-10 d).** Cross-sectoral and cross-border supervision; US and EU focus on conflicts, conduct, valuation, disclosure. Credit: bank-like monitoring by funds (insurers and pensions have no requirement to monitor loan creditworthiness, so rely on internal valuations and agencies). Liquidity: closed-end only, less frequent redemptions, longer notice and settlement, liquidity tools and stress tests, clear disclosure. Leverage: banks require PCFs to show stress tests (funding constraints, write-downs, credit-line draws); more reporting; caps if needed. Valuation: oversight of methods, more frequent external valuations where significant or error-prone, audits. Interconnectedness: data sharing, harmonisation against *regulatory arbitrage*. Conduct: disclosure of conflicts; more frequent redemptions can *raise* conduct risk (valuations manipulated to deter redemptions).

Rest of IM-10 and CI-3

- **Characteristics (IM-10 a).** Nonbank corporate credit from bilateral agreements, excluding public securities and commercial banks; historically closed-end funds lending to middle-market firms (revenues USD 100 million to 1 billion). Costlier than bank loans, more customisation (deferred interest) and covenants, better relationship management. Investors: pension funds, insurers, SWFs, family offices via closed-end funds (over 80% of the market), BDCs, CLOs. Borrowers: small and mid-sized firms (IT, healthcare), floating rate, riskier than high-yield or leveraged-loan issuers, usually PE-backed. Broadly syndicated loan: several lenders, one borrower; leveraged loan: syndicated loan to a highly leveraged, non-investment-grade firm.
- **Returns (IM-10 b).** Since 2020, private equity has *significantly outperformed* private credit; private credit is similar to natural resources, somewhat better than real estate, well ahead of the S&P 500 and venture capital. For about 10 years after the GFC, private credit beat PE and real estate. Growth about 20% a year in the US and Asia, 17% in Europe; share of corporate credit about 7%, 1.6%, 0.2%.
- **PCF structure (CI-3 a).** Loans held to maturity, closed-end (capital committed five to eight years), unlisted, so low liquidity and maturity transformation risk. BDCs are about one-fifth of the US market, often listed with open-end features for retail; a growing retail base could bring bank-like asset-liability mismatch. Strategies: mainly direct lending (floating rate, extensive covenants), also mezzanine, asset-based, distressed. AUM about USD 0.2 trillion to 2.5 trillion; US nearly 90% of originations. Individual PCFs are *highly concentrated* by industry (HHI), far less diversified than banks: specialised screening, but sector risk.

Private markets investing (IM-11)

- **Motivations (IM-11 a).** Six: performance and diversification; control and value creation; broader universe (over 300 million private firms vs about 52,000 listed); absolute returns; wider geographic reach (less home bias); specialised structures (funds of funds, co-investments). Asset types: PE, private debt, private real assets; PIPEs.
- **Structures (S6, IM-11 b).** Direct investment: high involvement and control, low diversification, low cost, concentration. Private funds: most common (closed-end LP), GP decides, moderate to high diversification. Funds of funds: very low involvement, very high diversification, extra fee layer, closest to passive, entry point for inexperienced investors. Managed accounts: discretionary or advisory, customised, large commitment. SPVs: one asset or deal, often real assets. Securitization: tranches, more liquid than funds, no control.
- **Performance metrics (S5, IM-11 c).** IRR: most used, time value, but not risk adjusted, timing sensitive (credit lines delaying calls inflate it), assumes reinvestment at the IRR; MIRR mitigates. MOIC: TVPI $=$ DPI $+$ RVPI; absolute, size independent, less manipulable than IRR, but ignores time and cannot be annualised. DPI uses realised cash only, so it is the least manipulable; TVPI and RVPI depend on NAV. PME: discount all fund cash flows at index returns; > 1 means outperformance; benchmark sensitive, best for fully realised funds; Cambridge Associates mPME. All three measure *deployed*, not committed, capital (cash drag; overcommitment, e.g. commit 110% expecting about 60% called).
- **Risks and measures (IM-11 d).** Systematic, illiquidity, credit (recovery and credit-event), settlement, information asymmetry, valuation (stale pricing, *denominator effect*, IPEV guidelines), SAA, TAA (procyclical; vintage diversification over 5 to 7 years), agency, operational, selection (wide top-5% vs bottom-5% dispersion), reporting. Measures: dispersion of pooled TVPI (macro and selection), DPI vs TVPI gap (reporting),

weights vs market-neutral (SAA), VaR and stress tests, vintage-year analysis, due diligence reviews, default and loss rates.

- **Co-investing (S6, IM-11 e).** Investing directly in a deal alongside a fund, usually by GP invitation. Pros: reduced or no management fee and carry (still share transaction costs), deal-level control, higher net returns, learning and access. Cons: adverse selection (procyclical flow, deals outside the GP's core), speed, resource needs, concentration, fewer legal protections and a passive role.
- **Intermediated vehicles (S6, IM-11 f).** Benefits: expertise, risk reduction through diversification, deal flow, cost sharing. Challenges: high fees (management fee plus carry create a hurdle), principal-agent risk, low transparency, *blind pool* risk (capital committed before assets are known; the GP decides), illiquidity (sales at a discount to NAV), due diligence demands, conflicts (equity and debt of the same company; co-investment allocation).
- **See also IM-13 Distress Symptoms and Remedies** (Ippolito Ch.20; Book 5 Reading 95), the workout side of private credit. Financial indicators (covenant breaches, late statements, creative accounting, earnings enhancements such as sale-leasebacks) vs nonfinancial (complex structure, resignations, fiscal year-end or auditor change). Excessive leverage with weak operations is the worst case (cash flow 70% or more below the default point). Remedies: restructure, raise cash (asset sales, secured financing, sale-leaseback, equity sponsor), cut debt (open market repurchases under about 20% of bonds, direct purchases, cash tender, exchange offers). Out-of-court restructurings are faster but face holdouts; Chapter 11 classes; legal risks: stay on claims, preference period, upstream guarantees.

Formulas

Multiples (paid-in = capital called, not committed):

$$\text{DPI} = \frac{\sum \text{Distributions}}{\text{Paid-in}}, \qquad \text{RVPI} = \frac{\text{Residual value (NAV)}}{\text{Paid-in}}, \qquad \text{TVPI} = \text{DPI} + \text{RVPI}$$

IRR: the rate r solving $\sum_t \dfrac{CF_t}{(1+r)^t} = 0$, with calls negative and distributions positive

PME (index levels I_t):

$$\text{PME} = \frac{\sum_t D_t\,(I_0/I_t)}{\sum_t C_t\,(I_0/I_t)}, \qquad \text{PME} > 1: \text{ fund outperformed the index}$$

BIS economic effects (one s.d.): FI index decline $\approx +33\%$; policy rate decline $\approx +12\%$; regulatory stringency rise $\approx +2\%$ to 4%; corporate leverage rise $< 1\%$

Exam traps

- **Wrong-input twin in multiples:** dividing by committed capital instead of paid-in (called) capital; all three metrics measure deployed capital only.
- **Dropped term in PME:** discounting only one leg (calls or distributions) at index returns; both legs must be index-adjusted. A TVPI above 1 or a high IRR does not mean the fund beat the market.
- **Sibling swap on drivers:** the FI index, not the policy rate, has the largest effect (33% vs 12%); the policy rate's coefficient (-0.03) is tiny yet its effect is second, so rank by economic effect, not coefficient size; leverage is weakest (under 1%).
- **Polarity flips on cost of capital and returns:** BDC CoE is *below* banks', BDC CoD marginally *above*; CoD played almost no role in the 250 bp erosion. Since 2020 PE beat private credit; private credit losses are below high yield, similar to leveraged loans.
- **Scope and absolute claims on policy:** more frequent redemptions add liquidity and conduct risk; closed-end structures mitigate liquidity risk; fund-level leverage figures miss investor and borrower layers; regulatory arbitrage concentrates risk.
- **Role misassignment in structures:** in a blind pool the GP decides which assets are bought; co-investors are passive with fewer protections; funds of funds are the most passive route with an extra fee layer.

Question A28.1

IM-11 c · S5, S3 · [self-gen]

A private markets analyst at a Canadian pension plan is evaluating a fully liquidated buyout fund before re-upping with the same manager. The plan committed CAD 150 million; the fund's cash flows to and from the plan (CAD million) and the level of the benchmark public equity index were:

Year	0	1	2	3	4
Capital called	80	40	0	0	0
Distributions	0	0	50	60	60
Index level	100	125	140	160	175

The manager's marketing deck reports a fund-level IRR of 17.5%, boosted by a subscription credit line that delayed capital calls to investors. No assets remain in the fund. Which of the following correctly states the fund's PME and TVPI, and the conclusion the analyst should draw?

A. PME 0.90 and TVPI 1.42x: the fund underperformed the index

B. PME 0.96 and TVPI 1.42x: the fund underperformed the index

C. PME 0.96 and TVPI 1.13x: the fund underperformed the index

D. PME 1.52 and TVPI 1.42x: the fund outperformed the index

Solution

Answer: B

Step 1 – Paid-in capital and distributions

Paid-in $= 80 + 40 = 120$; total distributions $= 50 + 60 + 60 = 170$; residual value $= 0$.
$\Rightarrow$ *The plan actually deployed CAD 120 million of its CAD 150 million commitment: the metrics are built on the 120.*

Step 2 – TVPI (equal to DPI here)

DPI $= 170/120 = 1.4167$; RVPI $= 0/120 = 0$; TVPI $= 1.4167 + 0 = 1.42$x.
$\Rightarrow$ *The plan got back CAD 1.42 per CAD 1 called. With no residual NAV, the multiple rests entirely on realised cash, so it is immune to valuation manipulation.*

Step 3 – Index discount factors I_0/I_t

Year 0: $100/100 = 1.0000$; Year 1: $100/125 = 0.8000$; Year 2: $100/140 = 0.7143$; Year 3: $100/160 = 0.6250$; Year 4: $100/175 = 0.5714$.

Step 4 – Index-discounted calls

$80 \times 1.0000 + 40 \times 0.8000 = 80.00 + 32.00 = 112.00$.

Step 5 – Index-discounted distributions

$50 \times 0.7143 + 60 \times 0.6250 + 60 \times 0.5714 = 35.71 + 37.50 + 34.29 = 107.50$.

Step 6 – PME

PME $= 107.50/112.00 = 0.96$.
$\Rightarrow$ *The same cash invested in the index on the call dates and withdrawn on the distribution dates would have done better: the fund underperformed the public benchmark despite a 1.42x multiple.*

Interpretation

B is correct. TVPI ignores time and benchmark; the plan's IRR on the tabled cash flows is 13.8%; the 17.5% headline is timing-sensitive (credit lines shorten the measured holding period) and neither IRR is risk adjusted or benchmarked. PME adds the benchmark: below 1, so the analyst should challenge the re-up case. A fully realised fund is exactly where PME is most reliable.

Why the other options are wrong

A is incorrect. Dropped term: discounts the distributions but leaves the calls undiscounted, $107.50/120 = 0.90$.
C is incorrect. Wrong-input twin: divides distributions by committed rather than called capital, $170/150 = 1.13$x; the metrics measure deployed capital only.
D is incorrect. Dropped term: discounts the calls but leaves the distributions undiscounted, $170/112.00 = 1.52$, which flips the conclusion to outperformance.

LO: IM-11 c: Apply and compare different methods to measure the performance of private equity investments and describe their advantages and disadvantages. · Reference: Demaria et al., Asset Allocation and Private Markets, Ch.3; SchweserNotes Book 5, Reading 93.2

Question A28.2 IM-10 c, d · S3, S4 · [Aug-2026] [Map #49]

A senior risk consultant is advising a Brazilian securities regulator on a draft framework for private credit funds, which are increasingly sold to retail investors through semiliquid evergreen structures with quarterly redemption windows. The funds' loans have no secondary market and are marked by the managers each quarter, and their reported NAVs have barely moved through a period in which listed leveraged-loan prices fell. Which of the following recommendations is consistent with the IMF's analysis of private credit risks?

A. Permit more frequent redemption windows for retail investors, since more frequent redemptions reduce both the liquidity risk and the conduct risk of these funds.

B. Where valuations are significant or prone to error, require more frequent external valuations and audits of valuations, since stale marks can defer losses and overstate fees.

C. Rely on leverage reported at the fund level, since most closed-end private credit funds use little leverage and leverage across the private credit system is therefore low.

D. Leave monitoring of the loans' credit quality to external rating agencies, so that funds holding large private credit exposures need not run bank-like monitoring of their own.

Solution **Answer: B**

Step 1 – Diagnose the symptom

Flat NAVs while public loan prices fall are stale valuations: few comparables, infrequent marks, no secondary market. Private credit values change less and respond more slowly than public debt, so price and NAV can diverge for several quarters.

⇒ *Stale marks may overstate asset values and returns, inflate manager fees, and defer losses that are later recognised all at once.*

Step 2 – Match the IMF policy fix

Valuation: greater regulatory oversight of valuation methods; more frequent external valuations where valuations are significant and/or prone to error; audits of valuations to fix deficiencies in methods and controls.

Step 3 – Check the retail angle

Liquidity fixes are closed-end structures, less frequent redemptions, longer notice and settlement, liquidity tools and stress tests, clear disclosure. Conduct fix: disclose conflicts and ensure retail investors understand credit, liquidity and redemption restrictions.

Interpretation

B is correct. The regulator's framework should pair the valuation fix with the liquidity and conduct fixes, because evergreen retail funds combine all three risks.

Why the other options are wrong

A is incorrect. Polarity flip: more frequent redemptions *increase* liquidity risk, and can raise conduct risk because managers may manipulate valuations to discourage redemptions.

C is incorrect. Scope/condition error: low leverage in closed-end funds does not extend to the system; leverage is layered at investor, fund (SPVs, collateralized fund obligations) and borrower level, and fund-level reports can obscure it.

D is incorrect. Role misassignment: the IMF wants funds with substantial private credit to run more rigorous, bank-like monitoring themselves; reliance on internal valuations and rating agencies is the weakness, not the fix.

LO: IM-10 d: Assess potential policy recommendations that could help mitigate the risks associated with private credit. · IM-10 c: Describe and assess the risks and vulnerabilities related to private credit, and explain how private credit can pose risks to financial stability. · Reference: IMF, Global Financial Stability Report, April 2024, Ch.2; SchweserNotes Book 5, Reading 92.2

Question A28.3 CI-3 b, c · S1, S2 · [Aug-2026] [Map #49]

The treasurer of a US regional bank is preparing a board briefing on why business development companies (BDCs) have been winning middle-market loans that the bank used to make, drawing on the BIS study of private credit. Which of the following statements is correct to include?

A. In the BIS regression across 45 countries, the policy rate has the largest economic effect: a one-standard-deviation decline is linked to about a 33% rise in private credit origination.

B. The banks' funding advantage over BDCs narrowed by about 250 bp in 2010 to 2019 mainly because BDCs' cost of debt fell below the banks' cost of debt.

C. The roughly 250 bp erosion of the banks' funding advantage came about equally from banks' higher cost of equity and BDCs' higher leverage, while BDCs' cost of debt stayed marginally above banks'.

D. Because private credit funds are funded at floating rates on both sides of the balance sheet, a sustained rise in policy rates would further widen their cost-of-capital advantage over banks.

Solution **Answer: C**

Step 1 – Rank the growth drivers by economic effect

FI index (bank efficiency): one s.d. decline, about $+33\%$ (largest). Policy rate: one s.d. decline, about $+12\%$ (second). Regulatory stringency: one s.d. increase, about $+2\%$ to 4%. Corporate leverage: under 1%.
$\Rightarrow$ *Private credit grows most where banks are least effective at meeting loan demand.*

Step 2 – Decompose the cost of capital

BDC cost of equity has stayed below banks' since the GFC; BDC cost of debt has been marginally above banks'. Bank CoE rose (market risk, stricter regulation); BDCs raised leverage, shifting the mix from expensive equity to cheaper debt, with only a modest rise in their CoE.

Step 3 – Attribute the 250 bp

The lower BDC CoE relative to banks and the increase in BDC leverage contributed roughly equally; the CoD played almost no role because the CoD gap stayed small while policy rates fell.

Interpretation

C is correct. For the board: the bank's lost advantage is an equity-cost and leverage story, not a deposit-cost story, and it may partly reverse if rates stay high.

Why the other options are wrong

A is incorrect. Sibling swap: the 33% effect belongs to the FI index; the policy rate is second, at about 12%.
B is incorrect. Polarity flip: BDCs' cost of debt has been marginally *above* banks', and the cost of debt played almost no role in narrowing the gap.
D is incorrect. Scope/condition error: floating-rate funding protects PCF margins when rates are low; when rates rise, BDC funding costs rise with policy rates faster than bank deposits, so BDCs could lose the cheap-debt benefit and the advantage may shift back.

LO: CI-3 c: Compare private credit loans with bank loans and assess how changes in cost of equity, cost of debt, and leverage have influenced the competitive dynamics between them. · CI-3 b: Discuss the potential drivers of private credit growth and interpret regression results about their impact. · Reference: BIS (February 2025), The Global Drivers of Private Credit; SchweserNotes Book 5, Readings 102.1 and 102.2

Question A28.4

IM-11 b, e, f · S6 · [self-gen]

The CIO of a Japanese life insurer is preparing a JPY 30 billion first allocation to private markets. Three routes are on the table: a fund of funds, a primary commitment as a limited partner in a blind-pool buyout fund, and co-investments offered by that fund's GP alongside its deals. An investment committee member summarises the trade-offs. Which of the following statements is correct?

A. The fund of funds gives the insurer the highest control and the lowest total fees of the three routes, making it the structure closest to a direct investment.

B. Co-investment deal flow tends to be countercyclical, arriving when valuations are low, which largely offsets the adverse selection risk of co-investing.

C. In the blind-pool fund, the limited partners choose which assets are acquired within the strategy guidelines, so blind pool risk falls mainly on the GP.

D. Co-investing reduces or removes management fees and carried interest and allows deal-level targeting, but brings adverse selection, compressed due diligence and concentrated positions.

Solution **Answer: D**

Step 1 – Place each route in the structure spectrum

Fund of funds: very low involvement, very high diversification, an additional layer of fees; the closest approximation to a passive approach and a common entry point for inexperienced investors. Primary fund: pooled capital, GP decides, diversification with limited LP control.

Step 2 – Intermediated vehicle trade-offs

Benefits: expertise, diversification, deal flow, cost sharing. Challenges: management fee plus carried interest create a hurdle, principal-agent risk, opacity, blind pool risk (capital committed before assets are known, the manager decides how it is deployed), illiquidity, due diligence demands, conflicts.

Step 3 – Co-investing trade-offs

Advantages: reduced or no management fees or carry (transaction and due diligence costs still shared), greater control at deal level, higher net returns, learning and access. Disadvantages: adverse selection (procyclical flow, deals outside the GP's core expertise), speed and complexity, resource needs, concentration, fewer legal protections and a passive role.

Interpretation

D is correct. For a first allocation the committee would typically weigh a fund of funds or a primary commitment for diversification, adding co-investments only once the insurer has the resources to evaluate deals quickly.

Why the other options are wrong

A is incorrect. Sibling swap: high control and low cost describe direct investment; a fund of funds is the most passive route with an extra fee layer.

B is incorrect. Polarity flip: co-investment deal flow can be procyclical, appearing when valuations are already high, which worsens rather than offsets adverse selection.

C is incorrect. Role misassignment: in a blind pool the fund manager (GP) ultimately decides how capital is deployed, and the LPs bear the risk of exposures they would not have chosen.

LO: IM-11 e: Describe the practice of co-investing in private markets and assess its advantages and disadvantages. · IM-11 f: Explain benefits and challenges of investing in private markets through an intermediated investment vehicle such as a private equity or private debt fund. · IM-11 b: Describe different investment structures that can be used to invest in private assets and compare their characteristics. · Reference: Demaria et al., Asset Allocation and Private Markets, Ch.3; SchweserNotes Book 5, Readings 93.1 and 93.3

A29 Money Laundering and Financing of Terrorism

Readings: ORR-10 a, b, c · Sound Management of Risks Related to Money Laundering and Financing of Terrorism (BCBS, revised July 2020, through para. 83; GARP ORR Ch.10); ORR-11 a, b · Case Study: Financial Crime and Fraud (GARP ORR Ch.11)

AUG 2026 RECALL | MAP TIER D · #50 · 2/16 SOURCES | TELEGRAM | VARDEEZ

Specifics reported

Your August 2026 recall: ML AFT (read as ML/FT) - 1Q.
Map #50: "Reported: 'ML/FT'. The governance question here is a role trap: the chief AML officer, not the CRO, is the actor who files reports and contacts regulators. Customer due diligence, correspondent banking risk, and the three stages of laundering are the definitional core."
Shapes covered here: S1 ML/FT governance role trap: who files suspicious transaction reports and contacts supervisors and FIUs, and what the chief AML/CFT officer may and may not combine (Q1); S2 customer due diligence, including reliance on another bank's CDD (Q1); S3 correspondent banking risk (Q1); S4 the three stages of laundering: placement, layering, integration (Q1).

Revision notes

The specific: governance and the chief AML/CFT officer (ORR-10 a)

- **Board and appointment.** The board approves and oversees the policies for risk, risk management and compliance, including ML/FT, and must understand ML/FT risks. The board and senior management appoint an appropriately qualified **chief AML/CFT officer** with overall responsibility for the AML/CFT function and the stature and authority to be heard by the board, senior management and business lines.
- **Three lines in AML/CFT.** Line 1: business units (front office, customer-facing activity) identify, assess and control the risks of their business; written policies and procedures, internal procedures for detecting and reporting suspicious transactions, staff screening and tailored ongoing training. Line 2: the chief AML/CFT officer, the compliance function, *and also* human resources or technology. Line 3: internal audit, reporting to the board audit committee. External auditors may also evaluate controls and compliance.
- **What the chief AML/CFT officer does.** Ongoing monitoring of the fulfilment of all AML/CFT duties (sample testing, exception reports, alerting senior management or the board); **contact point** for all AML/CFT issues for internal and external authorities, including supervisors and **FIUs**; responsible for **reporting suspicious transactions**; assesses alerts from the IT monitoring system; ensures prompt disclosure where suspected proceeds remain in an account.
- **What the role may and may not combine.** To avoid conflicts it should not have business line responsibilities and should not be entrusted with data protection or internal audit. It *may* also perform the function of the **CRO or the chief compliance officer**, with a direct reporting line to senior management or the board; where duties are separated, the relationship between these officers must be clearly defined. So the trap is about roles, not titles: the STR and FIU duties attach to the chief AML/CFT officer role, whoever holds it.
- **Risk assessment and monitoring.** Identify inherent and residual risk at country, sectoral, bank and business relationship level; understand risks in customers, products (including those under development), delivery channels, services and jurisdictions; measures proportional and risk-based and demonstrable to the supervisor. Transaction monitoring adequate to size and complexity, usually automated for internationally active banks (otherwise document an effective alternative), covering all accounts, with parameters reflecting the bank's own risk rather than vendor defaults. Internal audit evaluates the IT system and audits policies, staff effectiveness, compliance oversight (including alert parameters) and training.

Customer acceptance and due diligence (ORR-10 b)

- **Customer acceptance policy.** Identify higher-risk customer types by background, occupation (public or high-profile position), source of income and wealth, country of origin and residence, products, purpose of accounts, linked accounts, business activities. Basic due diligence for all; simplified measures for proven lower risk if law allows; **enhanced due diligence** for higher risk (large balances with regular cross-border wires, PEPs; required for *foreign* PEPs), with **senior management** approval to enter or continue the relationship. The policy must not deny access to the financially or socially disadvantaged (FATF Financial Inclusion

Guidance).

- **Identification and verification.** CDD covers customers, persons acting on their behalf and **beneficial owners** (natural persons who ultimately own or control a customer). Verify with reliable, independent source documents (best: hardest to obtain illicitly or counterfeit); a written declaration of beneficial ownership may be requested but not relied on solely; non-face-to-face is no excuse to skip verification. Generally no relationship or transaction until identity is verified; completion after onboarding only where essential not to interrupt business, with restrictions; if problems cannot be resolved, close or block and consider an STR.
- **Specific cases.** No anonymous or obviously fictitious-name accounts; **numbered accounts** get exactly the same CDD, with identity known to sufficient staff. Funds transferred from an account in the customer's name at another bank subject to the same CDD standard give some comfort, but the bank must still **conduct its own due diligence**; refusal by another bank over illicit activity points to higher risk, EDD, an STR or rejection.
- **Ongoing monitoring and records.** All relationships, intensity based on risk, enhanced for higher risk, plus cross-sectional product monitoring; flag transactions without economic sense, large cash deposits, deviations from the customer risk profile. Screen the database whenever sanction lists change, and periodically for foreign PEPs. Keep records at least **five years** after the relationship or occasional transaction ends, and until case closure if under investigation or litigation.
- **STRs, FT and freezing.** Report suspicious transactions promptly to the FIU; the customer must not be informed, even indirectly (no tipping off). FT differs from ML: funds may come from *legal* sources and transactions may be very small. Screen against terrorist lists before onboarding and on an ongoing basis; terrorist screening is **not risk-sensitive** and applies whatever the customer's risk profile; freeze without delay and without prior notice.

Group-wide, cross-border and correspondent banking (ORR-10 c)

- **Group-wide policies.** Consistently applied and supervised across the group; local policies consistent with group policy; if host requirements are stricter, the branch or subsidiary adopts them; where home and host minimums differ, apply the **higher standard of the two**. If the host country prevents proper implementation of FATF standards, the chief AML/CFT officer informs the **home supervisors**; closing host operations may be considered.
- **Information sharing and the group officer.** Branches and subsidiaries must be authorised to share customer information with head office and proactively send information on higher-risk customers; head office can require group-wide searches against lists; significant relationships monitored on a consolidated basis (on- or off-balance sheet, AUM, fiduciary). A bank operating nationally and abroad appoints a **group AML/CFT officer** who coordinates a single group strategy, can give orders to all branches and subsidiaries, and checks compliance including by regular on-site visits.
- **Introducers and correspondent banking.** Never rely on introducers subject to less strict standards; higher reliance on a same-group introducer subject to the same standards is possible, but customer information must still be obtained. The guidelines increased focus on third-party introducers (Annex 1) and **correspondent banking** (Annex 2); the June 2017 revision applies the risk-based approach to correspondent relationships, recognising that *not all correspondent banking relationships bear the same level of risk*, with a list of risk indicators. Mixed financial groups share customer information across banking, securities and insurance and apply sector-appropriate requirements when cross-selling.

Financial crime controls and USAA (ORR-11 a, b)

- **Scope.** FCA definition of financial crime: fraud or dishonesty; market misconduct or misuse of information; handling the proceeds of crime; financing of terrorism. ML and TF map to the Basel CPBP category. Internal fraud has two Level 2 types: unauthorised activities (now often called misconduct) and theft and fraud; external fraud: theft and fraud, systems security. "Zero tolerance" for internal fraud means every caught perpetrator is sanctioned, not that the risk cannot materialise. Fraud triangle: opportunity, incentive, rationalisation.
- **Control framework.** Four components for both internal fraud and AML: **selection** (screening, due diligence, tone and culture; for AML: CDD/KYC, identity and origin of funds), **prevention** (segregation of duties, access rights, delegation of authority; for AML: risk-based approach, training, allocation of responsibilities), **detection** (supervision, monitoring, reconciliations; for AML: transaction monitoring, alerts, reconciliation with OFAC and PEP lists), **deterrents** (sanctions, legal pursuits; for AML: escalation to the FIU, closure of accounts). A dedicated MLRO is mandatory in most jurisdictions. RegTech automates profiling and screening.

- **Three stages of laundering. Placement**: techniques that disguise the origins of funds (cash transfer to a business, false invoicing, trusts and offshore companies, **smurfing**: many small transactions below the reporting threshold instead of one large one, foreign accounts). **Layering**: combining placement and extraction strategies with varying amounts and patterns to make tracking as hard as possible. **Integration or extraction**: getting the money out to use (fake payments to employees, fake loans, dividends to accomplices).
- **USAA (2022).** USD 140 million from FinCEN and OCC consent orders for a "willful" failure to maintain a BSA/AML programme (January 2016 to April 2021): weak internal controls, suspicious activity identification and reporting, understaffing filled by untrained contractors, a new monitoring system creating about 90,000 unreviewed alerts and 6,900 unreviewed cases. Fined for a weak control environment, not for proven laundering. Lesson: "ignore regulatory findings at one's own peril". See also A22 for the general three lines model.

Exam traps

- **Role misassignment (the map's trap):** STRs and the FIU and supervisor contact point belong to the chief AML/CFT officer, not to the CRO as such, not to internal audit and not to the front office. The CRO may hold the chief AML/CFT officer role; then the duties come with it.
- **Forbidden combinations:** the chief AML/CFT officer must not have business line, data protection or internal audit responsibilities; combining with the CRO or CCO is allowed.
- **Scope error on reliance:** funds from another bank's account, or a same-group introducer, never remove the bank's own due diligence; never rely on introducers with less strict standards.
- **Absolute claims:** not all correspondent banking relationships bear the same risk; terrorist screening is the exception that is *not* risk-sensitive; verification after onboarding is allowed only where essential, with restrictions.
- **Polarity flip, cross-border:** apply the *higher* of home and host standards; stricter host rules are adopted locally.
- **Sibling swap on stages:** smurfing is placement; layering mixes amounts and patterns; integration extracts the funds for use.

Question A29.1 ORR-10 a, b, c; ORR-11 a · S1, S2, S3, S4 · [Aug-2026] [Map #50]

The board of a mid-size Singapore bank has appointed the bank's CRO as its chief AML/CFT officer, with a direct reporting line to the board. An internal auditor is reviewing the bank's revised AML/CFT manual, which covers governance, customer onboarding, correspondent banking and staff training. Which of the following statements in the manual is correct?

A. The training module classifies smurfing, splitting a large cash sum into many small deposits below the reporting threshold, as layering, the stage that mixes amounts and patterns.

B. In the combined role, the CRO as chief AML/CFT officer is responsible for reporting suspicious transactions and is the contact point for supervisors and the FIU on AML/CFT issues.

C. Where a new customer's first deposit comes from her own account at another bank subject to the same CDD standard, the bank may rely on that bank's due diligence instead of its own.

D. Because every correspondent banking relationship carries the same level of ML/FT risk, the manual applies identical enhanced due diligence to every respondent bank.

Solution **Answer: B**

Step 1 – Is the CRO allowed to hold the role?

Para. 24: "The chief AML/CFT officer may also perform the function of the chief risk officer or the chief compliance officer or equivalent. He/she should have a direct reporting line to senior management or the board." What is forbidden (para. 23) is business line responsibilities, data protection and internal audit.

⇒ *The appointment is permitted, and the board reporting line meets the requirement.*

Step 2 – Which duties come with the role

Para. 22: the chief AML/CFT officer "should be the contact point regarding all AML/CFT issues for internal and external authorities, including supervisory authorities or financial intelligence units (FIUs)". Para. 25: the officer "should also have the responsibility for reporting suspicious transactions".

⇒ *The map's trap ("the chief AML officer, not the CRO, files reports and contacts regulators") is about the role:*

a CRO acting only *as CRO does not carry these duties; a CRO appointed chief AML/CFT officer does.*

Step 3 – Customer due diligence and reliance

Para. 43: a transfer from an account in the customer's name at another bank subject to the same CDD standard "may provide some comfort", but the bank "should nevertheless conduct its own due diligence" and consider whether the previous bank closed the account over concerns about illicit activity.

Step 4 – Correspondent banking

Para. 9: the June 2017 revision of Annex 2 guides banks in applying the risk-based approach to correspondent banking, "recognising that not all correspondent banking relationships bear the same level of risk", with a list of risk indicators.

Step 5 – Stages of laundering (ORR-11)

Placement covers techniques that disguise the origin of funds, including smurfing (a series of small transactions instead of a single large one, to stay below the reporting threshold). Layering combines placement and extraction strategies with varying amounts and patterns. Integration or extraction gets the money out for use.

Interpretation

B is correct. The auditor should confirm the combined role is documented, that STR filing and FIU contact sit with the CRO in that capacity, that onboarding requires the bank's own CDD whatever the source of funds, that correspondent relationships are risk-rated individually, and that the training module places smurfing under placement.

Why the other options are wrong

A is incorrect. Sibling swap: smurfing is a placement technique; the definition of layering is correct but the technique is assigned to the neighbouring stage.

C is incorrect. Scope/condition error: the same-CDD-standard condition gives comfort only; it does not remove the bank's obligation to conduct its own due diligence.

D is incorrect. Absolute claim: the guidelines recognise that not all correspondent banking relationships bear the same level of risk, so due diligence is risk-based, not uniform.

LO: ORR-10 a: Explain best practices recommended by the Basel committee for the assessment, management, mitigation, and monitoring of money laundering and financing of terrorism (ML/FT) risks. · ORR-10 b: Describe recommended practices for the acceptance, verification, and identification of customers at a bank. · ORR-10 c: Explain practices for managing ML/FT risks in a group-wide and cross-border context. · ORR-11 a: Describe elements of a control framework to manage financial fraud risk and money laundering risk. · Reference: BCBS (July 2020), GARP ORR Ch.10, paras. 9, 22 to 25, 43; GARP ORR Ch.11, AML Risk Management

A30 Asset vs Borrowed Liquidity Management and the Investment Function

Readings: LTR-5 a, b, c · Liquidity and Reserves Management: Strategies and Policies (Rose & Hudgins, Bank Management & Financial Services 9e, Ch.11); LTR-4 a, b, c · The Investment Function in Financial-Services Management (Rose & Hudgins, Ch.10)

AUG 2026 RECALL | MAP TIER D · #63 · 1/16 SOURCES | VARDEEZ

Specifics reported

Your August 2026 recall: Asset liquidity management and borrowed liquidity management.
Map #63: 'Investment instruments available to banks, the factors affecting the choice of securities, tax-exposure considerations, and reserve-management strategy. Rated high by the provider; unconfirmed by candidate reports.'
Shapes covered here: S1 asset liquidity (asset conversion) vs borrowed liquidity (liability management) vs balanced strategy: pros, cons and who uses which, as seen in August (Q1); S2 reserve-management strategy: the net liquidity position as the starting point (Q1); S3 investment instruments available to banks and their advantages and disadvantages (Q2); S4 factors affecting the choice of securities, including tax exposure (Q2).

Revision notes

The specific first: liquidity strategies (LTR-5 b)

- **Asset liquidity management (asset conversion).** The oldest approach: store liquidity in assets, mainly cash and marketable securities, and convert them into cash as needed. A liquid asset has (1) a ready market, (2) a reasonably stable price (a deep market), (3) reversibility (principal recoverable with little risk of loss). Options: T-bills, Fed funds loans, purchase of securities under RP, correspondent deposits, municipals, agency securities, negotiable CDs, Eurocurrency loans. Used mainly by *smaller* institutions, which find it *less risky* than relying on borrowings.
- **Costs of asset conversion.** Lost future earnings on assets sold (opportunity cost), transactions costs (commissions), possible sale into a falling market, a weaker-looking balance sheet (the low-risk securities sold gave the impression of strength), and liquid assets carry the *lowest* returns. Sell the assets with the least profit potential first. Holding more liquid assets does not by itself make a firm liquid: the demands made against it matter too.
- **Borrowed liquidity (liability) management (purchased liquidity).** Borrow immediately spendable funds to cover demands: Fed funds, RPs, jumbo (USD 100,000+) negotiable CDs, Eurocurrency deposits, FHLB advances, the central bank's discount window. Used most extensively by the *largest* banks, which often borrow close to 100% of their liquidity needs.
- **Pros and cons of borrowing.** Pros: borrow only when needed (no storehouse of low-yield assets); the volume and composition of the asset portfolio stay unchanged (selling assets for liability-driven demands *shrinks* the firm); a control lever, the offer rate. Cons: the *most risky* approach (but the highest expected return) because of rate volatility and how fast credit availability can change; uncertain cost, so uncertain earnings; a firm in trouble needs borrowed liquidity most exactly when lenders are least willing.
- **Balanced liquidity management.** The compromise most firms use: some expected demands stored in assets (marketable securities), other anticipated needs backstopped by advance lines of credit; *unexpected* needs met from near-term borrowings; *longer-term* needs planned and parked in short- and medium-term assets.
- **Guidelines for liquidity managers.** Track all departments that use or supply funds; know in advance the plans of the biggest credit and funds customers; keep priorities clear (liquidity management now plays a *supporting* role to lending and fee services); analyse needs continuously to avoid both excess (lost income) and deficit (hurried borrowing or loss sales).

Net liquidity position and liquidity needs (LTR-5 a, c)

- **Net liquidity position** (LTR-5 a). L_t = supplies (deposit inflows, revenues from nondeposit services, loan repayments, asset sales, money market borrowings) − demands (deposit withdrawals, acceptable loan requests, repayments of borrowings, other operating expenses, dividends). $L < 0$: deficit, decide when and where to raise funds; $L > 0$: surplus, invest it until needed. A firm is liquid only with access to immediately spendable funds at reasonable cost at precisely the time they are needed. Two truths: supply rarely

equals demand, and there is a trade-off between liquidity and profitability. Costs: interest, transactions and opportunity costs; risks: interest rate and availability risk.

- **Why depositories face liquidity problems.** Maturity mismatch; a high share of liabilities payable immediately (demand deposits, money market borrowings); sensitivity to interest rates (rising rates bring withdrawals and faster drawings on low-rate credit lines); the need to protect public confidence. Most liquidity problems arise from *outside* the firm, from customers.
- **Sources and uses of funds** (LTR-5 c). Liquidity rises as deposits increase and loans decrease. Estimated deficit or surplus $=$ Δdeposits $-\Delta$loans, forecast with trend, seasonal and cyclical components; reserves have a planned and a protective component.
- **Structure of funds** (LTR-5 c). Hot money (e.g., 95% reserve), vulnerable funds (25 to 30% likely to leave), stable or core funds (e.g., 15% or less), each net of required legal reserves, plus 100% of (potential maximum loans $-$ actual loans), because all good loans should be made. Book: USD 63.57 million; probabilistic version gives an expected requirement of $+10$ million.
- **Liquidity indicators and market signals** (LTR-5 c). Cash position, liquid securities, net Fed funds and RP position, capacity ratio (negative indicator), pledged securities ratio (negative), hot money ratio, deposit brokerage index, core deposit ratio, deposit composition ratio (a decline means greater stability), loan commitments ratio (a rise means greater future needs); judge against peers and by changes. Market signals (the ultimate standard): public confidence, stock price, risk premiums on CDs and borrowings, loss sales of assets, meeting commitments to credit customers, central bank borrowings.

The investment function (LTR-4 a, b, c)

- **Why banks hold investments.** Loans are illiquid, risky, locally concentrated and taxable, so a fifth to a third of assets go into securities: the "crossroads account" between cash, loans and deposits. Functions: stabilise income, offset loan credit risk, geographic diversification, backup liquidity, reduce tax exposure, collateral for public deposits, hedge rate risk, flexibility, dress up the balance sheet.
- **Instruments** (LTR-4 a). Money market (within one year, low risk, ready marketability): T-bills (safety, liquidity, collateral; low yields), short-term T-notes and bonds, agency securities (implied support, yields close to Treasuries; taxable), CDs (insured to at least USD 100,000; limited resale for longer CDs), Eurocurrency deposits (million-dollar units, not insured, slightly higher yields than domestic time deposits), bankers' acceptances (among the safest; eligible ones discountable at the Fed), commercial paper (poor resale market), short-term municipals (TANs, RANs; tax-exempt interest). Capital market (beyond one year, higher returns and capital gains): Treasury notes and bonds, municipal GO and revenue bonds (tax-exempt interest, limited resale), corporate notes and bonds (higher pretax yields, more attractive to insurers and pension funds than banks), asset-backed securities (pass-throughs, CMOs, mortgage-backed bonds; prepayment risk). Newer: structured notes, securitised assets, stripped securities (POs more rate sensitive, IOs less).
- **Factors affecting the choice of securities** (LTR-4 b). Ten: expected rate of return (YTM or HPY), tax exposure, interest rate risk, credit or default risk (banks generally limited to investment grade, Baa/BBB or better), business risk (buy out-of-market securities to offset local loan risk), liquidity risk (Treasuries most liquid, then agencies, municipals, MBS; buying many liquid securities lowers average yield), call risk, prepayment risk (PSA model), inflation risk (TIPS), pledging requirements. Tax: after-tax yield $=$ before-tax yield $\times(1-t)$; TEY $=$ tax-exempt yield $/(1-t)$; bank-qualified municipals keep 80% interest deductibility; tax swapping and portfolio shifting.
- **Maturity strategies and tools** (LTR-4 c). Ladder (spaced maturity: steady income, little expertise), front-end load (all short term: liquidity), back-end load (5 to 10 years: income, relies on money market borrowing for liquidity), barbell (short and long, little in the middle; often smaller firms), rate expectations (shift short if rates expected to rise, long if expected to fall; largest firms; biggest potential gains and losses). Yield curve: implicit rate forecast, over/underpriced securities, carry trade, riding the curve. Duration: percentage price change $\approx -D \times \Delta r/(1+r)$; immunise by setting duration equal to the planned holding period. See also A16 (duration gap).
- **See also A27** for the nondeposit sources behind borrowed liquidity (Fed funds, RPs, CDs, Eurodollars, discount window, FHLB) and the factors in choosing among them.

Formulas

$$L_t = (\text{deposit inflows} + \text{nondeposit service revenues} + \text{loan repayments} + \text{asset sales} + \text{money market borrowings}) - (\text{withdrawals} + \text{acceptable loan requests} + \text{repaid borrowings} + \text{other operating expenses} + \text{dividends})$$

$$\text{Liquidity deficit } (-) \text{ or surplus } (+) = \Delta\text{deposits} - \Delta\text{loans}$$

$$\text{Total liquidity requirement} = 0.95(\text{Hot} - \text{LR}) + 0.30(\text{Vulnerable} - \text{LR}) + 0.15(\text{Stable} - \text{LR}) + 1.00(\text{Potential loans} - \text{Actual loans}) \qquad (\text{LR = required legal reserves})$$

$$\text{After-tax yield} = \text{before-tax yield} \times (1 - t), \qquad \text{TEY} = \frac{\text{after-tax return on tax-exempt investment}}{1 - t}$$

$$\text{Net after-tax return on qualified municipal} = (\text{nominal return} - \text{interest expense}) + t \times 0.80 \times \text{interest expense}$$

$$\frac{\Delta P}{P} \approx -D \times \frac{\Delta r}{1 + r}, \qquad \text{immunisation: } D = \text{planned holding period}$$

Exam traps

- **Strategy polarity flip:** asset conversion is the *less* risky approach used mainly by *smaller* institutions; borrowed liquidity is the *most* risky (highest expected return) and used most by the *largest* banks.
- **Strategy sibling swap:** leaving the asset portfolio's size and composition unchanged is a benefit of *borrowing*; selling assets *shrinks* the firm. The offer rate is the control lever of liability management, not of asset conversion.
- **Absolute claim:** more liquid assets do not guarantee liquidity; it depends on the demands made against the firm and on access at reasonable cost at the right time.
- **Tax direction:** after-tax yield *multiplies* by $(1 - t)$; TEY *divides* by $(1 - t)$. A 4.55% tax-exempt municipal is worth 7.00% taxable for a 35% taxpayer, not 2.96%.
- **Investment-risk sibling swaps:** call risk (issuer calls when rates fall) vs prepayment risk (specific to asset-backed securities); business risk (local economy, offset by out-of-market securities) vs credit risk (issuer default, investment grade only).
- **Sign and scope in liquidity estimates:** $L_t < 0$ is a deficit; sources and uses surplus is Δdeposits $-\Delta$loans; structure of funds reserves apply *net of* legal reserves and 100% of potential loan growth is held; capacity and pledged securities ratios are *negative* indicators.

Question A30.1

LTR-5 a, b · S1, S2 · [Aug-2026] [Map #63]

The treasurer of a CAD 900 million community savings bank in Manitoba is preparing next week's liquidity plan for the board. Projected flows (CAD million) are: deposit inflows 120, loan repayments 45, revenues from nondeposit services 8; deposit withdrawals 140, acceptable new loan requests 95, repayment of previous borrowings 10, other operating expenses 12, dividend payment 6. The bank holds CAD 150 million of marketable Treasury and agency securities. Which of the following recommendations to the board is consistent with Rose & Hudgins?

A. Cover the CAD 90 million deficit by borrowing, since liability management is the less risky strategy and the one smaller institutions mainly use.

B. Cover the CAD 90 million deficit by selling Treasury bills, since this leaves the size and composition of the bank's asset portfolio unchanged.

C. Treat the CAD 90 million deficit as no concern, since securities holdings above the deficit guarantee the bank is liquid whatever demands it faces.

D. Cover the CAD 90 million deficit with a balanced strategy: part from marketable securities, part from prearranged credit lines, unexpected needs from near-term borrowing.

Solution **Answer: D**

Step 1 – Net liquidity position

$$L_t = (120 + 45 + 8) - (140 + 95 + 10 + 12 + 6) = 173 - 263 = \text{CAD } -90 \text{ million}$$

$\Rightarrow$ *Demands exceed supplies, so the bank faces a liquidity deficit and must decide when and where to raise funds.*

Why D is correct

"Most financial firms compromise by using both asset and liability management": under a balanced strategy some expected demands are stored in assets (principally marketable securities), other anticipated needs are backstopped by advance lines of credit, and unexpected cash needs are met from near-term borrowings. This avoids both the storage cost of pure asset conversion and the risk of pure borrowing.

Interpretation

For a small bank, the securities portfolio is the natural first line, but selling CAD 90 million of it would forgo future earnings, incur transaction costs, risk selling into a falling market and shrink the balance sheet; borrowing the whole amount would expose earnings to rate volatility and availability risk. Mixing the two, and selling the securities with the least profit potential first, is what the book recommends.

Why the other options are wrong

A is incorrect. Polarity flip: borrowing liquidity is "the most risky approach" (with the highest expected return) and is used most by the *largest* banks; asset conversion is the less risky approach used mainly by smaller institutions.
B is incorrect. Sibling swap: leaving the asset portfolio's volume and composition unchanged is an advantage of *borrowed* liquidity. Selling assets to meet liability-driven demands shrinks the size of the firm.
C is incorrect. Absolute claim: a firm holding more liquid assets "will not necessarily be a liquid institution", because its position also depends on the demands made against it; liquidity means access at reasonable cost at precisely the time funds are needed.

LO: LTR-5 a: Calculate a bank's net liquidity position and explain factors that affect the supply and demand of liquidity at a bank. · LTR-5 b: Compare strategies that a bank can use to meet demands for additional liquidity. · Reference: Rose & Hudgins, Ch.11, 5.2 The Demand for and Supply of Liquidity (Equation 5.1); 5.4 Strategies for Liquidity Managers

Question A30.2

LTR-4 a, b · S3, S4 · [Map #63]

The investments officer of a regional US bank in the 35% federal tax bracket is drafting a memo for the asset-liability committee on securities purchases for the coming year. Loan demand in the bank's home market is concentrated in agriculture. Which of the following statements in the memo is correct?

A. A tax-exempt municipal note yielding 4.55% is worth the same to the bank as a fully taxable security yielding 2.96%, so taxable Treasuries above that yield are preferable.

B. Eurocurrency deposits are insured in the same way as domestic CDs, so they usually yield slightly less than time deposits of comparable-size US banks.

C. Adding a large volume of liquid, readily marketable securities raises the average yield on earning assets, because it lowers the bank's exposure to liquidity risk.

D. Buying investment-grade municipals issued outside the bank's home region helps offset business risk in the local loan book, and banks are generally limited to Baa/BBB or better.

Solution **Answer: D**

Why D is correct

Business risk, the risk that the economy of the market area served turns down, "can be reflected quickly in the loan portfolio", so many institutions rely on securities from borrowers *outside* the principal loan market to offset it ("bank examiners often encourage out-of-market security purchases"). Credit risk is controlled by regulation: US banks "generally are allowed to buy only investment-grade securities, rated at least Baa or BBB".

Interpretation

The choice of securities weighs ten factors at once (return, tax exposure, interest rate, credit, business, liquidity, call, prepayment and inflation risk, and pledging). For an agricultural lender, out-of-market investment-grade municipals address business risk and credit risk while also offering tax-exempt income.

Why the other options are wrong

A is incorrect. Polarity flip on the tax adjustment: multiplied by $(1 - t)$ instead of dividing. TEY $= 4.55\%/(1 -$

$0.35) = 7.00\%$, so a taxable security must yield 7.00% before tax to match the note; $4.55\% \times 0.65 = 2.96\%$ is not the equivalent yield.

B is incorrect. Sibling swap: deposit insurance belongs to domestic CDs. Eurocurrency deposits "are not insured, and due to their perceived higher credit risk, they often carry slightly higher market yields than domestic time deposits".

C is incorrect. Polarity flip: "the purchase of a large volume of liquid, readily marketable securities tends to *lower* the average yield" and reduce profitability; management faces a trade-off between profitability and liquidity.

LO: LTR-4 b: Identify and discuss various factors that affect the choice of investment securities by a bank. · LTR-4 a: Compare various money market and capital market instruments and discuss their advantages and disadvantages. · Reference: Rose & Hudgins, Ch.10, 4.3 Popular Money Market Investment Instruments; 4.7 Factors Affecting Choice of Investment Securities (Equations 4.3, 4.4)

A31 Basel IRB Capital

Readings: CR-10 d · Credit Value at Risk (Hull, RMFI 6e, Ch.19, 10.2 Vasicek's Model); ORR-21 e, f · Capital Regulation Before the Global Financial Crisis (Carey, GARP Risk Institute 2019: The IRB Approach; Bank, Corporate, and Sovereign Exposures Under IRB; Retail Exposures Under IRB); ORR-23 a, b, c · High-Level Summary of Basel III Reforms (BCBS, December 2017: Internal Ratings-Based Approaches; Output Floor); ORR-22 h · Solvency, Liquidity, and Other Regulation After the Global Financial Crisis (Carey: Basel III Finalizing the Post-Crisis Reforms, Key Changes, IRB)

AUG 2026 RECALL AUG ONLY · NOT IN MAP

Specifics reported

Your August 2026 recall: Credit risk IRB.
Shapes covered here:
S1 IRB capital for a single corporate loan under Advanced IRB, with the correlation formula and maturity adjustment: the plain form (Q1)
S2 retail IRB: fixed correlation, no maturity adjustment, reported as RWA (Q2)
S3 Foundation versus Advanced IRB inputs: supervisory LGD and the PD floor (Q3)
S4 December 2017 IRB revisions: A-IRB removed for certain asset classes, equities out of IRB, input floors (Q4)
S5 the 72.5% output floor and how input floors and the output floor bind (Q5).

Revision notes

- **Three Basel II credit risk options** (ORR-21 e), included because supervisors feared banks would manipulate internal risk measures to reduce capital. **Standardized:** like Basel I with risk weights somewhat more sensitive to credit quality, using *external ratings* (obligor type and rating, or asset type; retail 75%, mortgages 35%); intended for banks whose internal measures cannot support IRB. The United States chose not to implement it.
- **Foundation IRB** (ORR-21 e): the bank provides *only PD*; the accord specifies EAD and LGD (45% senior, 75% subordinated; collateral reduces LGD via the comprehensive approach); MA is set to 2.5 in most cases; PD can be no lower than 0.0003 for bank and corporate exposures (implicitly zero for sovereigns). Expected to be feasible for most large banks, which already had internal rating systems yielding a PD.
- **Advanced IRB** (ORR-21 e): the bank provides PD, LGD, EAD (and MA) from its own data, models, estimates and analysis. Limited EAD and LGD data meant fewer banks were expected to qualify.
- **The model** (ORR-21 f; CR-10 d): Gordy's (2003) asymptotic single risk factor model, i.e. a one-factor Gaussian copula. In large, well-diversified (infinitely granular) portfolios, each asset's contribution to capital depends only on its own PD, LGD and EAD; the Committee sets the percentile and the correlation ρ. The 2014 Large Exposure Framework was needed because real banks had sizable single-name exposures.
- **Calibration** (ORR-21 f): one-year horizon, 99.9th percentile. Carey writes capital $= \sum[\text{EAD}_i \times \text{LGD}_i \times \text{DR}_{99.9,i}] - \text{EL}$. Loan loss reserves were not Tier 1 but were thought to approximate EL, so capital covers *unexpected* loss only; a shortfall of reserves below EL reduces capital.
- **Vasicek in Hull** (CR-10 d): $\text{WCDR}(T, X)$ is the Xth percentile of the default rate; for one loan the Xth loss percentile is WCDR × EAD × LGD; Gordy extends this to a sum over a large portfolio (eq. 10.2), the basis of banking-book capital with $T = 1$ year and $X = 99.9\%$, sometimes multiplied by a maturity adjustment. ρ is roughly the asset (equity) return correlation. Weakness: very little tail correlation; another copula can fix it.
- **Wholesale correlation** (ORR-21 f): ρ *falls* from 0.24 to 0.12 as PD rises (Lopez 2004): very high-risk borrowers default for idiosyncratic reasons, middle-risk borrowers default together when the macroeconomy is distressed. Effect: DR rises somewhat less quickly with PD than at constant ρ (Table 21.4: PD 0.01 with $\rho = 0.2$ gives DR 0.146).
- **Maturity adjustment** (ORR-21 f): bank, corporate and sovereign exposures only; assets with more than one year of remaining maturity can deteriorate in credit quality without defaulting. Low-PD assets get the largest MA (more room to be downgraded).
- **Retail** (ORR-21 f): only an Advanced-IRB-like treatment (bank estimates PD, LGD, EAD), *no maturity adjustment*, and three correlations: 0.15 residential mortgages, 0.04 qualifying revolving (credit cards), for-

mula 0.03 to 0.16 for other retail. Retail correlations are *lower* than wholesale. Carey: USD 100 million mortgages, PD 0.01, LGD 30%: DR 0.09, RWA 30 (Basel I 50; Basel II standardized 35).

- **Credit mitigants** (ORR-21 f): credit substitution (guarantor's rating replaces the obligor's up to the amount covered); the 2005 double-default amendment multiplies capital without the mitigant by $0.15 + 160\,\mathrm{PD}_g$.
- **Why revise IRB in 2017** (ORR-23 a): restore credibility of RWAs and comparability of capital ratios. Shortcomings of IRB: excessive complexity, lack of comparability of internally modelled requirements, lack of robustness in modelling certain asset classes. Also: more risk-sensitive standardised approaches, internal models removed for CVA and operational risk, a G-SIB leverage ratio buffer, and a new output floor.
- **IRB revisions** (ORR-23 b): (i) A-IRB removed for asset classes that cannot be modelled robustly: large and mid-sized corporates (consolidated revenues above EUR 500 million) and banks and other financial institutions, which move to F-IRB (fixed LGD and EAD remove two sources of RWA variability) or SA; (ii) *all* IRB approaches removed for equities (to SA); specialised lending keeps A-IRB, F-IRB and slotting; (iii) input floors; (iv) greater specification of parameter estimation. F-IRB senior unsecured LGD to non-financial corporates cut from 45% to 40%; higher haircuts and lower LGDs for non-financial collateral. The 1.06 IRB scaling factor is removed. Carey adds: IRB must cover all assets in an asset class (no cherry picking).
- **Input floors** (ORR-23 b, Table 23.3): PD floors apply under *both* F-IRB and A-IRB (5 bp corporate, mortgages, QRRE transactors and other retail; 10 bp QRRE revolvers); LGD and EAD floors apply only under *A-IRB*. Unsecured LGD floors: corporate 25%, QRRE 50%, other retail 30%; secured LGD floors 0% financial, 10% receivables, 10% real estate, 15% other physical collateral (mortgages 5%). EAD floor: on-balance sheet plus 50% of off-balance sheet exposure using the SA credit conversion factor.
- **Output floor** (ORR-23 c): replaces the Basel II floor (80% of Basel I requirements, inconsistently applied). RWA = max(RWA under approved approaches, 72.5% of RWA under the standardised approaches only). It is a risk-based backstop that keeps a level playing field between modelled and standardised banks. Standardised approaches used in the floor: SA credit risk, SA-CCR for counterparty exposure, SA-CVA or BA-CVA (or 100% of CCR capital), SEC-ERBA, SEC-SA or 1250% for securitisation, SA market risk, SA operational risk. Phase-in: 50% (2022), 55%, 60%, 65%, 70%, 72.5% (1 January 2027). National discretion: transitional cap on the RWA increase at 25% of pre-floor RWA, removed in 2027.
- **See also** B1 (credit VaR and granularity), D20 (Vasicek versus CreditRisk+ versus CreditMetrics) and C5 (Basel III post-crisis reforms, SA credit, CVA and the SMA).

Formulas

Worst case default rate (Vasicek, CR-10 eq. 10.1; Carey's $\mathrm{DR}_{99.9}$).

$$\mathrm{WCDR}(T,X) = N\!\left(\frac{N^{-1}(\mathrm{PD}) + \sqrt{\rho}\,N^{-1}(X)}{\sqrt{1-\rho}}\right), \qquad N^{-1}(0.999) = 3.0902$$

Capital and RWA (ORR-21).

$$\text{Capital} = \sum_i \mathrm{EAD}_i \times \mathrm{LGD}_i \times (\mathrm{WCDR}_i - \mathrm{PD}_i) \times \mathrm{MA}_i, \qquad \mathrm{RWA} = 12.5 \times \mathrm{EAD} \times \mathrm{LGD} \times (\mathrm{DR} - \mathrm{PD}) \times \mathrm{MA}$$

so RWA $= 12.5\times$ capital and capital $= 8\%\times$ RWA. Carey's form $\sum \mathrm{EAD} \times \mathrm{LGD} \times \mathrm{DR}_{99.9} - \mathrm{EL}$ is the same thing without MA.

Wholesale correlation. $\rho = 0.12\,\dfrac{1-e^{-50\,\mathrm{PD}}}{1-e^{-50}} + 0.24\left[1 - \dfrac{1-e^{-50\,\mathrm{PD}}}{1-e^{-50}}\right]$

Maturity adjustment. $\mathrm{MA} = \dfrac{1+(M-2.5)\,b}{1-1.5\,b}, \quad b = [0.11852 - 0.05478\ln(\mathrm{PD})]^2$; at $M = 2.5$, MA $= 1/(1-1.5b)$.

Retail (no MA). $\rho = 0.15$ residential mortgages; 0.04 qualifying revolving; other retail $\rho = 0.03\,\frac{1-e^{-35\mathrm{PD}}}{1-e^{-35}} + 0.16\left[1 - \frac{1-e^{-35\mathrm{PD}}}{1-e^{-35}}\right]$.

Foundation IRB (Basel II). PD floor 0.0003 (bank, corporate); LGD 45% senior, 75% subordinated; MA at $M = 2.5$.

Revised floors (ORR-23). PD $\geq$ 5 bp (10 bp QRRE revolvers), both approaches; A-IRB unsecured LGD $\geq$ 25% corporate, 50% QRRE, 30% other retail.

Output floor. $\mathrm{RWA} = \max\left(\mathrm{RWA}_{\text{approved}},\ 72.5\% \times \mathrm{RWA}_{\text{SA only}}\right)$; optional transitional cap RWA $\leq 1.25 \times \mathrm{RWA}_{\text{approved}}$.

Book check. Carey: PD 0.01, LGD 30%, EAD 100, DR 0.14, MA 1.26: RWA $= 12.5\times100\times0.3\times0.13\times1.26 = 61.4$. (The book prints MA as $1/(1-1.5\times0.137)$, the $M=2.5$ form; with $M=3$ the full formula gives 1.35. This workbook uses the full formula whenever M is given.)

Exam traps

- **Dropped term ($-$PD or $-$EL):** WCDR alone is the stress loss; capital covers unexpected loss only.
- **Capital versus RWA:** RWA $= 12.5\times$ capital. Read which one the stem asks for.
- **Maturity adjustment misapplied:** omitted for a corporate loan, $M=2.5$ used when an A-IRB stem gives the maturity, or applied to retail (retail has none).
- **Correlation twin and polarity:** corporate ρ on mortgages, 0.04 on mortgages, or weights swapped so ρ *rises* with PD. Corporate ρ falls as PD rises.
- **Foundation inputs:** the bank's own LGD (A-IRB only) or a PD below the floor. In 2017, PD floors bind under both approaches; LGD and EAD floors only under A-IRB.
- **Output floor base:** 72.5% of *standardised-only* RWA, not of modelled RWA and not 100% of standardised RWA; it is a floor (a higher-of), never a cap.

Question A31.1

CR-10 d; ORR-21 f · S1 · [Aug-2026]

A credit risk analyst at a German universal bank using the Advanced IRB approach is computing the Basel II capital for a single corporate term loan. The data are:

- EAD: EUR 250 million
- One-year PD: 0.80%; LGD: 40%
- Remaining maturity: 4 years
- Correlation from the Basel corporate formula; confidence level 99.9%

Which of the following is closest to the required capital for this loan?

A. EUR 11.8 million

B. EUR 14.4 million

C. EUR 15.1 million

D. EUR 18.4 million

Solution

Answer: D

Step 1 – Correlation weight and corporate correlation

$$w=\frac{1-e^{-50\times0.008}}{1-e^{-50}}=0.3297,\qquad \rho=0.12(0.3297)+0.24(0.6703)=0.2004$$

$\Rightarrow$ *About 20% of each obligor's asset variance comes from the single systematic factor.*

Step 2 – Argument of $N(\cdot)$

$$z=\frac{N^{-1}(0.008)+\sqrt{0.2004}\times3.0902}{\sqrt{1-0.2004}}=\frac{-2.4089+1.3835}{0.8942}=-1.1468$$

Step 3 – WCDR

$$\text{WCDR}=N(-1.1468)=0.1257$$

$\Rightarrow$ *In the 1-in-1,000-year bad year, 12.57% of loans like this default, versus 0.80% on average.*

Step 4 – Maturity adjustment

$$b = [0.11852 - 0.05478 \ln(0.008)]^2 = 0.3830^2 = 0.1467$$

$$\text{MA} = \frac{1 + (4 - 2.5)(0.1467)}{1 - 1.5(0.1467)} = \frac{1.2201}{0.7799} = 1.5643$$

⇒ *A 4-year loan can be downgraded without defaulting, so capital is scaled up by about 56%.*

Step 5 – Capital

$$\text{Capital} = 250 \times 0.40 \times (0.1257 - 0.0080) \times 1.5643 = \text{EUR } 18.4 \text{ million}$$

Interpretation

RWA $= 12.5 \times 18.4 =$ EUR 230.2 million. Expected loss $250 \times 0.40 \times 0.008 =$ EUR 0.8 million is covered by reserves and pricing, not capital.

Why the other options are wrong

A is incorrect. Omitted the maturity adjustment: $250 \times 0.40 \times 0.1177 = 11.8$.

B is incorrect. Swapped the correlation weights so ρ rises with PD: $\rho = 0.24(0.3297) + 0.12(0.6703) = 0.1596$, WCDR $= 0.1001$, capital $= 100 \times 0.0921 \times 1.5643 = 14.4$.

C is incorrect. Used the Foundation default $M = 2.5$ (MA $= 1.2821$) although the A-IRB stem gives 4 years: $100 \times 0.1177 \times 1.2821 = 15.1$.

LO: CR-10 d: Describe the application of the Vasicek's model to estimate capital requirements under the Basel II internal-ratings-based (IRB) approach. ORR-21 f: Calculate credit risk capital under Basel II utilizing the IRB approach. · Reference: Hull, RMFI 6e, Ch.19, 10.2 eq. 10.1; Carey, Capital Regulation Before the Global Financial Crisis, Bank, Corporate, and Sovereign Exposures Under IRB

Question A31.2

ORR-21 f; CR-10 d · S2 · [Aug-2026]

A model validation analyst at a UK building society is checking the IRB risk-weighted assets reported for its residential mortgage book. The data are:

- EAD: GBP 400 million
- Average one-year PD: 1.50%; average LGD: 25%
- Average remaining contractual maturity: 18 years
- Confidence level 99.9%; Basel II retail treatment applies

Which of the following is closest to the risk-weighted assets for this book?

A. GBP 163.2 million

B. GBP 182.0 million

C. GBP 191.9 million

D. GBP 575.5 million

Solution

Answer: A

Step 1 – Correlation

Residential mortgages use the fixed retail correlation $\rho = 0.15$.

Step 2 – WCDR

$$z = \frac{N^{-1}(0.015) + \sqrt{0.15} \times 3.0902}{\sqrt{0.85}} = \frac{-2.1701 + 1.1968}{0.9220} = -1.0556, \qquad \text{WCDR} = N(-1.0556) = 0.1456$$

⇒ *In the 99.9th percentile year, 14.56% of the mortgage pool defaults.*

Step 3 – Capital (no maturity adjustment for retail)

$$\text{Capital} = 400 \times 0.25 \times (0.1456 - 0.0150) = \text{GBP } 13.06 \text{ million}$$

Step 4 – RWA

$$\text{RWA} = 12.5 \times 13.06 = \text{GBP } 163.2 \text{ million}$$

Interpretation

A risk weight of $163.2/400 = 40.8\%$. The 18-year maturity is a red herring: retail exposures carry no maturity adjustment.

Why the other options are wrong

B is incorrect. Dropped the $-$PD term: $12.5 \times 400 \times 0.25 \times 0.1456 = 182.0$.
C is incorrect. Applied the corporate correlation formula ($\rho = 0.1767$): WCDR $= 0.1685$, RWA $= 12.5 \times 100 \times (0.1685 - 0.015) = 191.9$.
D is incorrect. Applied a maturity adjustment with $M = 18$ ($b = 0.1215$, MA $= 3.526$), which retail does not have: $163.2 \times 3.526 = 575.5$.

LO: ORR-21 f: Calculate credit risk capital under Basel II utilizing the IRB approach. CR-10 d: Describe the application of the Vasicek's model to estimate capital requirements under the Basel II internal-ratings-based (IRB) approach. · Reference: Carey, Capital Regulation Before the Global Financial Crisis, Retail Exposures Under IRB; Hull, RMFI 6e, Ch.19, 10.2

Question A31.3

ORR-21 e, f · S3 · [self-gen]

A regulatory reporting manager at a Japanese bank that uses the Basel II Foundation IRB approach is computing the capital for a senior unsecured loan to a highly rated domestic corporate. The data are:

- EAD: JPY 50 billion
- Bank's internal one-year PD estimate: 0.01%
- Bank's internal LGD estimate: 30%
- Maturity treatment: the Foundation IRB default

Which of the following is closest to the required capital?

A. JPY 0.30 billion

B. JPY 0.39 billion

C. JPY 0.58 billion

D. JPY 7.22 billion

Solution

Answer: C

Step 1 – Foundation IRB inputs

PD is floored at 0.0003 for corporates, so PD $= 0.03\%$, not 0.01%. LGD is the supervisory 45% for senior claims; the internal 30% is not used. MA uses $M = 2.5$.

Step 2 – Correlation

$$w = \frac{1 - e^{-50 \times 0.0003}}{1 - e^{-50}} = 0.01489, \qquad \rho = 0.12(0.01489) + 0.24(0.98511) = 0.2382$$

$\Rightarrow$ *A very safe obligor gets close to the maximum correlation of 0.24.*

Step 3 – WCDR

$$z = \frac{-3.4316 + \sqrt{0.2382} \times 3.0902}{\sqrt{0.7618}} = \frac{-3.4316 + 1.5083}{0.8728} = -2.2037, \qquad \text{WCDR} = N(-2.2037) = 0.01377$$

Step 4 – Maturity adjustment at $M = 2.5$

$$b = [0.11852 - 0.05478 \ln(0.0003)]^2 = 0.5629^2 = 0.3168, \qquad \text{MA} = \frac{1}{1 - 1.5(0.3168)} = 1.9057$$

Step 5 – Capital

$$\text{Capital} = 50 \times 0.45 \times (0.01377 - 0.0003) \times 1.9057 = \text{JPY } 0.58 \text{ billion}$$

Interpretation

About 1.2% of EAD. Under the December 2017 revisions the corporate PD floor rises to 5 bp and F-IRB senior unsecured LGD to non-financial corporates falls to 40%; holding $M = 2.5$, capital would be about JPY 0.70 billion.

Why the other options are wrong

A is incorrect. Ignored the PD floor and used 0.01%: $\rho = 0.2394$, WCDR $= 0.00569$, MA $= 2.394$, capital $= 50 \times 0.45 \times 0.00559 \times 2.394 = 0.30$.

B is incorrect. Used the bank's own LGD of 30%, which only Advanced IRB permits: $50 \times 0.30 \times 0.01347 \times 1.9057 = 0.39$.

D is incorrect. Reported RWA instead of capital: $12.5 \times 0.578 = 7.22$.

LO: ORR-21 e: Compare the standardized internal ratings-based (IRB) approach, the foundation IRB approach, and the advanced IRB approach for the calculation of credit risk capital under Basel II. ORR-21 f: Calculate credit risk capital under Basel II utilizing the IRB approach. · Reference: Carey, Capital Regulation Before the Global Financial Crisis, The IRB Approach and Bank, Corporate, and Sovereign Exposures Under IRB

Question A31.4

ORR-23 a, b · S4 · [self-gen]

A regulatory consultant is briefing the credit risk committee of a European bank that currently uses the Advanced IRB approach for its wholesale, specialised lending and equity exposures. The committee wants to know how the December 2017 Basel III reforms change the bank's use of internal ratings. Which of the following statements should the consultant include?

A. Exposures to corporates with consolidated revenues above EUR 500 million must move to the standardised approach, since the advanced IRB option is removed for them.

B. Equity exposures can no longer be treated under any IRB approach, whereas specialised lending may remain on the advanced or foundation IRB approach.

C. Exposures to banks and other financial institutions may stay on the advanced IRB approach, since only corporate exposures lose the advanced option.

D. PD input floors apply only under the advanced IRB approach, so exposures moved to the foundation IRB approach face no PD floor.

Solution

Answer: B

Step 1 – What the reform targets

The Committee removed the A-IRB option "for asset classes that cannot be modelled in a robust and prudent manner", adopted input floors, and specified estimation practices more tightly, to reduce complexity and unwarranted RWA variability (ORR-23 a).

Step 2 – Table 23.2, revised scope

Large and mid-sized corporates (revenues above EUR 500 million): F-IRB or SA. Banks and other financial institutions: F-IRB or SA. Equities: SA only. Specialised lending: A-IRB, F-IRB, slotting or SA.

Interpretation

B is the only statement consistent with Table 23.2: "all IRB approaches are being removed for exposures to equities, which are typically a small component of the credit risk of banks", while banks "would be permitted to continue using the advanced and foundation IRB approaches" for specialised lending.

Why the other options are wrong

A is incorrect. Scope error. Losing A-IRB does not mean losing IRB: banks with supervisory approval use F-IRB for these corporates, which fixes LGD and EAD and so removes two sources of RWA variability.

C is incorrect. Sibling swap. Banks and other financial institutions are in the same position as large corporates: A-IRB is removed for both.

D is incorrect. Sibling swap. The LGD and EAD floors are A-IRB only; the PD floors apply under both F-IRB and A-IRB (5 bp for corporates).

LO: ORR-23 b: Summarize the December 2017 revisions to the Basel III framework in the following areas: the standardized approach to credit risk; the internal ratings-based (IRB) approaches for credit risk; the CVA risk framework; the operational risk framework; the leverage ratio framework. ORR-23 a: Explain the motivations for revising the Basel III framework and the goals and impacts of the December 2017 reforms to the Basel III framework. · Reference: BCBS, High-Level Summary of Basel III Reforms (December 2017), Internal Ratings-Based Approaches for Credit Risk, Tables 23.2 and 23.3

Question A31.5 ORR-23 b, c · S5 · [self-gen]

A capital manager at a eurozone bank is reporting under the fully phased-in December 2017 framework. The bank's A-IRB model estimates an LGD of 20% for unsecured loans to corporates with consolidated revenues below EUR 500 million. After all input floors are applied, the bank's total RWA under its approved approaches is EUR 60 billion; its total RWA calculated using only the standardised approaches is EUR 90 billion. Which of the following statements is correct?

A. The modelled LGD is raised to the 25% floor; the output floor does not bind, because EUR 60 billion exceeds 72.5% of the modelled RWA, EUR 43.5 billion.

B. The modelled LGD is raised to the 25% floor; the output floor binds, because 72.5% of standardised RWA is EUR 65.25 billion, so RWA become EUR 65.25 billion.

C. The modelled LGD of 20% stands, because LGD floors apply only under F-IRB; the output floor binds, so RWA become EUR 65.25 billion.

D. The modelled LGD is raised to the 25% floor; the output floor binds, so RWA become EUR 90 billion, the full standardised amount.

Solution **Answer: B**

Step 1 – Input floor (parameter level)

Table 23.3: unsecured corporate LGD floor 25% under A-IRB. The bank's 20% estimate is raised to 25% before modelled RWA are computed.

Step 2 – Output floor (aggregate level)

$$72.5\% \times 90 = \text{EUR } 65.25 \text{ billion}, \qquad \text{RWA} = \max(60,\ 65.25) = \text{EUR } 65.25 \text{ billion}$$

⇒ The floor adds EUR 5.25 billion of RWA: the modelled approaches may lower capital by at most 27.5% relative to the standardised approaches.

Interpretation

The two floors act at different levels: input floors constrain individual parameters inside the model; the output floor limits the total benefit of modelling. During the phase-in the floor would bind in 2026 ($70\% \times 90 = 63 > 60$) but not in 2025 ($65\% \times 90 = 58.5 < 60$); a supervisor using the transitional cap would limit RWA to $1.25 \times 60 = 75$, which does not bite here.

Why the other options are wrong

A is incorrect. Sibling swap of the base: 72.5% is applied to RWA under the *standardised* approaches, not to the bank's own modelled RWA.

C is incorrect. Sibling swap (F-IRB for A-IRB): LGD and EAD floors apply only under A-IRB (under F-IRB, LGD is supervisory anyway).

D is incorrect. Absolute claim: the floor does not require the full standardised amount; RWA are the higher of modelled RWA and 72.5% of standardised RWA.

LO: ORR-23 c: Describe the revised output floor introduced as part of the Basel III reforms and approaches to be used when calculating the output floor. ORR-23 b: Summarize the December 2017 revisions to the Basel III framework in the following areas: the standardized approach to credit risk; the internal ratings-based (IRB) approaches for credit risk; the CVA risk framework; the operational risk framework; the leverage ratio framework. · Reference: BCBS, High-Level Summary of Basel III Reforms (December 2017), Specification of Input Floors, Output Floor, Transitional Arrangements, Tables 23.3 and 23.5

A32 Net Stable Funding Ratio, LCR and Leverage Ratio

Readings: ORR-22 f · Solvency, Liquidity, and Other Regulation After the Global Financial Crisis (Carey, GARP Risk Institute 2019: Leverage Ratio; Liquidity Requirements; Example of LCR and NSFR); ORR-23 b · High-Level Summary of Basel III Reforms (BCBS, December 2017: Leverage Ratio Framework)

AUG 2026 RECALL AUG ONLY · NOT IN MAP

Specifics reported

Your August 2026 recall: NSFR numerical with factors given.
Shapes covered here:
S1 NSFR from a balance sheet with ASF and RSF factors given: the plain form (Q1)
S2 NSFR shortfall: new funding needed to reach 100%, with both the funding leg and the deployment leg (Q2)
S3 NSFR versus LCR with both parameter sets in one table: pick the right factors (Q3).

Revision notes

- **Why a leverage ratio** (ORR-22 f): minimum capital had been a percentage of RWA; during and after the crisis many felt RWA understated risk and left banks over-leveraged, and market participants focused on simple ratios of equity to unweighted assets.
- **Leverage ratio requirement** (ORR-22 f): a "simple" *supplement* to the risk-based requirements, not a replacement: Core Tier 1 capital / Leverage Exposure $\geq$ 3%. Leverage Exposure includes on-balance-sheet assets *and* fractions of off-balance-sheet assets (e.g. derivatives, potential future exposures), specified in detail for cross-country comparability because IFRS and GAAP differ.
- **2017 leverage ratio framework** (ORR-23 b): the leverage ratio complements risk-weighted requirements, guarding against unsustainable leverage and mitigating gaming and model risk across internal models and standardised approaches. G-SIB leverage buffer: met with Tier 1 capital, 50% of the G-SIB's risk-weighted higher-loss absorbency requirement (2% HLA gives a 1% buffer), divided into five ranges with distribution constraints. Refinements to the exposure measure (derivatives; off-balance-sheet items consistent with the SA credit conversion factors). National discretion to exempt central bank reserves temporarily in exceptional circumstances, with a commensurate recalibration of the minimum and disclosure. Earlier US rule: 2 and 3 percentage point leverage buffers (5% and 6% minimums).
- **Why liquidity requirements** (ORR-22 f): solvent institutions can fail because depositors and counterparties withdraw faster than assets can be sold. Northern Rock, dependent on securitisation and wholesale funding, lost retail deposits after news (Thursday 13 September) of Bank of England liquidity support; by the following Monday all UK deposits were guaranteed. Lesson: central bank funding can make funding stress *worse* and destabilise a system, so banks must survive funding stress with their *own* resources: a smaller proportion of illiquid assets and a larger proportion of stable liabilities.
- **LCR** (ORR-22 f): gives banks and authorities *a month* to manage a crisis by *selling liquid assets* while restoring confidence. LCR = HQLA / net cash outflows over 30 days $\geq$ 1. "Liquid" means a high likelihood of a quick sale with little price reduction.
- **HQLA** (ORR-22 f): central bank deposits and 0%-risk-weight sovereign securities with no haircut; corporate debt and equity 50% haircut; individual mortgage loans excluded. Level 2 assets (e.g. BBB corporate securities) may not exceed 40% of HQLA after haircut.
- **Net cash outflows** (ORR-22 f): stress run-off assumptions, e.g. 3% of insured retail deposits, 100% of most non-operational wholesale deposits, 30% of undrawn lines of credit to nonfinancial wholesale customers. "The definition of the LCR is simple but the implementation is complicated."
- **NSFR** (ORR-22 f): a *one-year* horizon, focused not on what can be sold but on what *funding would remain* after a stressful year. NSFR = available stable funding / required stable funding $\geq$ 1.
- **ASF factors** weight *liabilities and equity* and are "similar to haircuts": equity and 10-year bonds with none maturing within the year 100%, stable retail deposits 95%, 3-month wholesale CDs 0%. **RSF factors** weight *assets* and are *higher the more illiquid* the asset, since it cannot be sold easily when funding runs off: vault cash 0%, own-sovereign debt 5%, BBB corporate securities in the trading account 50%, business loans over one year with risk weights of 50% or more 85%. The NSFR categories differ from those of the LCR.

- **The book's example** (ORR-22 f): ASF $= 475 + 0 + 200 + 100 = 775$; RSF $= 0 + 5 + 50 + 595 = 650$; NSFR $= 1.19$. LCR: HQLA $= 100 + 100 + 50 = 250$ (corporate debt USD 50 is within the 40% Level 2 cap); outflows $= 5\% \times 500 + 100\% \times 200/3 = 25 + 67 = 92$; LCR $= 2.72$.
- **A major change in regulation** (ORR-22 f): before the crisis the presumption was that regulators would instantly know if a bank was solvent and could fund it until confidence returned. The new ratios constrain balance-sheet composition instead. The initial Basel III phase introduced the LCR and NSFR to mitigate "excessive liquidity risk and maturity transformation" (ORR-23 a).
- **Classic swaps:** LCR = 30 days, stock of liquid assets versus stressed outflows (short-term resilience); NSFR = one year, structural funding (ASF versus RSF). The leverage ratio is a *capital* measure on unweighted exposure, not a liquidity ratio, though it sits in the same LO.
- **See also** C5 (Basel III capital, buffers and the December 2017 reforms) and A26 (liquidity stress testing, TSECF and TSAA).

Formulas

Net stable funding ratio (one year).

$$\text{NSFR} = \frac{\text{ASF}}{\text{RSF}} = \frac{\sum_{\text{liabilities, equity}} \text{amount} \times \text{ASF factor}}{\sum_{\text{assets}} \text{amount} \times \text{RSF factor}} \geq 1$$

Factors in the book's example.

Funding	ASF	Asset	RSF
Stable retail deposits ($\leq$ 9 months)	95%	Vault cash	0%
3-month wholesale CDs	0%	Own-sovereign debt	5%
10-year senior bonds (none maturing in 1 yr)	100%	BBB corporate debt (trading)	50%
Common equity	100%	Business loans >1 yr, RW $\geq$50%	85%

Shortfall remedy. New funding x with ASF factor a, invested in an asset with RSF factor r: ASF$+ax =$ RSF$+rx \Rightarrow x = \dfrac{\text{RSF} - \text{ASF}}{a - r}$.

Liquidity coverage ratio (30 days). LCR $= \dfrac{\text{HQLA}}{\text{net cash outflows in 30 days}} \geq 1$; HQLA factors $(1 - \text{haircut})$: cash and 0%-RW sovereign 100%, corporate debt 50% (Level 2, $\leq 40\%$ of HQLA), loans 0%.

Leverage ratio. $\dfrac{\text{Core Tier 1 capital}}{\text{Leverage Exposure}} \geq 3\%$; G-SIB leverage buffer $= 50\% \times$ risk-weighted G-SIB HLA requirement (met with Tier 1).

Exam traps

- **Inversion:** NSFR is ASF/RSF. RSF/ASF gives a number below 1 for a compliant bank.
- **LCR factors in an NSFR question (sibling swap):** HQLA factors are not RSF factors (cash: HQLA 100%, RSF 0%); LCR run-off rates are not ASF haircuts; the LCR's "one-third of CDs maturing this month" has no role in the NSFR.
- **Dropped term:** omitting common equity (ASF 100%) or the 10-year bonds from ASF; using total assets or total funding as the denominator.
- **Two-leg remedies:** new funding adds ASF *and* its deployment adds RSF; dividing the gap by the ASF factor alone understates the amount.
- **Polarity:** RSF factors rise with illiquidity; ASF factors fall as funding becomes less stable (short wholesale CDs: 0%). Treating short CDs as stable inflates the NSFR.

Question A32.1

ORR-22 f · S1 · [Aug-2026]

A treasury analyst at a Canadian regional bank is computing the bank's Basel III net stable funding ratio for the ALCO pack. Balance sheet (CAD million) and the applicable factors are:

Funding	Amount	ASF	Asset	Amount	RSF
Stable retail deposits	620	95%	Vault cash	90	0%
3-month wholesale CDs	240	0%	Own-sovereign debt	150	5%
10-year senior bonds	180	100%	BBB corporate bonds	160	50%
Common equity	110	100%	Business loans (>1 yr)	750	85%

Which of the following is closest to the bank's NSFR?

A. 82%

B. 106%

C. 121%

D. 154%

Solution **Answer: C**

Step 1 – Available stable funding

$$\text{ASF} = 620(0.95) + 240(0) + 180(1.00) + 110(1.00) = 589 + 0 + 180 + 110 = \text{CAD } 879 \text{ million}$$

$\Rightarrow$ *Of CAD 1,150 million of funding, CAD 879 million is expected to remain through a stressful year.*

Step 2 – Required stable funding

$$\text{RSF} = 90(0) + 150(0.05) + 160(0.50) + 750(0.85) = 0 + 7.5 + 80 + 637.5 = \text{CAD } 725 \text{ million}$$

$\Rightarrow$ *The less liquid assets need CAD 725 million of stable funding behind them.*

Step 3 – NSFR

$$\text{NSFR} = \frac{879}{725} = 1.212 = 121\%$$

Interpretation

The bank complies with a cushion of CAD 154 million of ASF. The 3-month CDs contribute nothing, so replacing CDs with term bonds would raise the ratio.

Why the other options are wrong

A is incorrect. Inverted the ratio: $725/879 = 82\%$.
B is incorrect. Omitted common equity from ASF: $(879 - 110)/725 = 106\%$.
D is incorrect. Gave the 3-month wholesale CDs a 100% ASF factor, treating them as stable funding: $(879 + 240)/725 = 154\%$.

LO: ORR-22 f: Describe and calculate ratios intended to improve the management of liquidity risk, including the required leverage ratio, the liquidity coverage ratio, and the net stable funding ratio. · Reference: Carey, Solvency, Liquidity, and Other Regulation After the Global Financial Crisis, Liquidity Requirements and Example of LCR and NSFR

Question A32.2 ORR-22 f · S2 · [self-gen]

The head of balance sheet management at a Singapore bank finds that its NSFR is below the minimum. Balance sheet (SGD million) and factors are:

Funding	Amount	ASF	Asset	Amount	RSF
Stable retail deposits	400	95%	Vault cash	50	0%
3-month wholesale CDs	350	0%	Own-sovereign debt	100	5%
10-year senior bonds	150	100%	BBB corporate bonds	150	50%
Common equity	100	100%	Business loans (>1 yr)	700	85%

For its LCR, the bank applies a 5% run-off rate to stable retail deposits. The bank plans to raise additional stable retail deposits and invest all the proceeds in own-sovereign debt. Which of the following is closest to the minimum amount of new deposits needed to bring the NSFR to exactly 100%?

A. SGD 45.0 million

B. SGD 50.0 million

C. SGD 52.9 million

D. SGD 161.1 million

Solution **Answer: B**

Step 1 – Current ASF and RSF

$$\text{ASF} = 400(0.95) + 350(0) + 150(1.00) + 100(1.00) = 380 + 0 + 150 + 100 = 630$$
$$\text{RSF} = 50(0) + 100(0.05) + 150(0.50) + 700(0.85) = 0 + 5 + 75 + 595 = 675$$

Step 2 – Current NSFR and gap

$$\text{NSFR} = 630/675 = 93.3\%, \qquad \text{gap} = 675 - 630 = \text{SGD } 45 \text{ million}$$

Step 3 – Solve for new deposits x

Each SGD 1 of new deposits adds 0.95 of ASF and, once invested in sovereign debt, 0.05 of RSF:

$$630 + 0.95x = 675 + 0.05x \;\Rightarrow\; 0.90x = 45 \;\Rightarrow\; x = \text{SGD } 50.0 \text{ million}$$

Step 4 – Check

$$\frac{630 + 47.5}{675 + 2.5} = \frac{677.5}{677.5} = 100\%$$

Interpretation

Every remedy has two legs: where the funding comes from (ASF factor) and where it is deployed (RSF factor). The LCR run-off rate is a red herring: it belongs to the 30-day test.

Why the other options are wrong

A is incorrect. Reported the raw ASF gap, ignoring that deposits earn only 95% credit and that the new asset adds RSF: $675 - 630 = 45.0$.

C is incorrect. Added the 5% LCR run-off on the new deposits as a further stable funding requirement: $0.95x - 0.05x - 0.05x = 45$, $x = 45/0.85 = 52.9$.

D is incorrect. Omitted common equity from current ASF: gap $= 675 - 530 = 145$, $x = 145/0.90 = 161.1$.

LO: ORR-22 f: Describe and calculate ratios intended to improve the management of liquidity risk, including the required leverage ratio, the liquidity coverage ratio, and the net stable funding ratio. · Reference: Carey, Solvency, Liquidity, and Other Regulation After the Global Financial Crisis, Liquidity Requirements and Example of LCR and NSFR

Question A32.3 ORR-22 f · S3 · [self-gen]

An internal auditor at a US bank is checking the NSFR figure in a regulatory filing. The bank's balance sheet (USD million), with both NSFR and LCR parameters, is:

Item	Amount	ASF / RSF factor	LCR parameter
Stable retail deposits	800	ASF 95%	5% run-off
3-month wholesale CDs (1/3 mature each month)	300	ASF 0%	100% run-off on maturing third
10-year senior bonds	250	ASF 100%	0% run-off
Common equity	150	ASF 100%	0% run-off
Vault cash	150	RSF 0%	HQLA factor 100%
Own-sovereign debt	200	RSF 5%	HQLA factor 100%
BBB corporate bonds	250	RSF 50%	HQLA factor 50%
Business loans (>1 yr)	900	RSF 85%	HQLA factor 0%

Which of the following is closest to the bank's NSFR?

A. 129%

B. 150%

C. 162%

D. 339%

Solution **Answer: A**

Step 1 – ASF

$$\text{ASF} = 800(0.95) + 300(0) + 250(1.00) + 150(1.00) = 760 + 0 + 250 + 150 = \text{USD } 1{,}160 \text{ million}$$

Step 2 – RSF

$$\text{RSF} = 150(0) + 200(0.05) + 250(0.50) + 900(0.85) = 0 + 10 + 125 + 765 = \text{USD } 900 \text{ million}$$

Step 3 – NSFR

$$\text{NSFR} = \frac{1{,}160}{900} = 1.289 = 129\%$$

$\Rightarrow$ *The bank has 29% more stable funding than its one-year requirement.*

Interpretation

The LCR column answers a different, 30-day question about what can be sold; the NSFR asks what funding remains after a stressful year, and its categories differ from the LCR's.

Why the other options are wrong

B is incorrect. Treated the BBB bonds as liquid because they count as Level 2 HQLA and gave them a 0% RSF factor: RSF $= 10 + 765 = 775$, NSFR $= 1{,}160/775 = 150\%$.
C is incorrect. Gave the 3-month CDs a 100% ASF factor: $(1{,}160 + 300)/900 = 162\%$.
D is incorrect. Computed the LCR instead: HQLA $= 150 + 200 + 125 = 475$; outflows $= 0.05(800) + 300/3 = 140$; LCR $= 475/140 = 339\%$.

LO: ORR-22 f: Describe and calculate ratios intended to improve the management of liquidity risk, including the required leverage ratio, the liquidity coverage ratio, and the net stable funding ratio. · Reference: Carey, Solvency, Liquidity, and Other Regulation After the Global Financial Crisis, Liquidity Requirements and Example of LCR and NSFR

Answer key

A1.1	**B**	A1.2	**B**	A1.3	**D**	A1.4	**D**	A2.1	**C**	A2.2	**D**
A2.3	**B**	A2.4	**A**	A2.5	**A**	A2.6	**D**	A2.7	**C**	A2.8	**B**
A2.9	**C**	A3.1	**D**	A3.2	**A**	A3.3	**A**	A3.4	**D**	A4.1	**B**
A4.2	**D**	A4.3	**A**	A4.4	**D**	A5.1	**B**	A5.2	**C**	A5.3	**B**
A5.4	**B**	A5.5	**A**	A5.6	**A**	A6.1	**C**	A6.2	**A**	A6.3	**A**
A6.4	**A**	A6.5	**C**	A7.1	**D**	A7.2	**D**	A7.3	**C**	A7.4	**A**
A7.5	**C**	A8.1	**A**	A8.2	**A**	A8.3	**D**	A8.4	**A**	A8.5	**C**
A9.1	**C**	A9.2	**C**	A9.3	**D**	A9.4	**A**	A9.5	**B**	A9.6	**C**
A9.7	**A**	A10.1	**C**	A10.2	**D**	A10.3	**A**	A11.1	**B**	A11.2	**A**
A11.3	**C**	A11.4	**B**	A11.5	**C**	A12.1	**D**	A12.2	**B**	A12.3	**C**
A12.4	**D**	A13.1	**D**	A13.2	**B**	A13.3	**B**	A13.4	**B**	A14.1	**C**
A14.2	**D**	A14.3	**D**	A15.1	**A**	A15.2	**C**	A15.3	**A**	A15.4	**C**
A15.5	**C**	A15.6	**C**	A15.7	**B**	A16.1	**C**	A16.2	**B**	A16.3	**A**
A16.4	**C**	A17.1	**B**	A17.2	**C**	A18.1	**B**	A18.2	**A**	A18.3	**B**
A18.4	**B**	A18.5	**D**	A18.6	**C**	A19.1	**B**	A19.2	**B**	A19.3	**A**
A19.4	**C**	A19.5	**B**	A20.1	**C**	A20.2	**C**	A21.1	**D**	A21.2	**A**
A21.3	**A**	A21.4	**B**	A21.5	**B**	A21.6	**D**	A22.1	**D**	A22.2	**A**
A22.3	**B**	A23.1	**D**	A23.2	**C**	A24.1	**B**	A24.2	**A**	A24.3	**B**
A24.4	**C**	A25.1	**A**	A25.2	**C**	A25.3	**B**	A25.4	**D**	A25.5	**B**
A26.1	**A**	A26.2	**C**	A26.3	**B**	A26.4	**A**	A27.1	**C**	A27.2	**C**
A27.3	**A**	A28.1	**B**	A28.2	**B**	A28.3	**C**	A28.4	**D**	A29.1	**B**
A30.1	**D**	A30.2	**D**	A31.1	**D**	A31.2	**A**	A31.3	**C**	A31.4	**B**
A31.5	**B**	A32.1	**C**	A32.2	**B**	A32.3	**A**				

Formula index

Sections with a formula box, for fast lookup.

A1	Unexpected Loss, UL Contribution and Economic Capital	13
A2	RAROC, Adjusted RAROC and Hurdle Rate	19
A3	Volatility Smiles, Skews and Put-Call Parity	30
A4	Fundamental Review of the Trading Book (FRTB)	36
A5	Merton Model, Distance to Default and Risk-Neutral PD	43
A6	CVA, DVA, BCVA and CVA as a Spread	51
A7	Ho-Lee, Time-Dependent Drift and Volatility	58
A8	Correlation Swaps, Quanto Options and Concentration Ratio	65
A9	Securitisation: Tranches, Waterfall, Prepayment and Performance Ratios	72
A10	Altman Z-Score and Linear Discriminant Analysis	82
A11	Liquidity-Adjusted VaR and Cost of Liquidation	87
A12	Incremental, Marginal and Component VaR	94
A13	Lognormal VaR	101
A15	Hazard Rates, CDS Spreads and Conditional vs Unconditional PD	113
A16	Duration Gap and Interest-Sensitive Gap	121
A17	Extreme Value Theory: GEV vs POT	127
A18	Performance Measures and Attribution	132
A19	Non-Parametric VaR: Weighted HS, Bootstrap and FHS	140
A21	Collateral, Two-Way CSA, Margin Period of Risk and Netting	152
A24	Hedge Funds: Strategies, Survivorship and Other Biases, and Madoff	172
A25	Gauss+ and Vasicek	178
A26	Monitoring Liquidity: TSECF, TSAA and Liquidity Stress Testing	185
A27	Non-Deposit Liabilities, Pooled Cost of Funds and Liquidity Transfer Pricing	192
A28	Private Credit, Private Markets and Their Drivers	198
A30	Asset vs Borrowed Liquidity Management and the Investment Function	209
A31	Basel IRB Capital	214
A32	Net Stable Funding Ratio, LCR and Leverage Ratio	221

Index

Adjusted RAROC, 20
Advanced IRB, 214
Age-weighted historical simulation, 140
AI/ML models vs sophisticated AI, 107
Altman Z-score, 82
  cut-off zones, 83
Arbitrage-free models, 59
Asset liquidity management, 209
Asymptotic single risk factor model, 214
Available funds gap, 193
Available stable funding, 221

Backtesting
  FRTB desk-level, 37
Balanced liquidity management, 209
Barbell strategy, 210
Basel event-type taxonomy, 162
  Level 3 examples, 163
Basel III
  December 2017 reforms, 215
BCVA
  survival adjustment, 52
Benchmark
  naive, 133
Best hedge, 95
Beta distribution
  credit losses, 14
Binomial variance of default, 13
Blind pool risk, 200
Board risk committee, 162
Bootstrap
  historical simulation, 140
Borrowed liquidity management, 209
Business development companies (BDCs), 198

Capital multiplier, 14
Cascade approach, 37
Cash flow at risk, 186
CCP
  wrong-way risk, 148
CDS-bond basis, 114
Chief AML/CFT officer, 205
Close-out netting, 154
Co-investing, 200
Component VaR, 95
Concentration ratio, 65
Concentration risk
  AI service providers, 108
Constant prepayment rate, 73
Convertible arbitrage, 172
Corporate operational risk function (CORF), 161
Correlation risk, 66
Correlation swap, 65
Correlation-weighted historical simulation, 141
Correspondent banking, 206
Cost of liquidation, 87

Crashophobia, 31
Credit default swap, 113
Credit ratings
  through the cycle, 82
Credit support annex, 152
CSA
  one-way vs two-way, 153
Curvature risk charge, 37
Customer due diligence, 205
CVA, 51
  as a spread, 51
  collateralised, 154
  incremental, 52
  marginal, 52
  netting, 52
Cyber information sharing, 168
Cyber-resilience, 167

Data poisoning, 107
Deepfakes, 107
Default probability
  by rating, 83
  conditional, 113
  cumulative, 113
  unconditional, 113
Delinquency ratio, 73
Distance to default, 43
  short form, 43
Diversification
  capital allocation, 20
Dollar-weighted return, 132
Due diligence, 173
Duration gap
  change in net worth, 121
  leverage-adjusted, 121
DVA, 52

Economic capital
  credit risk, 13
Equifax, 168
Equity as a call option, 43
Equity tranche, 72
EVA (NIACC), 20
Excess spread, 72
Expected loss, 13
Expected shortfall
  average of tail VaRs, 101
  stressed, 97.5%, 36

FI index (IMF), 198
Filtered historical simulation, 141
Fisher-Tippett theorem, 127
Flighty liquidity under stress, 108
Foundation IRB, 214
Fraud control framework, 206
Fund of funds, 199
Fundamental law of active management, 133

Gauss+ model, 179
estimation, 179
Generalised extreme value distribution, 127
Generalised Pareto distribution, 128
Generative AI (GenAI), 107
Geometric return, 101
Ghost effects, 140

Half-life, 178
Hallucination, 107
Hazard rate, 113
from CDS spread, 114
from Merton PD, 44
Hedge fund strategies, 172
High-quality liquid assets, 221
Hill estimator, 128
Historical average cost approach, 192
Historical simulation
VaR, 101
Ho-Lee model, 58
Hurdle rate, 20

Immunisation, 121
Implied distribution, 30
Incremental VaR, 94
shortcut, 95
Information ratio, 132
Initial margin, 152
Input floors, 215
Interest-sensitive gap, 121
Internal audit
third line, 161
Internal credit ratings, 82
IRB
99.9th percentile, 214

Jensen's alpha, 132
Jump approach, 149

Kamakura, 44

Leverage multiplier, 43
Leverage ratio, 221
Linear discriminant analysis, 82
Liquid asset buffer, 186
Liquidation horizon, 87
Liquidity coverage ratio, 221
Liquidity generation capacity, 186
Liquidity horizon, 36
Liquidity transfer pricing, 193
Liquidity-adjusted VaR, 87
Lognormal VaR, 101
versus normal VaR, 101

Madoff
red flags, 174
split-strike conversion, 173
Margin period of risk, 153
Marginal VaR, 94
Market correlations (AI), 108
Master trust, 73
Master-feeder structure, 173
Matched-maturity marginal cost of funds, 193
Maturity adjustment, 214
Mean reversion, 178
Merger arbitrage, 172
Merton model
simultaneous equations, 43
Minimum transfer amount, 152
Minimum variance delta, 31
Model 1 (no drift), 58
Model 2 (constant drift), 58
Model 3 (time-dependent volatility), 59
Money laundering
three stages, 207
Money market instruments, 210
Monthly payment rate, 73
Moody's KMV
default point, 44
Morningstar risk-adjusted rating, 133
Multivariate EVT, 128

Negotiable CD, 193
Net interest margin, 121
Net liquidity position, 209
Net stable funding ratio, 221
NIST Cybersecurity Framework, 168
Non-parametric density estimation, 140
Northern Rock, 221

Operational resilience, 168
Output floor, 215
Overcollateralization account, 72

P&L attribution, 37
Peaks over threshold, 128
threshold choice, 128
Performance attribution, 133
Pooled-funds approach, 192
Private credit
financial-stability risks, 198
growth drivers, 198
Prompt injection, 107
PSA benchmark, 73
Public market equivalent (PME), 199
Put-call parity
implied volatility, 30

QQ plot, 102
Quantile standard error, 102
Quanto CDS, 149
Quanto option, 65

RAROC, 19
horizon, 20
Rating migration matrix, 113
Real-world default probability, 43, 114
Recovery rate, 113
RegTech, 108
Required stable funding, 221
Right-way risk, 147
Risk appetite

operational risk, 162
Risk capital, 19
Risk-neutral default probability, 43

Shadow rate, 58
Sharpe ratio, 132
from marginal VaR, 95
Sources and uses of funds approach, 185
Spread risk factor, 88
Stale valuations, 198
Standardized approach (FRTB), 36
Strategic risk capital, 19
Style analysis, 133
Supply-side drivers (AI), 108
SupTech, 108
Survivorship bias, 174
Systemic risk, 66
hedge funds, 173

Tail index, 127
Tax-equivalent yield, 210
Term structure of available assets, 185
Term structure of expected cash flows, 185
Three lines of defence
AML/CFT, 205
cyber risk, 167
first line, 161
Threshold, 152
TIBER-EU, 167
Time to escape, 88
Trade compression, 154
Trading book
boundary with banking book, 37
Transactions liquidity risk, 88
Treynor measure, 132
TVPI and DPI, 199
Type I error
Altman, 82

Undiversified VaR, 94
Unexpected loss, 13
Unexpected loss contribution, 13
identical loans, 13

Valuation agent, 153
Value at risk
and correlation, 66
Vasicek model
expected rate, 178
recombining tree, 178
terminal standard deviation, 178
Volatility regime, 101
Volatility skew, 30
Volatility smile, 30
Volatility term structure, 30
Volatility-weighted historical simulation, 141

Waterfall, 73
Weighted average life, 73
Weighted expected shortfall, 37
Worst case default rate, 214
Wrong-way collateral, 148
Wrong-way risk, 147
collateral, 148
credit derivatives, 148
general vs specific, 147
models, 148